

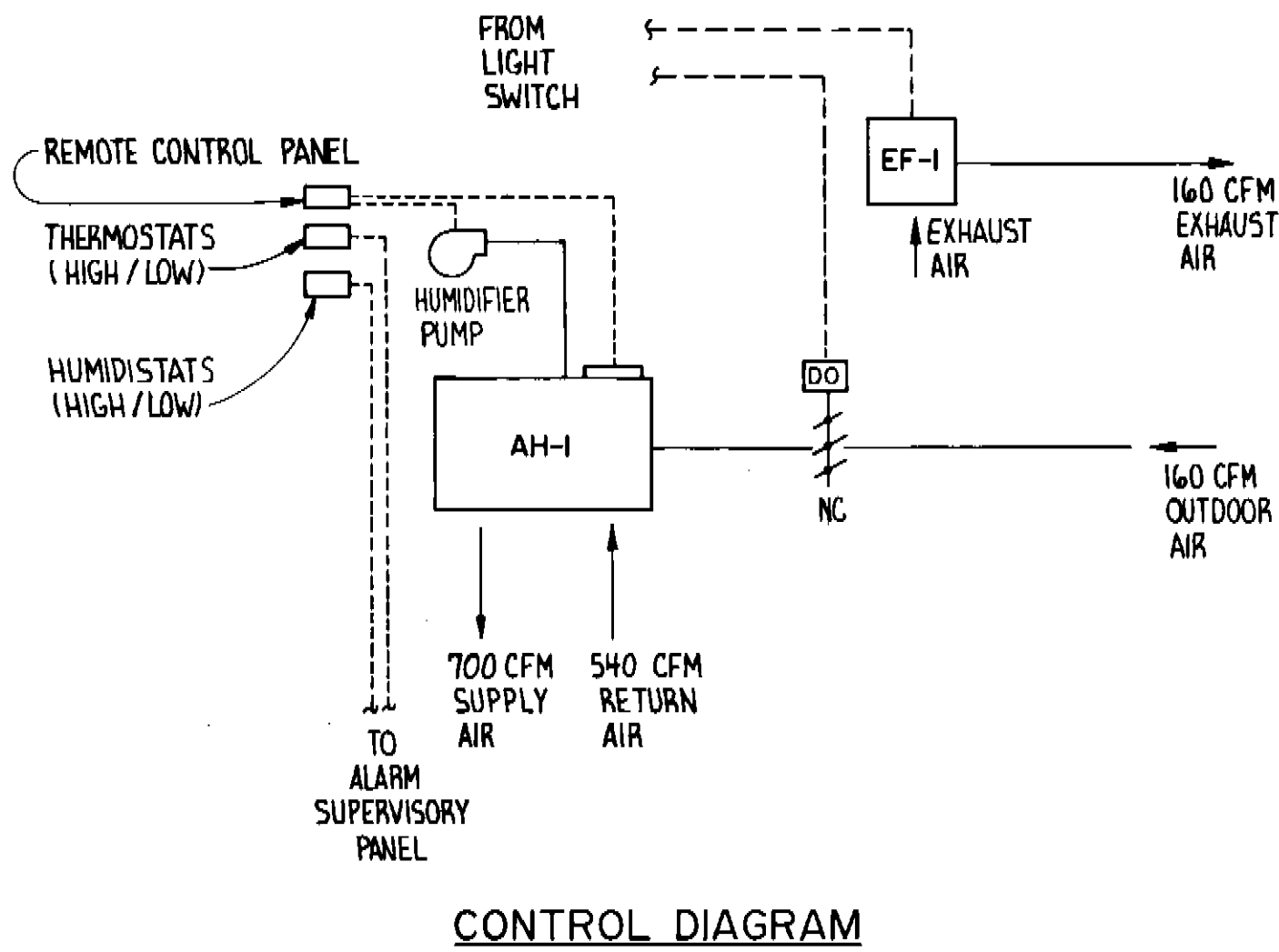
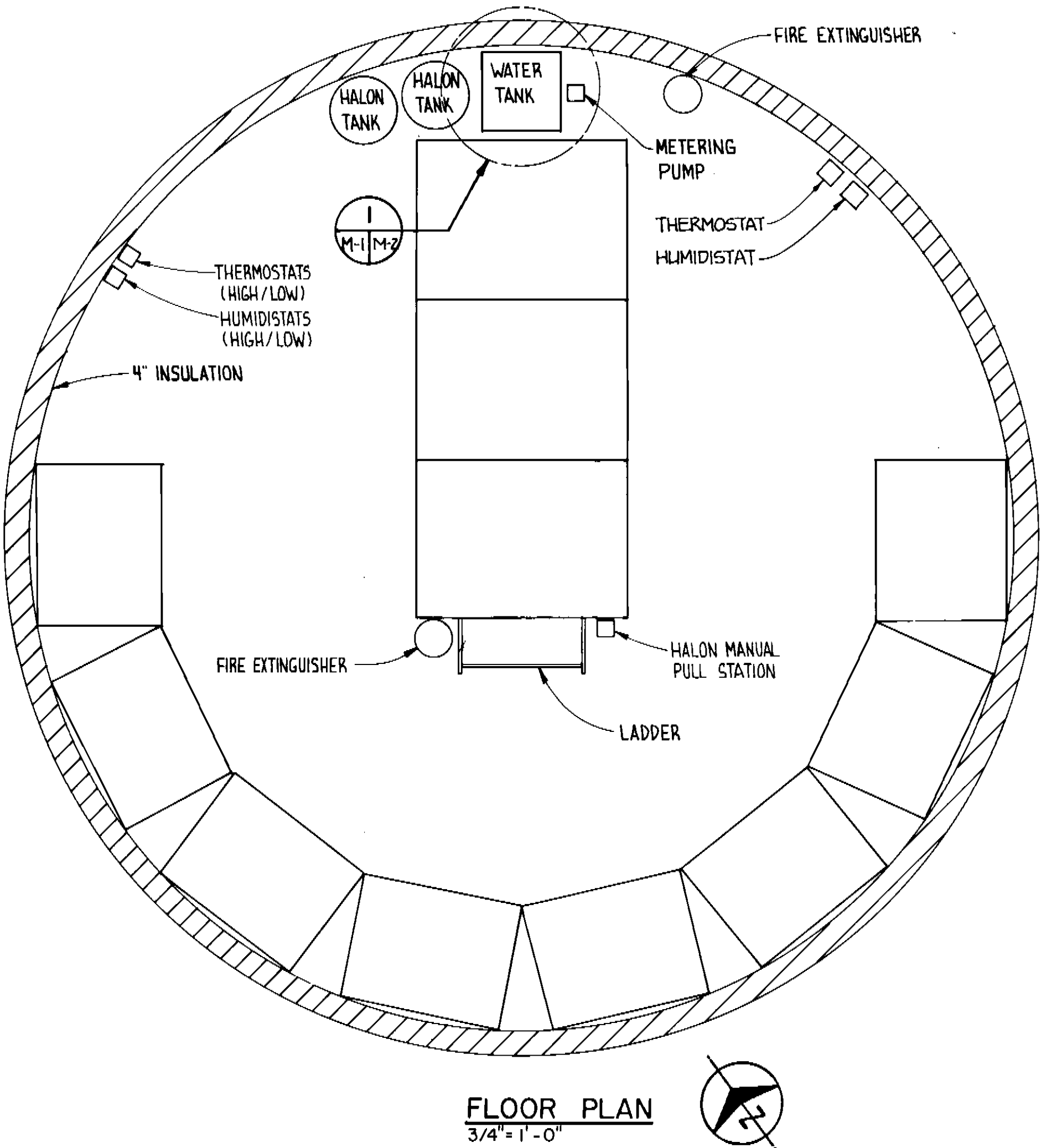
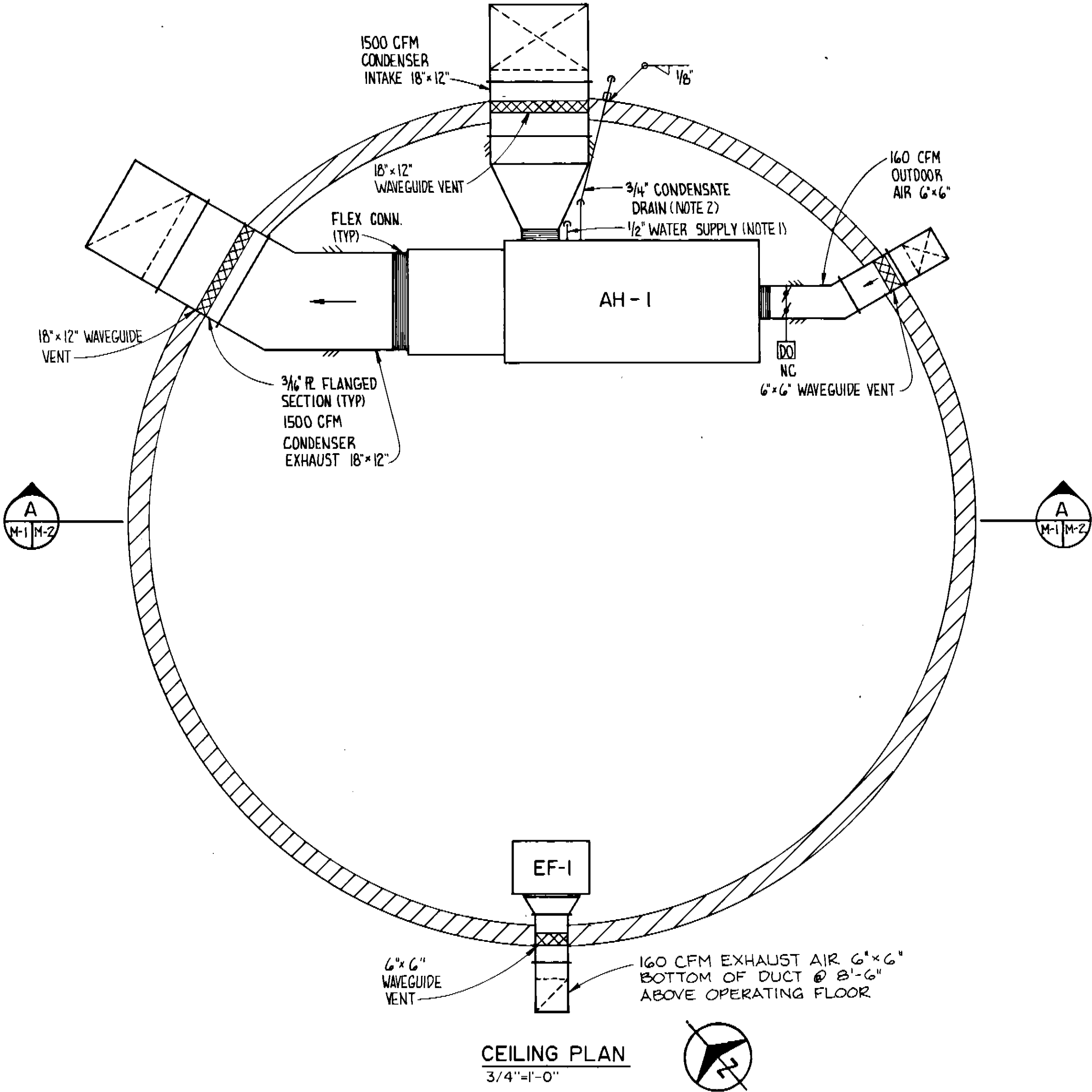
MAGNETIC SILENCING FACILITY TYPE III DEGAUSSING RANGE



NORFOLK NAVAL STATION
NORFOLK, VIRGINIA

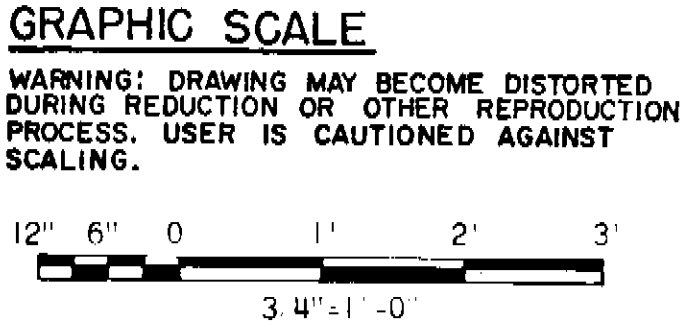
REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED

- NOTES:
1. WATER SUPPLY PIPE SHALL BE COPPER (TYPE AS SPECIFIED IN SECTION 15895)
 2. CONDENSATE DRAIN SHALL BE GALVANIZED STEEL (TYPE AS SPECIFIED IN SECTION 15895)
 3. REFER TO SPECIFICATION SECTION 07232 FOR INSULATION REQUIREMENTS.
 4. PROVIDE BIRD SCREENS ON ALL OPENINGS.
 5. HANGERS AND SUPPORTS FOR EQUIPMENT AND DUCTWORK SHALL BE PROVIDED BY THE CONTRACTOR IN ACCORDANCE WITH THE MANUFACTURERS RECOMMENDATIONS FOR THE EQUIPMENT PURCHASED.

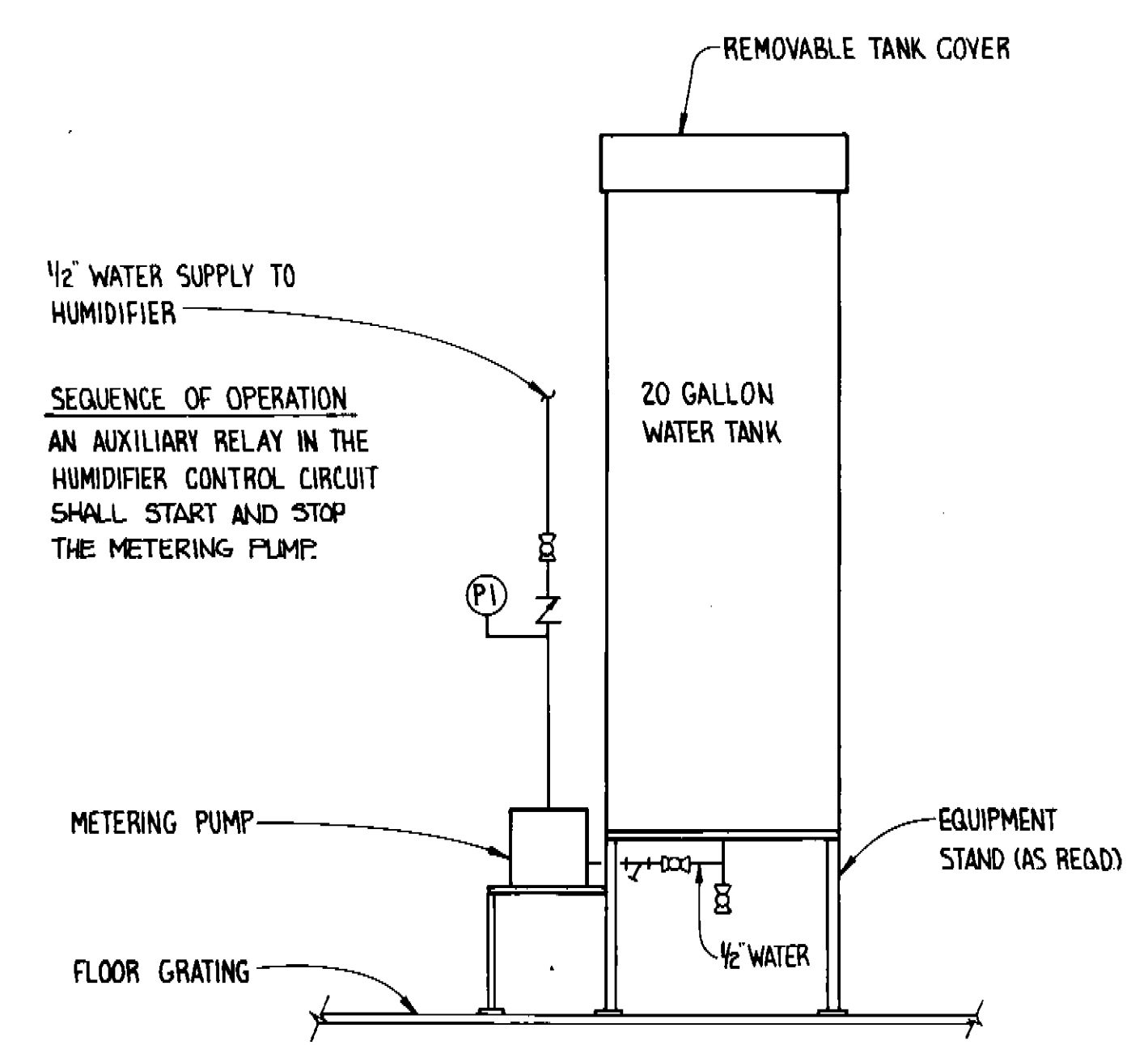
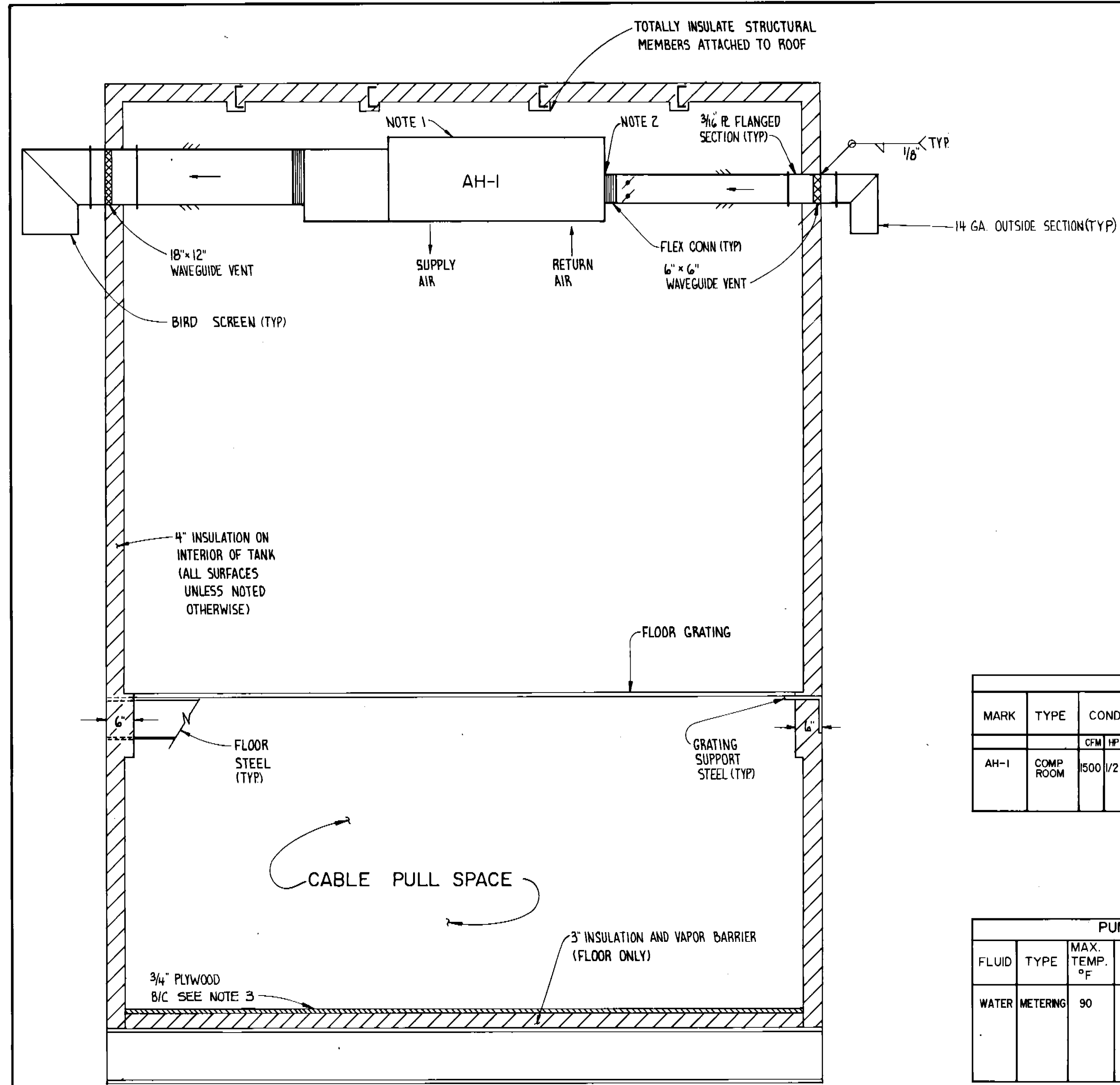


SEQUENCE OF OPERATION

OPERATION OF AH-1 IS TO BE CONTROLLED BY A REMOTE CONTROL PANEL. THE PANEL WILL MAINTAIN THE REQUIRED TEMPERATURE AND HUMIDITY SET POINTS WITHIN THE SPECIFIED LIMITS. THE OUTDOOR AIR DAMPER AND EF-1 WILL BE CONTROLLED BY THE DCS LIGHT SWITCH.



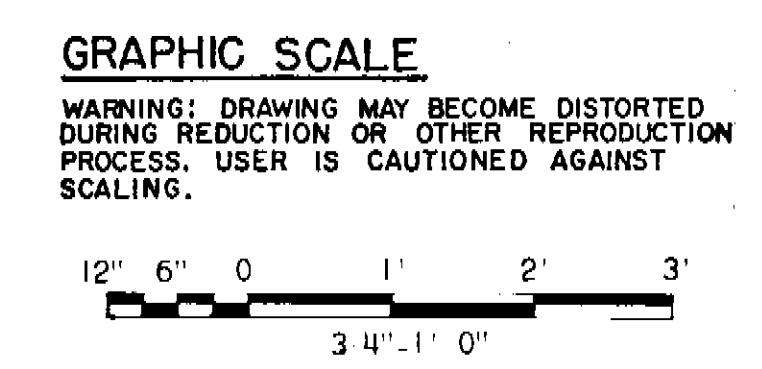
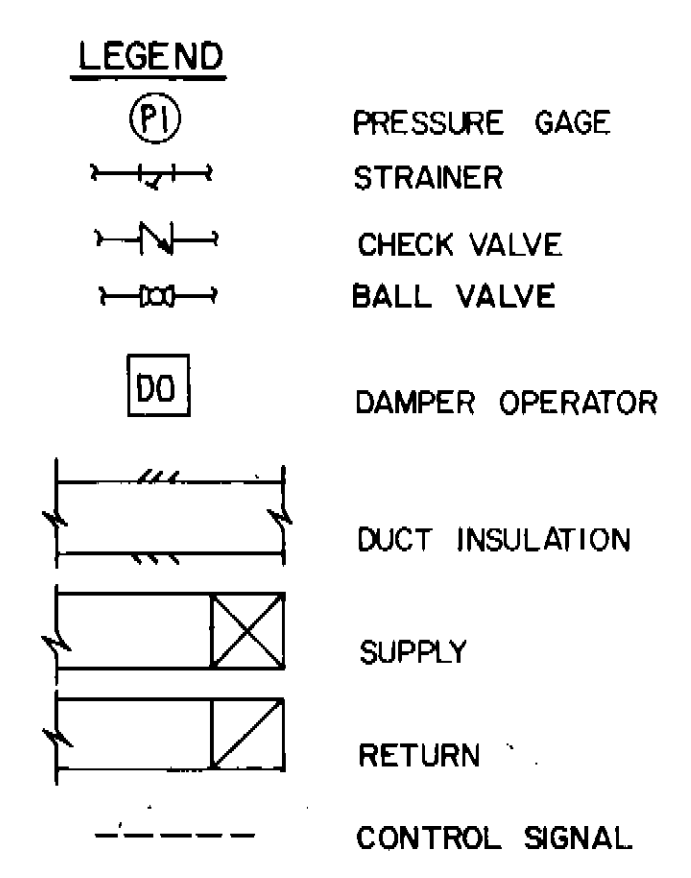
SATISFACTORY TO ROICC DATE		APPROVED HEAD FPO-1 DATE		SIZE D		CODE IDENT NO 80091		NAVJAC DRAWING NO. 3020692		M-1	
SATISFACTORY TO COMNAVSEASYS COM DATE		APPROVED DATE		EFD FOR COMMANDER, NAVJAC		SCALE AS SHOWN		CONSTR. CONTR. NO. N62477-89-C-0053		SHEET 19 OF 40	



AIR HANDLING UNIT SCHEDULE											
MARK	TYPE	COND. FAN		EVAP. FAN		EXT. STATIC IN. W.G.	CLG. CAP BTUH	EDB / EWB °F	VOLTS PHASE HZ	HTG. CAP	HUMIDIFIER CAP
		CFM	HP	CFM	HP						
AH-1	COMP ROOM	1500	1/2	700	1/4	0.5	16,000	75 / 62.5	208 1 60	4.4	2.0 .5

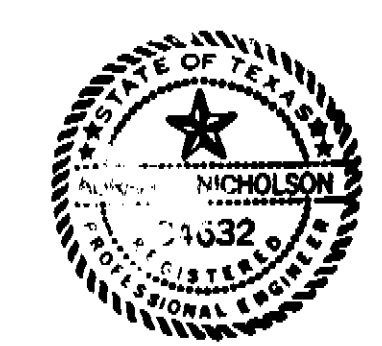
PUMP SCHEDULE					
FLUID	TYPE	MAX. TEMP. °F	FLOW GPH	TOTAL HEAD PSI	VOLTS PHASE HZ
WATER	METERING	90	1.5	15	115 1 60
FURNISH SKID MOUNTED WATER TANK. (SEE SPECIFICATIONS)					

FAN SCHEDULE				
MARK	TYPE	FAN CFM	STATIC PRESS. IN. W.G.	FAN RPM
EF-1	CEILING EXHAUSTER	160	0.250	1150
				115 1 60



REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED
A	REVISED TO REFLECT AS BUILT CONDITION		

- NOTES:
- HANG AH-1 PER MANUFACTURERS RECOMMENDATIONS. USE VIBRATION ISOLATING SPRING HANGERS.
 - CONTRACTOR SHALL MAKE REQUIRED MODIFICATIONS TO AH-1 HOUSING TO CONNECT OUTDOOR AIR DAMPER SECTION.
 - PLYWOOD SHALL CONFORM TO U.S. DEPARTMENT OF COMMERCE PRODUCT STANDARD PS-1.
 - REFER TO SPECIFICATION SECTION 07232 FOR INSULATION REQUIREMENTS.

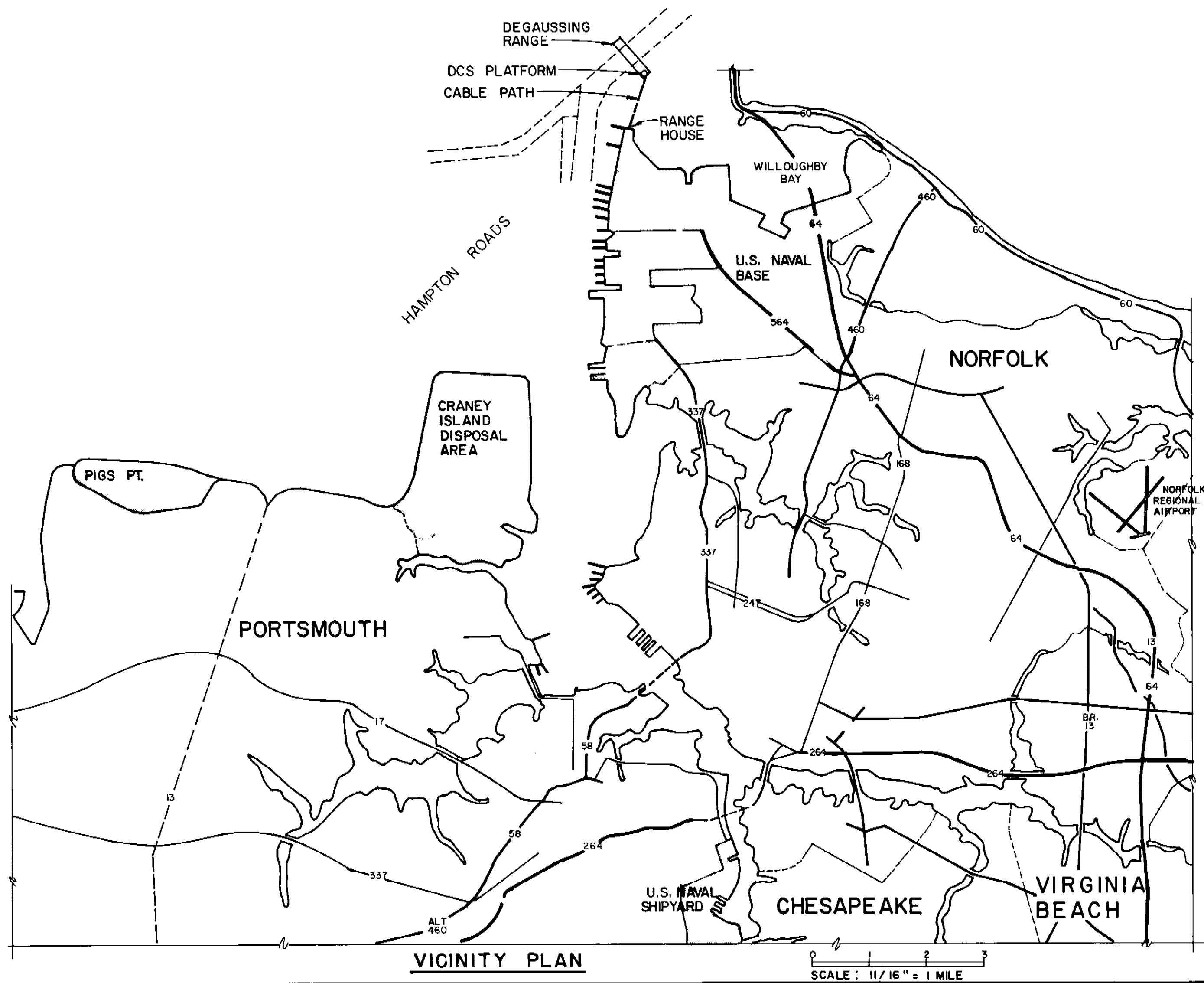
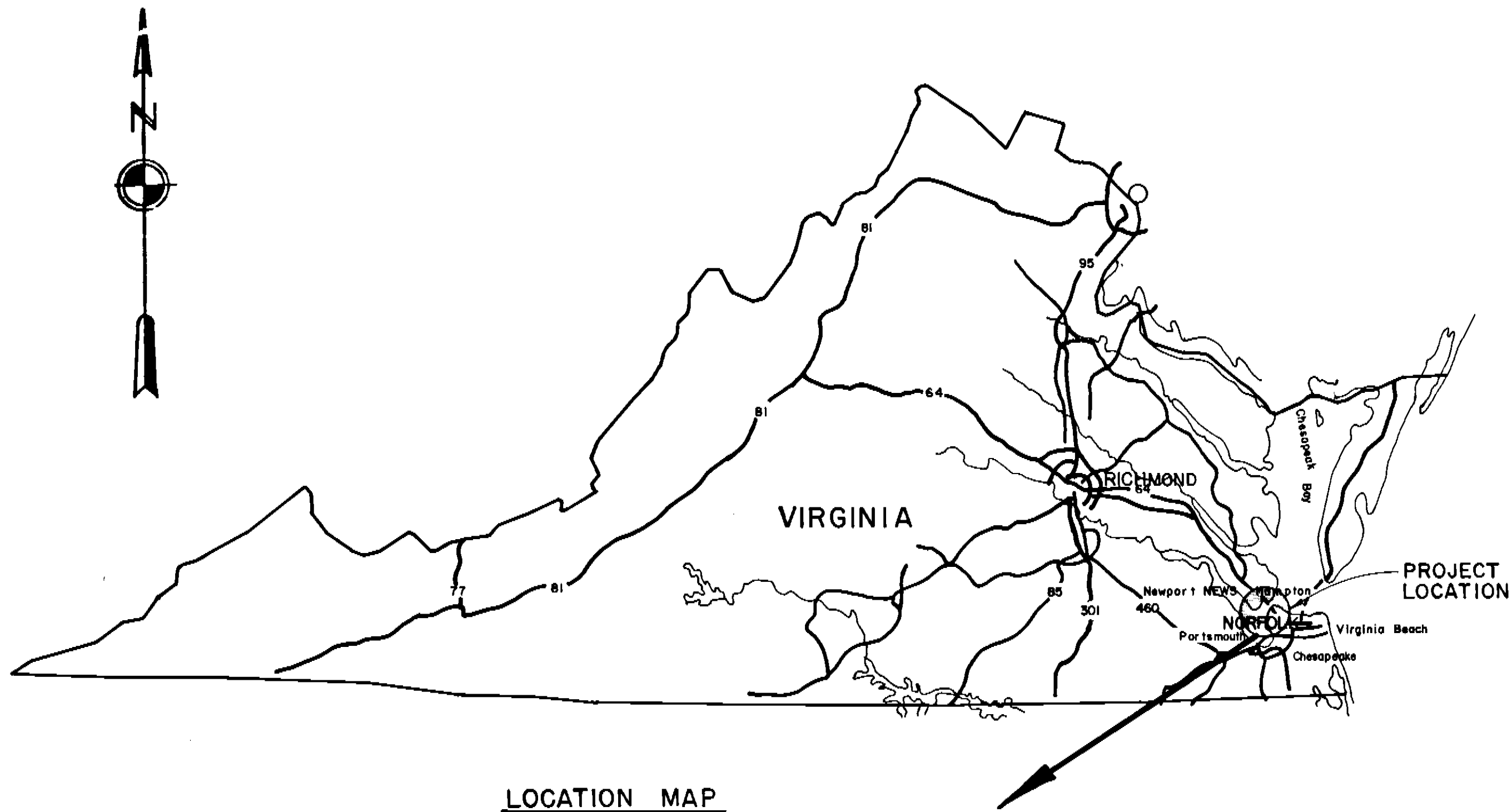


WARNING: DRAWING MAY BECOME DISTORTED DURING REDUCTION OR OTHER REPRODUCTION PROCESS. USER IS CAUTIONED AGAINST SCALING.

SATISFACTORY TO	DATE
ROICC	5-15-89
SATISFACTORY TO	DATE
COMNAVSEASYS	

Black & Veatch Engineers-Architects		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DES. BY	DR. MA	CHK. LAC	
SUBMITTED BY	DATE	TITLE	
OCEAN ENGR. B. CONST. PROJ. OFF. PRO-1	5/15/89		
EIC	RDV		
APPROVED HEAD	DATE	SIZE	CODE IDENT. NO.
5-15-89		D	80091
APPROVED	DATE	SCALE	AS SHOWN
EFD FOR COMMANDER, NAVFAC		CONSTR. CONTR. NO.	N62477-89-C-0053
		SHEET	20 OF 40

MAGNETIC SILENCING FACILITY
NORFOLK NAVAL STATION
MECHANICAL SECTIONS, DETAILS AND SCHEDULES
3020693
M-2



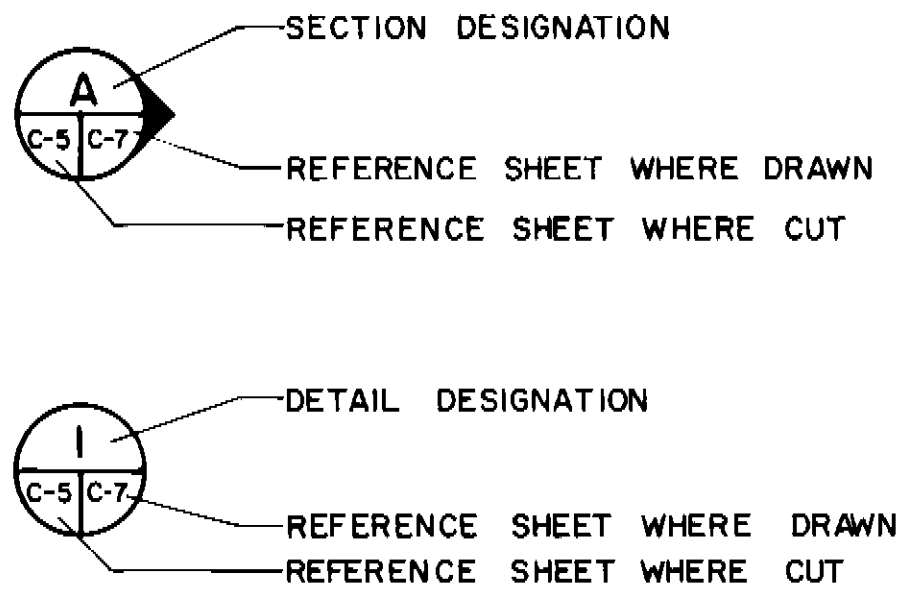
DRAWING INDEX		
DRAWING NO.	NAVFAC NO.	DRAWING TITLE
	3020673	TITLE SHEET
C - 1	3020674	VICINITY PLAN, DRAWING INDEX, AND GENERAL NOTES
C - 2	3020675	SITE PLAN
C - 3	3020676	DREDGING PLAN PROFILE AND SECTION FOR MINESWEEPER SENSOR RANGE
C - 4	3020677	DREDGING PLAN PROFILE AND SECTION FOR MEDIUM / DEEP SENSOR RANGE
C - 5	3020678	PLAN & PROFILE MEDIUM / DEEP SENSOR RANGE
C - 6	3020679	PLAN & PROFILE MINESWEEPER SENSOR RANGE
C - 7	3020680	CROSS SECTIONS AT MEDIUM / DEEP AND MINESWEEPER SENSOR RANGES
C - 8	3020681	RANGEHOUSE SITE PLAN
C - 9	3020682	MISCELLANEOUS DETAILS
C - 10	3020683	ELECTRICAL TRENCH PROFILE AND MISCELLANEOUS DETAILS
C - 11	3020684	SENSOR TUBE INSTALLATION DETAILS
C - 12	3020685	SENSOR TUBES AND DETAILS
S - 1	3020686	DATA CONCENTRATOR SHED
S - 2	3020687	STRUCTURAL ROOF PLAN & ELEVATION
S - 3	3020688	DATA CONCENTRATOR SHED
S - 4	3020689	STRUCTURAL FLOOR FRAMING PLANS
S - 5	3020690	DATA CONCENTRATOR SHED
S - 6	3020691	SECTIONS & DETAILS
S - 6	3020691	SUPPORT STRUCTURE
S - 6	3020691	DATA CONCENTRATOR SHED
S - 6	3020691	SUPPORT STRUCTURE SHED DETAILS
M - 1	3020692	MECHANICAL CEILING PLAN AND FLOOR PLAN
M - 2	3020693	MECHANICAL SECTIONS, DETAILS, AND SCHEDULES
E - 1	3020694	EQUIPMENT FLOOR LEVEL PLANS
E - 2	3020695	CABLE SPREADING AREA AND ROOF TOP PLANS
E - 3	3020696	TERMINAL CABINET LAYOUT AND DETAILS
E - 4	3020697	INTERFACE WIRING DIAGRAM TERMINAL
E - 5	3020698	CABINET NO. 1
E - 5	3020698	INTERFACE WIRING DIAGRAM TERMINAL
E - 6	3020699	CABINET NO. 2
E - 6	3020699	INTERFACE WIRING DIAGRAM TERMINAL
E - 7	3020700	CABINET NO. 3
E - 7	3020700	INTERFACE WIRING DIAGRAM TERMINAL
E - 8	3020701	CABINET NO. 4
E - 8	3020701	INTERFACE WIRING DIAGRAM TERMINAL
E - 9	3020702	CABINET NO. 5
E - 9	3020702	INTERFACE WIRING DIAGRAM TERMINAL
E - 10	3020703	CABINET NO. 6
E - 10	3020703	GROUNDING AND LIGHTNING PROTECTION
E - 11	3020704	MISCELLANEOUS DETAILS SHEET 1
E - 12	3020705	MISCELLANEOUS DETAILS SHEET 2
E - 13	3020706	ALARM SUPERVISORY CIRCUIT AND REMOTE
E - 14	3020707	DATA POWER CONTROL
E - 14	3020707	CONDUIT & CASING ASSEMBLY DETAILS
E - 15	3020708	COMPOSITE CABLE TERMINATION DETAILS
E - 16	3020709	INTERFACE CABLE ASSEMBLY AND DATA
E - 17	3020710	HIGHWAY CABLE TERMINATIONS
E - 17	3020710	TUBE & CABLE TAGS AND SENSOR
E - 18	3020711	INSTALLATION DETAILS
E - 18	3020711	SENSOR CABLE LENGTHS
E - 19	3020712	RANGE HOUSE ELECTRICAL POWER AND CONTROL
E - 19	3020712	PLAN AND DETAILS
E - 20	3020713	POWER AND CONTROL ONE-LINE DIAGRAMS

REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED

GENERAL NOTES

- GRIDS AND COORDINATES ARE REFERENCED TO THE VIRGINIA STATE GRID (SOUTH ZONE)
- ALL ELEVATIONS AND DEPTHS ARE IN FEET REFERENCED TO NATIONAL OCEAN SERVICE (NOS), MEAN LOW WATER (MLW) AS DETERMINED AT SEWELLS POINT.
- MEAN LOW WATER AS DETERMINED BY NOS IS 1.28 FT. BELOW NATIONAL GEODETIC VERTICAL DATUM (NGVD), 1972 ADJUSTMENT.
- MEAN RANGE OF TIDE: 2.5 FT. BASED ON NOS OBSERVATIONS

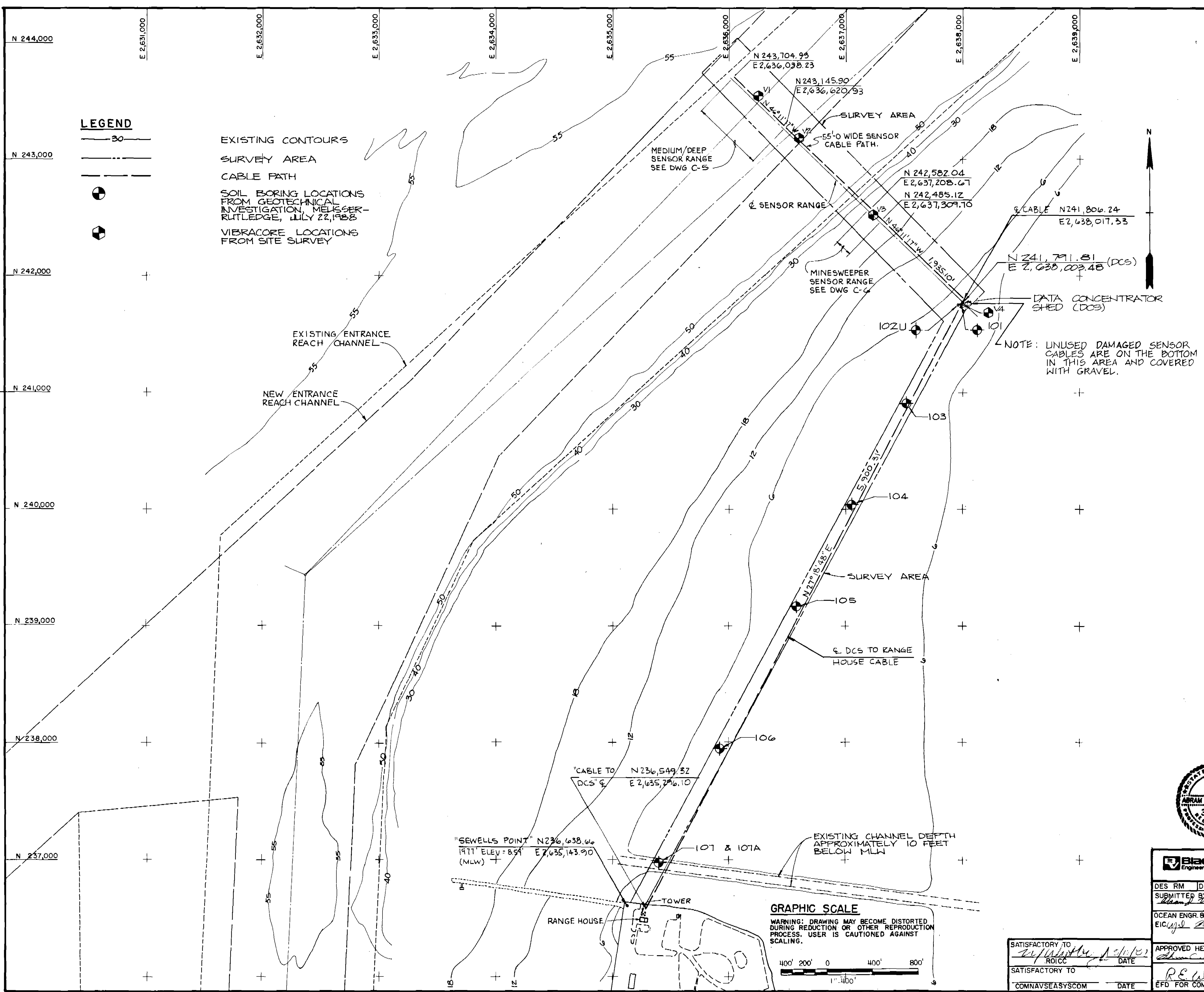
SYMBOLS:



ABBREVIATIONS

PVC	POLYVINYL CHLORIDE
O.D.	OUTSIDE DIAMETER
I.D.	INSIDE DIAMETER
FRP	FIBERGLASS REINFORCED PLASTIC
S.S.	STAINLESS STEEL
MLW	MEAN LOW WATER
GALV	GALVANIZED
PL	PLATE
O.C.	ON CENTER
PLCS	PLACES
CNTRS	CENTERS
REINF	REINFORCING / REINFORCEMENT
INV. EL	INVERT ELEVATION
FDN	FOUNDATION
EXIST.	EXISTING
CL	CENTER LINE
DCS	DATA CONCENTRATOR SHED
C.B.	CATCH BASIN
MIN.	MINIMUM
Ø	DIAMETER
MED.	MEDIUM
CLR	CLEARANCE
SLV	SHORT LEG VERTICAL
U.N.O.	UNLESS NOTED OTHERWISE
RFI	RADIO FREQUENCY INTERFERENCE
N.I.C.	NOT IN CONTRACT

	DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.
	MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION
DES. BY: ABRAM J. NICHOLSON SUBMITTED BY: [Signature] OCEAN ENGR. & CONST. PROJ. OFF. FPO-1 EIC: [Signature] RVD: [Signature] 5/1/89	VICINITY PLAN, DRAWING INDEX AND GENERAL NOTES
SATISFACTORY TO ROICC [Signature] DATE 5/18/89 SATISFACTORY TO COMNAVSE&SYSCOM [Signature] DATE 5/18/89	SIZE: D CODE IDENT. NO.: 80091 NAVFAC DRAWING NO.: 3020674 C-1 CONSTR. CONTR. NO. N62477-89-C-0053 SPEC. 21-89-0053 SHEET 1 OF 40



LEGEND

- EXISTING CONTOURS
- SURVEY AREA
- CABLE PATH
- SOIL BORING LOCATIONS FROM GEOTECHNICAL INVESTIGATION, MELISSER-RUTLEDGE, JULY 22, 1988
- VIBRACORE LOCATIONS FROM SITE SURVEY

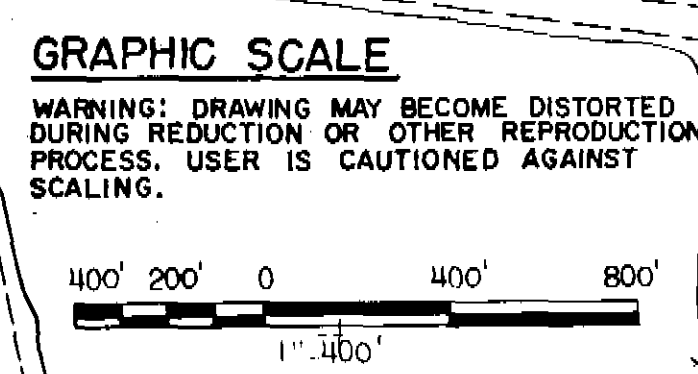
REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED
A	REVISED TO REFLECT AS BUILT CONDITION		

NOTES:

- ALL CONTOURS SHOWN ARE BASED ON DATA OBTAINED FROM NAVIGATION CHART NO. 12245 DATED JUNE 30, 1984 (HAMPTON ROADS) AND CORPS OF ENGINEERS SURVEY DRAWINGS FOR NORFOLK HARBOR AND CHANNEL DEEPENING PROJECT DATED FEBRUARY 24, 1986. FOR DEPTHS WITHIN THE SURVEY AREA REFER TO OCEAN SURVEY INC. DWG. 88E5036-A&B DATED JUNE 24, 1988.
- ALL COORDINATES SHOWN ARE ϕ OF CABLE PATH UNLESS OTHERWISE NOTED.
- ALL DEPTHS ARE REFERENCED TO MEAN LOW WATER (MLW)

BORING COORDINATES

BORING NO.	COORDINATES	
	EAST	NORTH
101	2,637,998	241,757
102U	2,637,971	241,757
103	2,637,518	240,904
104	2,637,050	240,039
105	2,636,582	239,168
106	LOCATED 1100' NORTH ALONG ALIGNMENT FROM BORING NO. 107A.	
107 & 107A	LOCATED AT NORTH EDGE OF 10 FT. DEEP CHANNEL ALONG ALIGNMENT.	
V1	2,636,250	243,539
V2	2,636,600	243,181
V3	2,637,240	242,518
V4	2,637,990	241,757



Black & Veatch Engineers-Architects		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON D.C.	
DES. RM. DR. FS. LT. CHK. JW. SUBMITTED BY: <i>[Signature]</i> DATE: 8/10/89 TITLE: PROGRAM MGR. OCEAN ENGR. & CONST. PROJ. OFF. FPO-1 EIC: <i>[Signature]</i> RVO: <i>[Signature]</i>		MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION SITE PLAN	
SATISFACTORY TO: <i>[Signature]</i> DATE: 8/10/89 ROICC	APPROVED HEAD FPO-1 DATE: 8/10/89 <i>[Signature]</i>	SIZE: D	CODE IDENT. NO.: 80091
SATISFACTORY TO: <i>[Signature]</i> DATE: 8/10/89 COMNAVSEASYS COM	FOR COMMANDER, NAVFAC	SCALE: AS NOTED	NAVFAC DRAWING NO.: 3020675 CONSTR. CONTR. NO.: N62477-89-C-0053 SHEET 2 OF 40

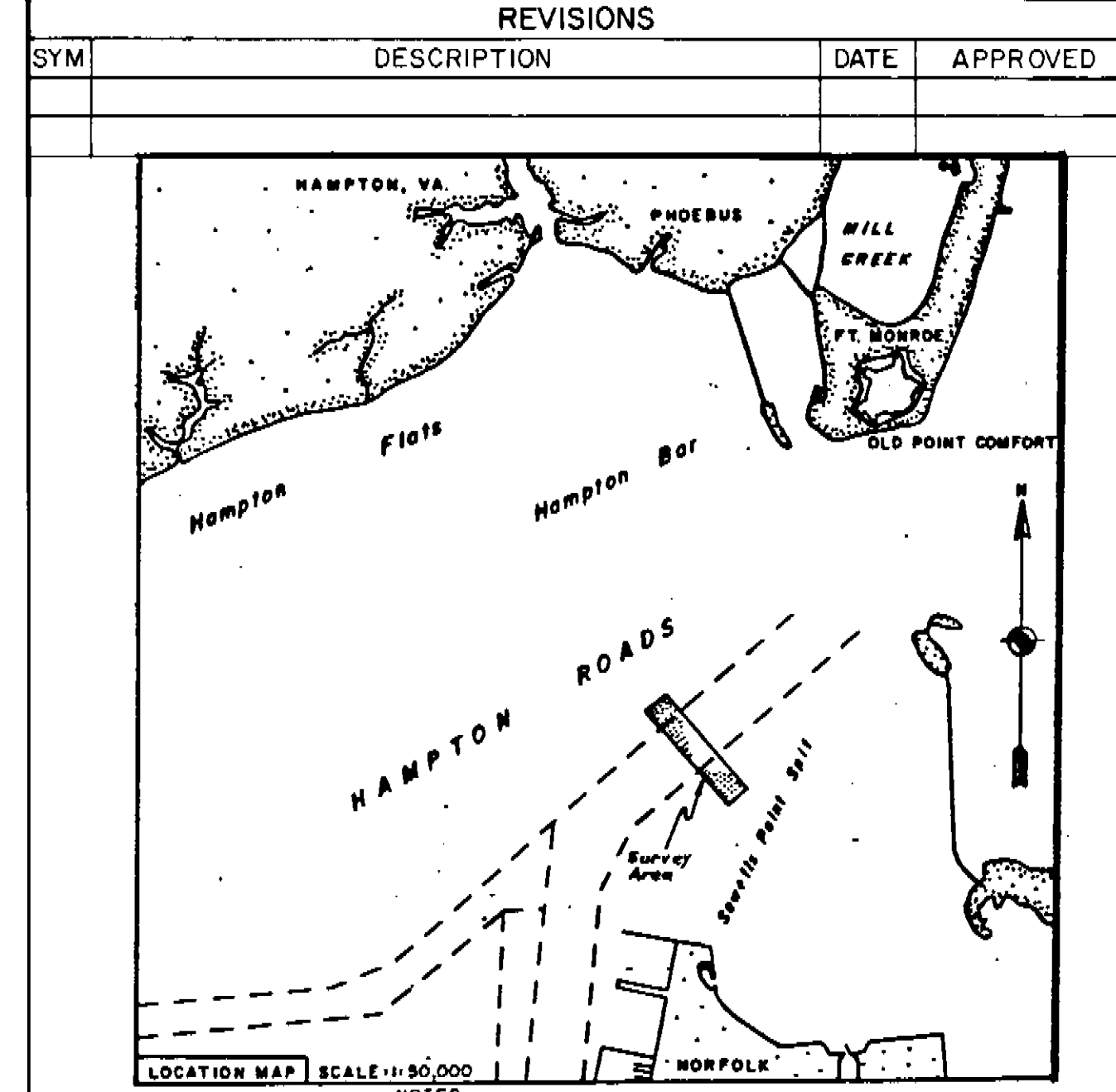
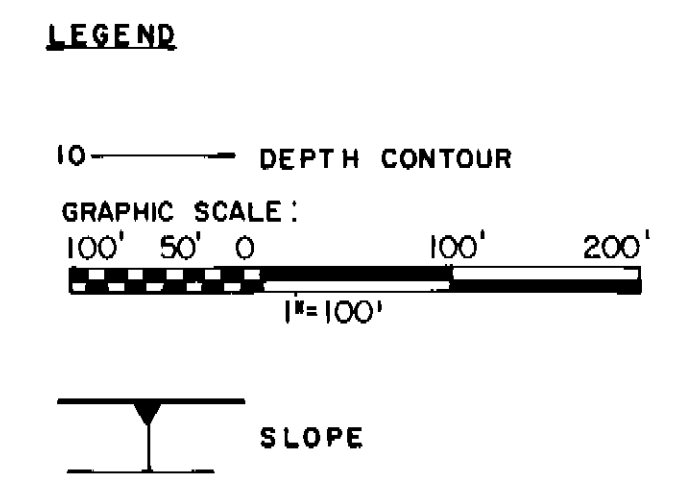
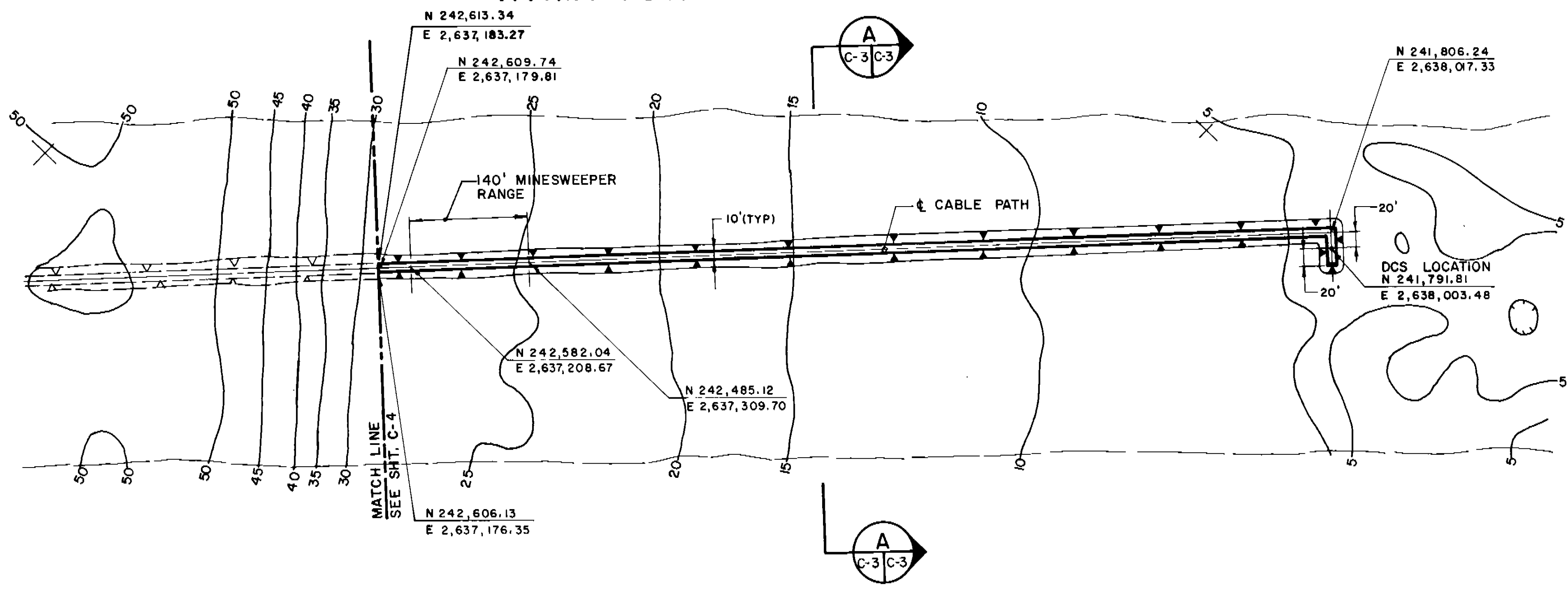
E 2,637,000

N 243,000

E 2,638,000

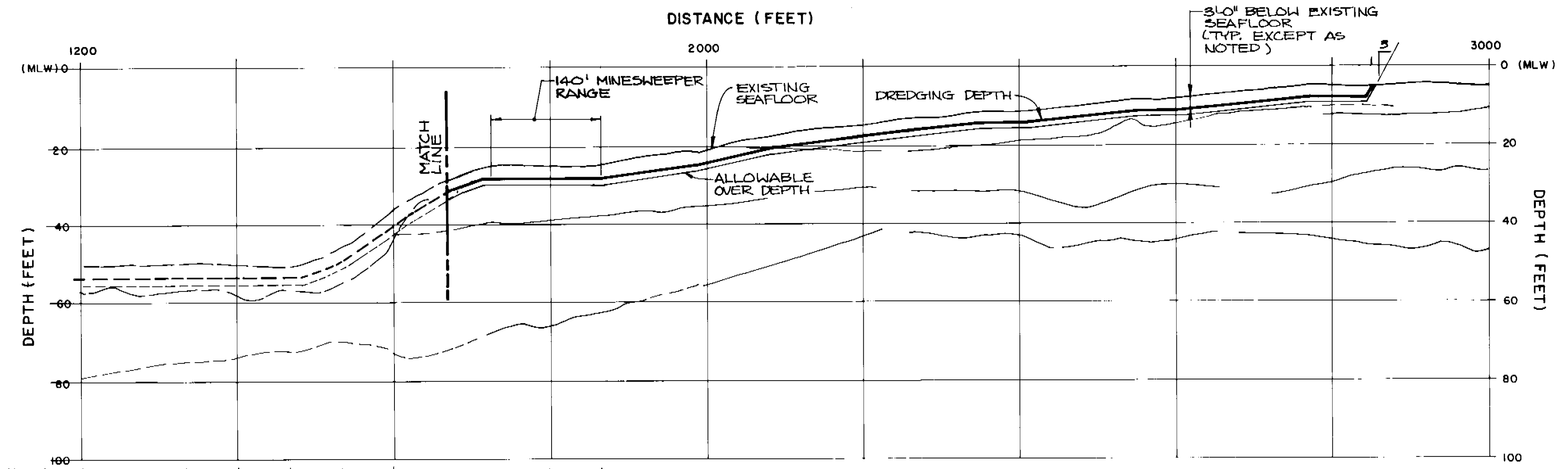
N 242,000

HAMPTON ROADS

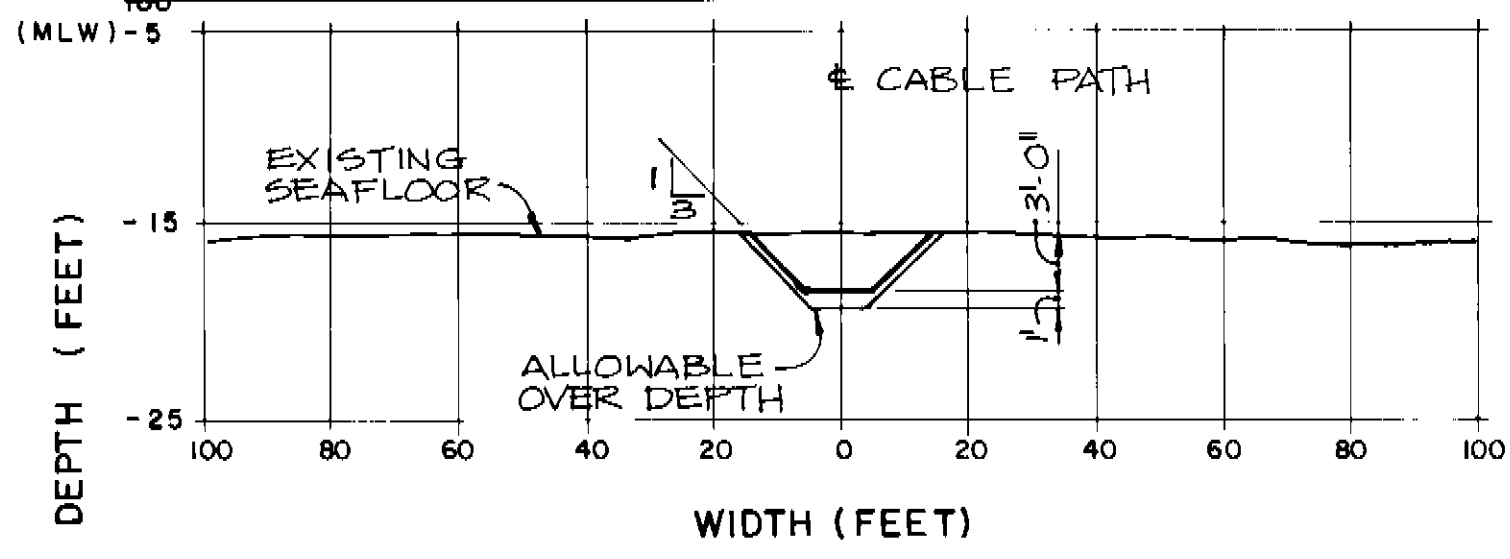
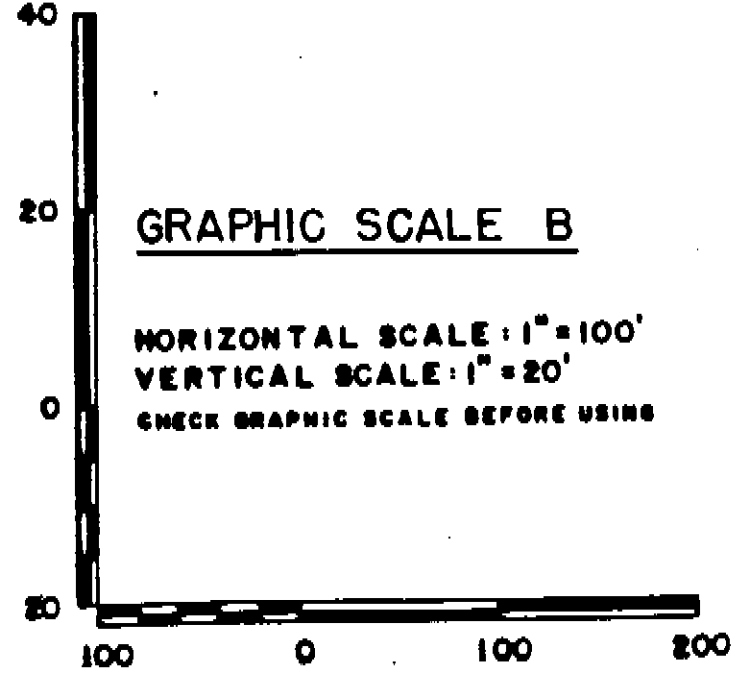
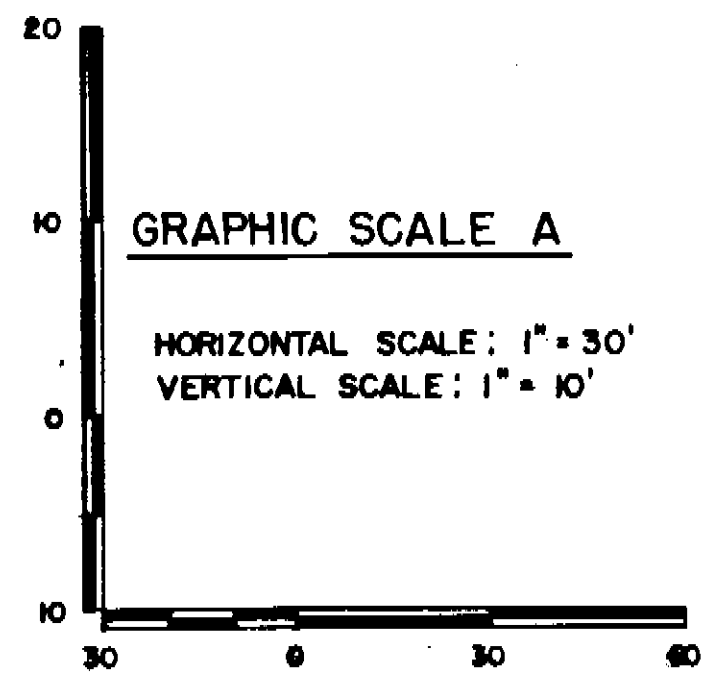


NORFOLK VIRGINIA

PANEL A



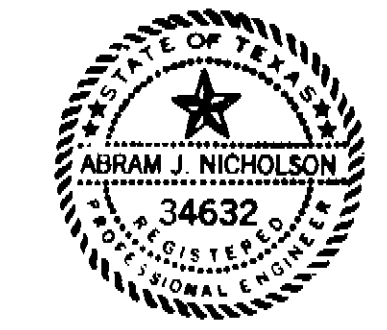
GRAPHIC SCALE
WARNING: DRAWING MAY BECOME DISTORTED DURING REDUCTION OR OTHER REPRODUCTION PROCESS. USER IS CAUTIONED AGAINST SCALING.



PROFILE & CABLE PATH

SCALE: "B"

SECTION
SCALE: "A"
C-3, C-4

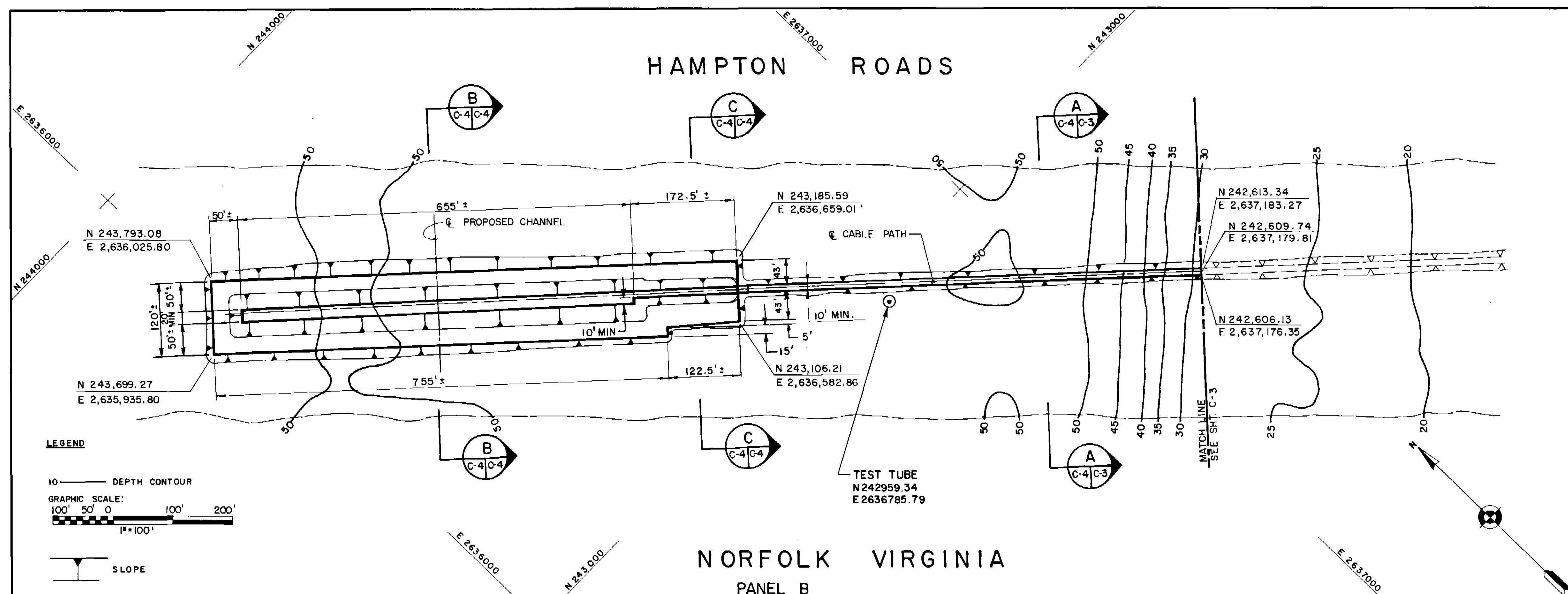


Black & Veatch Engineers-Architects	
DES. RM. DR. <i>[Signature]</i>	CHK. JW.
SUBMITTED BY <i>[Signature]</i> DATE <i>5/17/89</i>	
TITLE: PROGRAM MGR	
OCEAN ENGR. & CONST. PROJ. OFF. FPO-1	
EIC <i>[Signature]</i> RVD <i>[Signature]</i>	

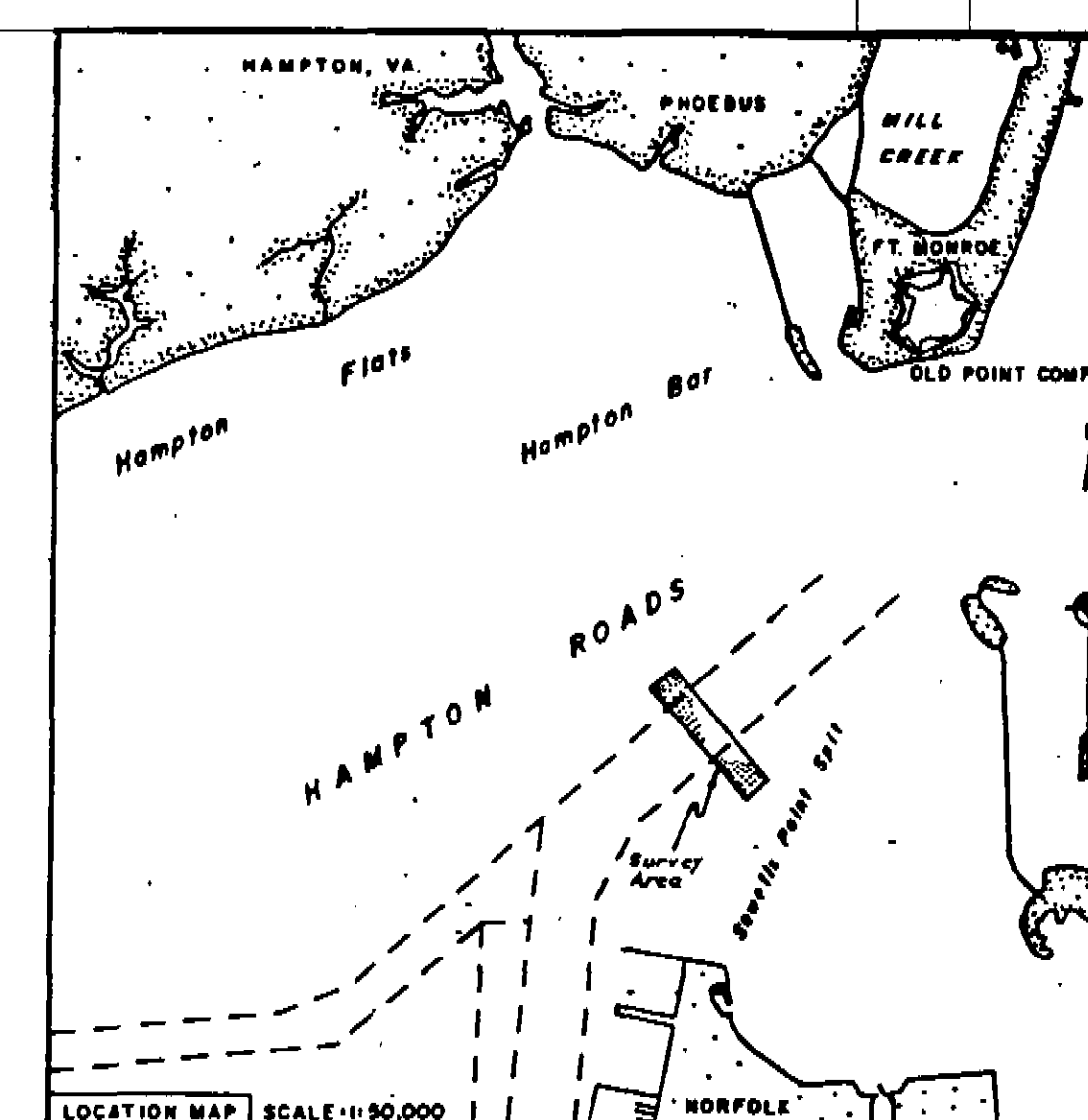
DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION DREDGING PLAN PROFILE AND SECTION FOR MINESWEEPER SENSOR RANGE	

SATISFACTORY TO <i>[Signature]</i> DATE <i>5/17/89</i>	APPROVED HEAD FPO-1 DATE <i>5/17/89</i>
SATISFACTORY TO <i>[Signature]</i> DATE <i>5/17/89</i>	APPROVED DATE <i>5/17/89</i>
COMNAVSEASYS.COM	EFD FOR COMMANDER, NAVFAC

SIZE	CODE	IDENT. NO.	NAVFAC DRAWING NO.	C-3
D	80091	3020676	CONSTR. CONTR. NO. N62477-89-C-0053	
SCALE AS SHOWN		SPEC. 21-89-0053	SHEET 3 OF 40	



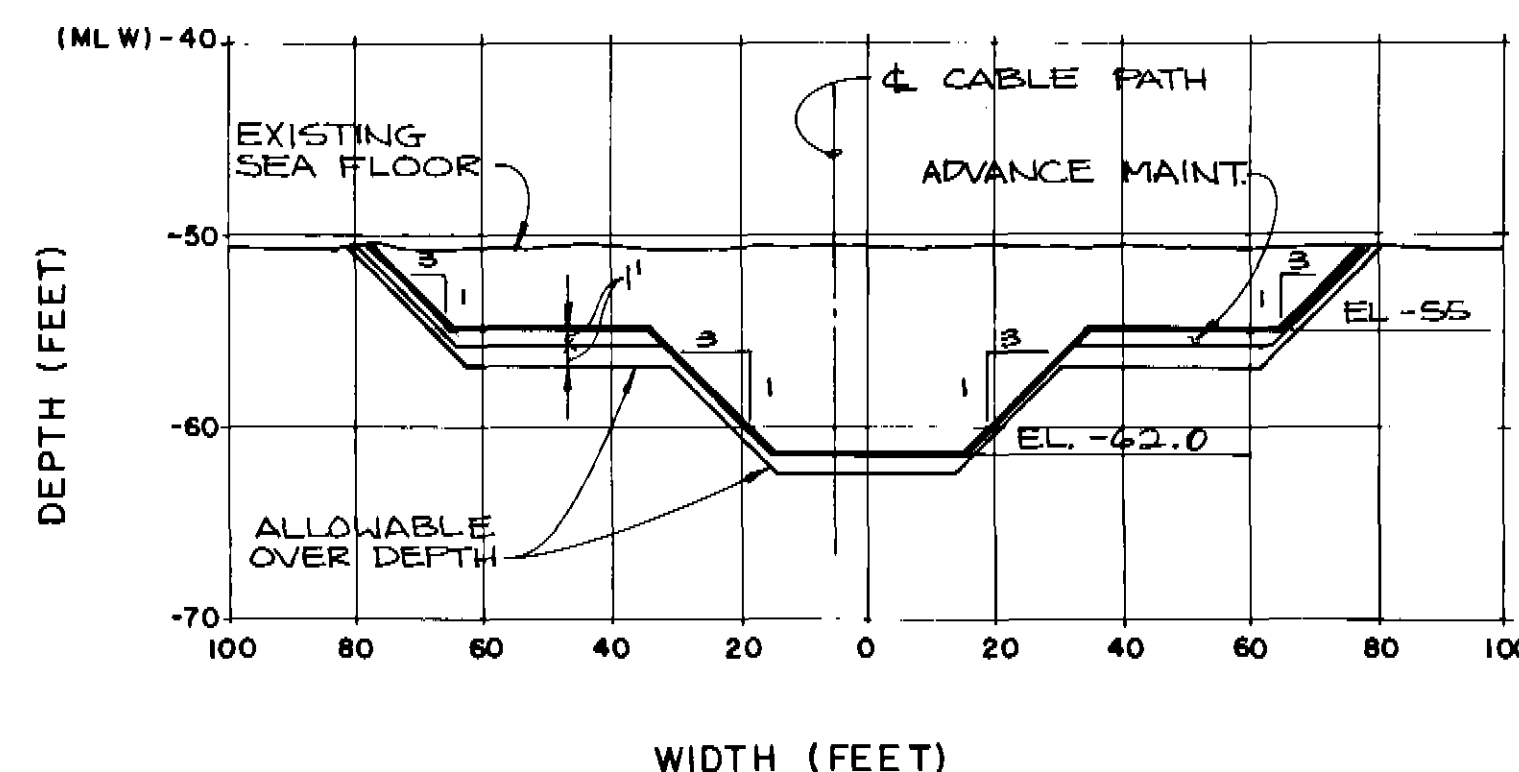
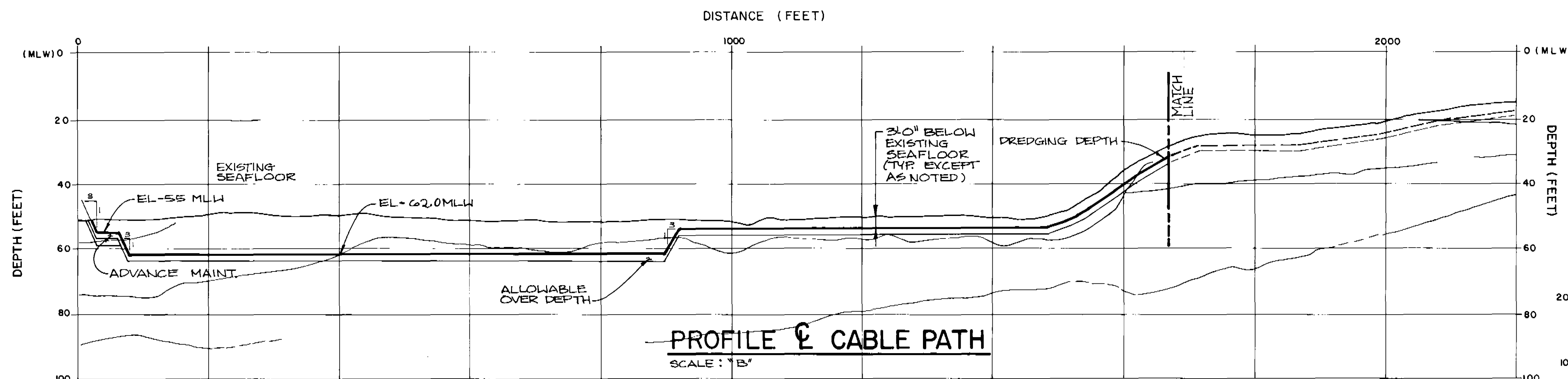
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SYM	DESCRIPTION	DATE	APPROVED
A	REVISED TO REFLECT AS BUILT CONDITION		



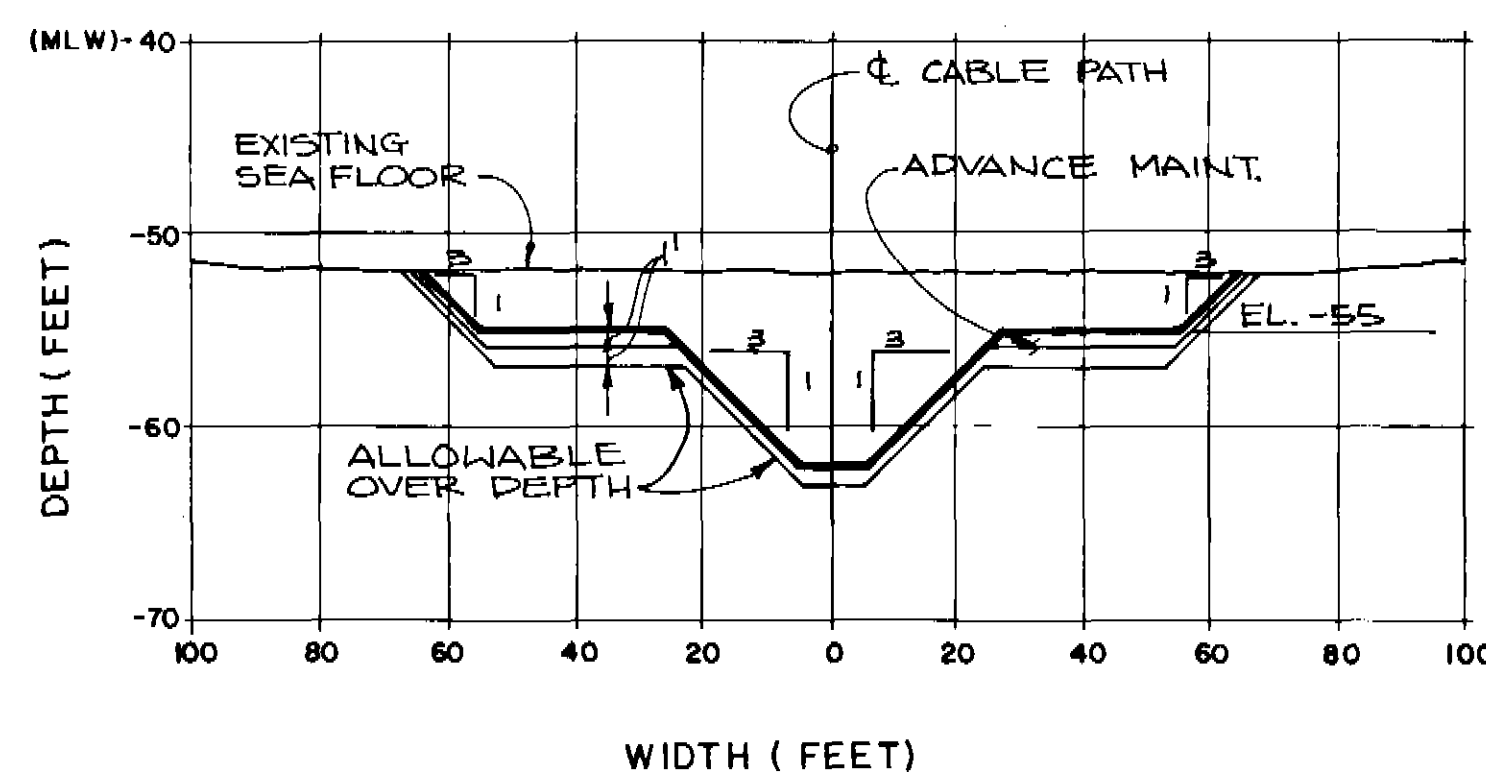
- NOTES**
1. PROFILE COORDINATES ARE IN FEET AND ARE REFERENCED TO THE VIRGINIA STATE PLANE COORDINATE SYSTEM (SOUTH ZONE).
 2. COMPUTATION OF SUBBOTTOM THICKNESSES ARE BASED ON AN ACOUSTIC PROPAGATION VELOCITY OF 4800 FT./SEC.
 3. THE INTERPRETED SUBBOTTOM REFLECTORS ARE INFERRED FROM SEISMIC DATA AND HAVE NOT BEEN VERIFIED BY PHYSICAL SAMPLING.
 4. ELEVATIONS ARE BASED ON U.S. ARMY CORPS OF ENGINEERS BENCHMARK "NIT" ELEVATION 8.67 FT. MEAN LOW WATER (MLW), LOCATED AT PIER 1, NORFOLK INTERNATIONAL TERMINAL.
 5. THE INFORMATION PRESENTED ON THIS CHART REPRESENTS THE RESULTS OF A SURVEY PERFORMED BY OCEAN SURVEYS, INC. ON 24-26 MAY 1988 AND CAN ONLY BE CONSIDERED AS INDICATING THE CONDITIONS EXISTING AT THAT TIME. REFER TO SURVEY REPORT FOR ADDITIONAL INFORMATION.

LEGEND TO PROFILES

- DREDGED ELEVATION
- STRONG SUBBOTTOM REFLECTOR
- - - WEAK SUBBOTTOM REFLECTOR



SECTION B
SCALE: "A"



SECTION C
SCALE: "A"

GRAPHIC SCALE

WARNING: DRAWING MAY BECOME DISTORTED DURING REDUCTION OR OTHER REPRODUCTION PROCESS. USER IS CAUTIONED AGAINST SCALING.

GRAPHIC SCALE A

HORIZONTAL SCALE: 1"=30'
VERTICAL SCALE: 1"=10'

GRAPHIC SCALE B

HORIZONTAL SCALE: 1"=100'
VERTICAL SCALE: 1"=20'
CHECK GRAPHIC SCALE BEFORE USING



Black & Veatch
Engineers-Architects

DES. RM. DR. LT. CHK. JW.
SUBMITTED BY: *John J. Nicholson* DATE: *5/1/88*
TITLE: PROGRAM MGR.
OCEAN ENGR. & CONST. PROJ. OFF. FPO-1
EIC: *JPL* RVD: *5/1/88*

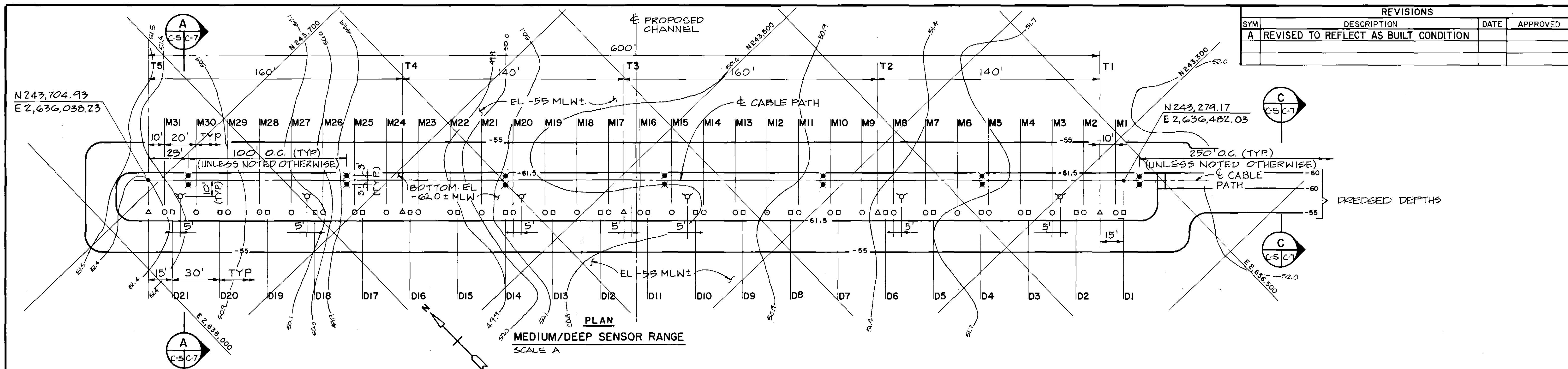
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ROICC: *[Signature]* DATE: *5/1/88*
SATISFACTORY TO: *[Signature]* DATE: *5/1/88*
COMNAVSEASYS/COM DATE: *5/1/88*

APPROVED HEAD FPO-1 DATE: *5/1/88*
APPROVED: *[Signature]* DATE: *5/1/88*
EFD FOR COMMANDER, NAVFAC

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND
CHESAPEAKE DIVISION
WASHINGTON, D.C.

**MAGNETIC SILENCING FACILITY
NORFOLK NAVAL STATION**
DREDGING PLAN PROFILE AND SECTION FOR
MEDIUM/DEEP SENSOR RANGE

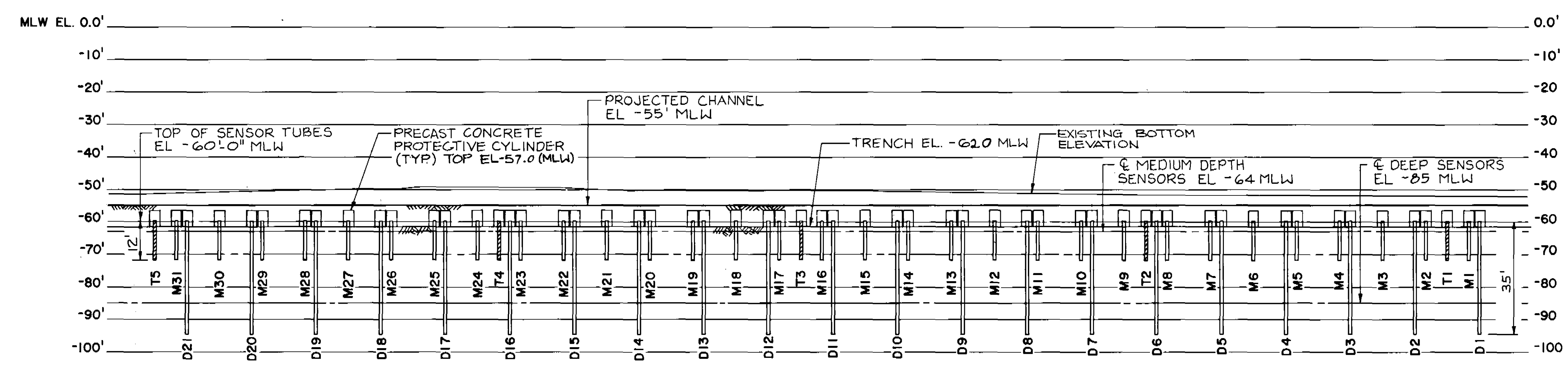
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D 80091 **3020677** **C-4**
CONSTR. CONTR. NO. **N62477-89-C-0053**
SCALE: AS SHOWN SPEC 21-89-0053 SHEET 4 OF 40



REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED
A	REVISED TO REFLECT AS BUILT CONDITION		

LEGEND

- o MEDIUM SENSOR TUBE (M1-M31)
- DEEP SENSOR TUBE (D1-D21)
- Y STAKE PILE TUBE FOR SPARE MEDIUM/DEEP SENSOR CABLE (SMD1-SMD6)
- * GUIDE PILE
- 50- EXISTING CONTOUR
- 55- FINISHED CONTOUR

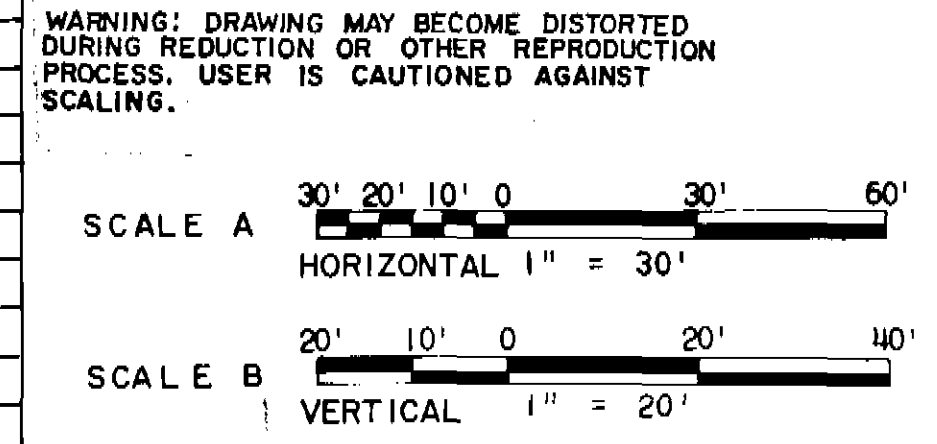


**PROFILE
MEDIUM/DEEP SENSOR RANGE
SENSOR TUBE LOCATION
SCALE A & B**

NOTES

- FOR SENSOR TUBE INSTALLATION TOLERANCES, SEE DWG C-11
- FOR SENSOR TUBE DETAILS, SEE DWG C-12
- FOR STAKE PILE DETAILS SEE DWG C-9. FOR PRECAST CYLINDER DETAILS SEE C-9

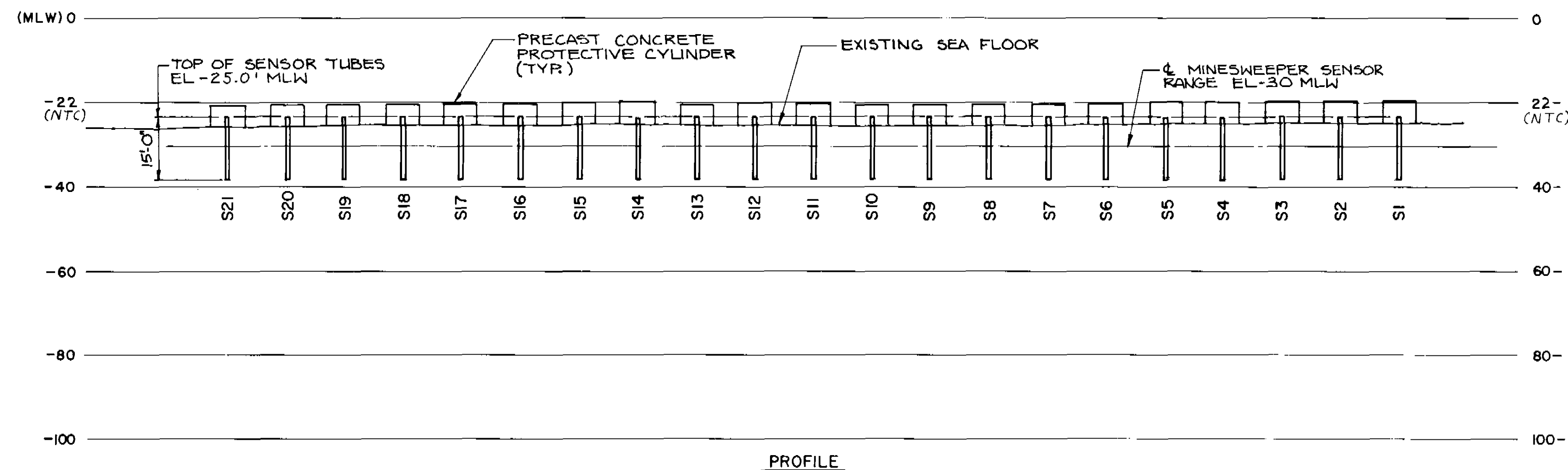
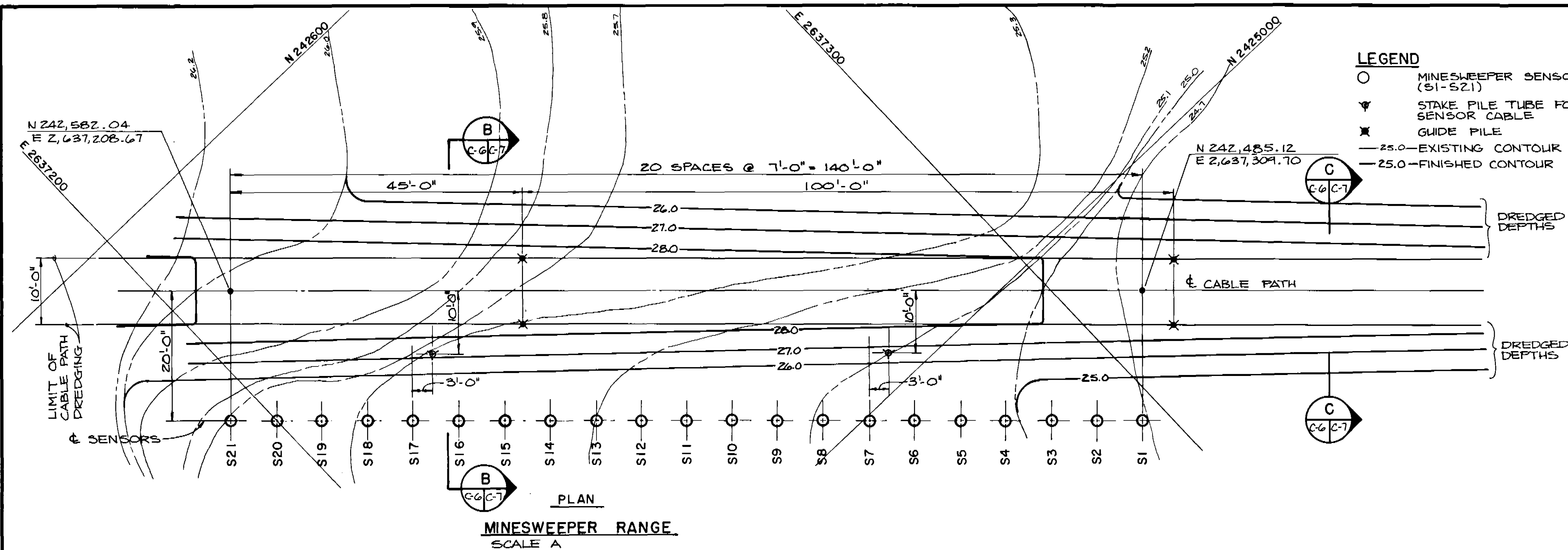
GRAPHIC SCALE



COORDINATE			COORDINATE			COORDINATE			COORDINATE		
TAG NO.	N	E	TAG NO.	N	E	TAG NO.	N	E	TAG NO.	N	E
M1	243,268.20	2,636,464.57	M18	243,503.58	2,636,219.22	D4	243,327.04	2,636,403.24	D21	243,680.11	2,636,035.21
M2	243,282.04	2,636,450.14	M19	243,517.42	2,636,204.19	D5	243,347.81	2,636,381.59	T1	243,275.12	2,636,457.36
M3	243,295.89	2,636,435.71	M20	243,531.27	2,636,190.36	D6	243,368.58	2,636,359.94	T2	243,312.04	2,636,356.33
M4	243,309.73	2,636,421.29	M21	243,545.11	2,636,175.93	D7	243,389.35	2,636,338.29	T3	243,482.81	2,636,240.87
M5	243,323.58	2,636,406.84	M22	243,558.96	2,636,161.49	D8	243,410.12	2,636,316.64	T4	243,579.73	2,636,139.85
M6	243,337.43	2,636,392.41	M23	243,572.81	2,636,147.06	D9	243,430.89	2,636,294.99	T5	243,690.50	2,636,024.39
M7	243,351.27	2,636,377.98	M24	243,586.65	2,636,132.63	D10	243,451.65	2,636,273.34			
M8	243,365.12	2,636,363.55	M25	243,600.50	2,636,118.20	D11	243,472.42	2,636,251.70			
M9	243,378.96	2,636,349.11	M26	243,614.34	2,636,103.77	D12	243,493.19	2,636,230.05			
M10	243,392.81	2,636,334.68	M27	243,628.19	2,636,089.33	D13	243,513.96	2,636,208.40			
M11	243,406.66	2,636,320.25	M28	243,642.04	2,636,074.90	D14	243,534.73	2,636,186.75			
M12	243,420.50	2,636,305.82	M29	243,655.88	2,636,060.47	D15	243,555.50	2,636,165.10			
M13	243,434.35	2,636,291.38	M30	243,669.73	2,636,046.04	D16	243,576.27	2,636,143.45			
M14	243,448.19	2,636,276.95	M31	243,683.57	2,636,031.60	D17	243,597.04	2,636,121.81			
M15	243,462.04	2,636,262.52	D1	243,264.73	2,636,468.18	D18	243,617.81	2,636,100.16			
M16	243,475.88	2,636,248.09	D2	243,285.50	2,636,446.93	D19	243,638.57	2,636,078.51			
M17	243,489.73	2,636,233.66	D3	243,306.27	2,636,424.88	D20	243,659.34	2,636,056.86			



Black & Veatch ENGINEERS-ARCHITECTS		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DES. RM. [] DR. LT. [] CHK. JW. [] SUBMITTED BY: [Signature] DATE: 5-15-89 TITLE: PROGRAM MGR. OCEAN ENGR. & CONST. PROJ. OFF. FPO-1 EIC: [Signature] RVD: [Signature]		MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION PLAN & PROFILE MEDIUM / DEEP SENSOR RANGE	
SATISFACTORY TO: [Signature] DATE: 5-15-89 SATISFACTORY TO: ROICC DATE: 5-15-89 COMMNAVSEASYS COM DATE:		SIZE: CODE IDENT. NO. NAVFAC DRAWING NO. D 80091 3020678 CONSTR. CONTR. NO. N62477-89-C-0053 SCALE: AS SHOWN SPEC. 21-89-0053 SHEET 5 OF 40	



SENSOR TUBE HORIZONTAL LOCATIONS (DESIGN)

TAG NO.	COORDINATE	
	N	E
S1	242,470.69	2,637,295.85
S2	242,475.54	2,637,290.80
S3	242,480.38	2,637,285.75
S4	242,485.23	2,637,280.70
S5	242,490.07	2,637,275.65
S6	242,494.92	2,637,270.60
S7	242,499.77	2,637,265.55
S8	242,504.61	2,637,260.49
S9	242,509.46	2,637,255.44
S10	242,514.30	2,637,250.39
S11	242,519.15	2,637,245.34
S12	242,524.00	2,637,240.29

TAG NO.	COORDINATE	
	N	E
S13	242,528.84	2,637,235.24
S14	242,533.69	2,637,230.19
S15	242,538.54	2,637,225.14
S16	242,543.38	2,637,220.08
S17	242,548.23	2,637,215.03
S18	242,553.07	2,637,209.98
S19	242,557.92	2,637,204.93
S20	242,562.77	2,637,199.88
S21	242,567.61	2,637,194.83

NOTES

- FOR SENSOR TUBE INSTALLATION TOLERANCES SEE DWG. C-11
- FOR SENSOR TUBE DETAILS SEE DWG. C-12
- FOR STAKE PILE TUBE DETAILS SEE DWG. C-1. SEE DWG. C-9 FOR PRECAST CYLINDER DETAILS.
- CONTRACTOR SHALL EXCAVATE AS REQUIRED FOR PLACEMENT OF THE PRECAST CONCRETE CYLINDERS AND CABLES SUCH THAT THE ELEVATION OF THE TOP OF EVERY CYLINDER IN THE RANGE SHALL FALL BETWEEN -22.0 AND -22.5 MLW.
- THE CONTRACTOR SHALL MEET A SPECIFIC COLLECTIVE TOLERANCE LIMIT BETWEEN THE TOP OF THE CONCRETE CYLINDER AND THE TOP OF THE SENSOR TUBE. THIS DIMENSION SHALL BE NOT LESS THAN 30 INCHES AND NOT MORE THAN 42 INCHES.

GRAPHIC SCALE

WARNING: DRAWING MAY BECOME DISTORTED DURING REDUCTION OR OTHER REPRODUCTION PROCESS. USER IS CAUTIONED AGAINST SCALING.

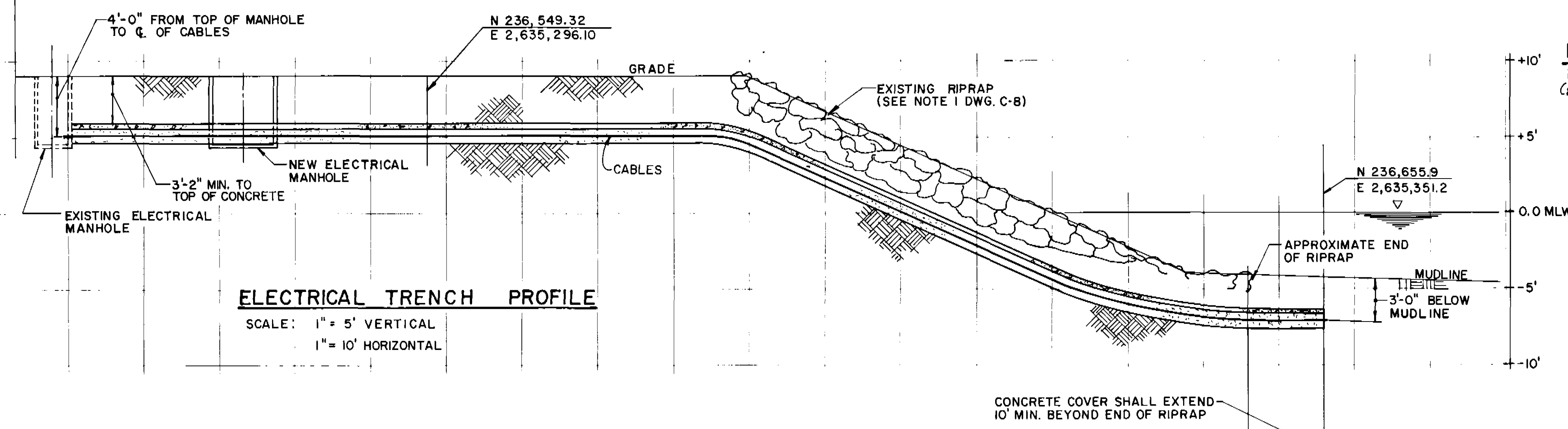
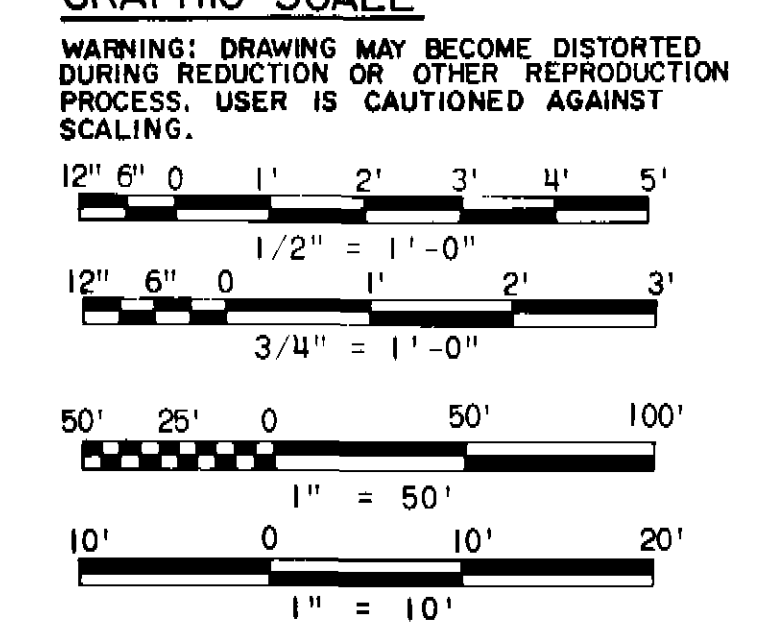
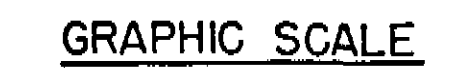
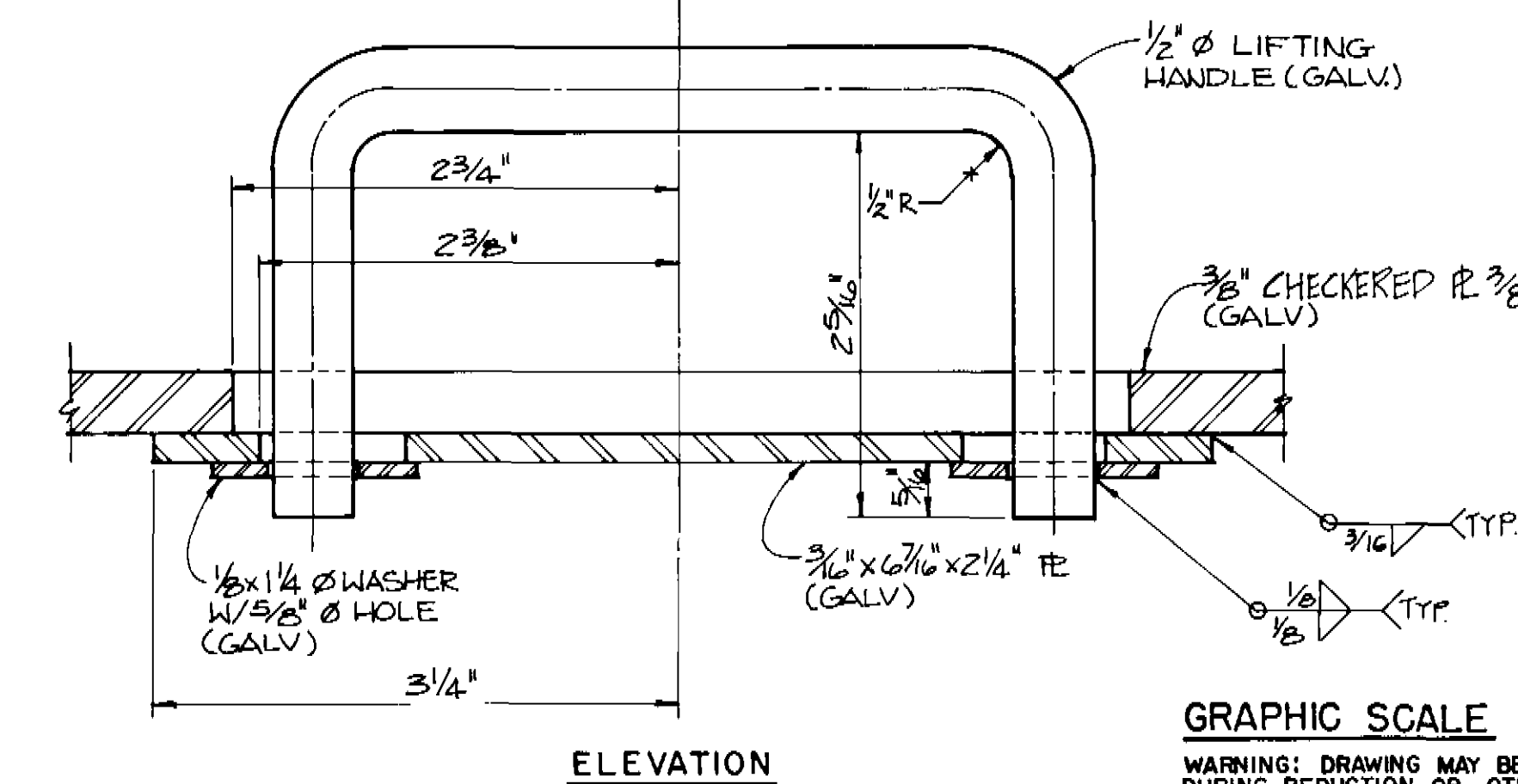
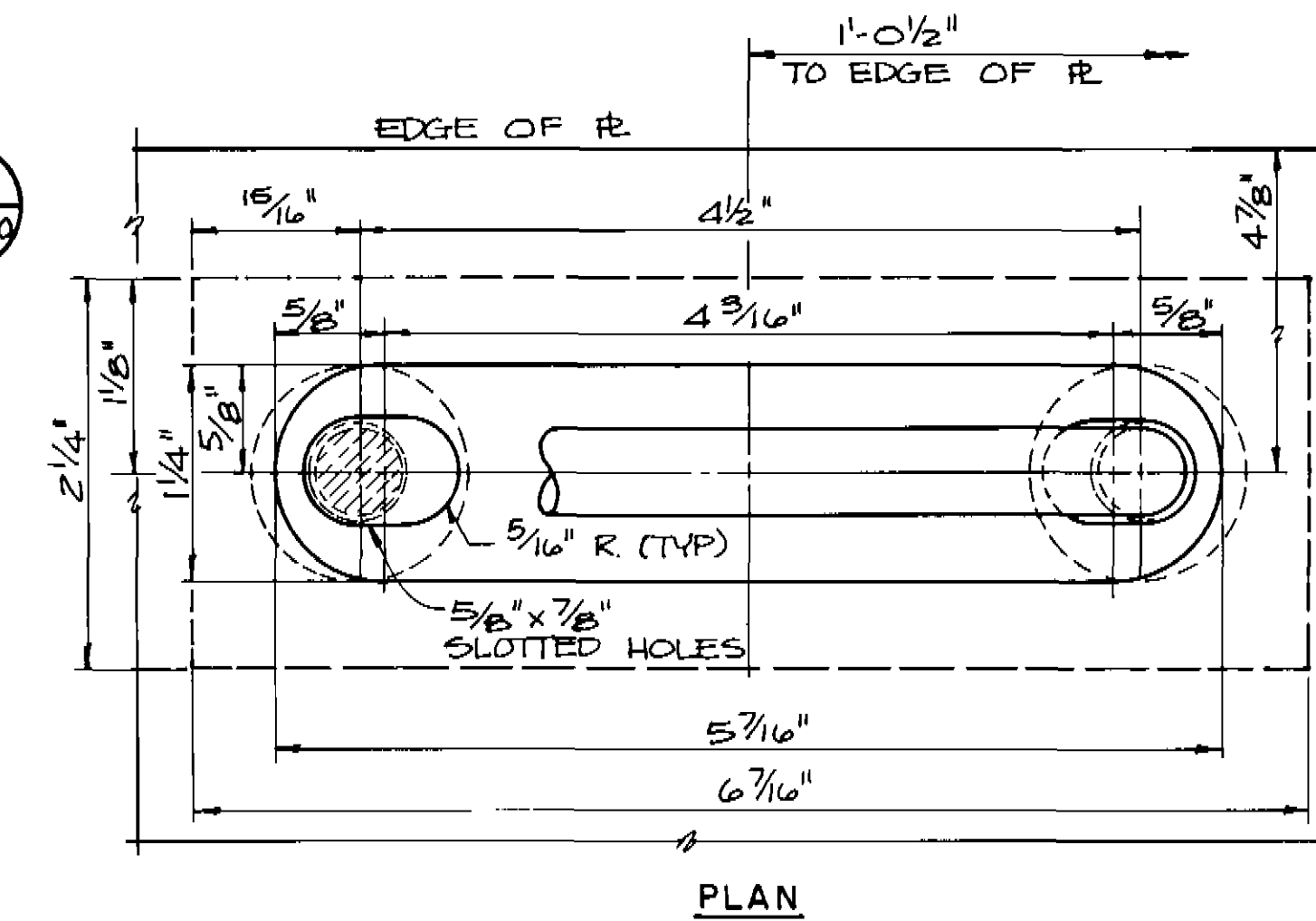
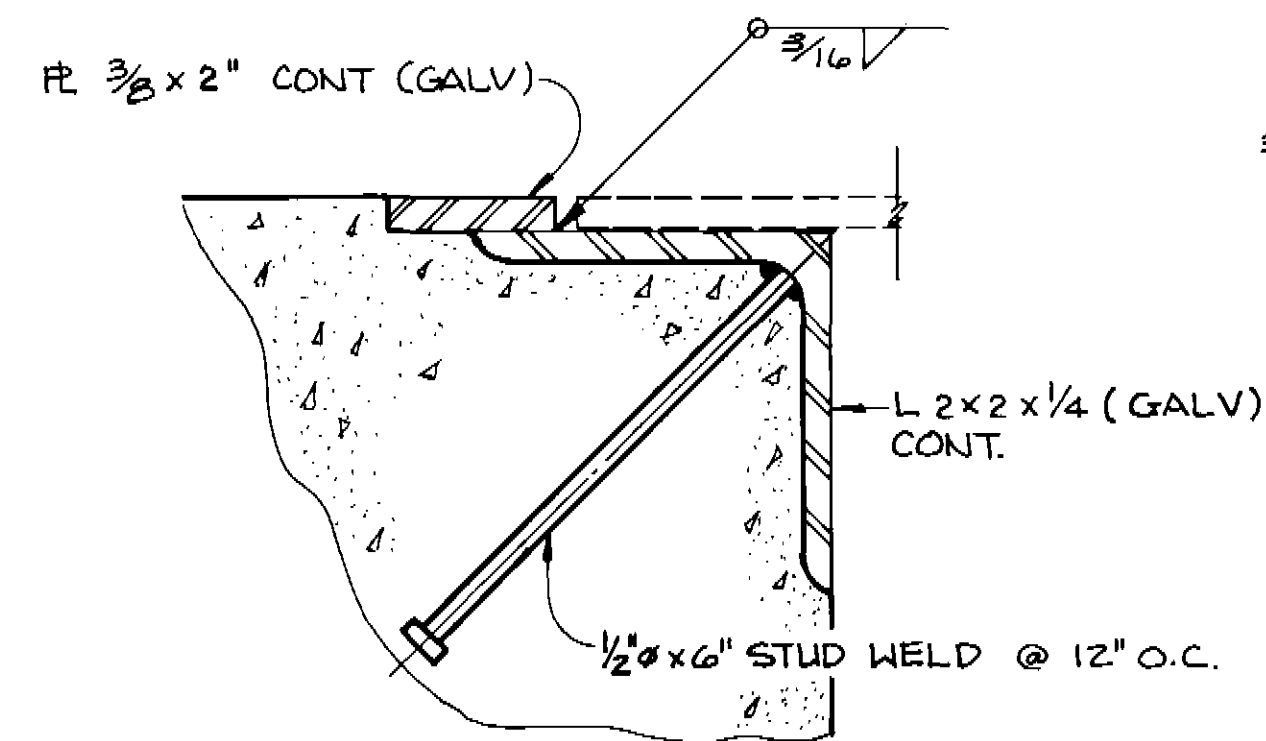
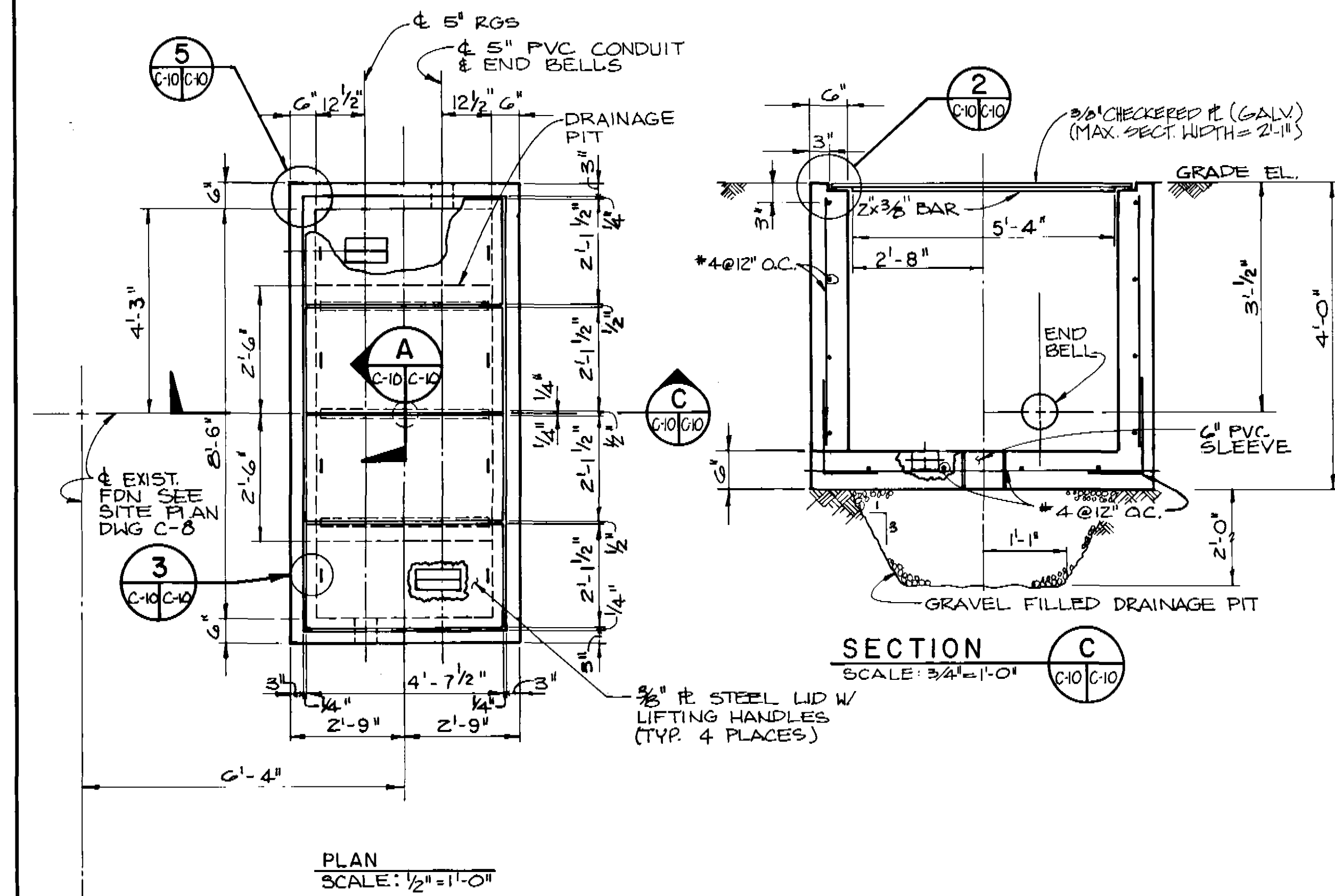
SCALE A
HORIZONTAL 1"=10'

SCALE B
VERTICAL 1"=20'



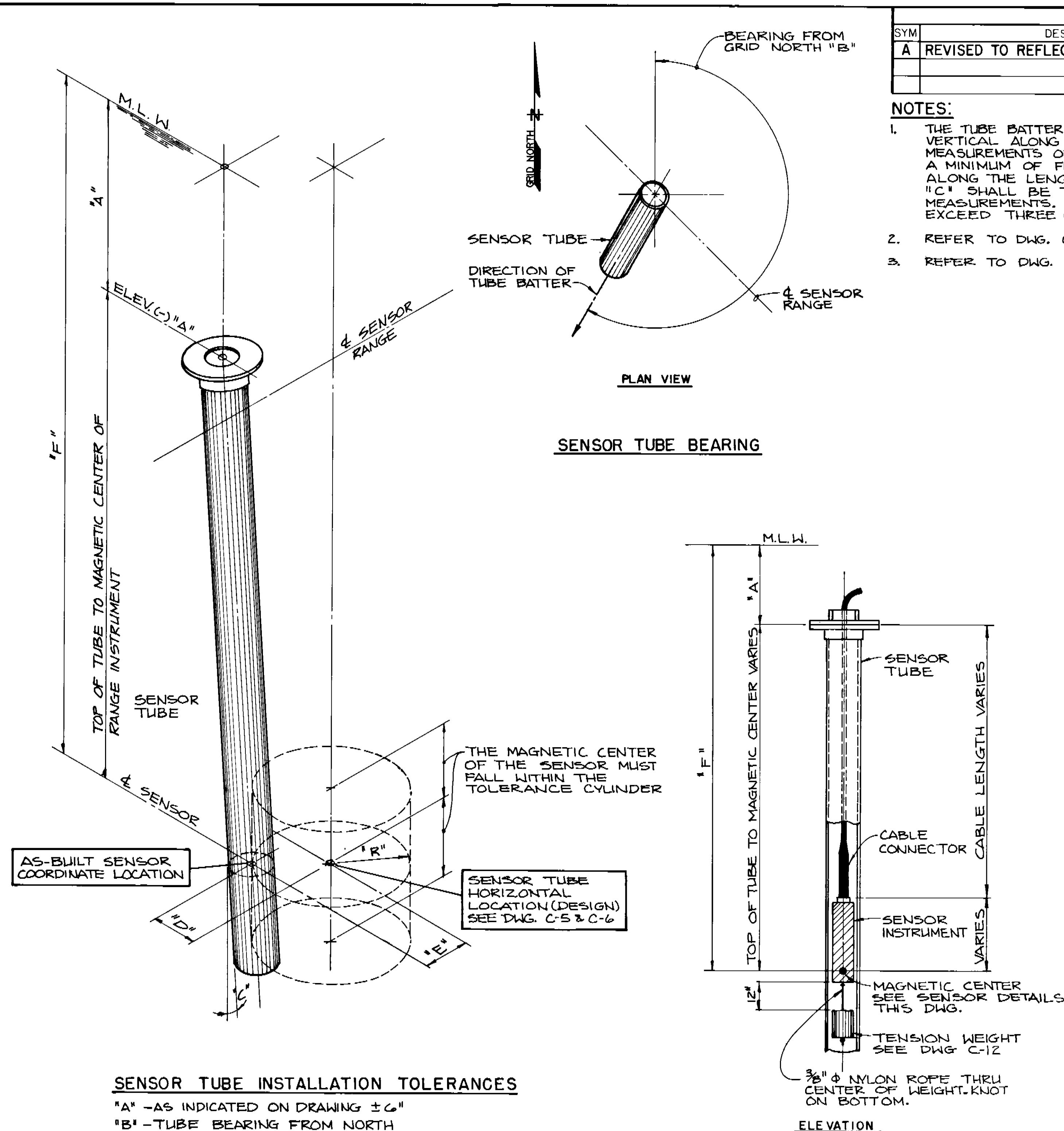
Black & Veatch Engineers-Architects		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DES. RM. DR. LT. CHK. JW. SUBMITTED BY: <i>[Signature]</i> DATE: 6-15-89 TITLE: PROGRAM MGR. OCEAN ENGR. & CONST. PROJ. OFF. FPO-1 EIC: <i>[Signature]</i> RVD: <i>[Signature]</i> 5/1/89		MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION PLAN & PROFILE MINESWEEPER SENSOR RANGE	
SATISFACTORY TO: <i>[Signature]</i> DATE: 6-15-89 ROICC: <i>[Signature]</i> DATE: 6-15-89 SATISFACTORY TO: <i>[Signature]</i> DATE: 6-15-89 COMNAVSEASCOM: <i>[Signature]</i> DATE: 6-15-89	APPROVED HEAD, FPO-1: <i>[Signature]</i> DATE: 6-15-89 APPROVED: <i>[Signature]</i> DATE: 6-15-89 EFD FOR COMMANDER, NAVFAC: <i>[Signature]</i> DATE: 6-15-89	SIZE: D CODE IDENT. NO.: 80091 NAVFAC DRAWING NO.: 3020679 CONSTR. CONTR. NO.: N62477-89-C-0053 SCALE AS SHOWN: SPEC. 21-89-0053	SHEET 6 OF 40 C-6

REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED
A	REVISED TO REFLECT AS BUILT CONDITION		



 Black & Veatch Engineers-Architects	DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.		
	MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION ELECTRICAL TRENCH PROFILE AND MISCELLANEOUS DETAILS		
DES RM DR LT CHK JW SUBMITTED BY <i>W. H. Jones</i> DATE <i>5/12/59</i> TITLE: PROGRAM, MGR. OCEAN ENGR. B. CONST PROJ. OFF. FPO-1 EIC <i>(J)</i> <i>11</i> RVD <i>100</i> <i>5/14/59</i>			
APPROVED HEAD FPO-1 DATE <i>5-19-59</i> <i>W. H. Jones</i> APPROVED DATE <i>5/17/59</i> <i>W. H. Jones</i> EFD FOR COMMANDER, NAVFAC	SIZE CODE IDENT. NO. NAVFAC DRAWING NO. C-10 D 80091 3020683	CONSTR. CONTR. NO. N62477-89-C-0053 SCALE AS SHOWN SPEC. 21-89-0053 SHEET 10 OF 40	

AS-BUILT DIMENSIONS									
TUBE TAG NUMBER	TOP OF TUBE ELEVATION "A"	INSTALLED BEARING "B"	INSTALLED ANGLE "C"	HORIZONTAL OFFSETS		SENSOR DEPTH "F"	INSTALLED COORDINATES		
				"D"	"E"		NORTH	EAST	
M-1	-60.3	S75.27°E	.59°	-.24'	-.30'	-64.0	243,267.96	2,636,464.87	
M-2	-60.3	S46.23°E	.40°	-.09'	+.16'	-64.0	243,381.95	2,636,450.30	
M-3	-60.3	S24.53°W	.49°	+.14'	+.04'	-64.0	243,296.03	2,636,435.74	
M-4	-60.3	N71.43°W	.57°	-.11'	+.01'	-64.0	243,309.63	2,636,421.28	
M-5	-60.3	S81.56°W	1.72°	+.16'	-.73'	-64.0	243,323.74	2,636,406.11	
M-6	-60.3	N1.73°W	.79°	+.32'	-.17'	-64.0	243,337.75	2,636,392.24	
M-7	-60.3	N29.27°W	.77°	-.12'	+.05'	-64.0	243,351.15	2,636,378.03	
M-8	-60.3	N50.38°W	.52°	+.20'	0	-64.0	243,365.32	2,636,363.54	
M-9	-60.3	N71.28°E	.34°	+.27'	+.27'	-64.0	243,379.23	2,636,349.38	
M-10	-60.3	S63.32°E	.51°	-.14'	+.01'	-64.0	243,392.67	2,636,334.69	
M-11	-60.3	S6.18°W	.11°	-.03'	-.58'	-64.0	243,406.63	2,636,319.67	
M-12	-60.3	S18.45°E	.53°	-.13'	-.11'	-64.0	243,420.37	2,636,305.70	
M-13	-60.3	N64.03°W	1.16°	-.62'	+.12'	-64.0	243,433.73	2,636,291.50	
M-14	-60.3	N43.3°W	.67°	+.02'	-.14'	-64.0	243,448.21	2,636,276.81	
M-15	-60.3	S17.37°W	.47°	-.03'	-.03'	-64.0	243,462.01	2,636,262.49	
M-16	-60.3	S0.40°E	.48°	+.67'	+.05'	-64.0	243,476.56	2,636,248.13	
M-17	-60.3	S18.98°W	.06°	-.34'	+.31'	-64.0	243,489.39	2,636,233.96	
M-18	-60.3	N38.57°E	.89°	-.36'	-.20'	-64.0	243,503.22	2,636,219.02	
M-19	-60.3	S16.85°E	.38°	-.03'	-.05'	-64.0	243,517.39	2,636,204.74	
M-20	-60.3	N76.08°W	.64°	-.13'	+.22'	-64.0	243,531.14	2,636,190.58	
M-21	-60.3	S73.45°E	.52°	0	-.08'	-64.0	243,545.12	2,636,175.84	
M-22	-60.3	S27.3°E	.51°	+.08'	+.01'	-64.0	243,559.04	2,636,161.50	
M-23	-60.3	N6.03°E	.62°	-.09'	-.08'	-64.0	243,572.72	2,636,146.98	
M-24	-60.3	S4.63°W	.50°	+.04'	+.07'	-64.0	243,586.69	2,636,132.70	
M-25	-60.3	S8.40°W	.89°	+.58'	-.04'	-64.0	243,601.08	2,636,118.15	
M-26	-60.3	N55.07°E	.43°	+.15'	+.13'	-64.0	243,614.50	2,636,103.89	
M-27	-60.3	N28.87°E	.64°	-.04'	-.02'	-64.0	243,628.15	2,636,089.31	
M-28	-60.3	N53.73°E	.36°	-.24'	+.02'	-64.0	243,641.80	2,636,074.92	
M-29	-60.3	N84.93°W	.59°	+.13'	-.48'	-64.0	243,656.01	2,636,059.98	
M-30	-60.3	S44.80°E	1.33°	+.11'	-.05'	-64.0	243,669.84	2,636,045.98	
M-31	-60.3	S1.10°E	.80°	+.38'	+.24'	-64.0	243,683.95	2,636,031.84	
D-1	-60.3	N71.12°E	1.80°	-.05'	-.100'	-85.0	243,264.69	2,636,467.18	
D-2	-60.3	S86.07°E	1.35°	+.05'	-.41'	-85.0	243,285.55	2,636,446.12	
D-3	-60.3	S24.53°E	1.04°	+.09'	-.18'	-85.0	243,306.36	2,636,424.70	
D-4	-60.3	S5.33°E	1.46°	+.18'	-.17'	-85.0	243,327.22	2,636,403.06	
D-5	-60.3	N47.45°E	.65°	+.11'	+.49'	-85.0	243,347.92	2,636,382.07	
D-6	-60.3	S20.32°E	1.06°	+.14'	+.08'	-85.0	243,368.72	2,636,360.01	
D-7	-60.3	S12.17°W	.43°	-.20'	-.17'	-85.0	243,389.15	2,636,338.46	
D-8	-60.3	S79.80°E	.79°	0	-.42'	-85.0	243,410.12	2,636,316.22	
D-9	-60.3	S21.73°E	1.08°	+.73'	+.06'	-85.0	243,431.62	2,636,295.05	
D-10	-60.3	S15.47°W	1.75°	+.03'	+.27'	-85.0	243,451.69	2,636,273.61	
D-11	-60.3	S27.45°W	1.41°	+.13'	+.22'	-85.0	243,472.55	2,636,251.47	
D-12	-60.3	S25.42°W	.99°	+.43'	+.25'	-85.0	243,493.62	2,636,230.29	
D-13	-60.3	S0.28°E	.53°	+.69'	+.06'	-85.0	243,514.65	2,636,208.46	
D-14	-60.3	S53.73°E	1.26°	+.60'	-.09'	-85.0	243,535.33	2,636,186.66	
D-15	-60.3	N82.05°W	.68°	+.02'	-.16'	-85.0	243,555.52	2,636,164.94	
D-16	-60.3	N62.58°E	.38°	+.19'	+.08'	-85.0	243,576.46	2,636,143.53	
D-17	-60.3	N65.97°W	.78°	-.15'	+.25'	-85.0	243,596.89	2,636,122.05	
D-18	-60.3	N30.67°W	.16°	-.36'	+.18'	-85.0	243,617.45	2,636,100.33	
D-19	-60.3	N76.15°E	.38°	+.45'	-.05'	-85.0	243,639.03	2,636,078.46	
D-20	-60.3	S10.6°W	.59°	+.72'	-.59'	-85.0	243,660.06	2,636,056.27	
D-21	-60.3	S7.92°E	.42°	-.54'	-.02'	-85.0	243,679.57	2,636,035.19	
S-1	-23.6	N29.4°W	.25°	-.02'	+.01'	-30.0	242,470.67	2,637,295.86	
S-2	-23.6	N50.22°W	.61°	-.06'	+.05'	-30.0	242,475.47	2,637,290.85	
S-3	-23.6	S25.00°W	.64°	-.02'	-.03'	-30.0	242,480.36	2,637,285.72	
S-4	-23.6	S3.82°E	.91°	-.08'	+.03'	-30.0	242,485.15	2,637,280.73	
S-5	-23.6	N75.00°W	.48°	+.01'	+.06'	-30.0	242,490.08	2,637,275.71	
S-6	-23.6	S18.88°W	.33°	+.03'	+.02'	-30.0	242,494.95	2,637,270.62	
S-7	-23.6	S42.87°W	.56°	+.06'	+.08'	-30.0	242,499.83	2,637,265.62	
S-8	-23.6	S22.93°E	.92°	+.10'	+.09'	-30.0	242,504.71	2,637,260.58	
S-9	-23.6	S77.37°W	.35°	+.08'	-.04'	-30.0	242,509.54	2,637,255.40	
S-10	-23.6	S74.03°W	.72°	+.03'	+.07'	-30.0	242,514.33	2,637,250.46	
S-11	-23.6	S49.93°W	.61°	+.03'	-.01'	-30.0	242,519.88	2,637,245.35	
S-12	-23.6	N33.57°W	.79°	0	-.03'	-30.0	242,524.00	2,637,240.26	
S-13	-23.6	S81.17°W	.52°	-.03'	+.10'	-30.0	242,528.81	2,637,235.34	
S-14	-23.6	S52.35°W	.48°	0	+.06'	-30.0	242,533.69	2,637,230.25	
S-15	-23.6	N87.52°W	.32°	+.13'	-.05'	-30.0	242,538.66	2,637,225.08	
S-16	-23.6	S73.60°W	.32°	+.01'	+.05'	-30.0	242,543.39	2,637,220.13	
S-17	-23.6	S86.55°W	.62°	-.03'	+.05'	-30.0	242,548.20	2,637,215.08	
S-18	-23.6	N86.80°W	.51°	+.01'	+.10'	-30.0	242,553.08	2,637,210.08	
S-19	-23.6	S12.83°W	.40°	+.02'	-.01'	-30.0	242,557.94	2,637,204.92	
S-20	-23.6	S84.80°W	.65°	-.01'	+.06'	-30.0	242,562.75	2,637,199.94	
S-21	-23.6	S62.58°W	.75°	+.02'	+.08'	-30.0	242,567.63	2,637,194.91	

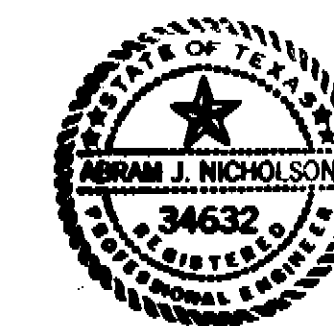


SENSOR TUBE INSTALLATION TOLERANCES

- "A" - AS INDICATED ON DRAWING $\pm 6"$
- "B" - TUBE BEARING FROM NORTH
- "C" - MAXIMUM DEVIATION FROM VERTICAL (SEE NOTE #1)
- "R" - MAXIMUM DEVIATION FROM HORIZONTAL:
 - SENSORS D-1 THRU D-21 = 12"
 - SENSORS M-1 THRU M-31 = 12"
 - SENSORS T-1 THRU T-5 = 12"
 - SENSORS S-1 THRU S-21 = 12"
- "F" - ELEVATION OF MAGNETIC CENTER OF INSTRUMENT MAXIMUM DEVIATION FROM DESIGN ELEVATION:
 - SENSORS D-1 THRU D-21 (EL. (-) 85' MLW) = $\pm 6"$
 - SENSORS M-1 THRU M-31 (EL. (-) 64' MLW) = $\pm 6"$
 - SENSORS T-1 THRU T-5 (EL. (-) 64' MLW) = $\pm 6"$
 - SENSORS S-1 THRU S-21 (EL. (-) 30' MLW) = $\pm 6"$

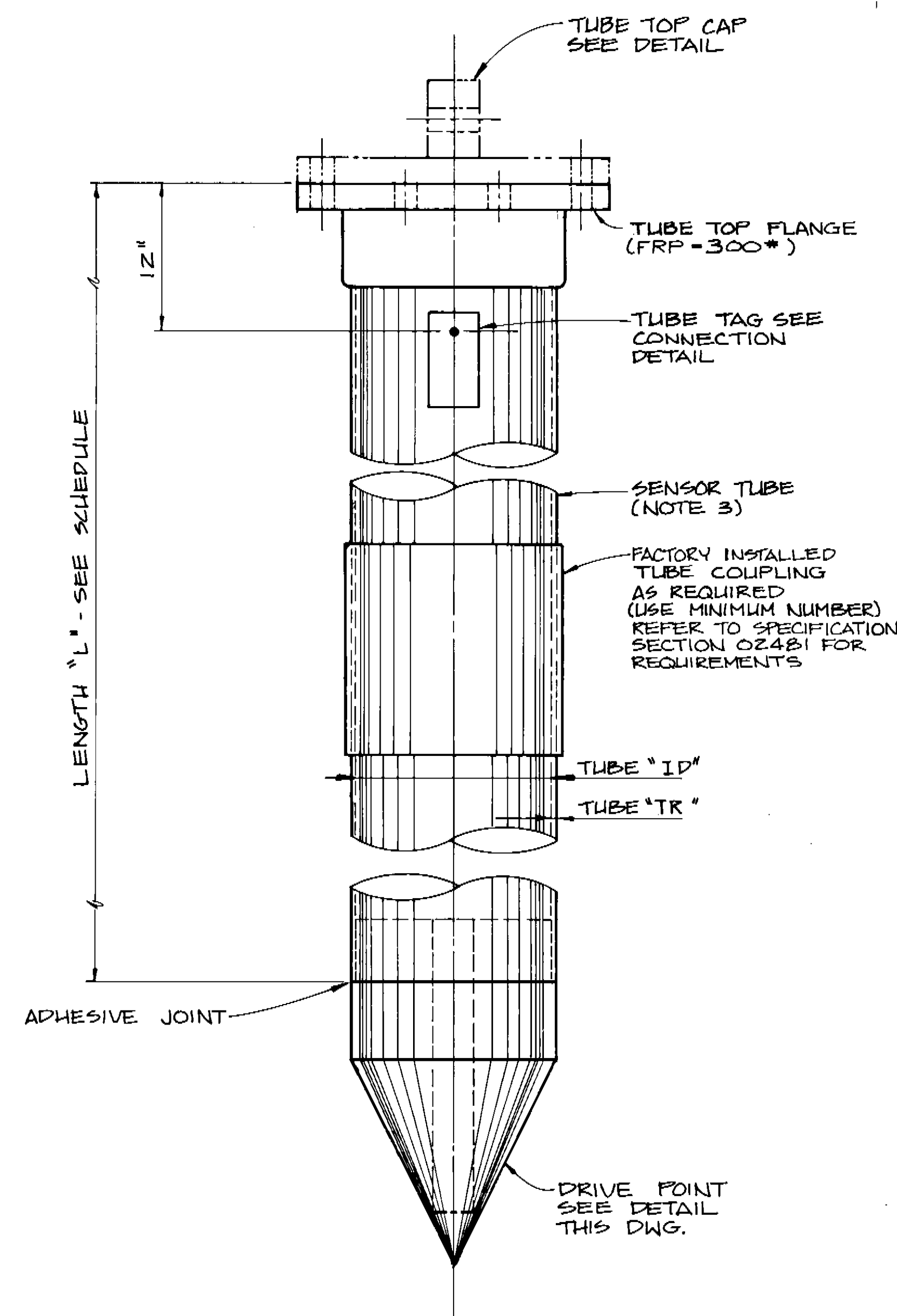
COLLECTIVE TOLERANCES

THE CONTRACTOR SHALL MEET SPECIFIC COLLECTIVE TOLERANCES BETWEEN THE TOP OF THE SENSOR TUBE AND THE TOP OF THE CONCRETE CYLINDER. AT BOTH RANGES, THIS DIMENSION MEASURED VERTICALLY, SHALL BE NOT LESS THAN 30 INCHES AND NOT MORE THAN 42 INCHES.

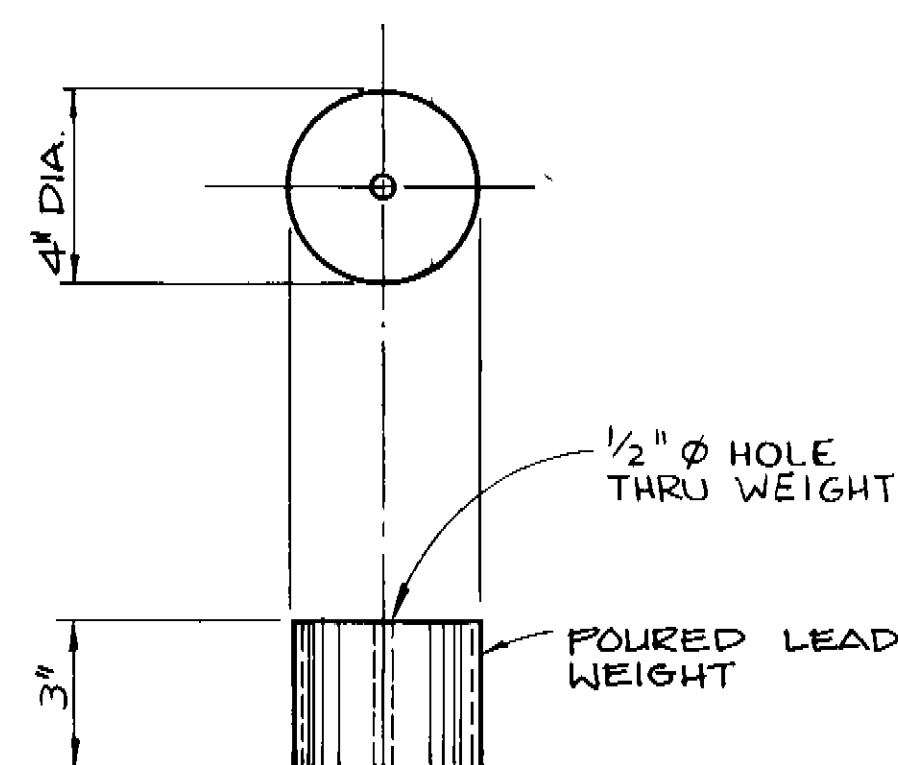


SENSOR INSTALLATION

DES. RM. DR. LT. CHK. JW SUBMITTED BY: <i>William J. Nicholson</i> TITLE: PROGRAM M33 OCEAN ENGR. & CONST. PROJ. OFF. FPO-1 EIC: <i>WJN</i> RVD: <i>11/22/99</i>		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
APPROVED HEAD FPO-1 DATE: <i>11/22/99</i> APPROVED: <i>WJN</i> DATE: <i>11/22/99</i> EFD FOR COMMANDER, NAVFAC		MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION SENSOR TUBE INSTALLATION DETAILS	
SIZE: D CODE: 80091 IDENT. NO.: 3020684 CONSTR. CONTR. NO.: N62477-89-C-0053 SCALE: AS NOTED		NAVFAC DRAWING NO.: SHEET 11 OF 40	

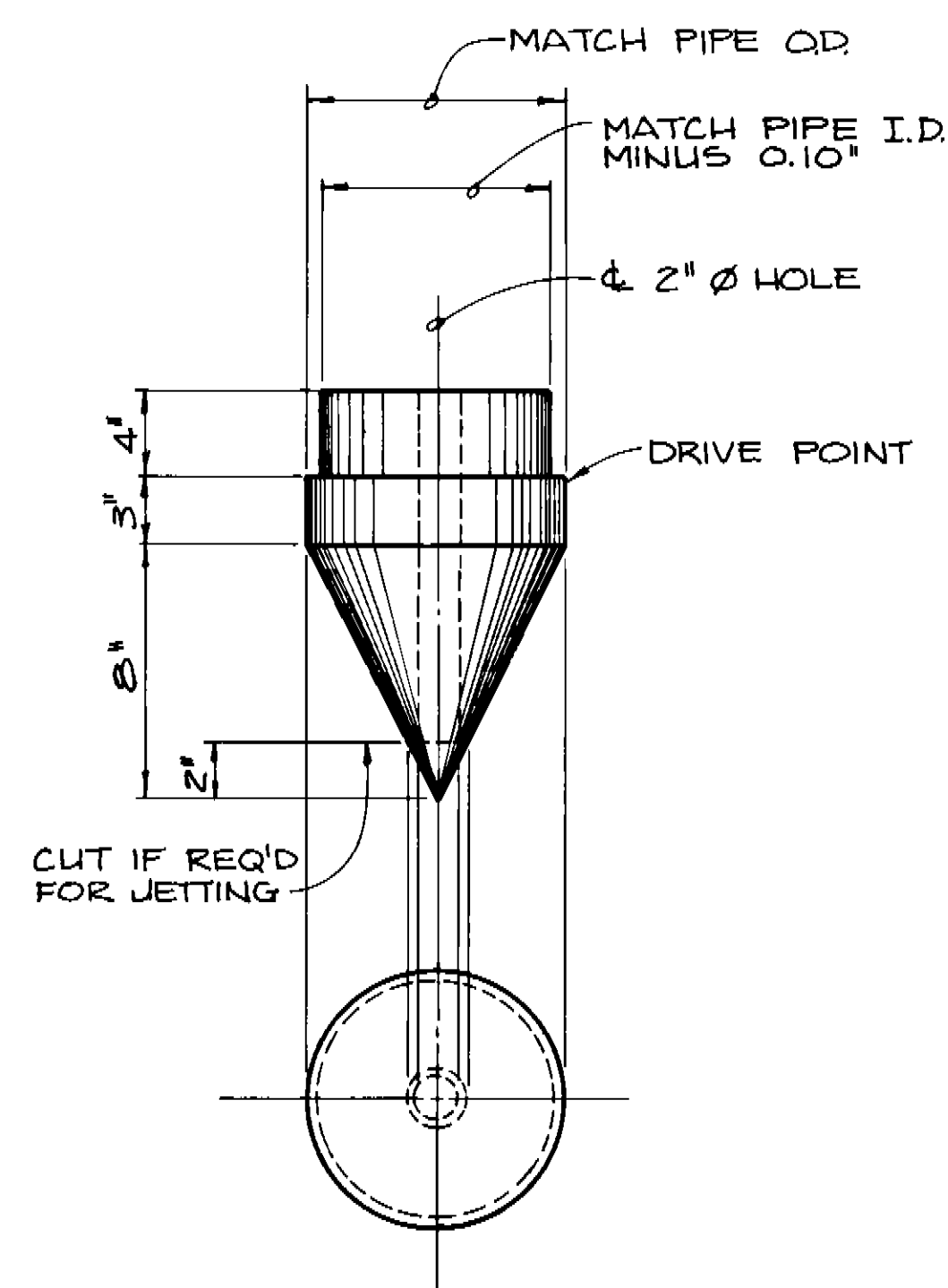
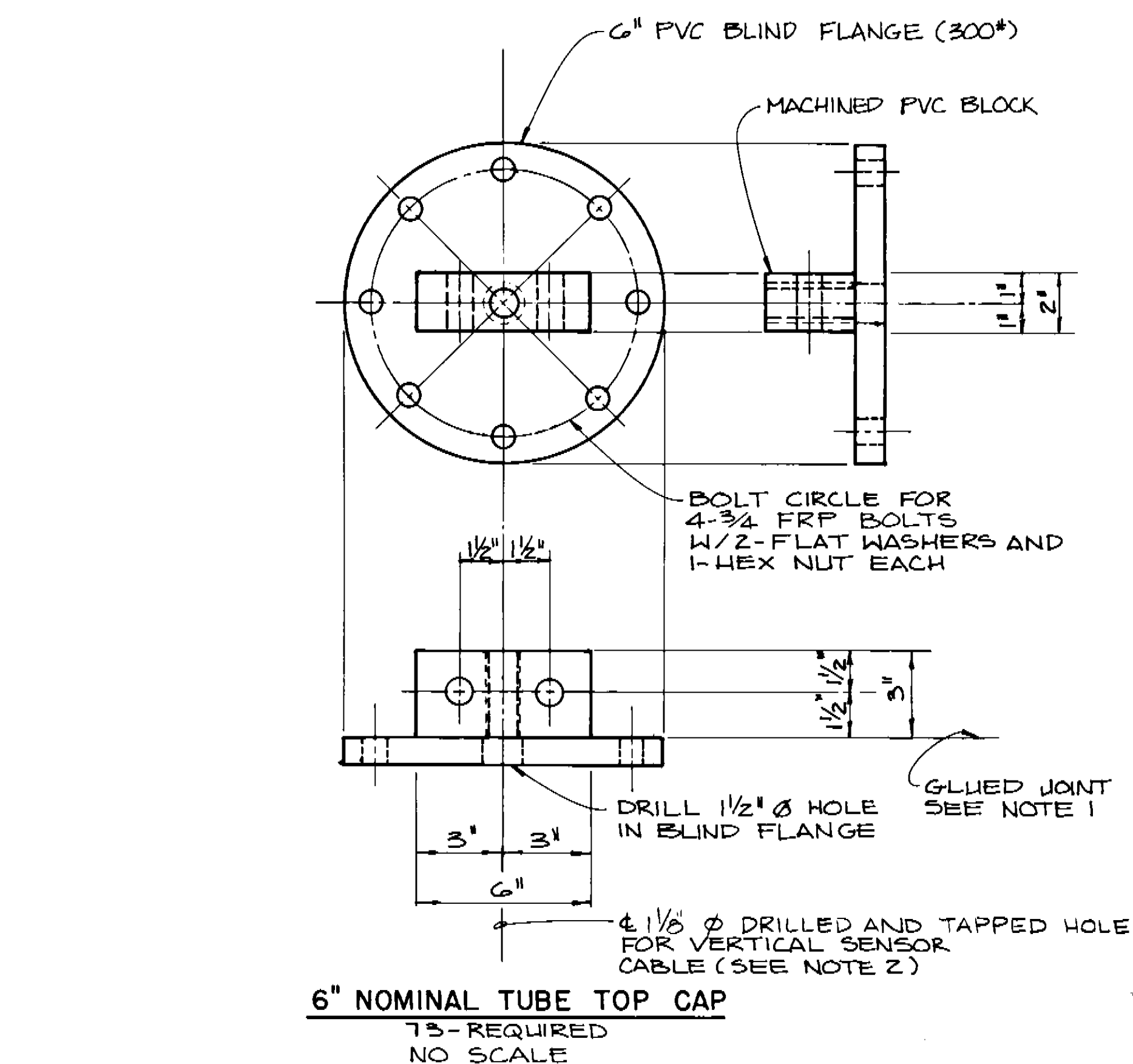


SENSOR TUBE ASSEMBLY
NO SCALE

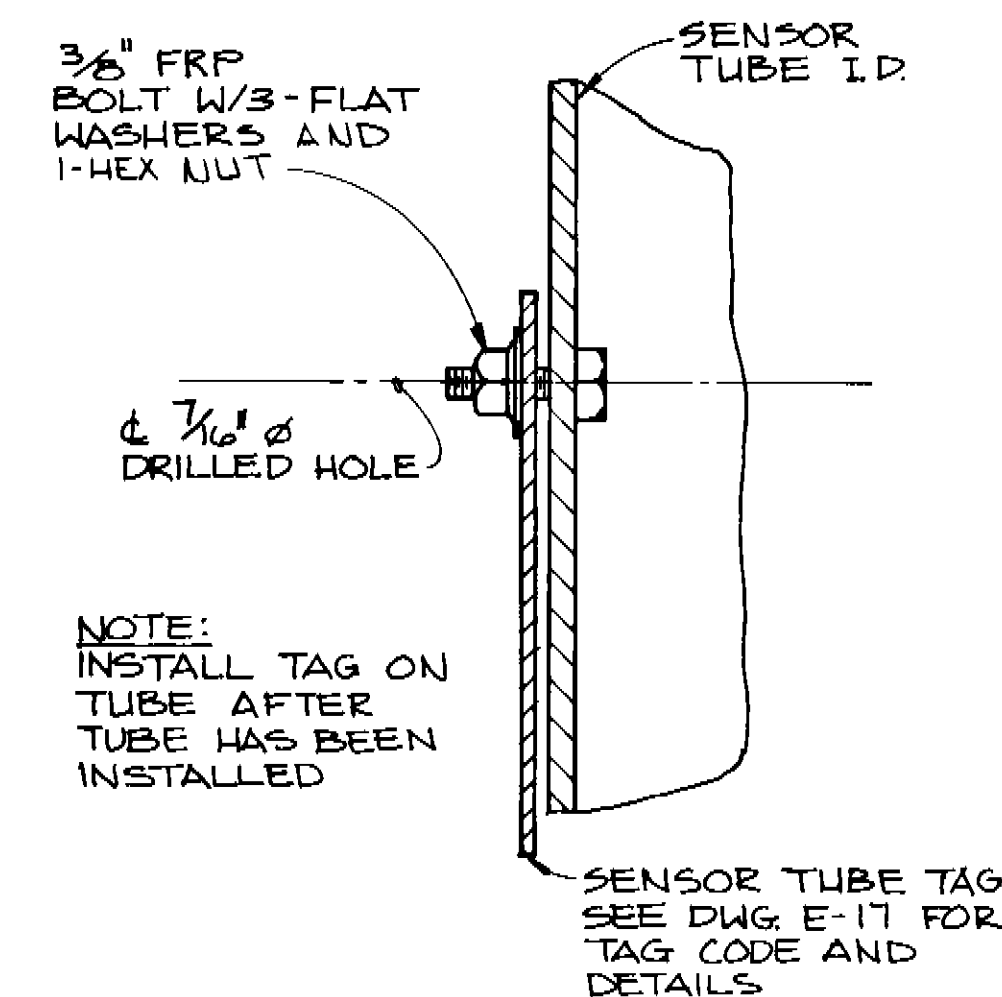


SENSOR TENSION WEIGHT
7/8 REQUIRED
NO SCALE

TUBE NUMBER	LENGTH "L"	INSIDE DIAMETER	THICKNESS RANGE	MATERIAL
M1	12'-0"	6" ± 0.5"	0.2 - 0.35	FRP
M2				
M3				
M4				
M5				
M6				
M7				
M8				
M9				
M10				
M11				
M12				
M13				
M14				
M15				
M16				
M17				
M18				
M19				
M20				
M21				
M22				
M23				
M24				
M25				
M26				
M27				
M28				
M29				
M30				
M31	12'-0"	6" ± 0.5"	0.2 - 0.35	FRP
D1	35'-0"	6" ± 0.5"	0.2 - 0.35	FRP
D2				
D3				
D4				
D5				
D6				
D7				
D8				
D9				
D10				
D11				
D12				
D13				
D14				
D15				
D16				
D17				
D18				
D19				
D20				
D21	35'-0"	6" ± 0.5"	0.2 - 0.35	FRP
T1	12'-0"	8" ± 0.5"	0.2 - 0.35	FRP
T2				
T3				
T4				
T5	12'-0"	8" ± 0.5"	0.2 - 0.35	FRP
S1	15'-0"	6" ± 0.5"	0.2 - 0.35	FRP
S2				
S3				
S4				
S5				
S6				
S7				
S8				
S9				
S10				
S11				
S12				
S13				
S14				
S15				
S16				
S17				
S18				
S19				
S20				
S21	15'-0"	6" ± 0.5"	0.2 - 0.35	FRP



DRIVE POINT DETAIL
NO SCALE



TUBE TAG CONNECTION
7/8 REQUIRED
NO SCALE

- NOTES:
- PVC BLOCK TO BE GLUED TO BLIND FLANGE ACCORDING TO THE FOLLOWING:
 - DRILL HOLE AFTER BLOCK IS CUT IN HALF GRIND EDGES OF HOLE TO PREVENT CABLE DAMAGE AS CAP IS CLAMPED AROUND CABLE.
 - ALL SENSOR TUBES AND FITTINGS ARE TO BE FABRICATED FROM FRP TUBING.
 - ALL BOLTS, NUTS AND WASHERS TO BE FRP.

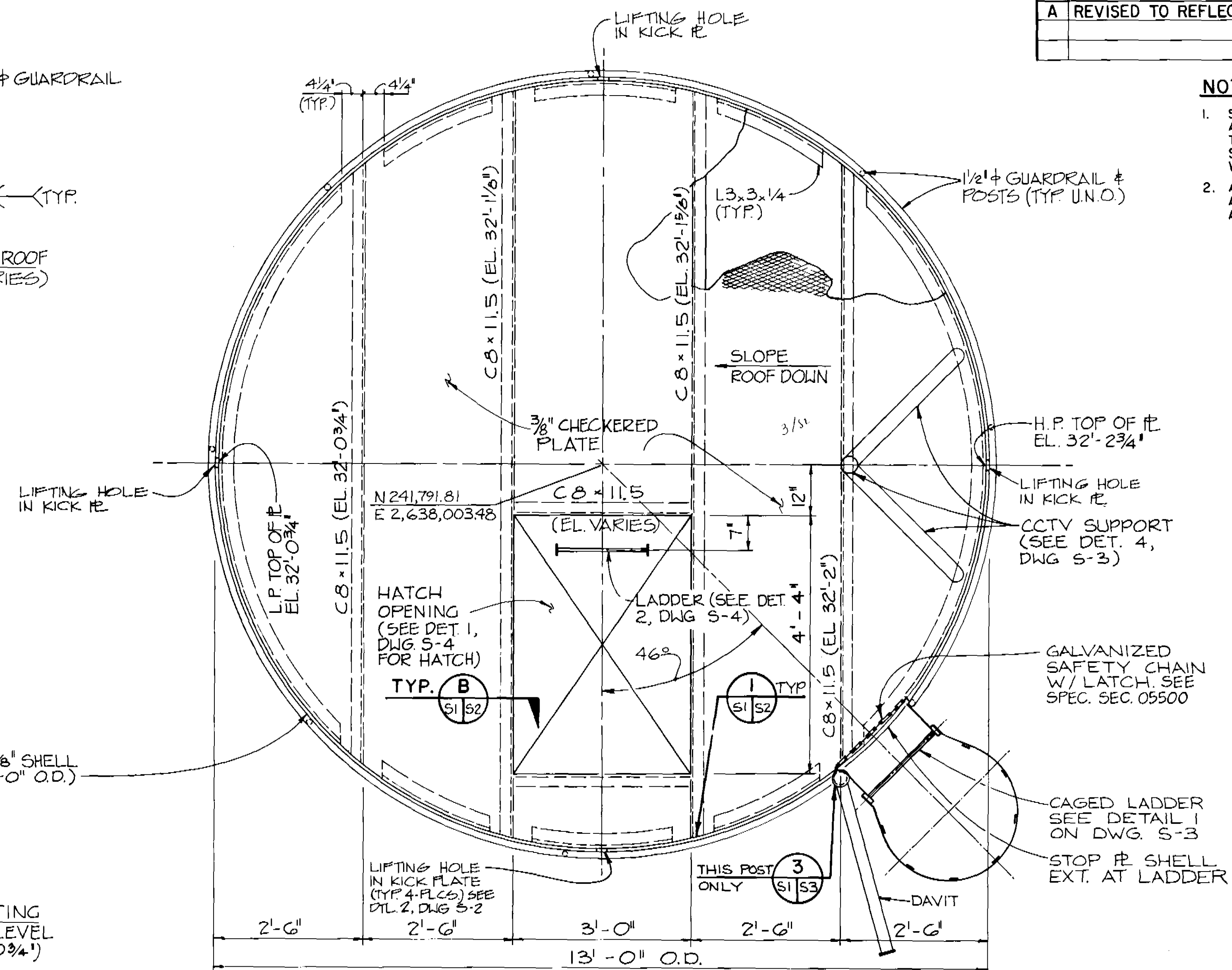
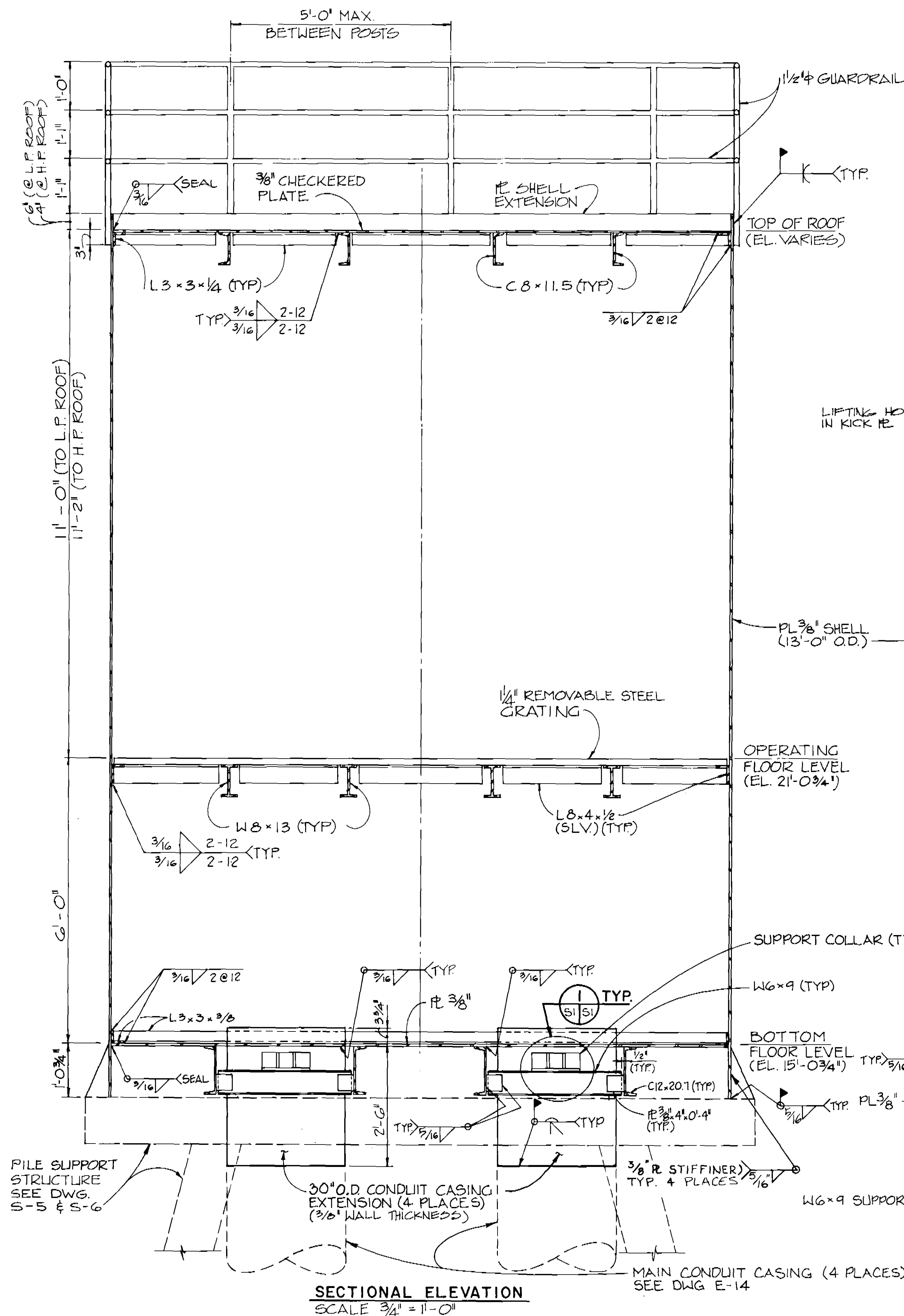
REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED
A	REVISED TO REFLECT AS BUILT CONDITION		

SATISFACTORY TO ROICC DATE COMNAVSEASCOM		APPROVED HEAD, FPO-1 DATE APPROVED DATE EFD FOR COMMANDER, NAVFAC		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C. MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION SENSOR TUBES AND DETAILS	
SIZE D		CODE IDENT. NO. 80091		NAVFAC DRAWING NO. 3020685	
SCALE AS NOTED		CONSTR. CONTR. NO. N62477-89-C-0053		SHEET 12 OF 40	

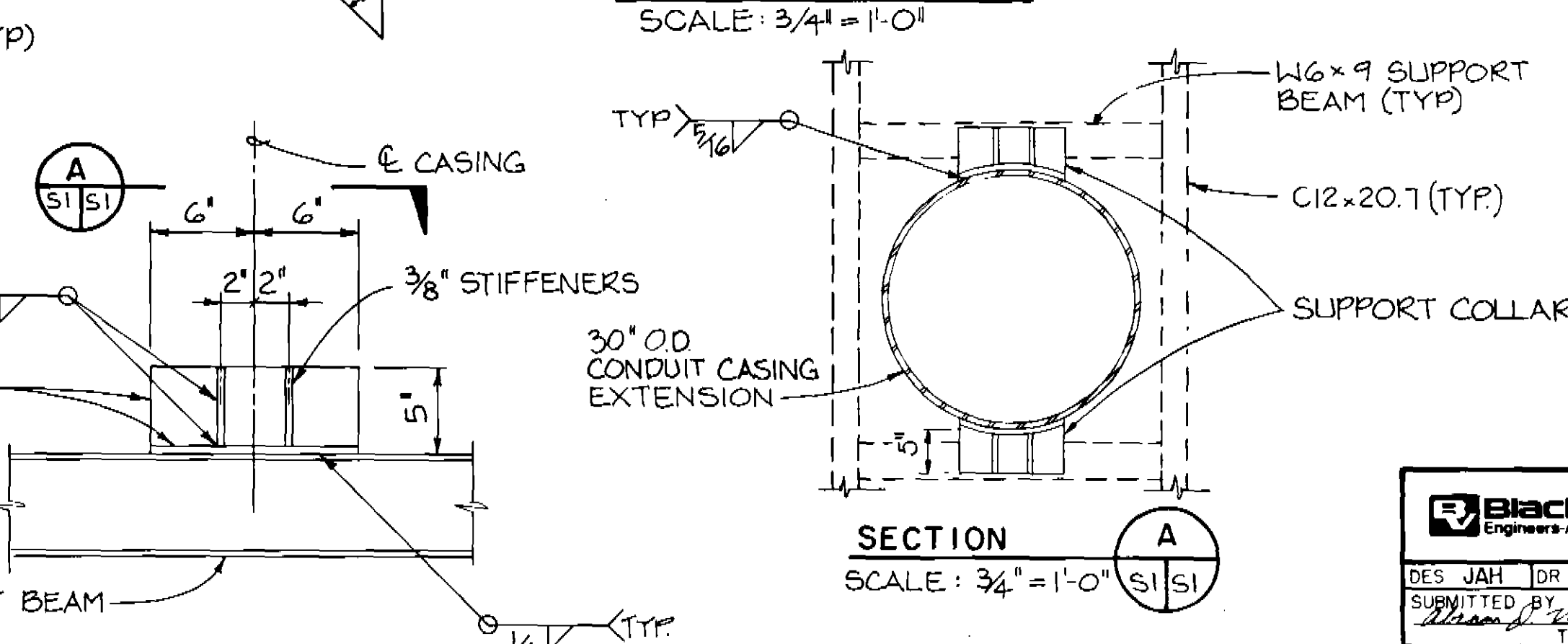
REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED
A	REVISED TO REFLECT AS BUILT CONDITION		

NOTES

- SEE MECHANICAL DRAWINGS FOR SIZES AND LOCATIONS OF DUCT PENETRATIONS THRU WALL OF THE STRUCTURE. CONTRACTOR SHALL COORDINATE LOCATION OF OPENINGS WITH MECHANICAL EQUIPMENT FURNISHED.
- ALL BUTT JOINTS IN THE DCS SHELL, ROOF, AND FLOOR PLATES SHALL BE CONTINUOUS AND FORM A WATERTIGHT JOINT.



ROOF FRAMING PLAN
T.O.S. EL. VARIES
SCALE: 3/4" = 1'-0"

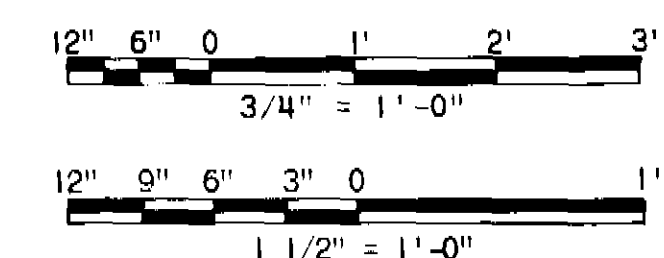


SECTION
SCALE: 3/4" = 1'-0"

DETAIL
SCALE: 1 1/2" = 1'-0"

GRAPHIC SCALE

WARNING: DRAWING MAY BECOME DISTORTED DURING REDUCTION OR OTHER REPRODUCTION PROCESS. USER IS CAUTIONED AGAINST SCALING.

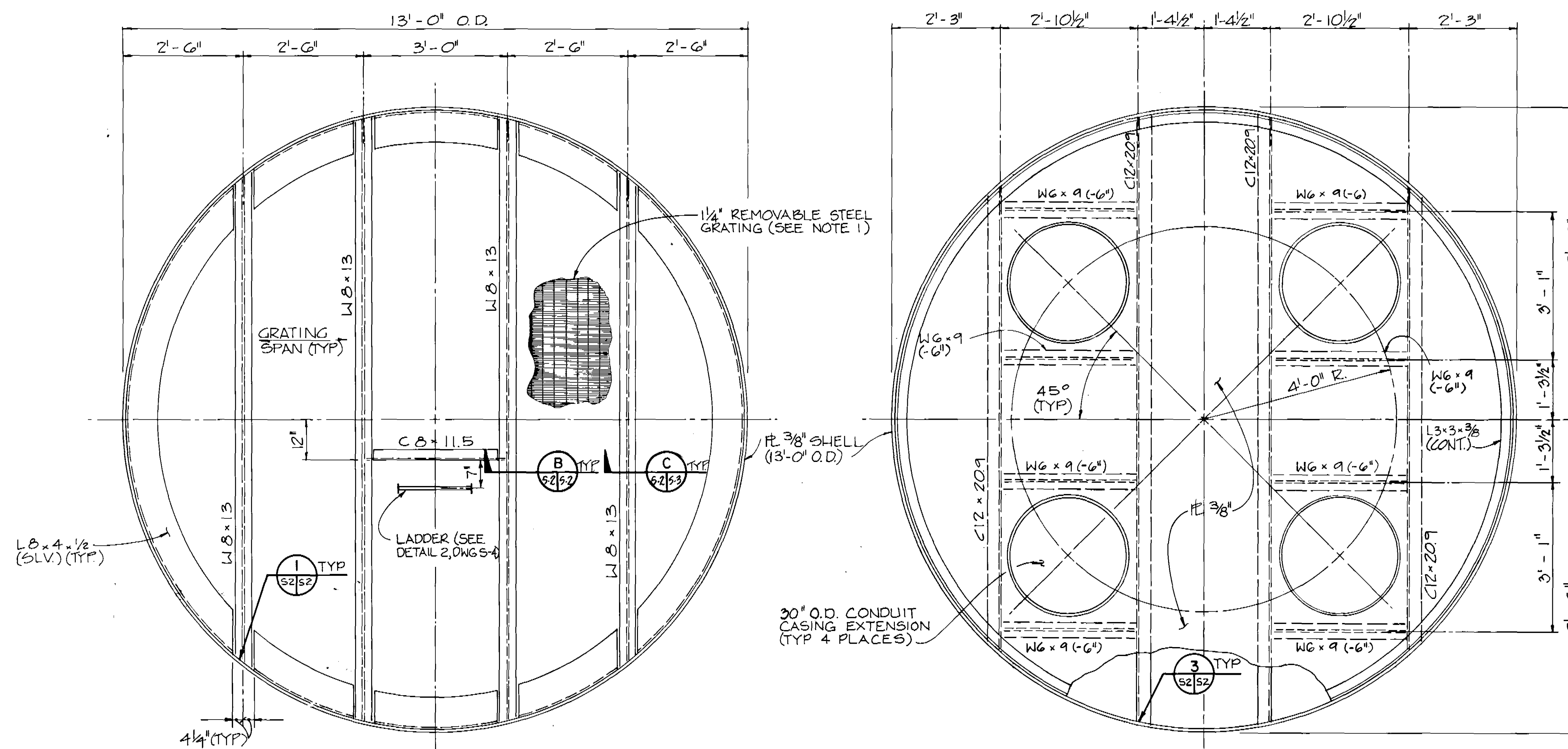


Black & Veatch Engineers-Architects		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DES. JAH DR. GIC CHK. MFV SUBMITTED BY: <i>William C. King</i> DATE: 5-15-83 TITLE: PROGRAM: MBS OCEAN ENGR. & CONST. PROJ. OFF FPO-1 EIC: <i>W. C. King</i> RVD: <i>W. C. King</i> 5/17/83		MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION DATA CONCENTRATOR SHED STRUCTURAL ROOF PLAN & ELEVATION	
SATISFACTORY TO: <i>W. C. King</i> ROICC DATE: 5-15-83 COMNAVSEASYS COM DATE:	APPROVED HEAD FPO-1: <i>W. C. King</i> DATE: 5-15-83 APPROVED: <i>W. C. King</i> DATE: 5/17/83 EFD FOR COMMANDER, NAVFAC	SIZE: D CODE IDENT. NO.: 80091 NAVFAC DRAWING NO.: 3020686 CONSTR. CONTR. NO.: N62477-89-C-0053 SPEC. 21-89-0053 SHEET 13 OF 40	S-1

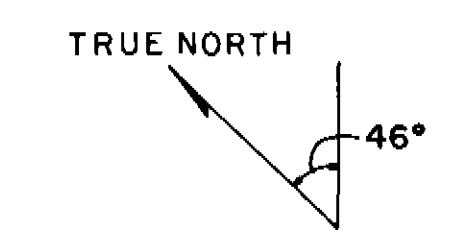
REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED
A	REVISED TO REFLECT AS BUILT CONDITION		

NOTES

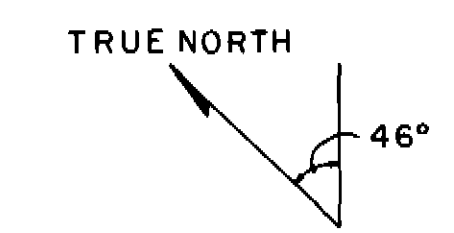
1. THE 1/4" STEEL GRATING AT THE OPERATING FLOOR SHALL BE REMOVABLE AND NOT BE ANY LONGER THAN 3'-0".
2. SUPPLY HOLES IN THE OPERATING FLOOR GRATING AS REQUIRED FOR CABLE ACCESS TO THE ELECTRICAL EQUIPMENT. HOLES SHALL BE FRAMED WITH L3x3x1/4 ANGLES SUPPORTED BY THE W8 BEAMS. HOLES SHALL BE Banded WITH STEEL PLATE TO ELIMINATE SHARP EDGES. LOCATION AND SIZE OF OPENINGS TO BE DETERMINED BY THE CONTRACTOR.



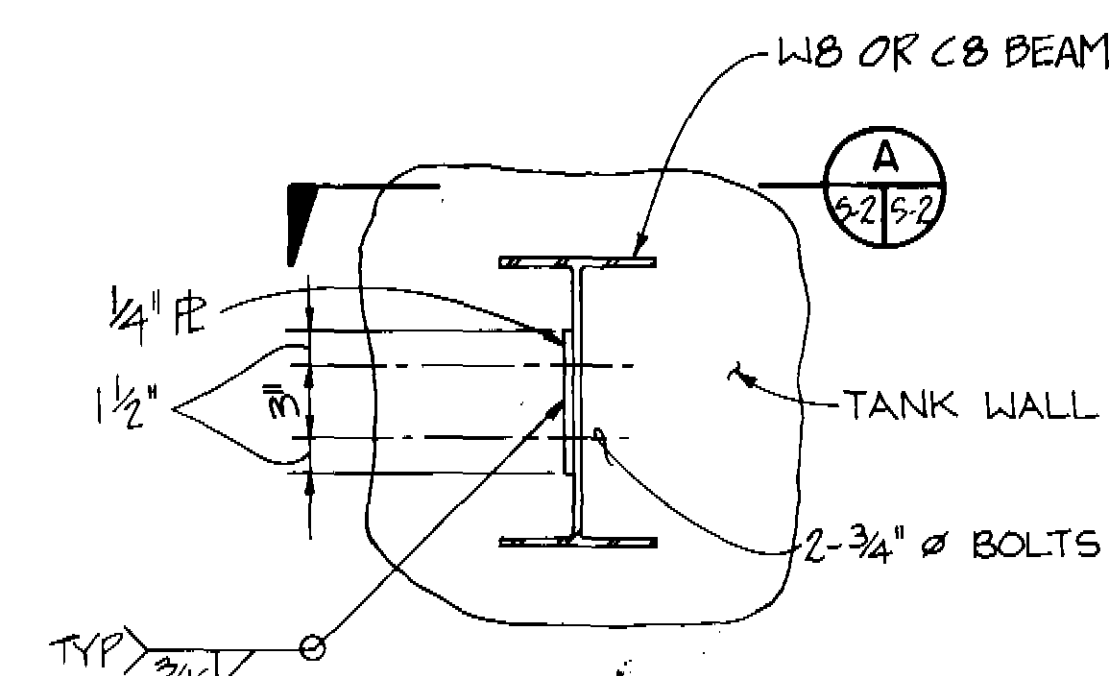
NOTE: DIMENSIONS IN PARENTHESES INDICATE THE DISTANCE THE TOP OF STEEL IS INSTALLED ABOVE OR BELOW THE REFERENCED T.O.S. ELEVATION.



OPERATING FLOOR FRAMING PLAN
T.O.S. EL. 20'-11 3/8"
SCALE: 3/4" = 1'-0"

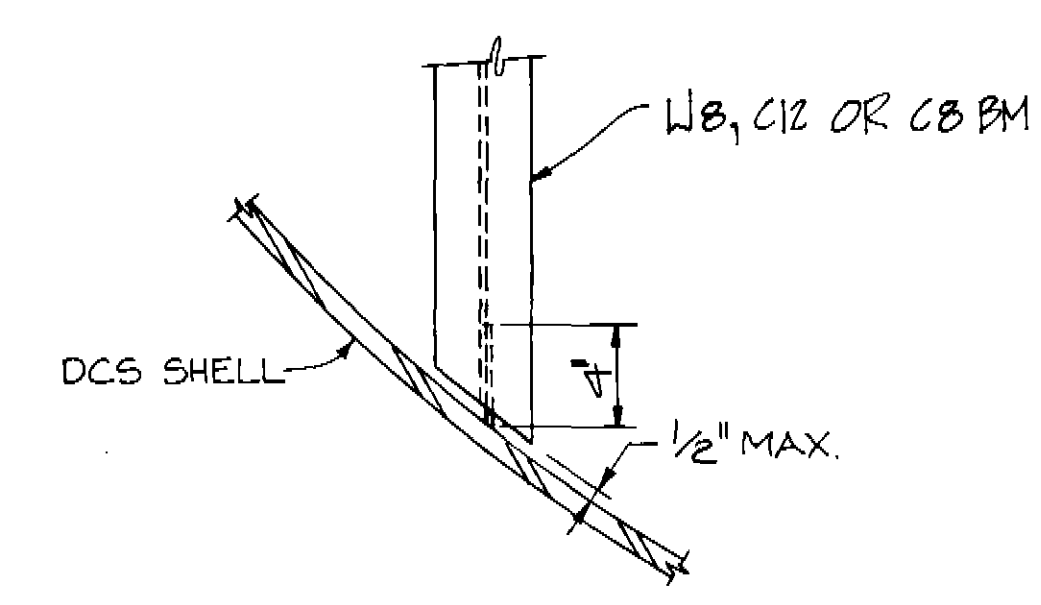


BOTTOM FLOOR FRAMING PLAN
T.O.S. EL. 15'-0 3/8" (U.N.O.)
SCALE: 3/4" = 1'-0"

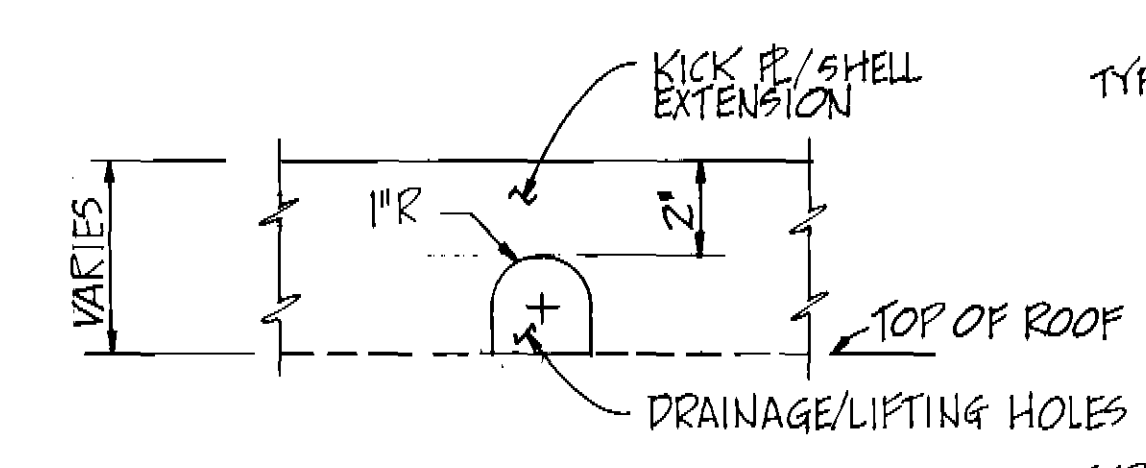


BEAM TO WALL DETAIL

DETAIL 1
NOT TO SCALE 5/1 5/2

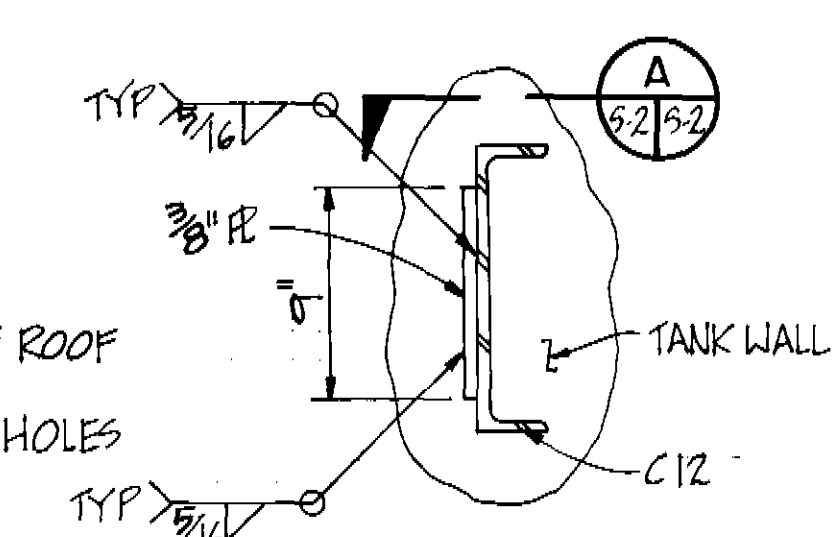


SECTION A
NOT TO SCALE 5/2 5/2



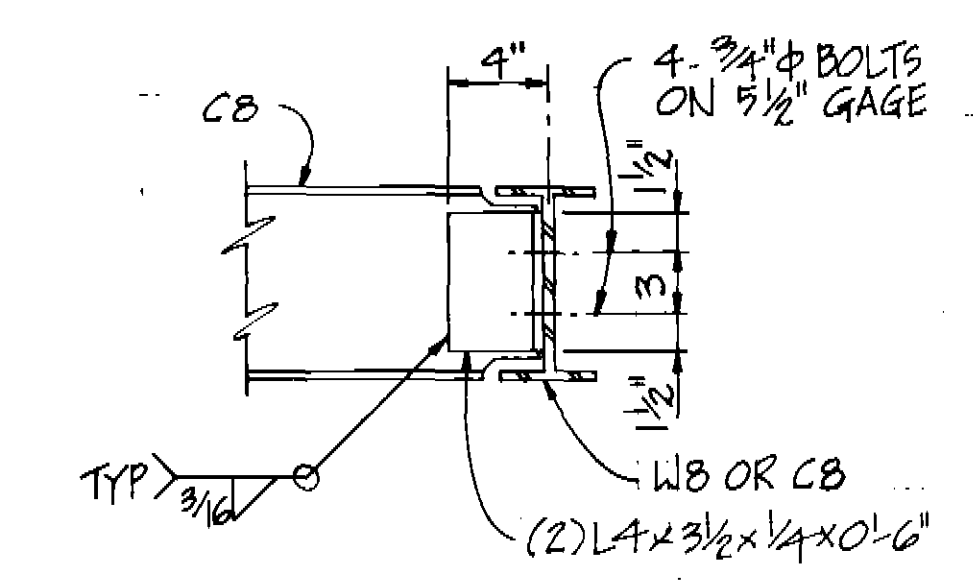
ELEVATION
DRAINAGE/LIFTING HOLE DETAIL

DETAIL 2
NOT TO SCALE 5/1 5/2

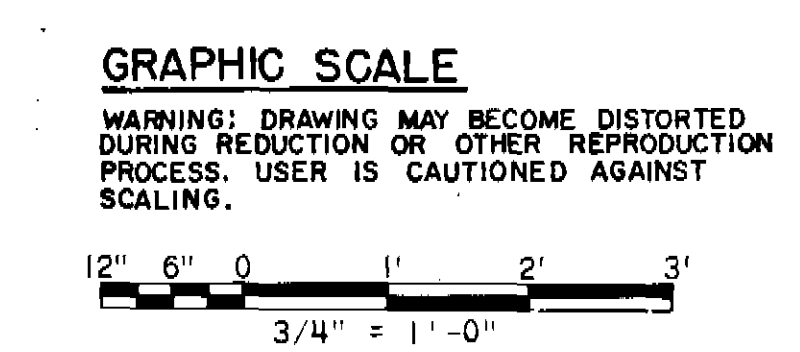


BEAM TO WALL DETAIL

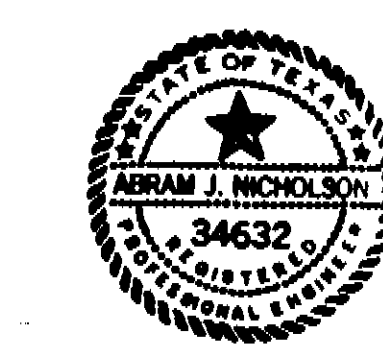
DETAIL 3
NOT TO SCALE 5/2 5/2



SECTION B
NOT TO SCALE 5/1 5/2



GRAPHIC SCALE
WARNING: DRAWING MAY BECOME DISTORTED DURING REDUCTION OR OTHER REPRODUCTION PROCESS. USER IS CAUTIONED AGAINST SCALING.



SATISFACTORY TO	DATE
ROICC	5/1/89
SATISFACTORY TO	DATE
COMNAVSEASYS/COM	DATE

Black & Veatch	DES. JAH DR. GIC CHK. MEV
	SUBMITTED BY DATE
	TITLE PROGRAM MGR
	OCEAN ENGR. A CONST. PROJ. OFF. FPO-1
	EIC 1/16 RVD 5/1/89
APPROVED HEAD FPO-1	DATE
APPROVED	DATE
EFD FOR COMMANDER, NAVFAC	DATE

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND			
CHESAPEAKE DIVISION WASHINGTON, D.C.			
MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION			
DATA CONCENTRATOR SHED STRUCTURAL FLOOR FRAMING PLANS			
SIZE	CODE IDENT NO.	NAVFAC DRAWING NO.	
D	80091	3020687	S-2
CONSTR. CONTR. NO.	N62477-89-C-0053		
SCALE AS SHOWN	SPEC 21-89-0053	SHEET 14 OF 40	

NOTES

 Black & Veatch Engineers-Architects		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DES JAH DR JHC CHK MFV SUBMITTED BY DATE <i>William C. Long</i> <i>2/17/69</i> TITLE PROGRAM MOD OCEAN ENGR & CONST PROJ OF F-P-1 EIC <i>WJH</i> RWD <i>5/10/69</i>		MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION DATA CONCENTRATOR SHED SECTIONS & DETAILS	
APPROVED HEAD FPO-1 DATE <i>William C. Long</i> <i>2-15-69</i> APPROVED DATE <i>R. G. Jones</i> <i>2/1/69</i> EFD FOR COMMANDER, NAVFAC		SIZE CODE IDENT. NO. NAVFAC DRAWING NO. S-3 D 80091 3020688 CONSTR CONTR. NO. N62477-89-C-0053 SCALE AS SHOWN SPEC. 21-89-0053 SHEET 15 OF 40	

NOTES

1. SUPPLY A SIGN ON TOP OF THE MANWAY PLATE WITH 3/4" BLACK LETTERS ON A YELLOW BACKGROUND. SIGN SHALL READ "CAUTION: OPEN HATCH AND TURN ON EXHAUST FAN FOR 15 MINUTES BEFORE ENTERING STRUCTURE. (SWITCH LOCATED INSIDE HATCH)."
2. ALL MATERIAL DESIGNATED AS STAINLESS STEEL (SS) SHALL BE TYPE 316L
3. GASKET MATERIALS SHALL NOT BE ATTACHED UNTIL ALL WELDING IS COMPLETE.



LADDER DETAIL

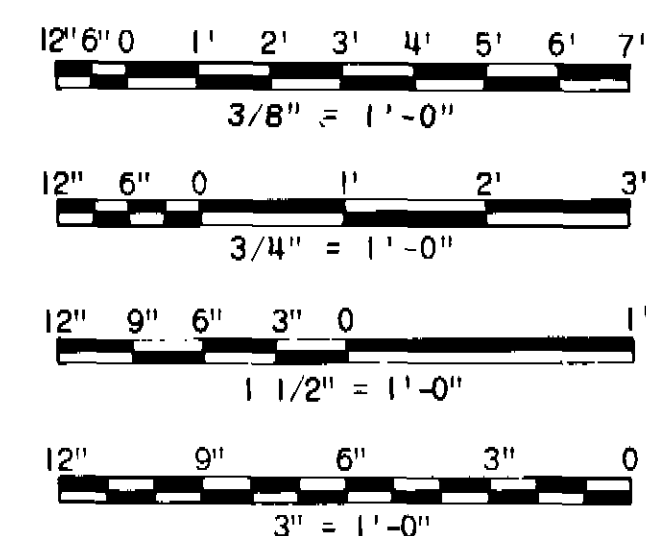
PLAN



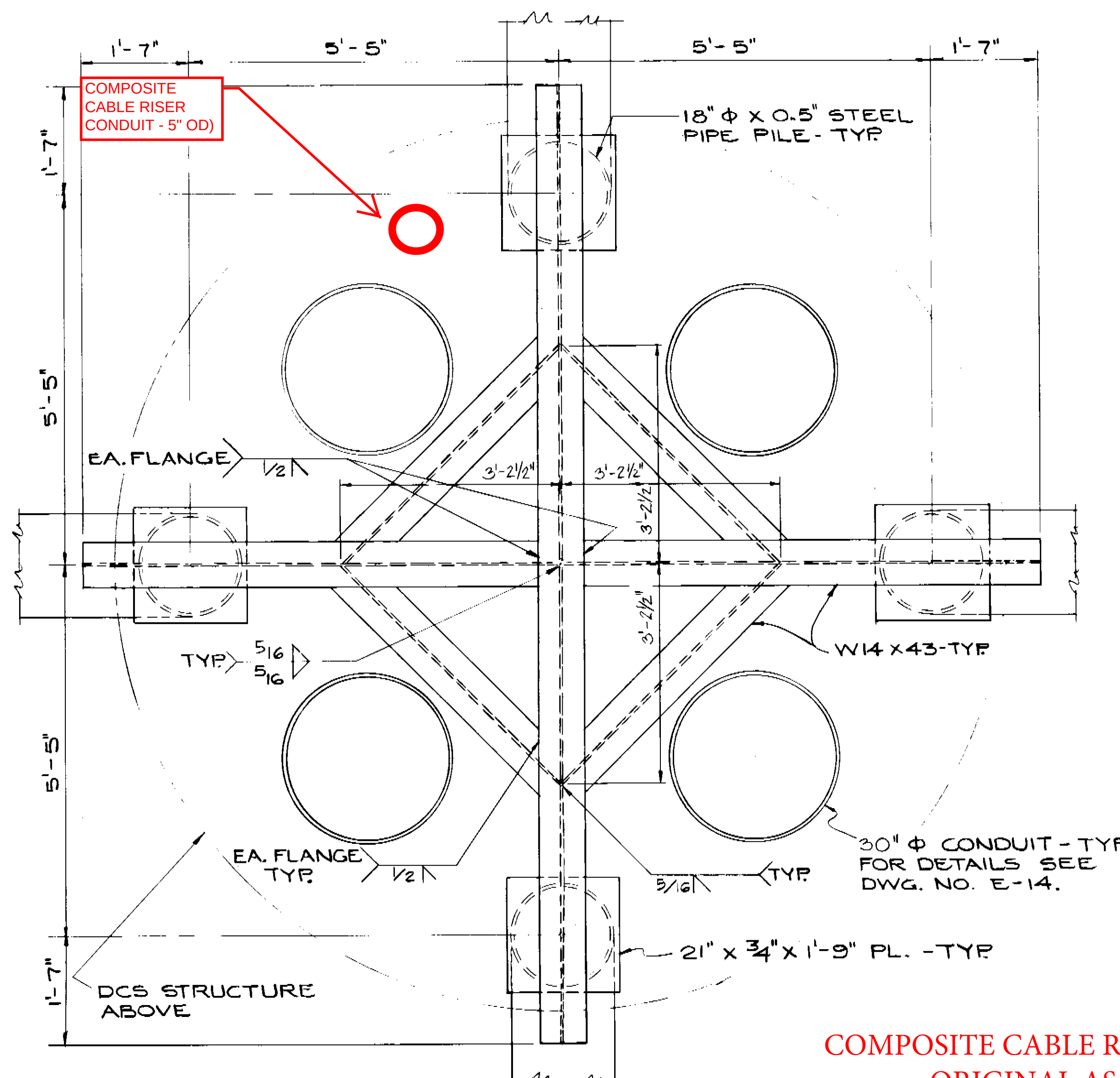
HATCH (SEE DETAIL
1, THIS DWG) —

GRAPHIC SCALE

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PROCESS. USER IS CAUTIONED AGAINST
SCALING.



 Black & Veatch Engineers-Architects	DEPARTMENT OF THE NAVY CHESAPEAKE DIVISION WASHINGTON, D.C.	
	MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION DATA CONCENTRATOR SHED SECTIONS & DETAILS	
	DES JAH DR JHC CHK MFV SUBMITTED BY DATE <i>John J. Veatch</i> 5/22/59 TITLE PROGRAM MFP OCEAN ENGR B CONST PROJ OFF FPO-1 EIC <i>(Signature)</i> RVD <i>(Signature)</i> 5/16/59	
	APPROVED HEAD FPO-1 DATE <i>(Signature)</i> 5-15-59 APPROVED DATE <i>(Signature)</i> 5/15/59 EFD FOR COMMANDER, NAVFAC	
SIZE D	CODE IDENT. NO. 80091	NAVFAC DRAWING NO. 3020689 CONSTR. CONTR. NO. N62477-89-C-0053 SPEC. 21-89-0063 SHEET 16 OF 40
SCALE AS SHOWN		S-4

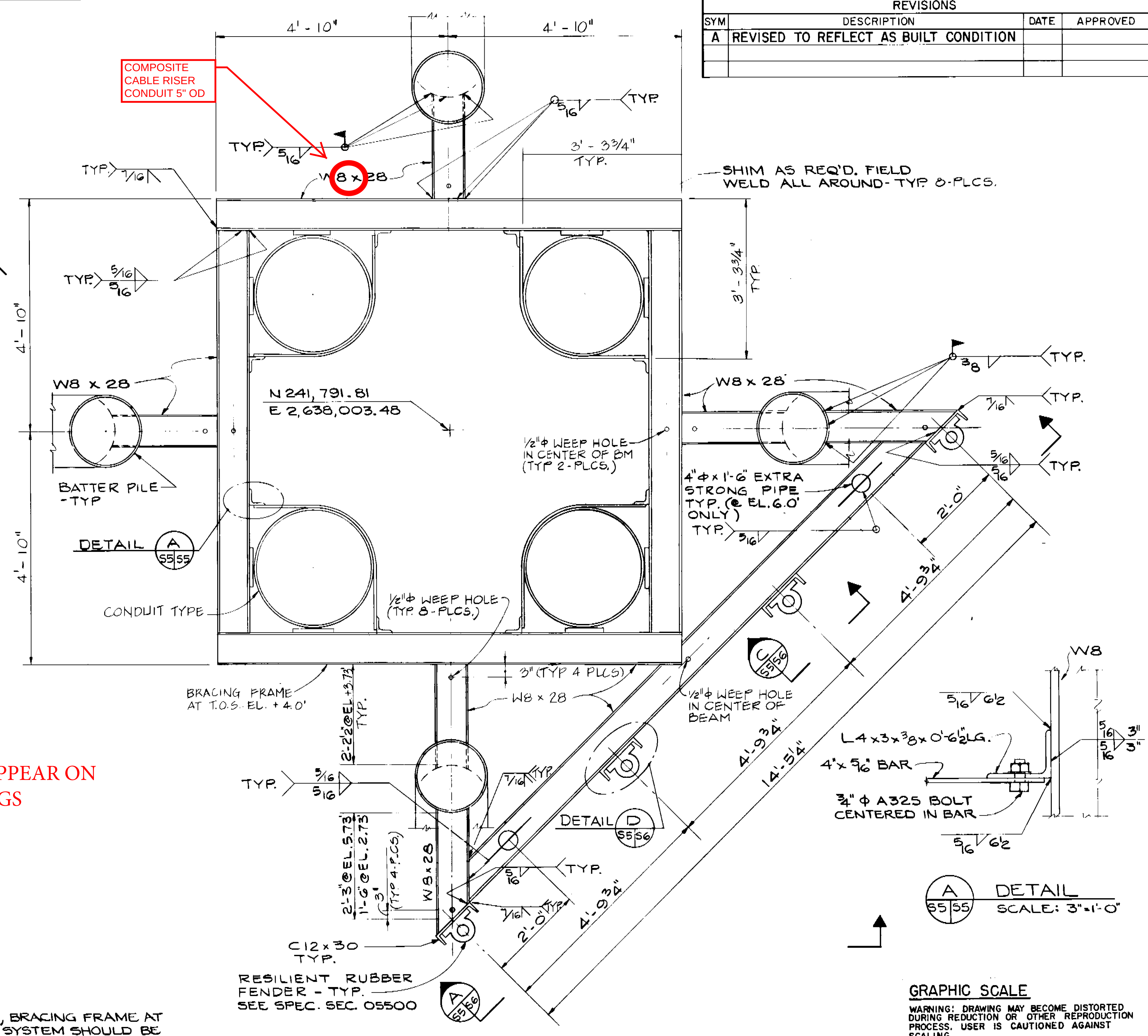
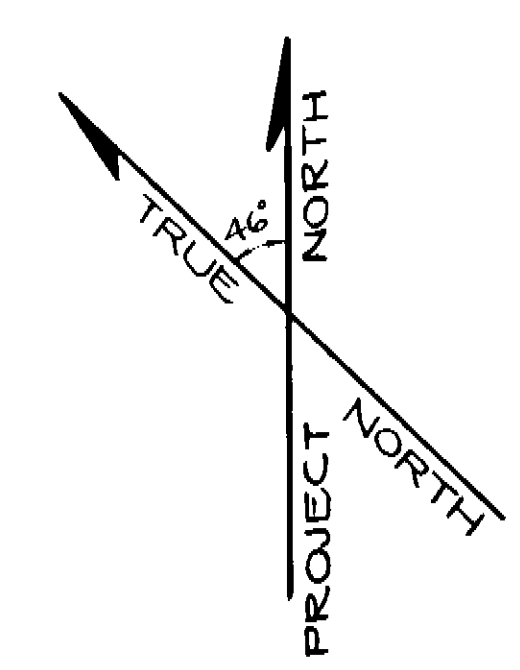


PLAN - AT T.O.S. EL. +14.0'
SCALE: 3/4" = 1'-0"

COMPOSITE CABLE RISER DID NOT APPEAR ON ORIGINAL AS-BUILT DRAWINGS

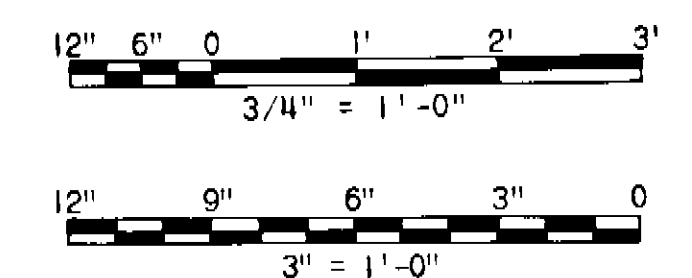
GENERAL NOTES

- FILES ARE API 5L, GRADE B, PLAIN ENDS. SEE SPECIFICATION SECTION 02363.
- ALL STRUCTURAL SHAPES, PLATES ETC. ARE ASTM A36 STEEL. SEE SPECIFICATION SECTION 05120.
- ALL STEEL ELEMENTS OF SUPPORT STRUCTURE SHALL HAVE CORROSION PROTECTION ACCORDING TO THE DESIGN SPECIFICATION SECTION 05915. COATING DAMAGED DURING PILE HANDLING AND INSTALLATION AS WELL AS LOCALLY REMOVED FOR WELDING SHALL BE RESTORED.
- ALL HARDWARE, INCLUDING BOLTS, NUTS, WASHERS AND FLAT BARS SHALL BE HOT-DIP GALVANIZED ACCORDING TO ASTM A153 CLASS C.
- ALL WELDING SHALL BE PERFORMED USING E70 LOW HYDROGEN ELECTRODES IN ACCORDANCE WITH AWS D.1.1.
- TOP SUPPORT FRAME, BRACING FRAME AT EL. 4 FT. AND FENDER SYSTEM SHOULD BE SHOP-FABRICATED AS MUCH AS POSSIBLE. DIMENSIONS FOR FABRICATION ARE TO BE VERIFIED BASED ON THE SURVEY OF THE ACTUAL POSITION OF DRIVEN PILES.
- CAST-IN-PLACE REINFORCED CONCRETE SHALL HAVE A MINIMUM 28 DAY COMPRESSIVE STRENGTH f'_c OF 3000 PSI.
- REINFORCING STEEL SHALL CONFORM TO THE REQUIREMENTS OF ASTM A615 GRADE 60.
- DETAILS OF CONNECTION BETWEEN THE DCS AND SUPPORT STRUCTURES ARE SHOWN ON DWG. NO. S-1



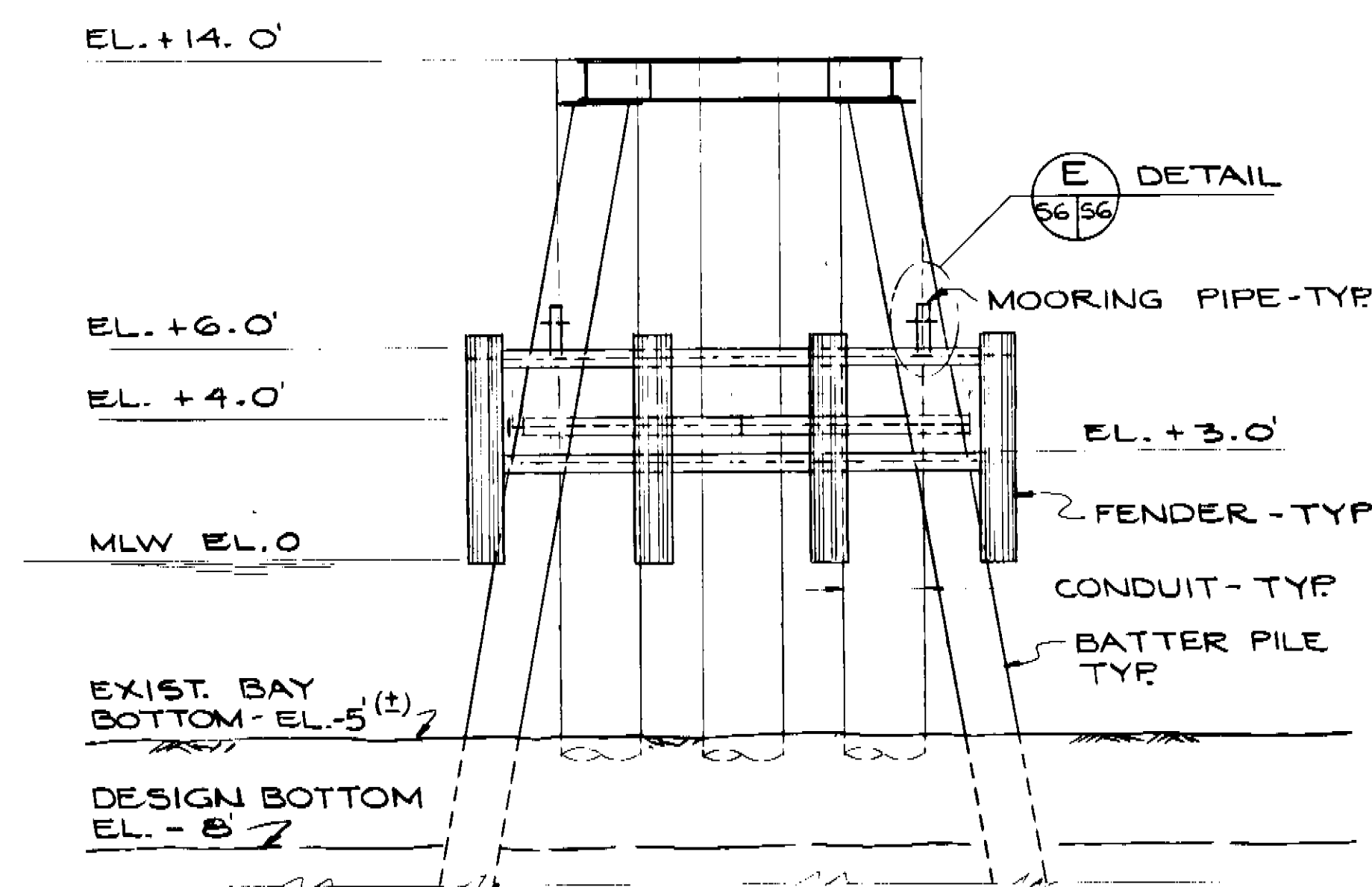
PLAN - AT T.O.S. EL. +6.0' & +3.0' (U.N.O.)
SCALE: 3/4" = 1'-0"

GRAPHIC SCALE
WARNING: DRAWING MAY BECOME DISTORTED DURING REDUCTION OR OTHER REPRODUCTION PROCESS. USER IS CAUTIONED AGAINST SCALING.



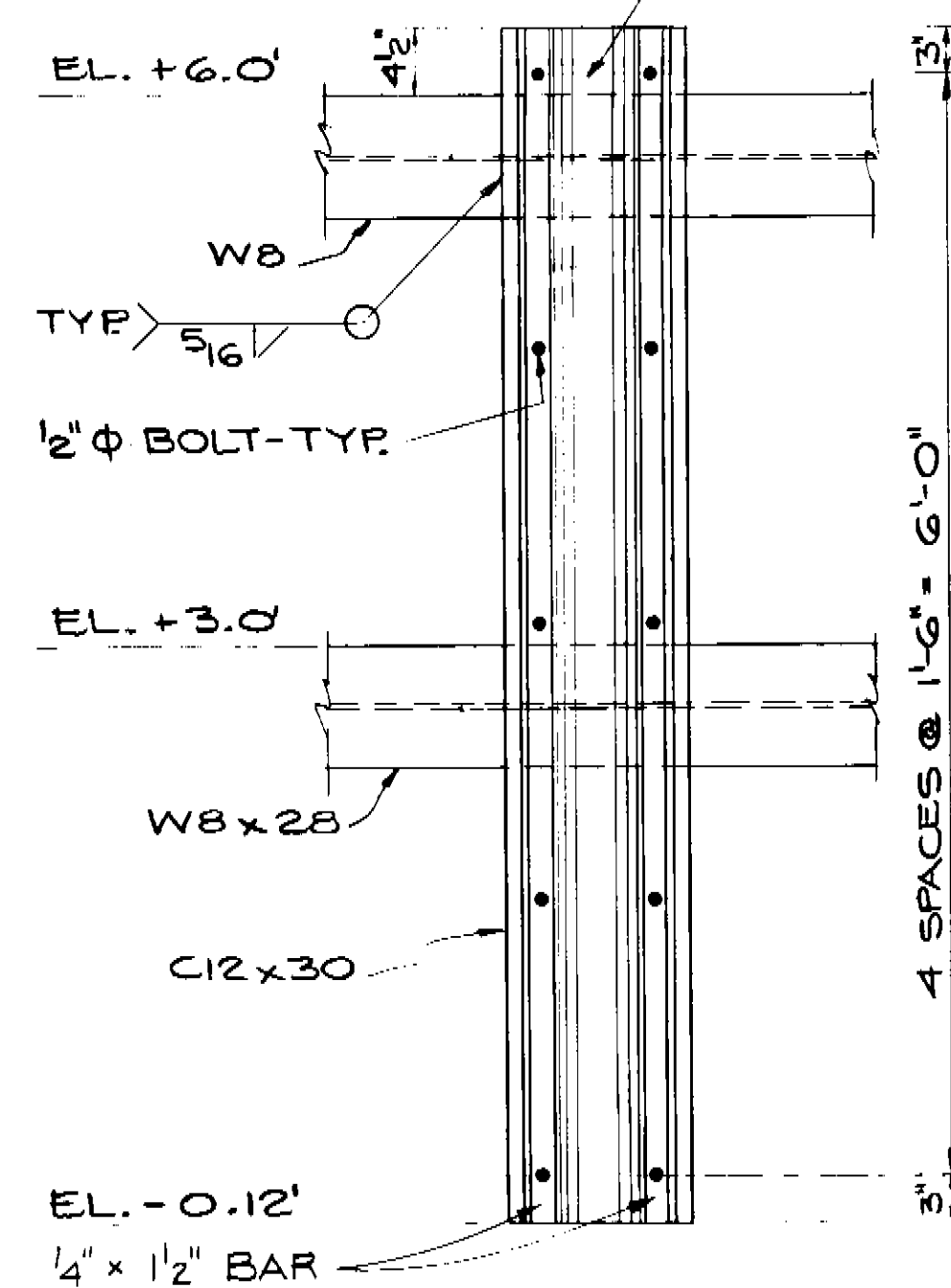
MUESER RUTLEDGE CONSULTING ENGINEERS SATISFACTORY TO ROICC DATE 5/1/99 SATISFACTORY TO EFD FOR COMMANDER, NAVFAC DATE		Black & Veatch DES A.A. DR W.R. CHK V.B. SUBMITTED BY DATE 5-15-99 TITLE: PROGRAM MGR OCEAN ENGR. & CONST. PROJ. OFF FPO-1 EIC. WJD RVD 5/1/99		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C. MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION DATA CONCENTRATOR SHED SUPPORT STRUCTURE PLANS AND DETAIL	
SIZE CODE IDENT. NO. NAVFAC DRAWING NO. D 80091 3020690		CONSTR. CONTR. NO. N62477-89-C-0053 SPEC. 21-89-0053 SHEET 17 OF 40		S-5	

REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED
A	REVISED TO REFLECT AS BUILT CONDITION		

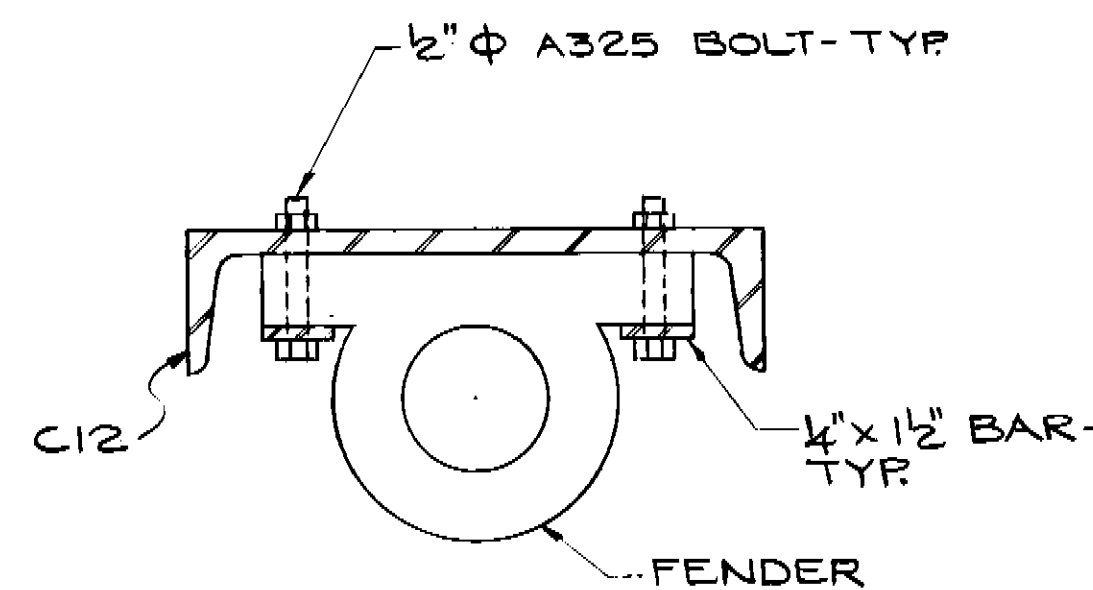


A ELEVATION
SCALE: 1/4"=1'-0"

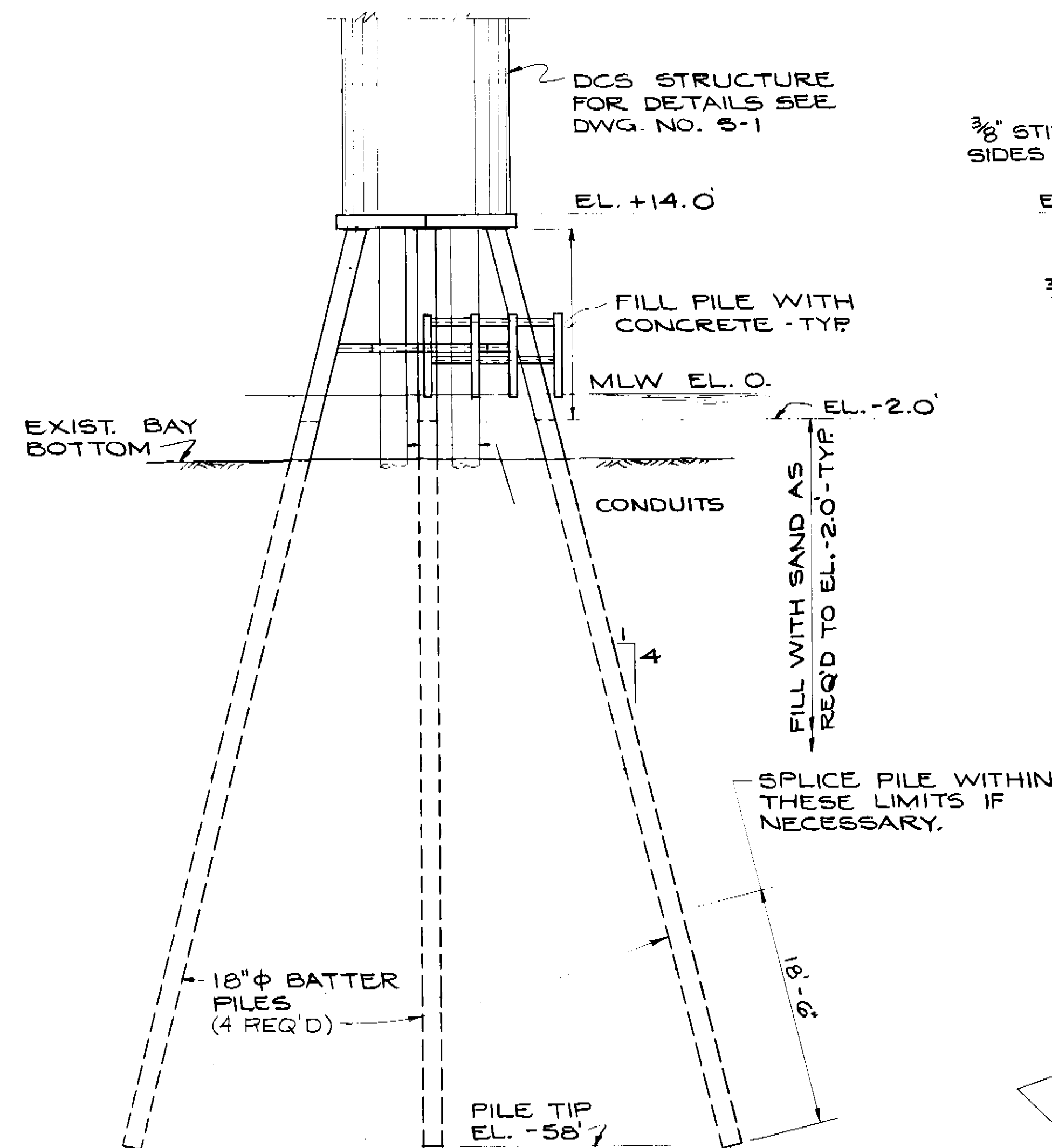
6"x9" WING-TYPE RUBBER FENDER SEE SPEC. SEC. 05300



C ELEVATION
SCALE: 1"=1'-0"



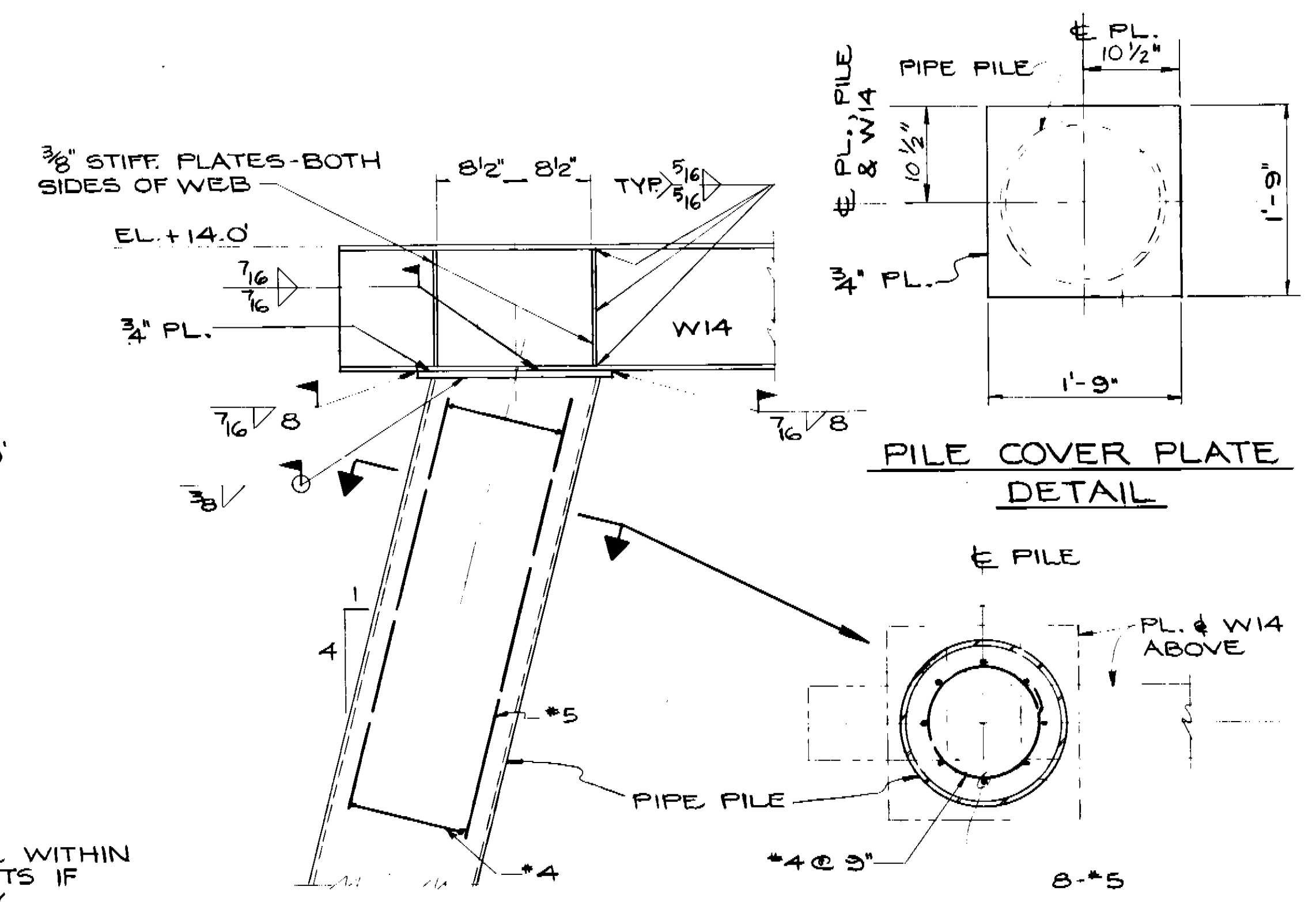
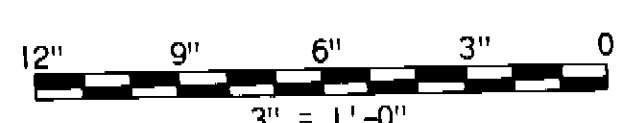
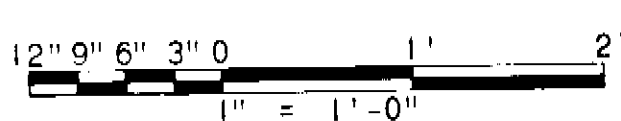
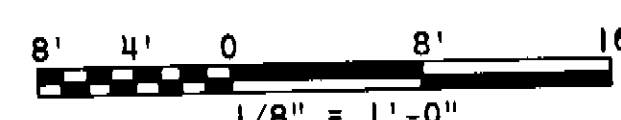
D DETAIL
SCALE: 3"=1'-0"



B ELEVATION
SCALE: 1/8"=1'-0"

GRAPHIC SCALE

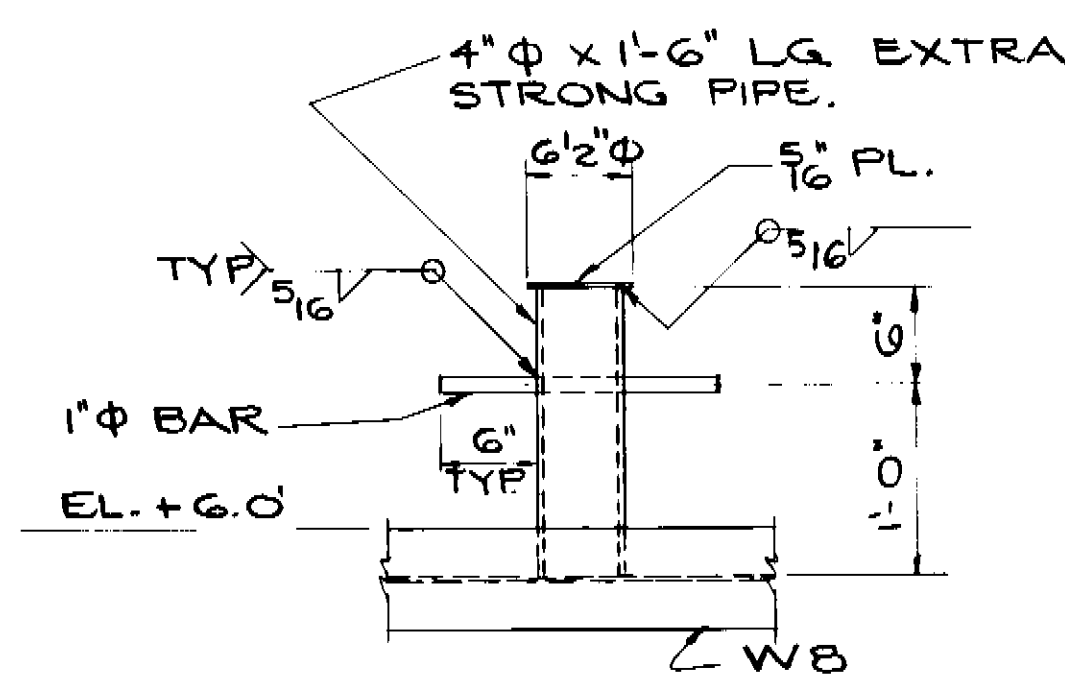
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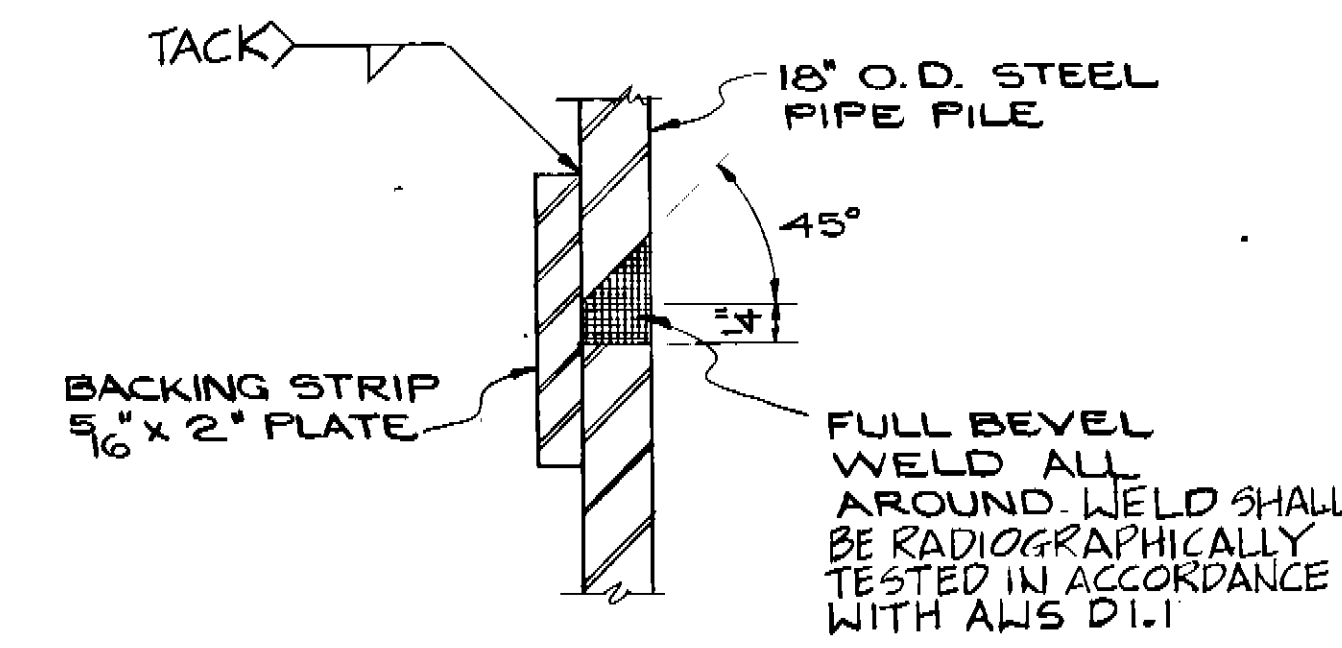
E ELEVATION

SECTION

DETAIL AT PILE HEAD
SCALE: 1"=1'-0"



E DETAIL
SCALE: 1"=1'-0"



PILE SPLICE DETAIL
N.T.S.



MUESER RUTLEDGE CONSULTING ENGINEERS	
SATISFACTORY TO	DATE
ROICC	5/13/89
SATISFACTORY TO	DATE
COMNAVSEASYS/COM	DATE

Black & Veatch Engineers-Architects		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DES. AA	DR. WR	CHK. VB	
SUBMITTED BY	DATE	TITLE: PROGRAM MGR	
OCEAN ENGR. & CONST. PROJ. OFF. FPO-1	DATE	EIC	5/13/89
APPROVED HEAD FPO-1	DATE	APPROVED	DATE
5-15-89		5/13/89	
EFD FOR COMMANDER, NAVFAC			
SIZE	CODE	IDENT. NO.	NAVFAC DRAWING NO.
D	80091		3020691
SCALE AS SHOWN		SPEC. 21-89-0053	SHEET 18 OF 40

W14

NOTES

- ABBREVIATIONS:

EQUIPMENT AND LIGHTING PLAN

SCALE: 3/4"=1'-0"

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H      -HUMIDIFIER PUMP
HAR    -HUMIDIFICATION AUXILIARY RELAY
HCV    -HALON CONTROL VALVE
HU     -HUMIDISTAT
IG     -ISOLATED GROUND
IN     -ISOLATED NEUTRAL
JB     -JUNCTION BOX
NFGS   -NAVAL FACILITIES GUIDE SPECIFICATIONS
N.T.S. -NOT TO SCALE
P      -POLE
PC     -POWER CONDITIONER
PH     -PHASE
PP1    -POWER PANEL FOR UTILITIES
PP2    -POWER PANEL FOR DATA
PS     -PHOTOCELL SWITCH
RFI    -RADIO FREQUENCY INTERFERENCE
SA     -SURGE ARRESTER
SD     -SMOKE DETECTOR
SHLD   -SHIELDED (CABLE)
SW     -SWITCH
TH     -THERMOSTAT
TR     -TRANSFORMER
TYP    -TYPICAL
V      -VOLT
XFMR   -TRANSFORMER

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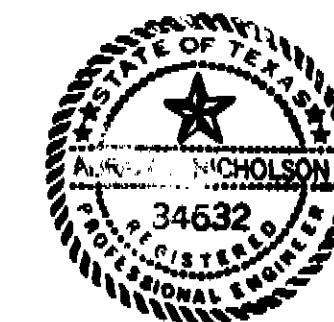
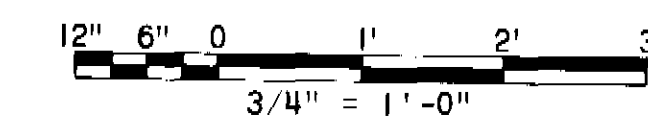
POWER AND RACEWAY PLAN

SCALE: 3/4"=1'-0"

LEGEND:

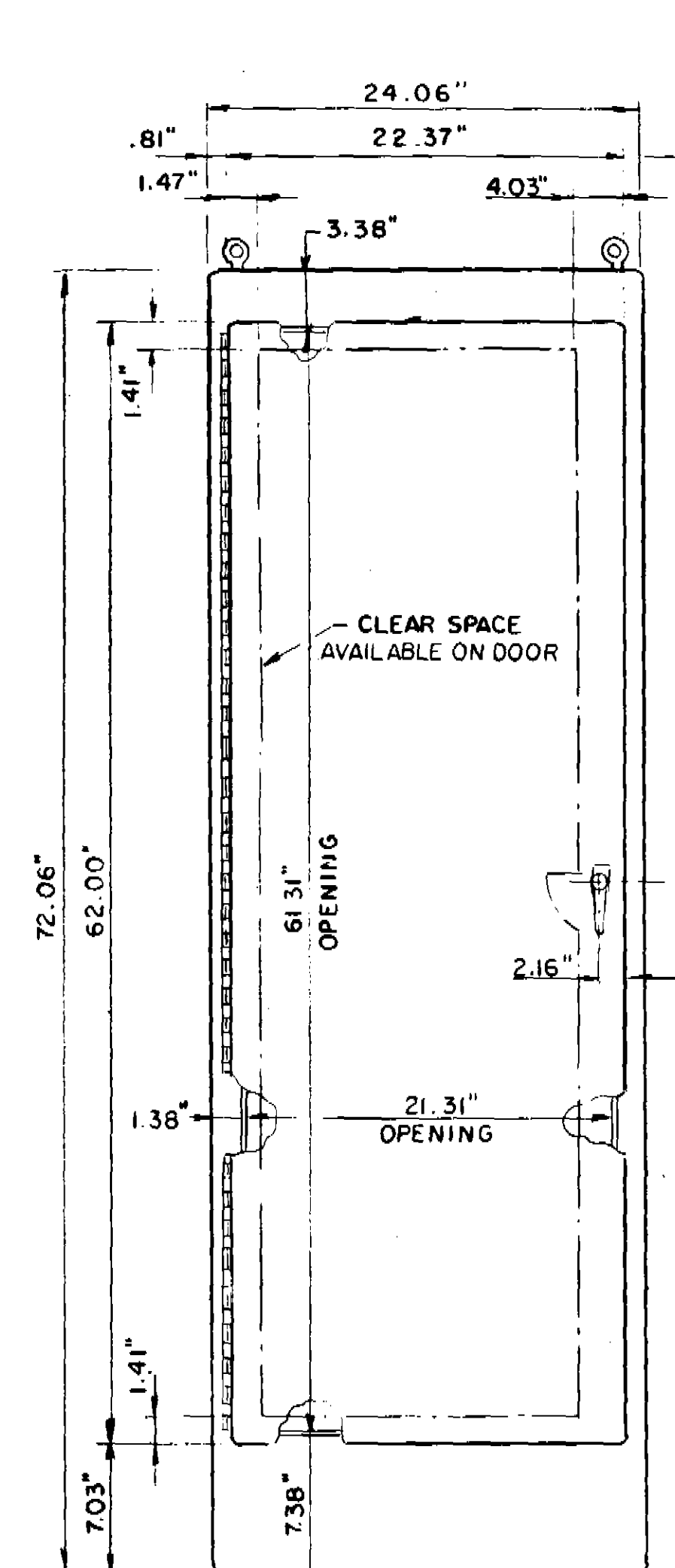
- GRAPHIC SCALE

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PROCESS. USER IS CAUTIONED AGAINST
SCALING.

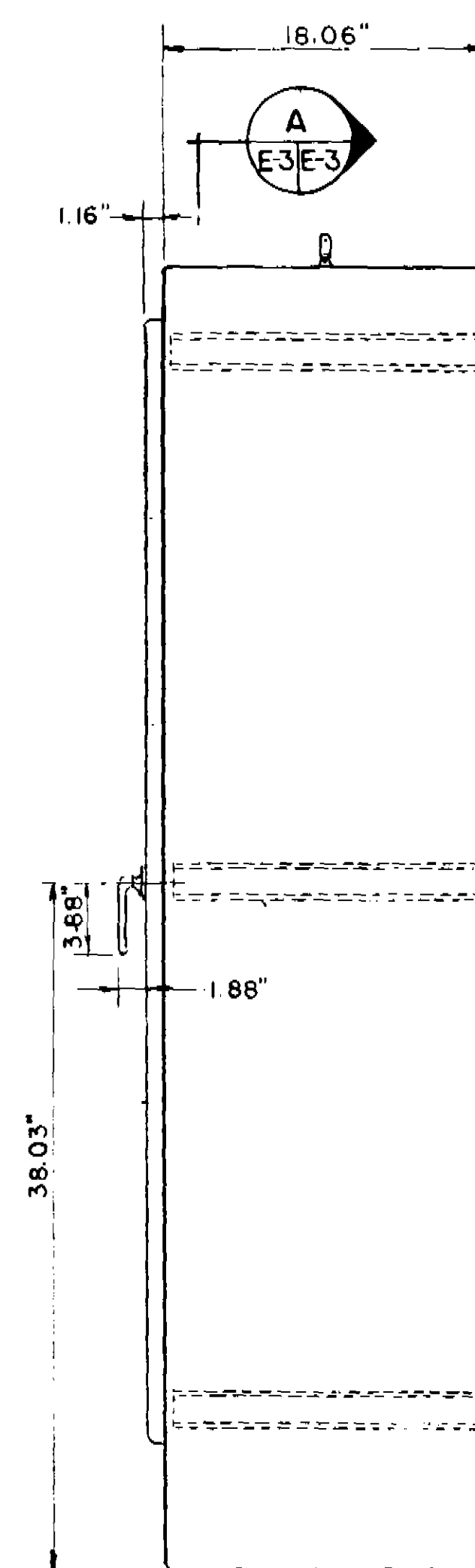


 Black & Veatch Engineers-Architects	DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND	
	CHESAPEAKE DIVISION WASHINGTON, D.C.	
	MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION EQUIPMENT FLOOR LEVEL PLANS	
DES RM <u>DR D1G</u> CHK <u>HRA</u> SUBMITTED BY <u>William C. Tucker</u> DATE <u>02/07/89</u> TITLE PROGRAM MGR OCEAN ENGR & CONST PROJ. OFF FPO-1 EIC <u>CJG</u> <u>PI</u> RVD <u>TIP</u> <u>5/6/89</u>	SIZE CODE IDENT. NO. NAVFAC DRAWING NO. E-I D 80091 3020694 CONSTR. CONTR. NO. N62477-89-C-0053	
APPROVED HEAD FPO-1 DATE <u>5-15-89</u> <u>William C. Tucker</u> APPROVED DATE <u>8/7/89</u> EFF FOR COMMANDER, NAVFAC	SCALE: AS SHOWN SPEC. 21-89-0053 SHEET 21 OF 40	

REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED
A	REVISED TO REFLECT AS BUILT CONDITION		

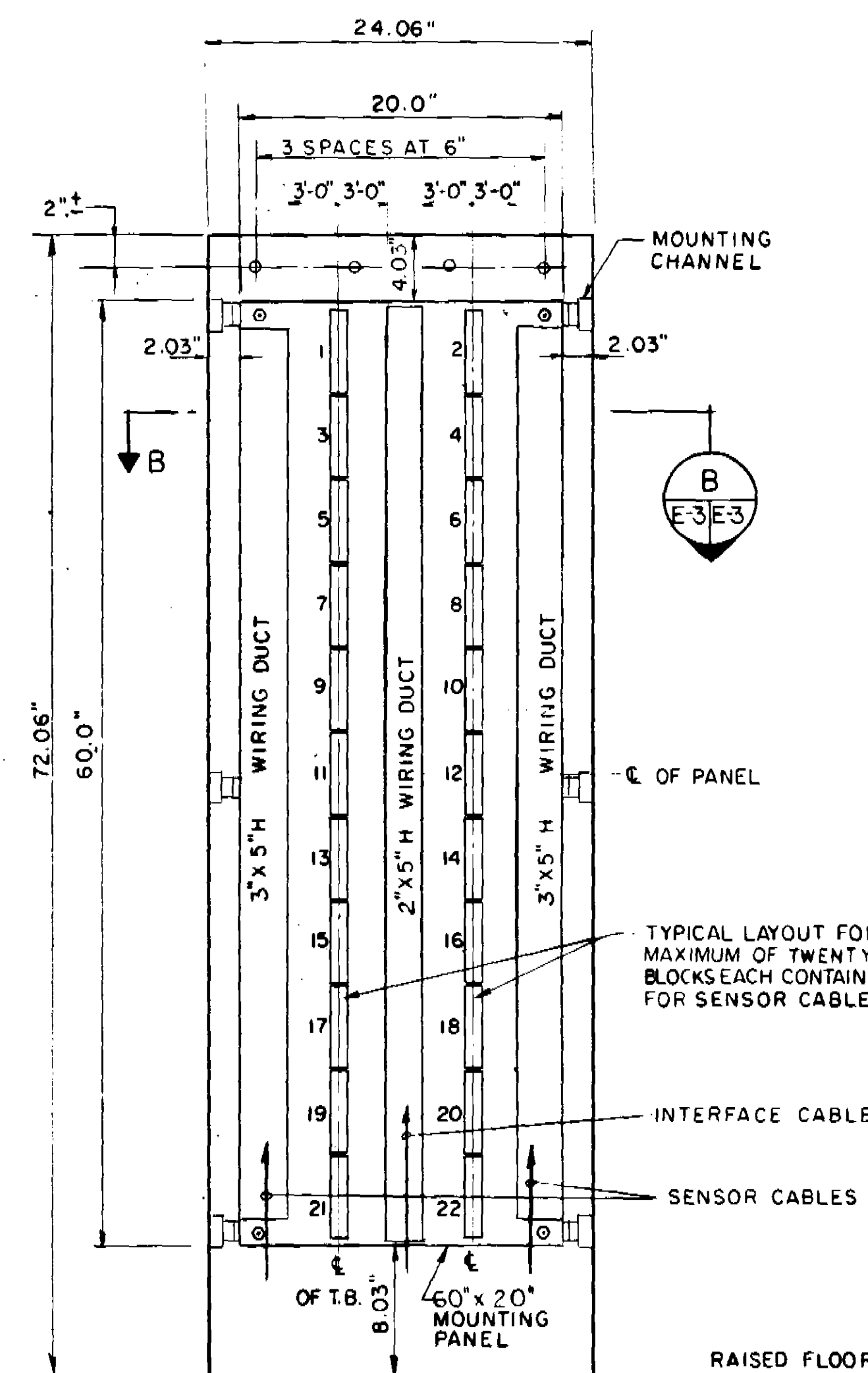


FRONT VIEW

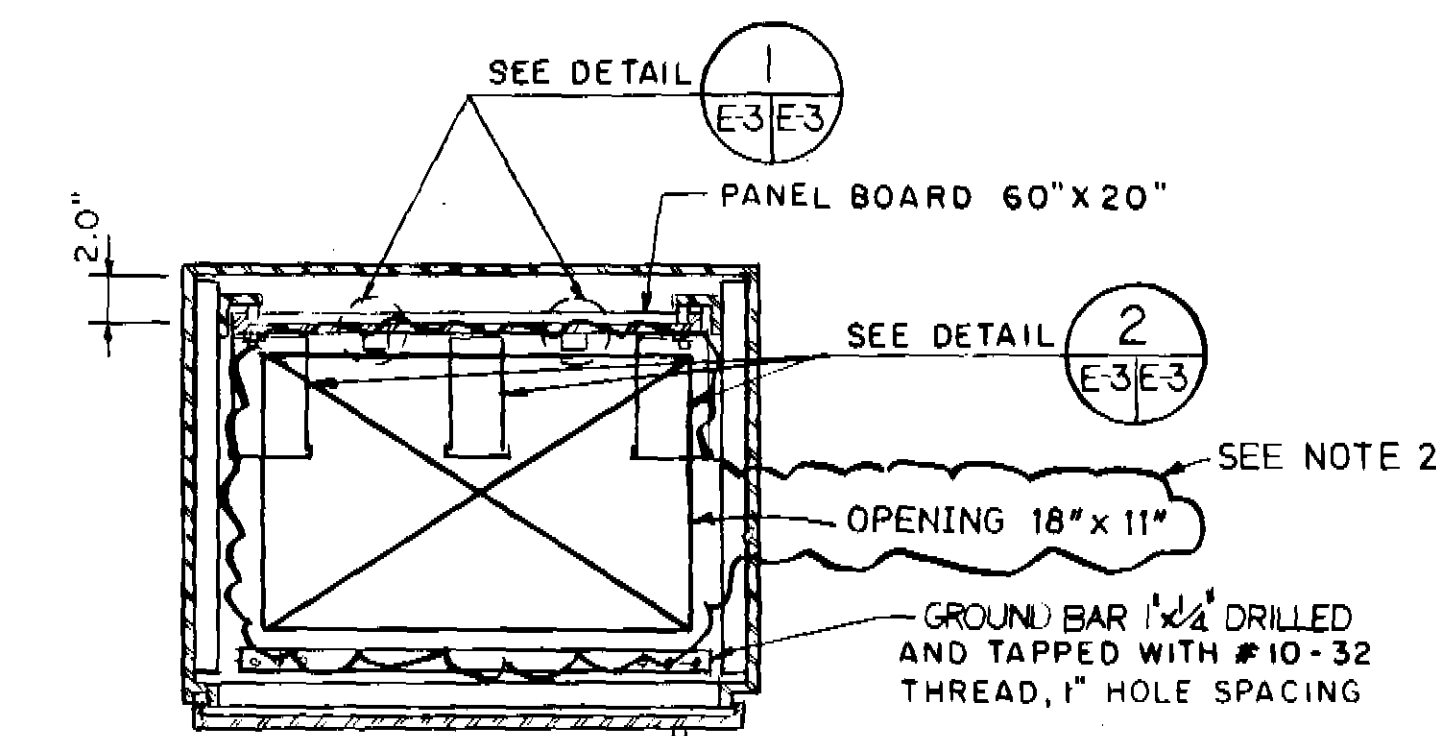


SIDE VIEW

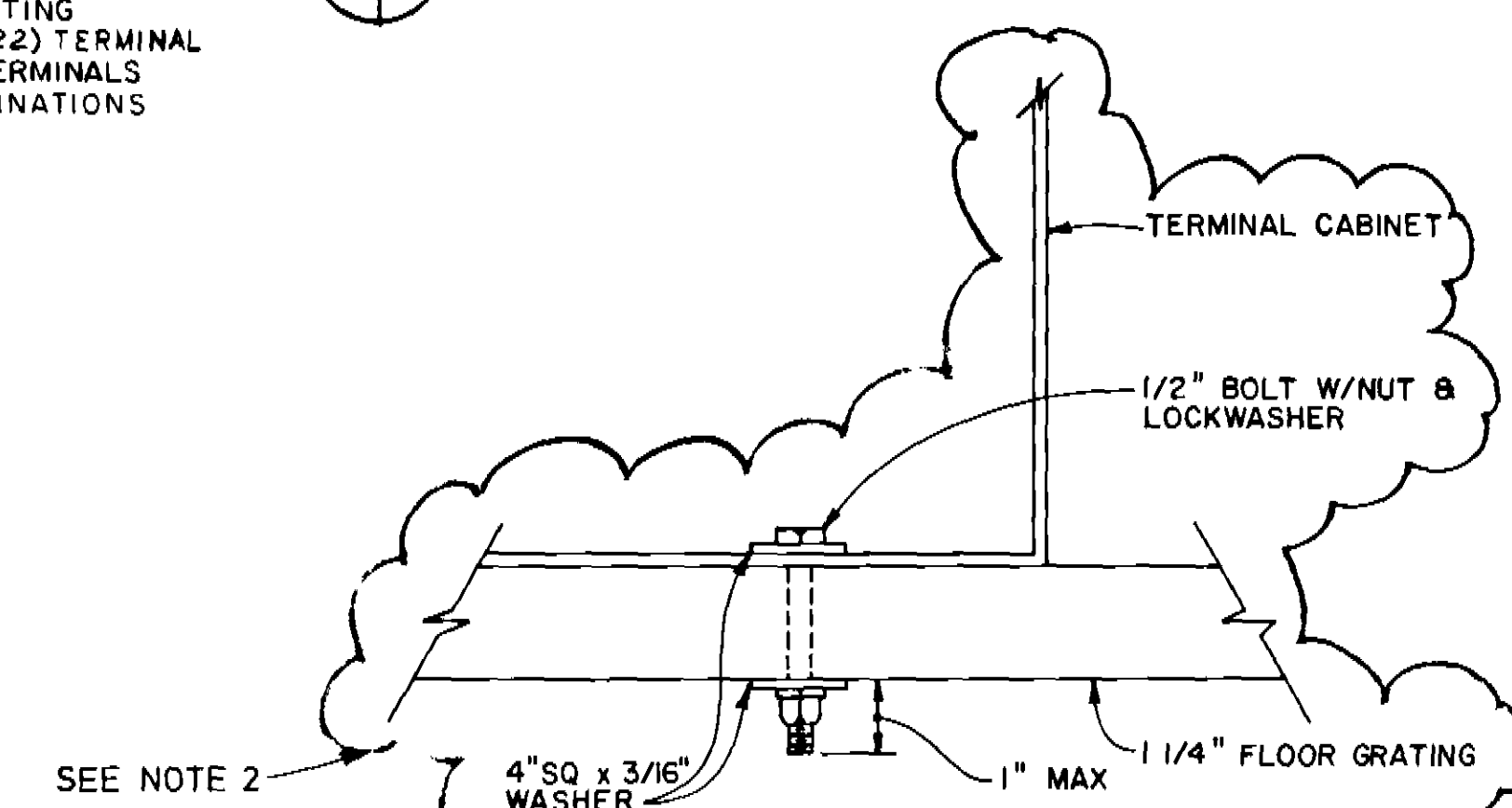
TERMINAL CABINET (TYPICAL)
SCALE: 1/2" = 1'-0"



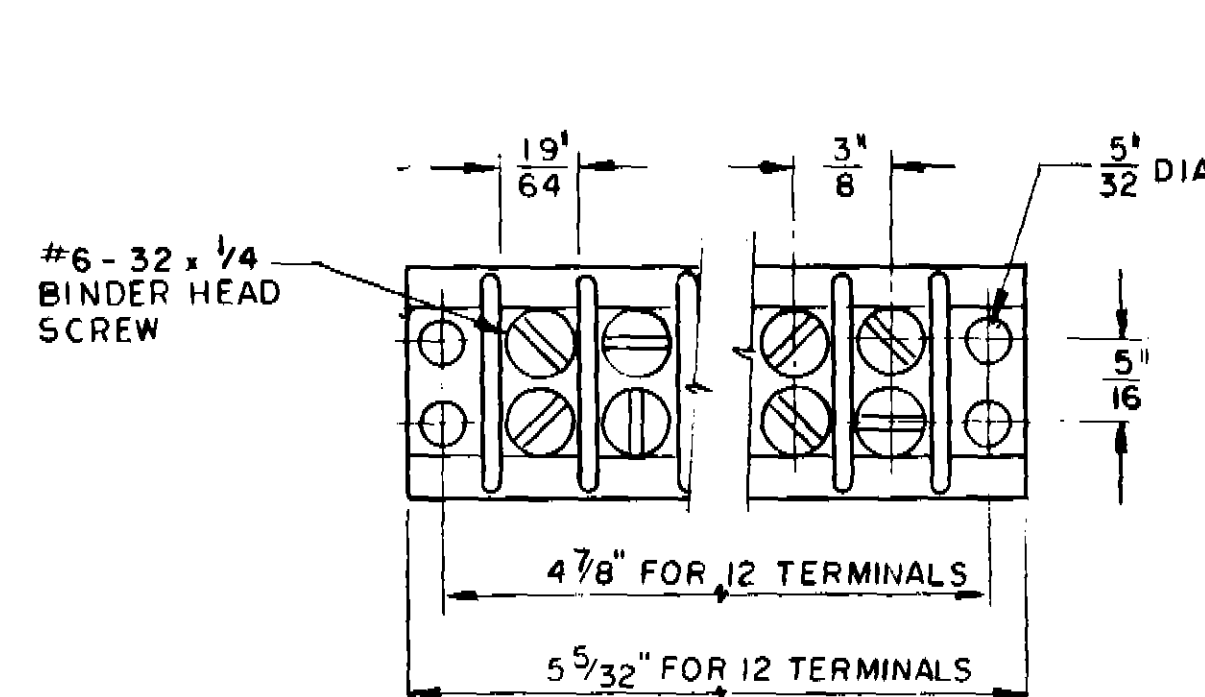
A SECTION
E-3/E-3 SCALE: 1/2" = 1'-0"



B SECTION
E-3/E-3 SCALE: 1/2" = 1'-0"

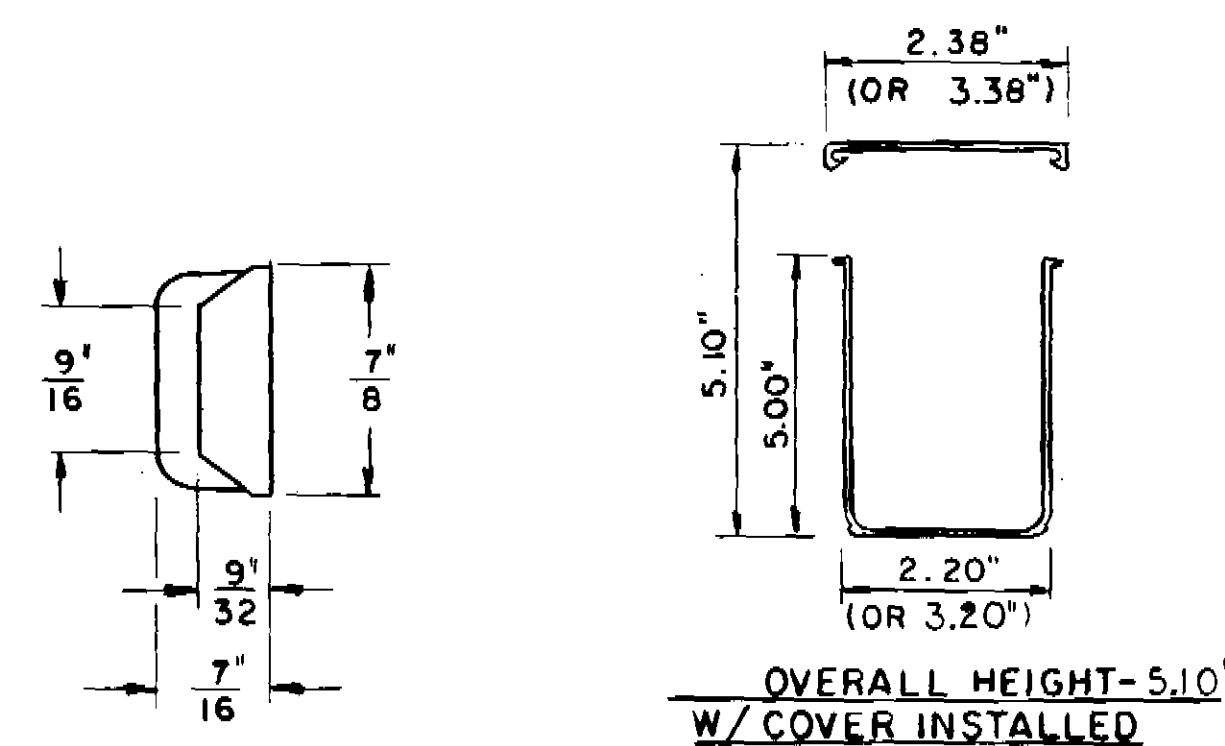


DATAc AND TERMINAL CABINET MTG. DETAIL
TYP 4 PLC EACH CABINET
NO SCALE



1 DETAIL
E-3/E-3

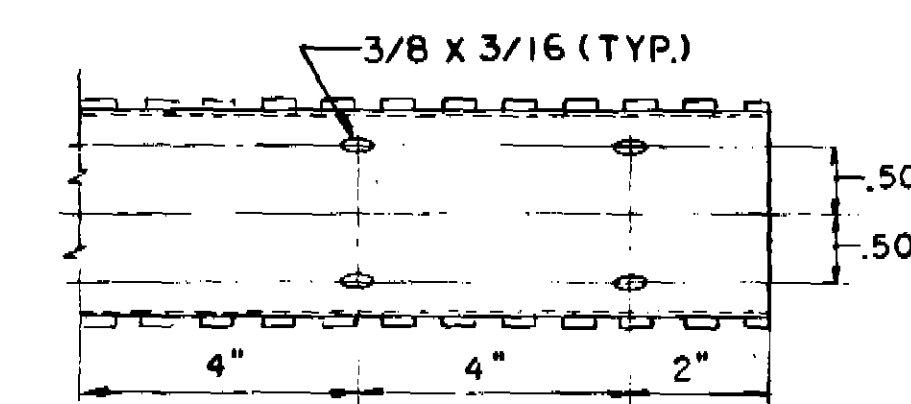
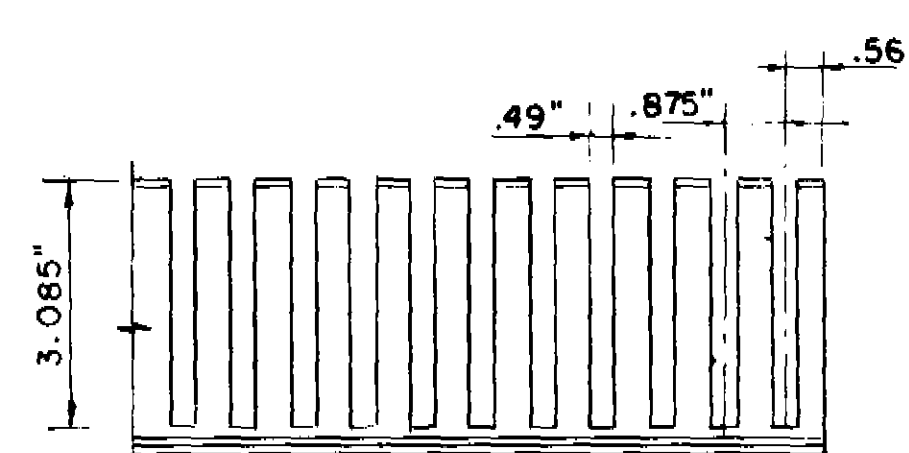
TERMINAL BLOCK
N.T.S.



TYPICAL SIDE VIEW

2 DETAIL
E-3/E-3

WIRING DUCT
N.T.S.

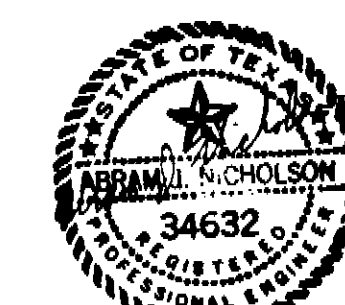
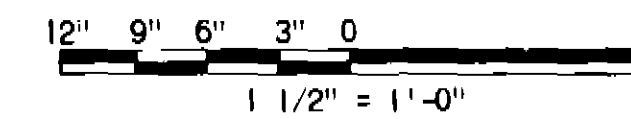


BOTTOM VIEW

- NOTES:
- FOR SENSOR WIRING DIAGRAMS SEE DRAWINGS E-4 THRU E-9.
 - ALL WORK SHOWN ON THIS DRAWING IS INCLUDED IN THIS CONSTRUCTION CONTRACT. ALL WORK SHOWN CLOUDED IS A REVISION BY BLACK & VEATCH.

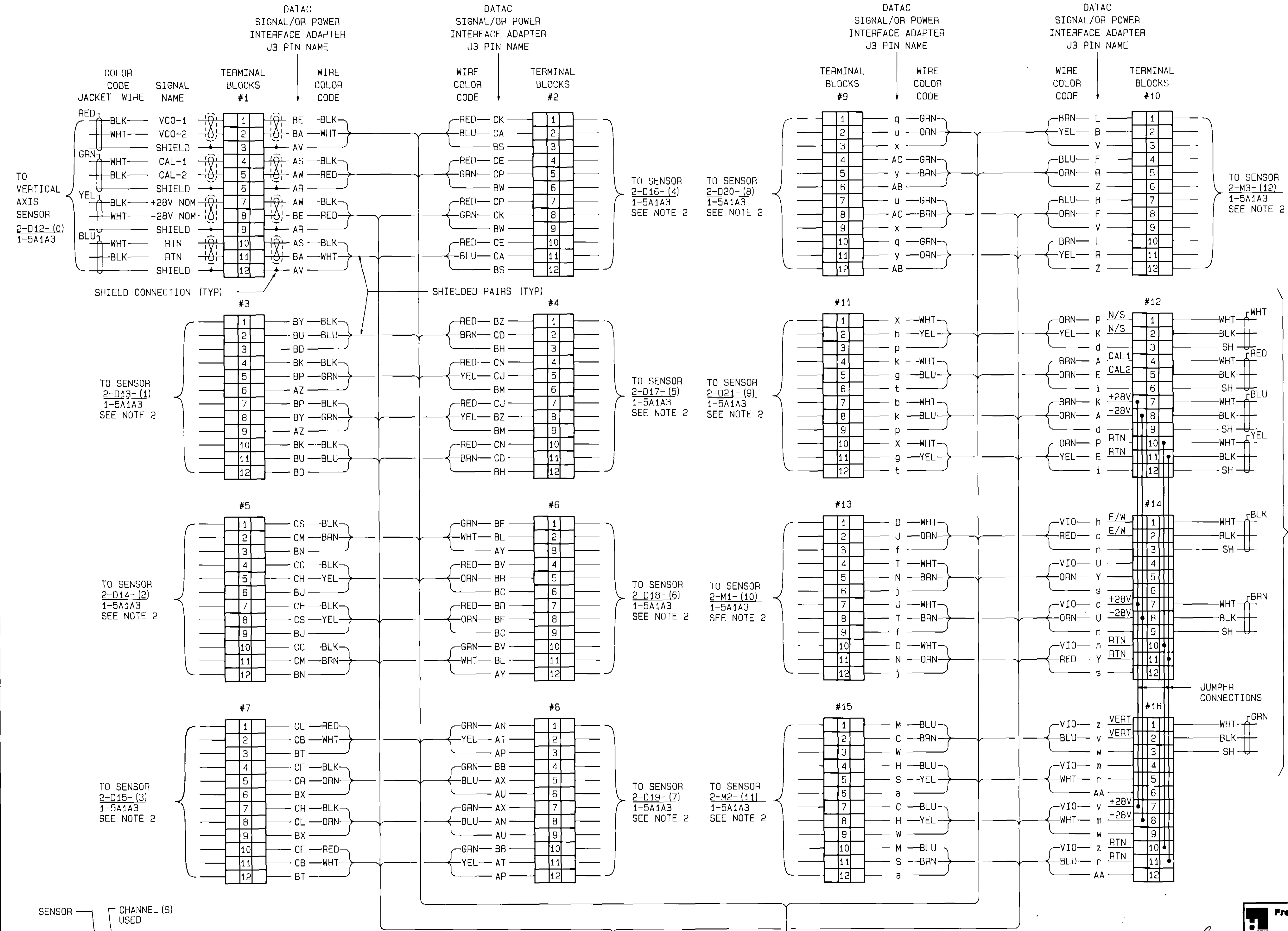
GRAPHIC SCALE

WARNING: DRAWING MAY BECOME DISTORTED DURING REDUCTION OR OTHER REPRODUCTION PROCESS. USER IS CAUTIONED AGAINST SCALING.



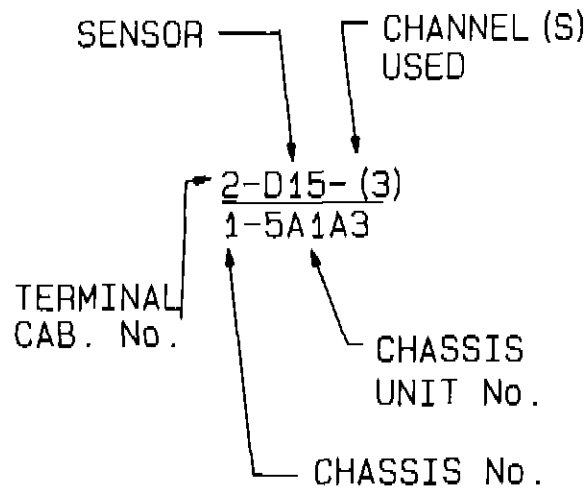
Frederic R. Harris, Inc. 300 East 42nd Street New York, NY 10017		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DESIGNED BY: [Signature] SUBMITTED BY: [Signature] OCEAN ENGR. & CONST. PROJ. OFF. FPO-1 ETC. [Signature] RVD [Signature]		DATE: 4/25/80 TITLE: PROGRAM MGR. DATE: 4/25/80	
SATISFACTORY TO: [Signature] DATE: 5/15/80 SATISFACTORY TO: [Signature] DATE: 5/15/80 COMNAVSEASYS COM		APPROVED HEAD FPO-1: [Signature] DATE: 5/15/80 APPROVED: [Signature] DATE: 5/15/80 EFD FOR COMMANDER, NAVFAC	
SIZE: D CODE: 80091	IDENT. NO.: 3020696	NAVFAC DRAWING NO.: 3020696	E-3
SCALE: AS NOTED		CONSTR. CONTR. NO. N62477-89-C-0053 SPEC. 21-89-0053 SHEET 23 OF 40	

REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED
A	REVISED TO REFLECT AS BUILT CONDITION		



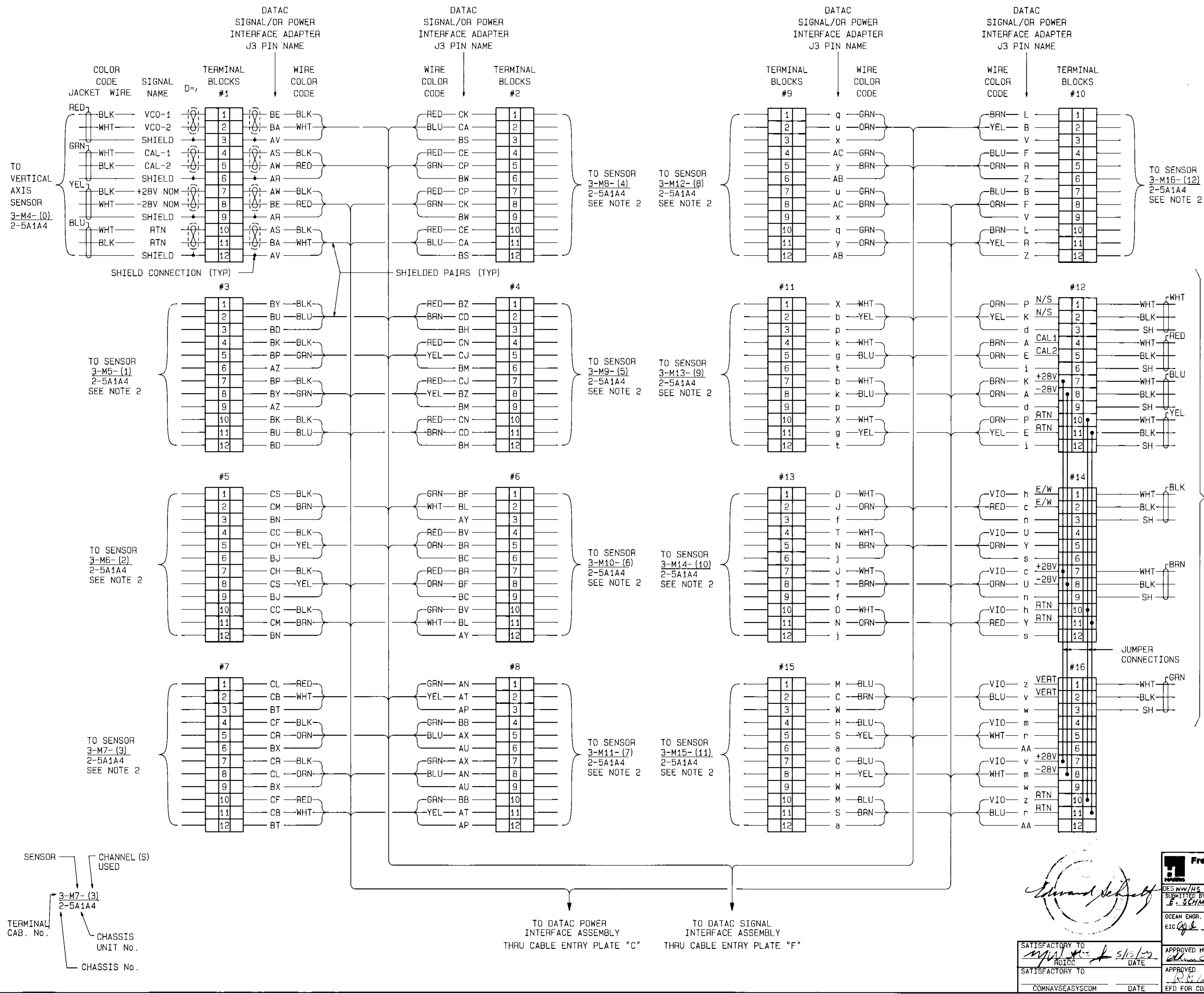
NOTES:

- FOR REFERENCE SEE NAVSEA DRAWING FSCM #53711, DWG. 6163251, SH. 1 THROUGH 8.
- ALL VERTICAL SENSOR CONNECTIONS ARE SIMILAR TO THE CONNECTIONS OF SENSOR #D12.
- FOR INTERCONNECTIONS BETWEEN TERMINAL CABINET #2 AND DATAC CABINET SEE INTERCONNECTION WIRING DIAGRAM ON DWG. E-4.
- FOR INTERFACE CABLE ASSEMBLY DETAILS SEE DWG. E-16



Frederic R. Harris, Inc. 300 East 42nd Street New York, NY 10017		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DESIGNED BY: E. SCHMELTZ CHECKED BY: CH. MK DATE: 4-15-88 TITLE: INTERFACE WIRING DIAGRAM OCEAN ENGR. & CONST. PROJ. OFF. - FPD-1 EIC: AP RVD: LD 5/1/89		INTERFACE WIRING DIAGRAM TERMINAL CABINET No. 2	
SATISFACTORY TO: NAVSEA DATE: 5/1/89		APPROVED HEAD FPD-1: CH. MK DATE: 5-15-89	
SATISFACTORY TO: NAVSEA DATE: 5/1/89		APPROVED: CH. MK DATE: 5/1/89	
COMMNAVSEASYS/COM		EFD FOR COMMANDER, NAVFAC	
SIZE: D CODE: 80091		NAVFAC DRAWING NO.: 3020698	
SCALE: NONE		CONSTR. CONTR. NO.: N62477-89-C-0053	
SHEET: 25 OF 40		SPEC: 21-89-0053	

REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED
A	REVISED TO REFLECT AS BUILT CONDITION		

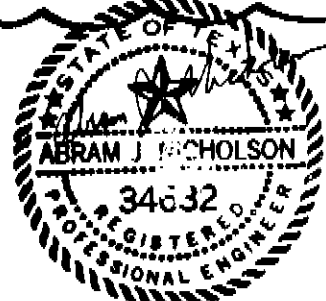
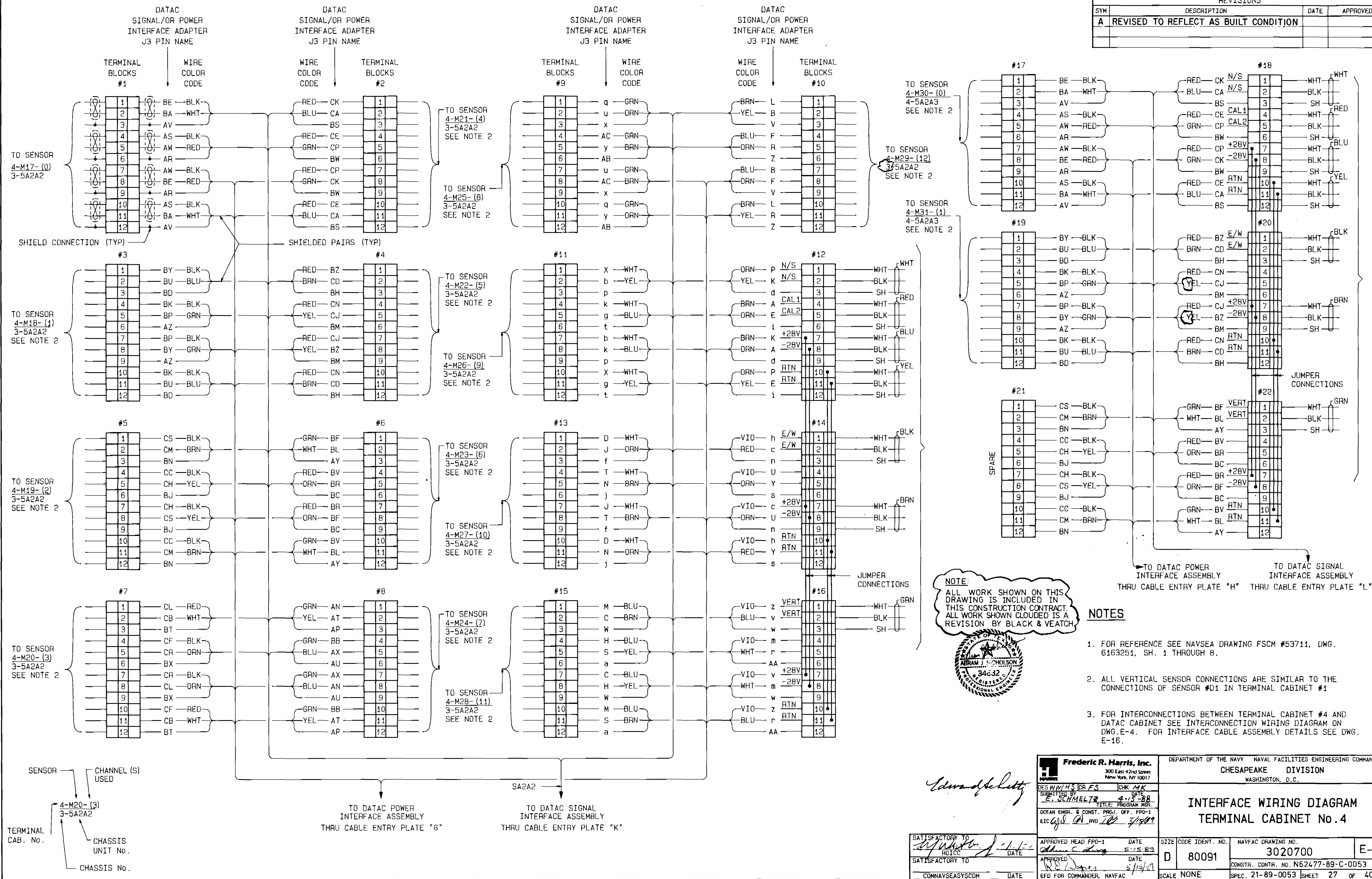


NOTES:

- FOR REFERENCE SEE NAVSEA DRAWING FSCM #53711, DWG. 6163251, SH. 1 THROUGH 8.
- ALL VERTICAL SENSOR CONNECTIONS ARE SIMILAR TO THE CONNECTIONS OF SENSOR #M4.
- FOR INTERCONNECTIONS BETWEEN TERMINAL CABINET #3 AND DATAC CABINET SEE INTERCONNECTION WIRING DIAGRAM ON DWG. E-4.
- FOR INTERFACE CABLE ASSEMBLY DETAILS SEE DWG.E-16

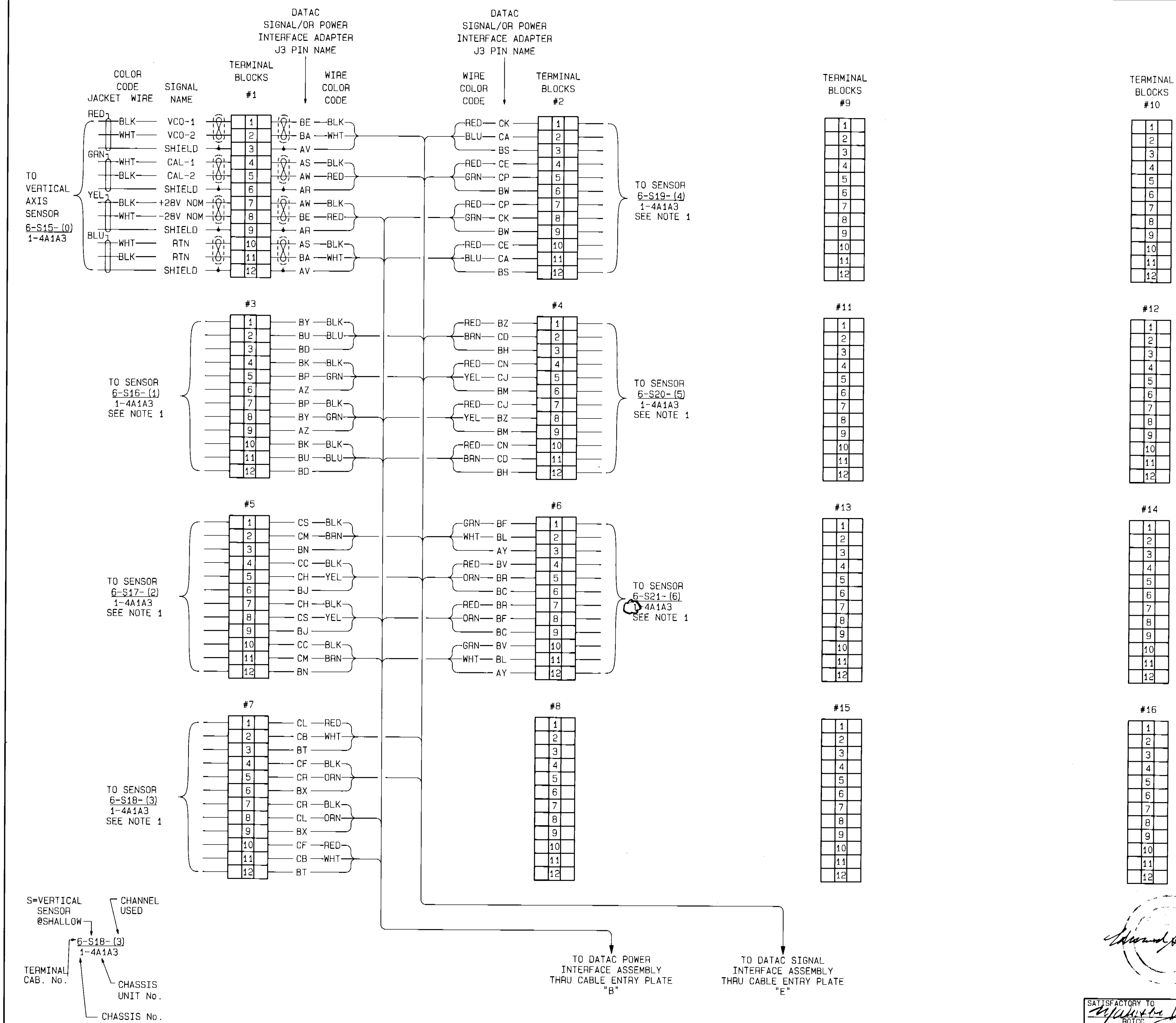
Frederick R. Harris, Inc. 300 East 42nd Street New York, NY 10017		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DESIGNED BY DR ES DATE 4-15-88 CHECKED BY CHK MK OCEAN ENGR. & CONST. PROJ. OFF. FPO-1 EIC DR RVD DR		INTERFACE WIRING DIAGRAM TERMINAL CABINET No.3	
APPROVED HEAD FPO-1 DATE 5-15-89 APPROVED DATE 5-15-89 EFD FOR COMMANDER, NAVFAC		SIZE CODE IDENT. NO. D 80041	NAVFAC DRAWING NO. 3020699
SATISFACTORY TO ROICC DATE 5/15/89 SATISFACTORY TO COMNAVSEASYSOM DATE		CONSTR. CONTR. NO. N62477-89-C-0053	E-6 SPEC. 21-89-0053 SHEET 26 OF 40

REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED
A	REVISED TO REFLECT AS BUILT CONDITION		



Frederic R. Harris, Inc. 300 East 42nd Street New York, NY 10017		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DESIGNED BY: DR FS DRAWN BY: E. SCHMELTZ CHECKED BY: 4-18-88 DATE: 5/15/88 TITLE: PROGRAM MGR.		INTERFACE WIRING DIAGRAM TERMINAL CABINET No. 4	
OCEAN ENGR. & CONST. PROJ. OFF. FPO-1 EIC WJ RD 10 5/15/88		SIZE: D CODE: 80091 NAVFAC DRAWING NO.: 3020700 E-7	
APPROVED HEAD FPO-1 DATE: 5/15/88		CONSTR. CONTR. NO. N62477-89-C-0053	
APPROVED DATE: 5/15/88		SCALE: NONE SPEC. 21-89-0053 SHEET 27 OF 40	

REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED



NOTES:

- ALL VERTICAL SENSOR CONNECTIONS ARE SIMILAR TO THE CONNECTIONS OF SENSOR #S15.
- FOR REFERENCE SEE CONTRACT FSCM# 53711, DWG. 6163251, SH. 1 THROUGH 8
- FOR INTERFACE CABLE ASSEMBLY DETAILS SEE DWG.E-16
- FOR INTERCONNECTIONS BETWEEN TERMINAL CABINET #6 AND DATAC CABINET SEE INTERCONNECTION WIRING DIAGRAM ON DWG. E-8.
- UNUSED PAIRS OF INTERFACE CABLES SHALL BE TAPED, LABELED AND SECURED IN WIRING DUCT. THE LENGTH OF UNUSED PAIR SHALL BE SUFFICIENT TO REACH ANY TERMINAL BLOCK FOR FUTURE USE.



NOTE: ALL WORK SHOWN ON THIS DRAWING IS INCLUDED IN THIS CONSTRUCTION CONTRACT ALL WORK SHOWN CLOUD IS A REVISION BY BLACK & VEATCH.

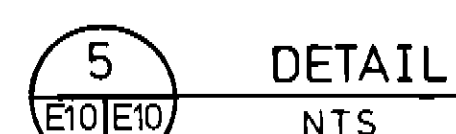
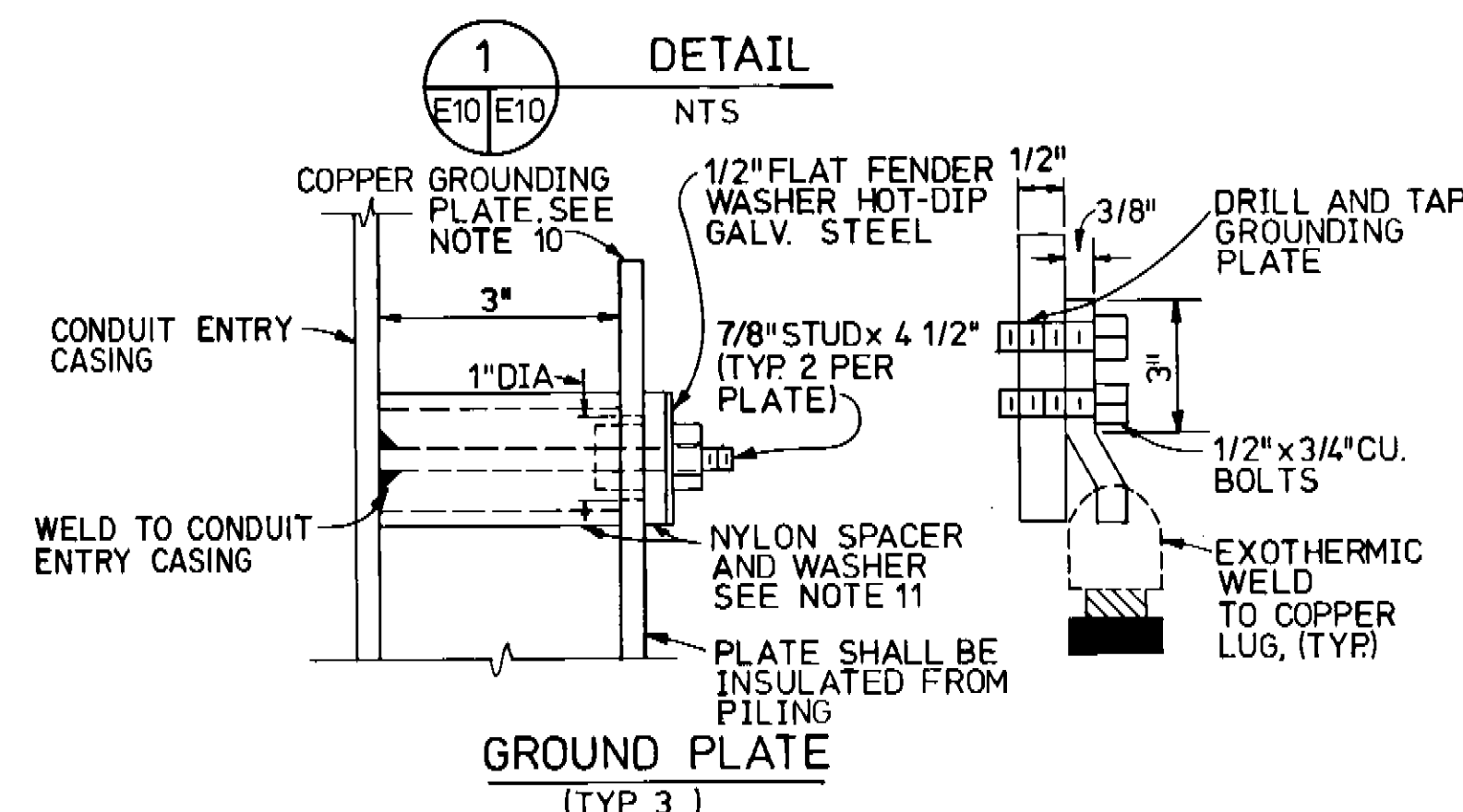
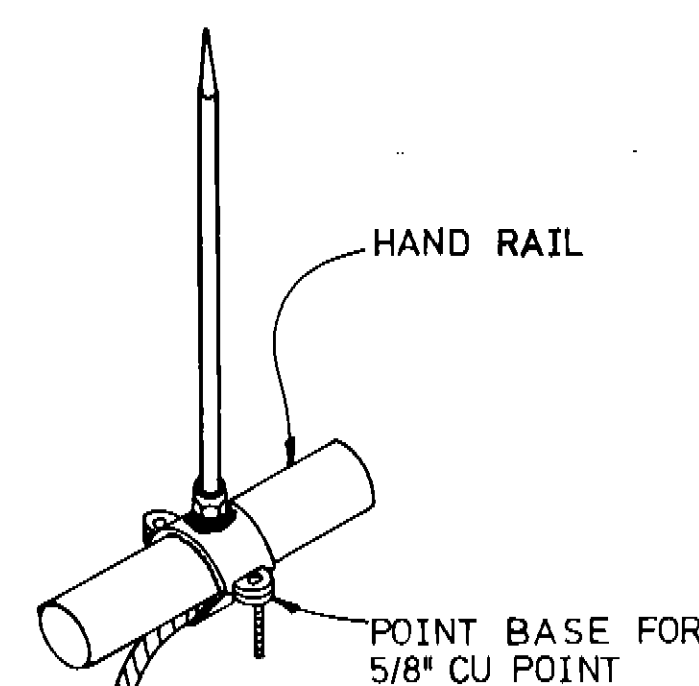
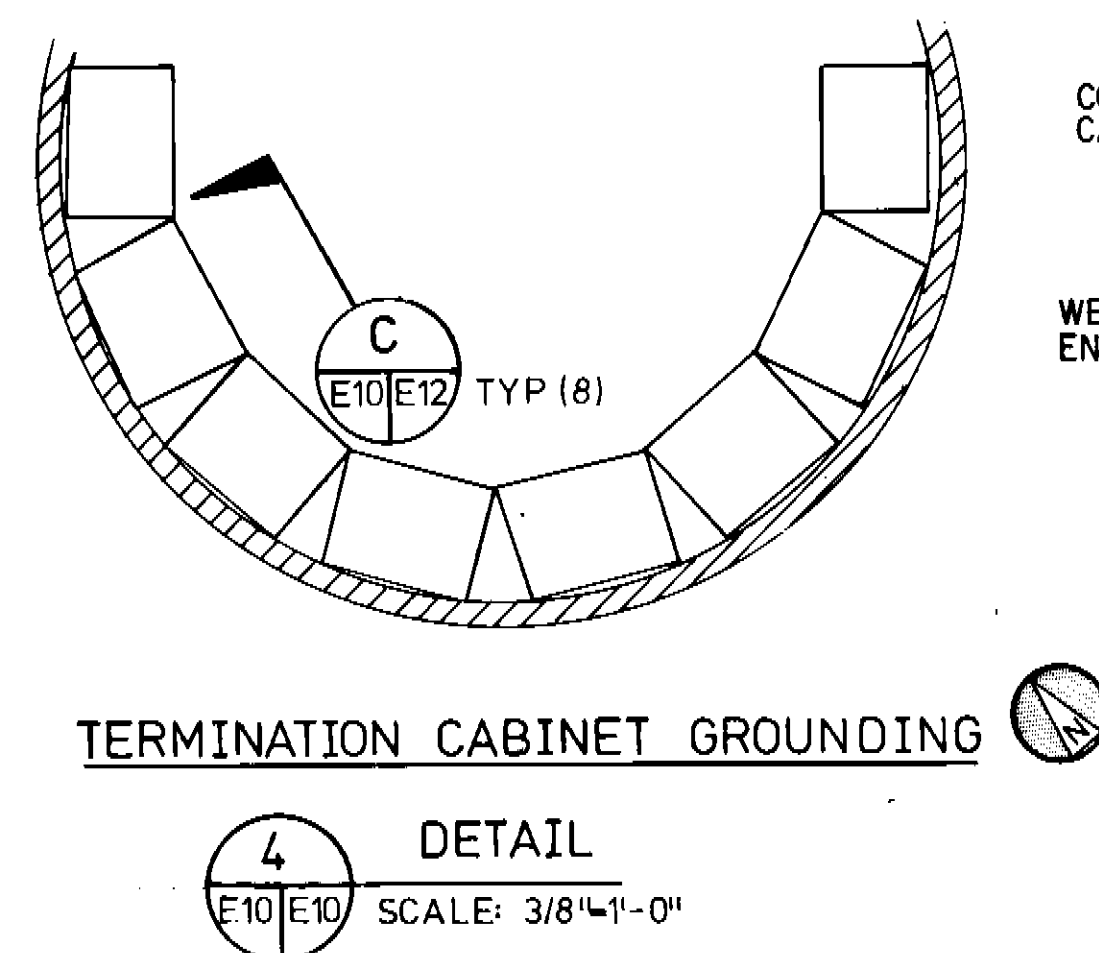
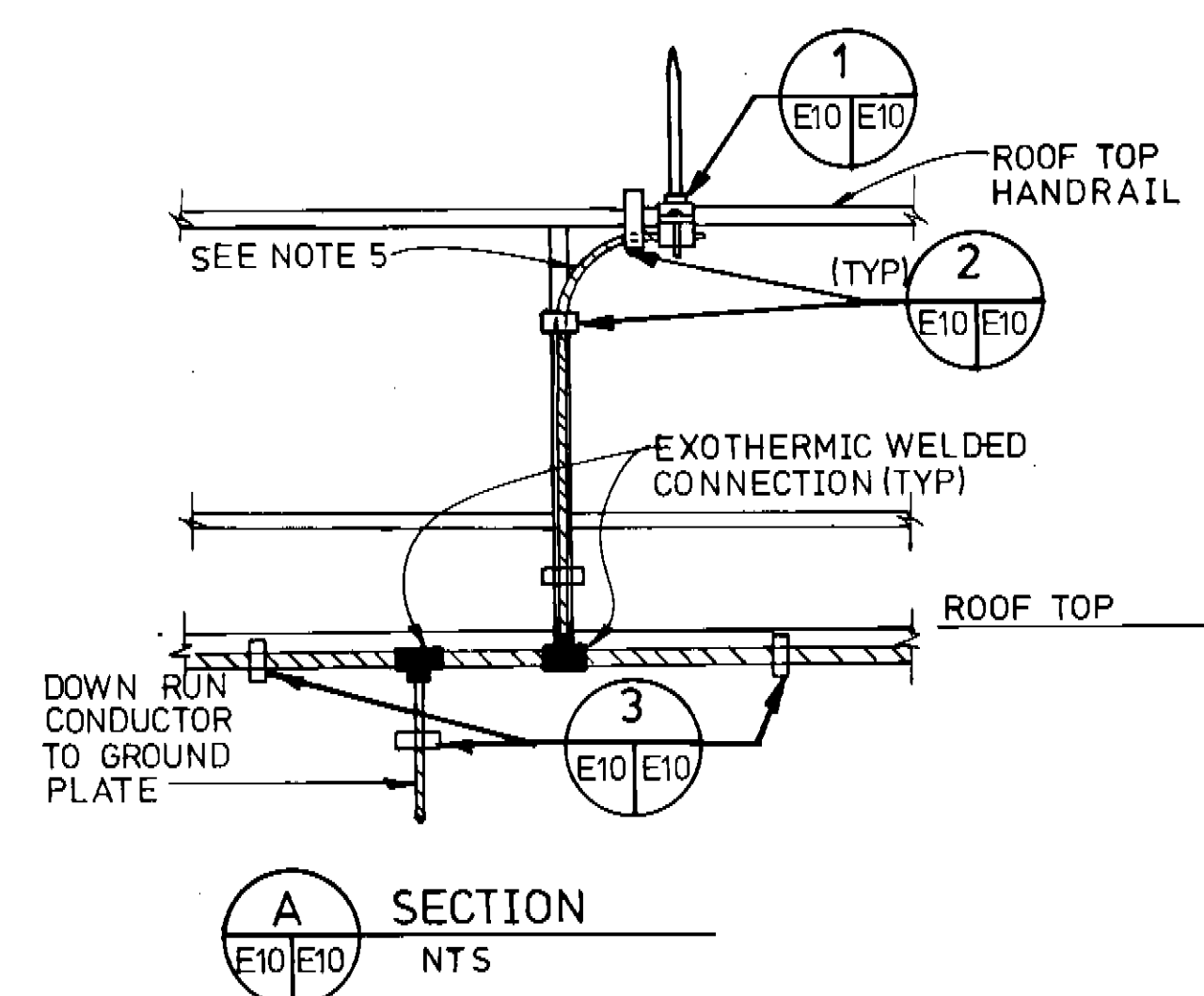
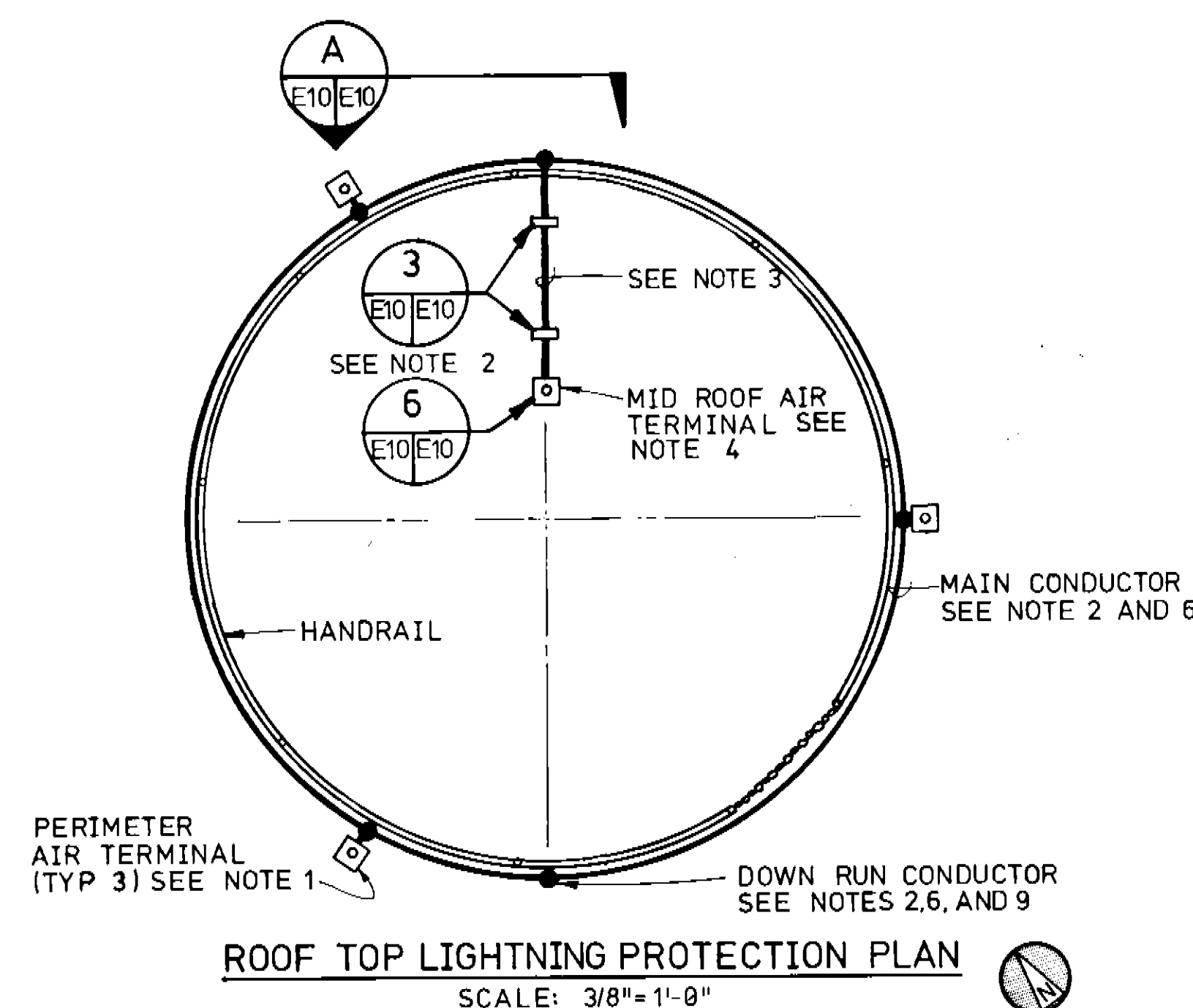
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SATISFACTORY TO ROICC DATE 5/15/89
 APPROVED DATE 5/15/89
 COMNAVSEASYS COM DATE

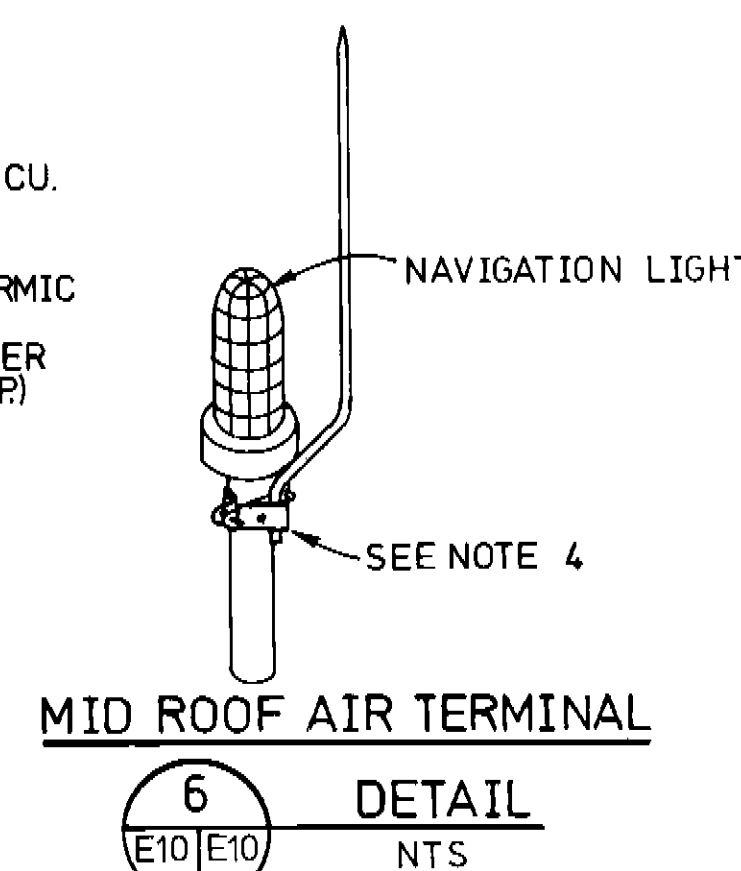
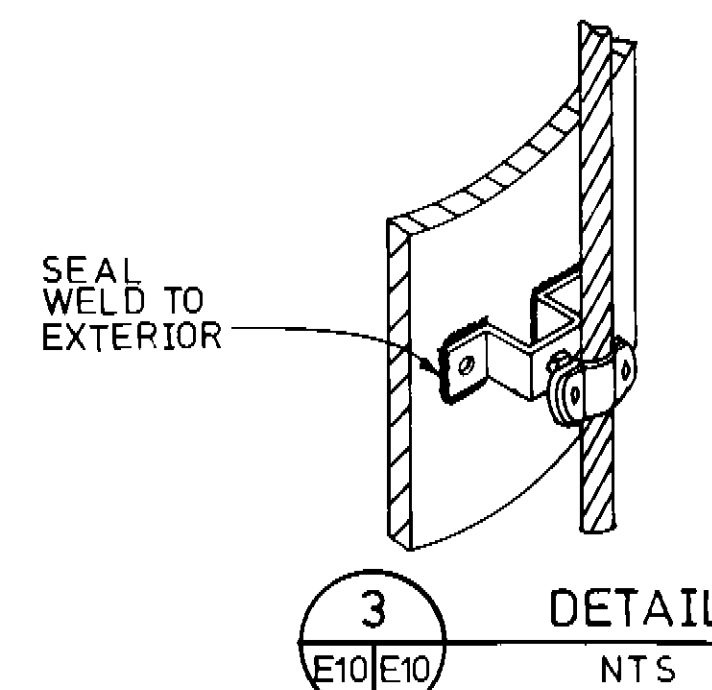
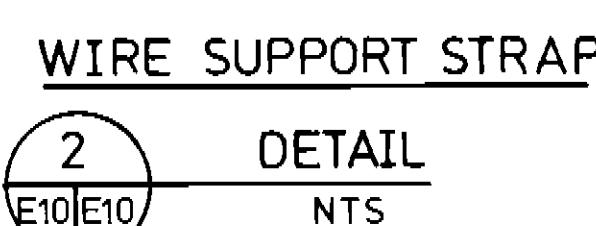
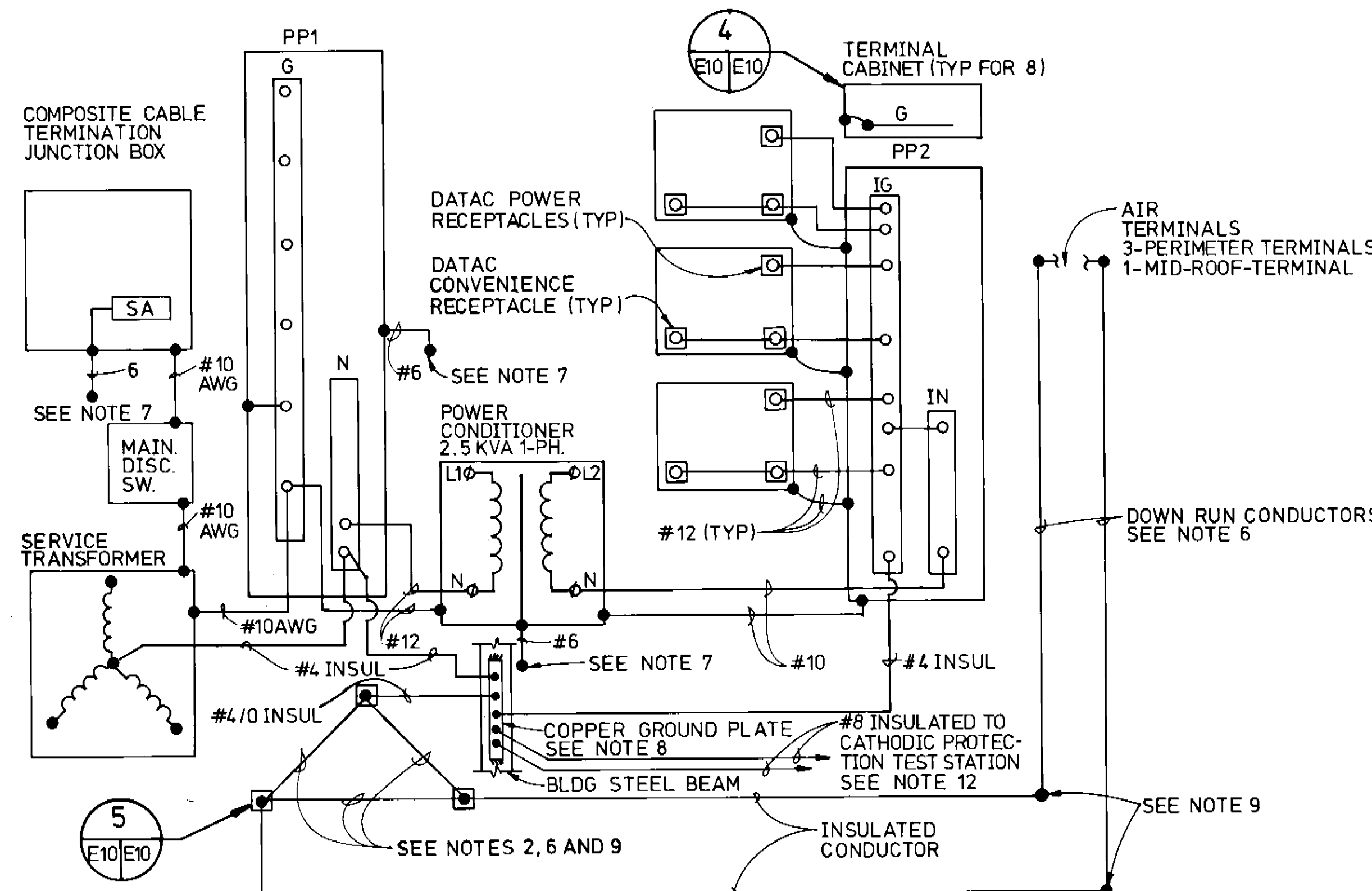
Frederic R. Harris, Inc.
 300 East 42nd Street
 New York, NY 10017
 DES OR CHK
 SUBMITTED BY DATE 5/15/89
 TITLE: PROGRAM WGR
 OCEAN ENGR. & CONST. PROJ. OFF. FPD-1
 EIC: ajf RVD TD 5/15/89
 APPROVED HEAD FPD-1 DATE 5-15-89
 APPROVED DATE 5/15/89
 EFO FOR COMMANDER, NAVFAC

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND
 CHESAPEAKE DIVISION
 WASHINGTON, D.C.
INTERFACE WIRING DIAGRAM
TERMINAL CABINET No. 6
 SIZE CODE IDENT. NO. NAVFAC DRAWING NO.
 D 80091 3020702 E-9
 CONSTR. CONTR. NO. N62477-89-C-0053
 SCALE NONE SPEC. 21-89-0053 SHEET 29 OF 40

REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED



GROUNDING DIAGRAM

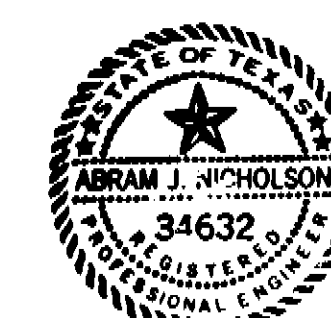
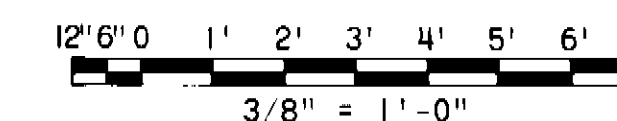


NOTES:

- PERIMETER AIR TERMINALS SHALL BE EVENLY SPACED AROUND THE PERIMETER OF THE ROOF.
- ALL LIGHTNING PROTECTION CONDUCTORS SHALL BE SECURED AT INTERVALS NOT TO EXCEED THREE FEET.
- CONDUCTORS CONNECTING AIR TERMINALS TO THE MAIN CONDUCTOR AS A SINGLE RETURN PATH SHALL NOT EXCEED 16 FEET IN TOTAL LENGTH.
- MID ROOF AIR TERMINAL SHALL BE SECURED TO NAVIGATION LIGHT SUPPORT. FOR POLE DETAIL SEE DWG. E-12 DETAIL 3.
- NO BEND OF A CONDUCTOR SHALL FORM AN INCLUDED ANGLE OF LESS THAN 90 DEGREES NOR SHALL IT HAVE A RADIUS OF BEND LESS THAN 8 INCHES EXCEPT WHERE APPROVED CLAMPS OR CONNECTORS ARE USED TO ACCOMPLISH CABLE BENDS.
- CONDUCTORS SHALL CONTAIN 37 STRANDS OF .0756 BARE COPPER WIRE WITH A CIRCULAR MIL CROSS SECTION OF 211,600 CM AND A WEIGHT OF 653 POUNDS PER 1,000 FEET.
- BOND EQUIPMENT GROUNDING CONDUCTORS TO NEAREST CONVENIENT STRUCTURAL STEEL WHERE REQUIRED OR INDICATED.
- COPPER PLATE FOR GROUNDING CONDUCTORS SHALL BE SECURED BY EXOTHERMIC WELDS TO STRUCTURAL STEEL INSIDE THE DCS. IN AN ACCESSIBLE LOCATION. GROUNDING CONDUCTOR CONNECTIONS TO PLATE SHALL ALSO BE EXOTHERMIC WELDED.
- LIGHTNING PROTECTION DOWN CONDUCTORS SHALL CHANGE FROM A BARE CONDUCTOR TO INSULATED BEFORE ENTERING THE WATER. EXOTHERMIC WELDED CONNECTIONS SHALL BE USED. BONDING CONDUCTOR BETWEEN PLATES SHALL ALSO BE INSULATED.
- GROUNDING PLATE SHALL HAVE A THICKNESS OF NOT LESS THAN 1/2 INCH AND SHALL PROVIDE A SURFACE AREA OF NOT LESS THAN 144 SQ. IN. PER SIDE.
- PLATE MOUNTING STUDS SHALL BE ATTACHED TO PILING BEFORE PAINTING. INSULATING ATTACHMENT SHALL BE AS SHOWN OR APPROVED EQUAL. GROUNDING PLATE SHALL BE MOUNTED 2 FEET BELOW MEAN LOW WATER LEVEL AND PROTECTED FROM IMPACT FROM FLOATING OR SUBMERGED OBJECTS BY LOCATING ON THE INNER SIDE OF PILING. NYLON SHALL BE 1/4 INCH THICK MINIMUM.
- TWO #8 INSULATED GROUNDING CONDUCTORS SHALL BE INSTALLED IN A 3/4" CONDUIT FROM THE CATHODIC PROTECTION TEST STATION TO THE COPPER GROUND PLATES. CONDUIT SHALL TERMINATE NEAR GROUND PLATE WITH A GROUNDING TYPE BUSHING AND SHALL BE EFFECTIVELY BONDED TO GROUND PLATE BY A BONDING JUMPER NOT SMALLER THAN #8 AWG. FOR TEST STATION LOCATION SEE DWG. E-1. FOR TEST STATION DETAILS SEE DWG. E-11.

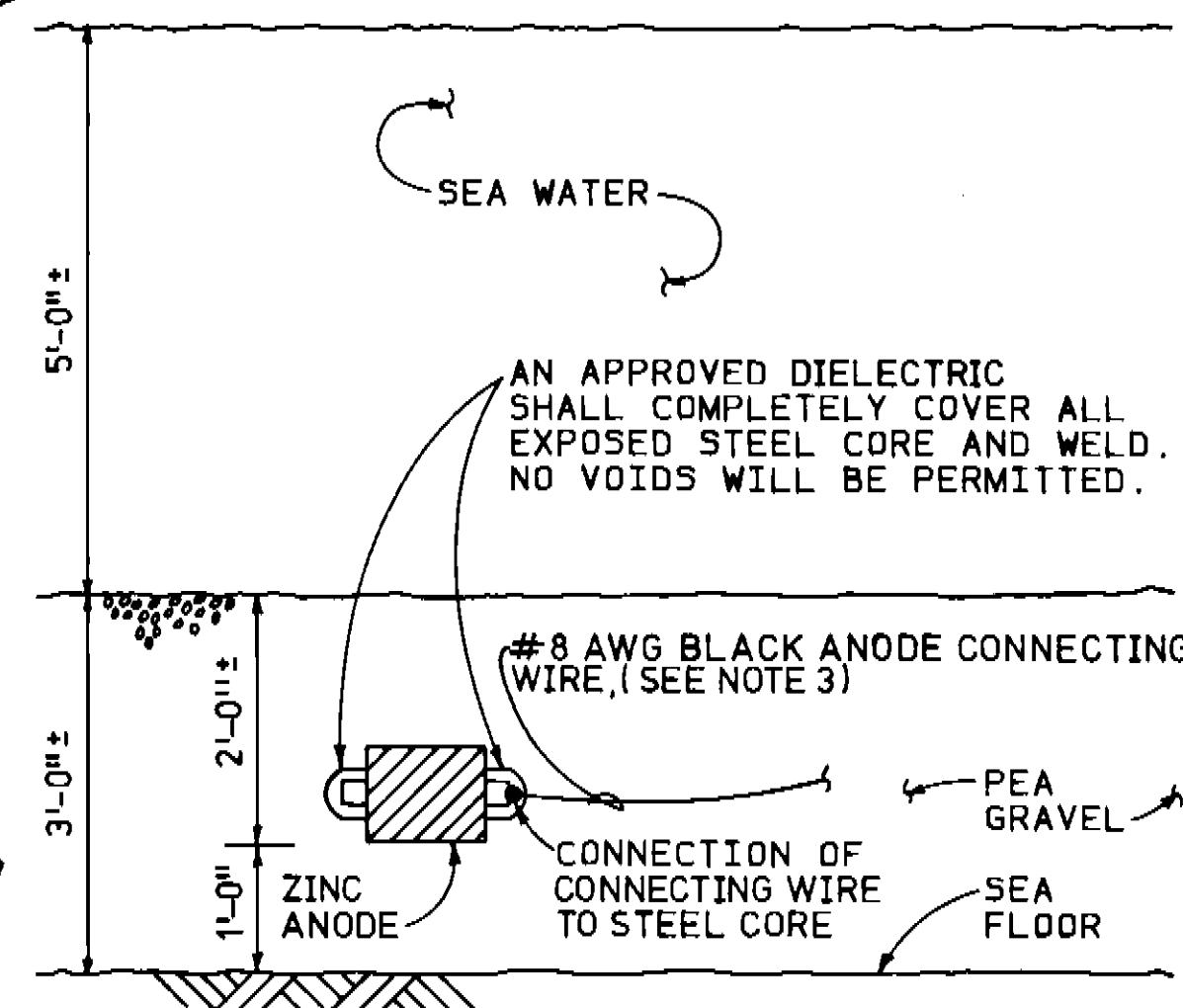
GRAPHIC SCALE

WARNING: DRAWING MAY BECOME DISTORTED DURING REDUCTION OR OTHER REPRODUCTION PROCESS. USER IS CAUTIONED AGAINST SCALING.

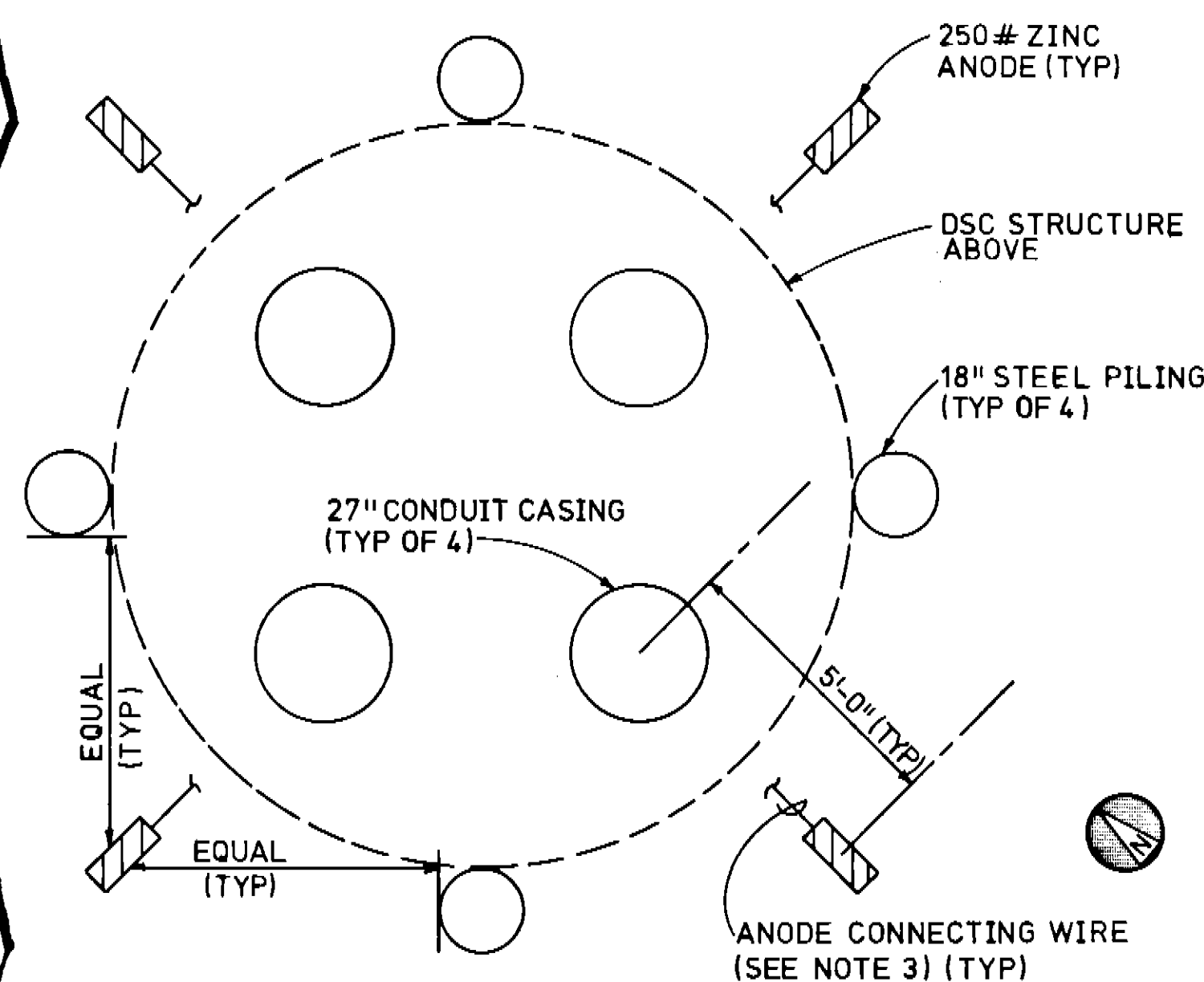


SATISFACTORY TO ROICC DATE 5/1/89	APPROVED HEAD FPO-1 DATE 5-15-89
SATISFACTORY TO COMNAVSEASYS COM DATE	APPROVED DATE 5/12/89

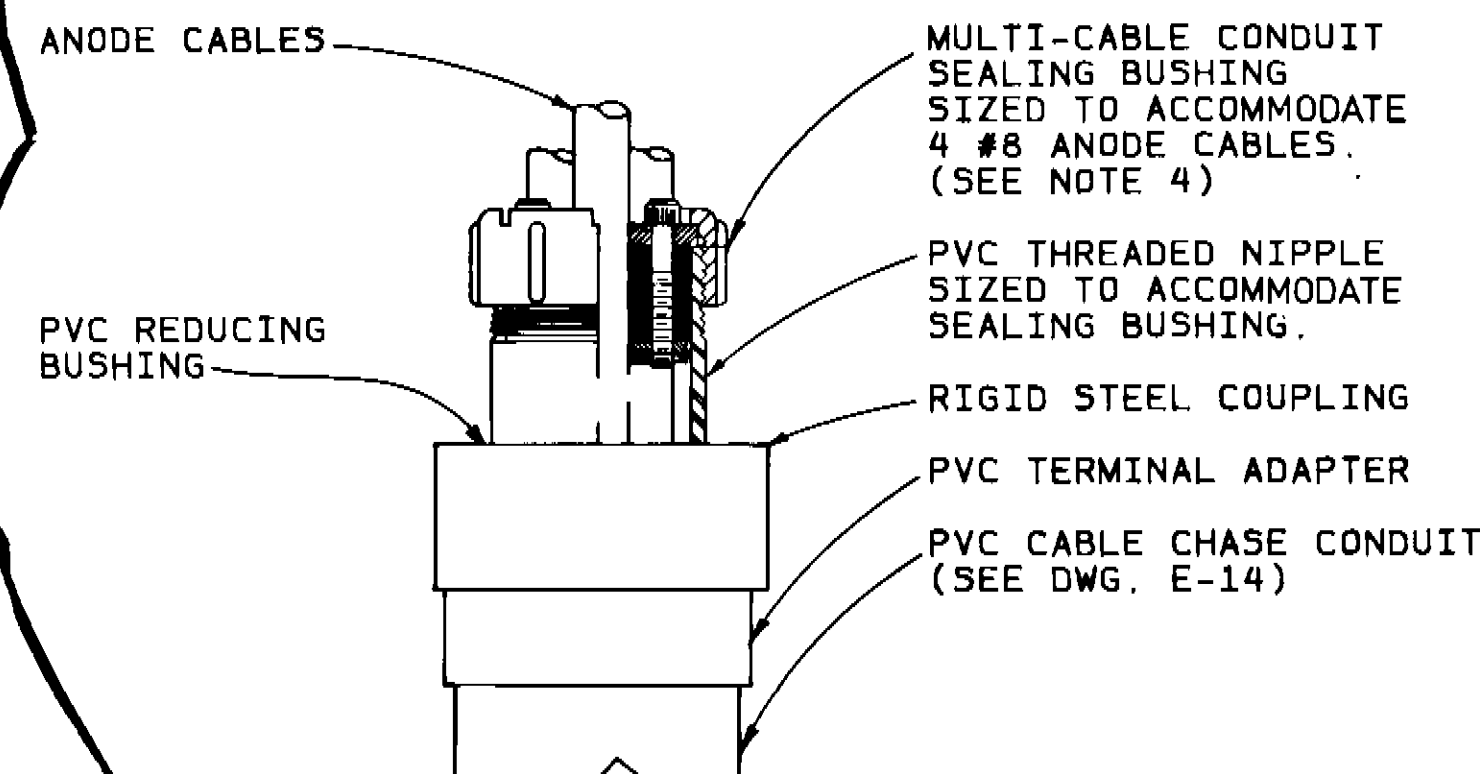
Black & Veatch Engineers Architects		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DES RM/AV DR DG CHK HRA SUBMITTED BY DATE 5/1/89 TITLE: PROGRAM MGR OCEAN ENGR. & CONST. PROJ. OFF FPO-1 EIC 4/8 11 RVO 7/8 3/4/89		MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION	
GROUNING AND LIGHTNING PROTECTION		NAVFAC DRAWING NO. 3020703	
SIZE CODE IDENT. NO. D 80091		CONSTR. CONTR. NO. N62477-89-C-0053	
SCALE AS SHOWN		SPEC. 21-89-0053 SHEET 30 OF 40	



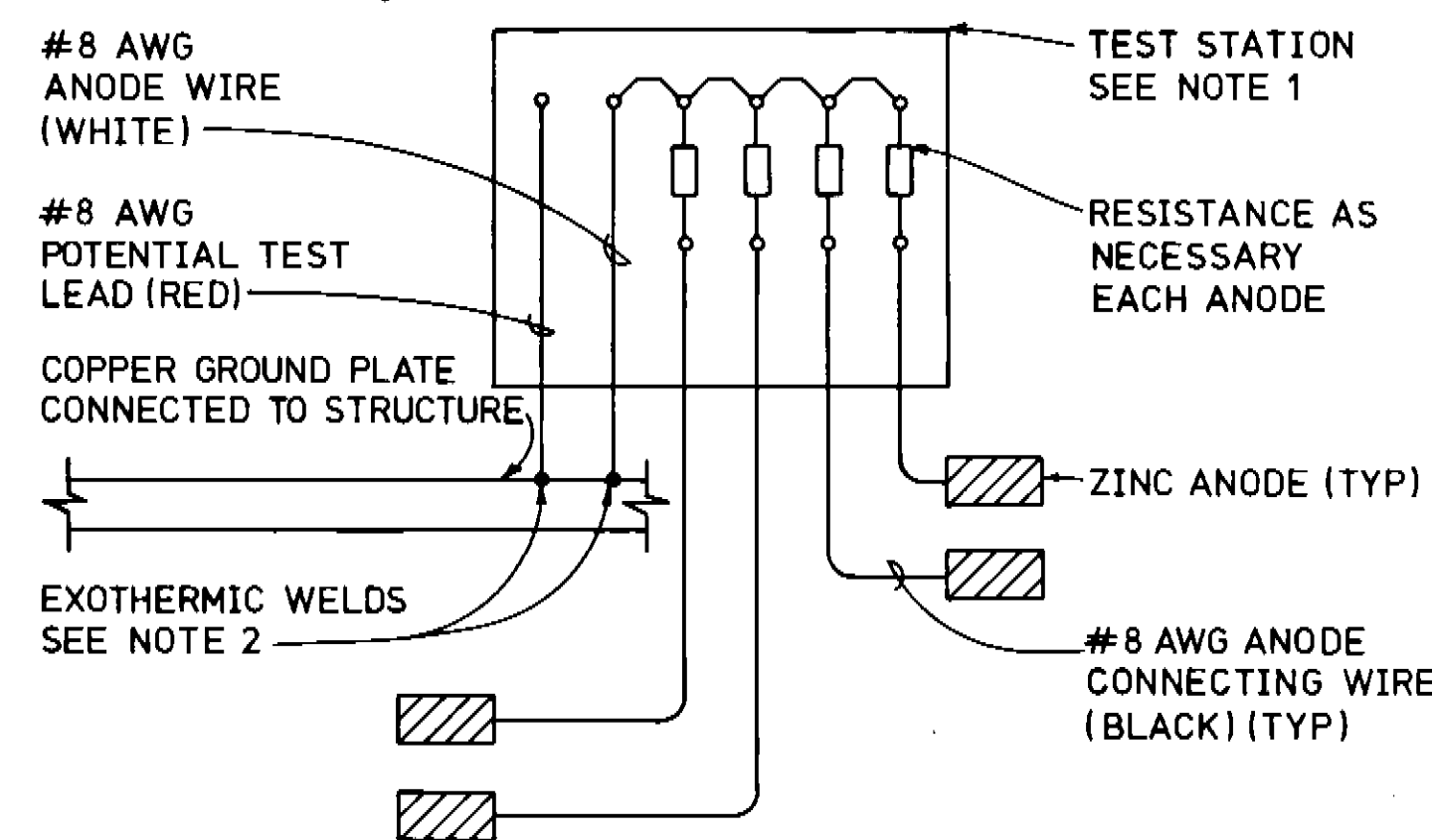
TYPICAL
ANODE INSTALLATION DETAIL
N.T.S.



ANODE PLACEMENT PLAN
N.T.S.



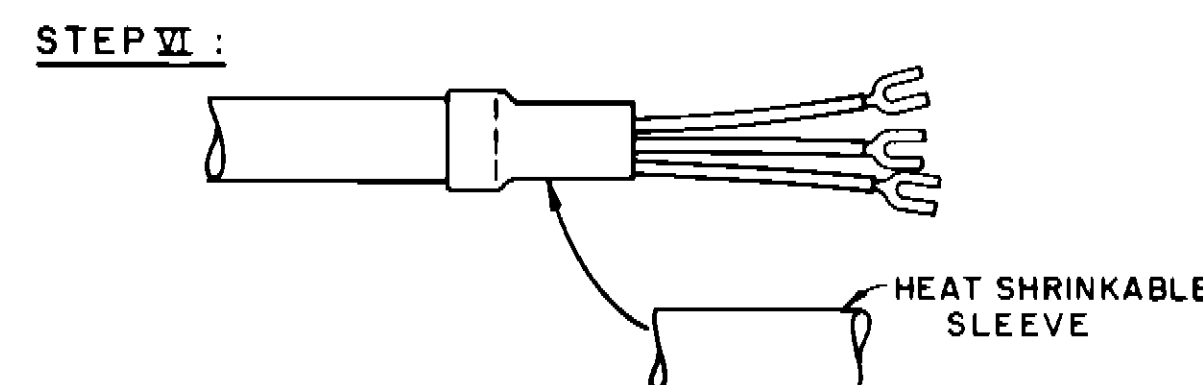
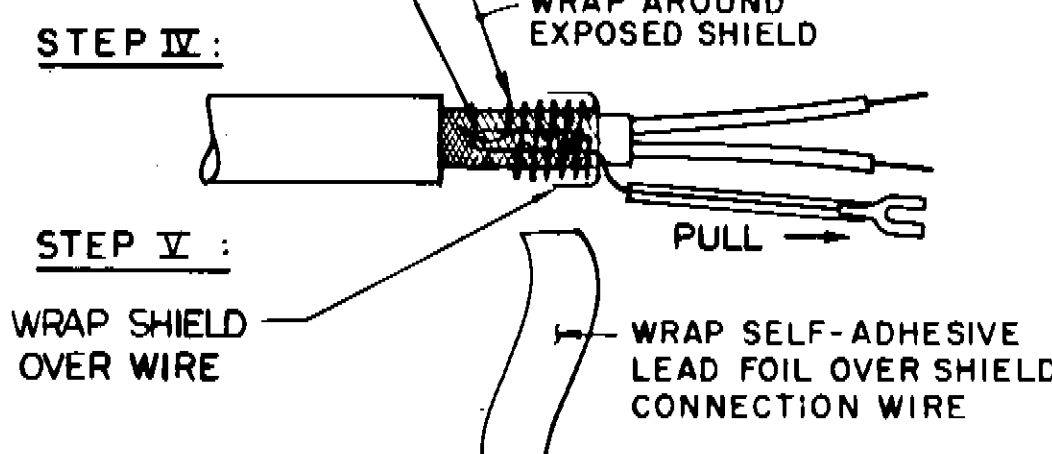
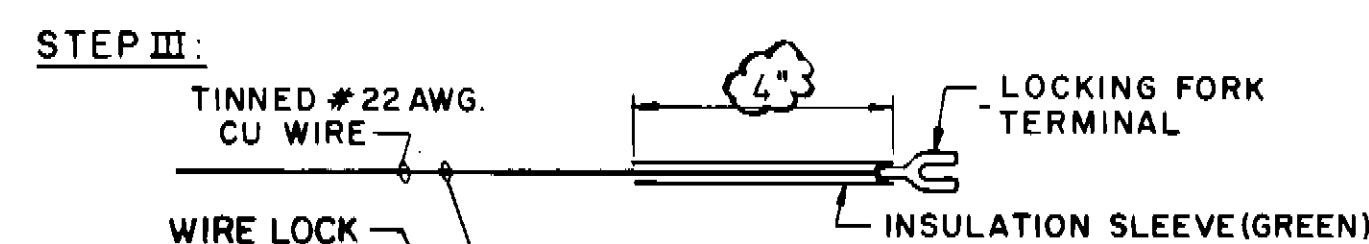
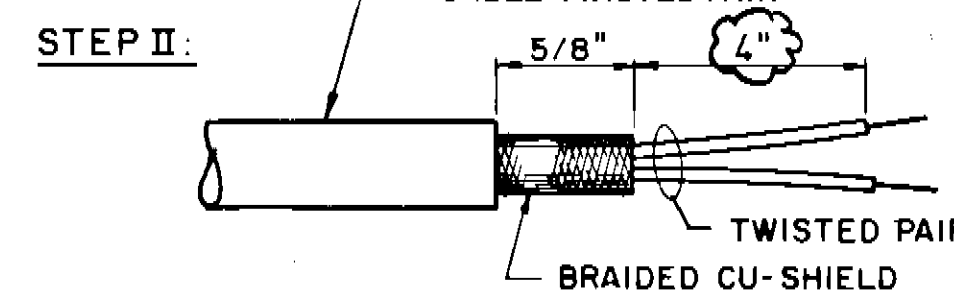
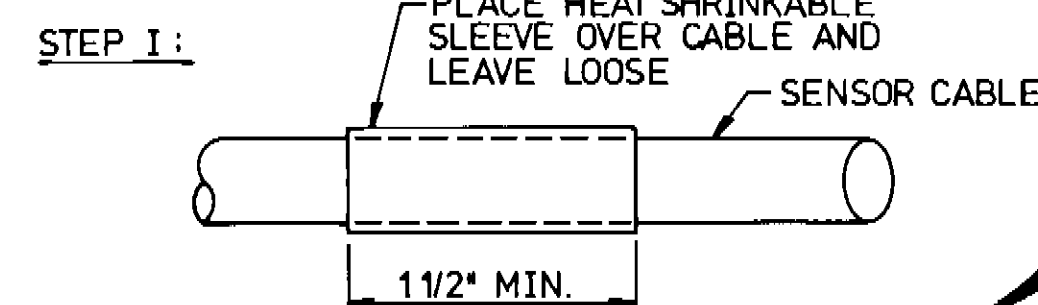
ANODE CABLE ENTRY
N.T.S.



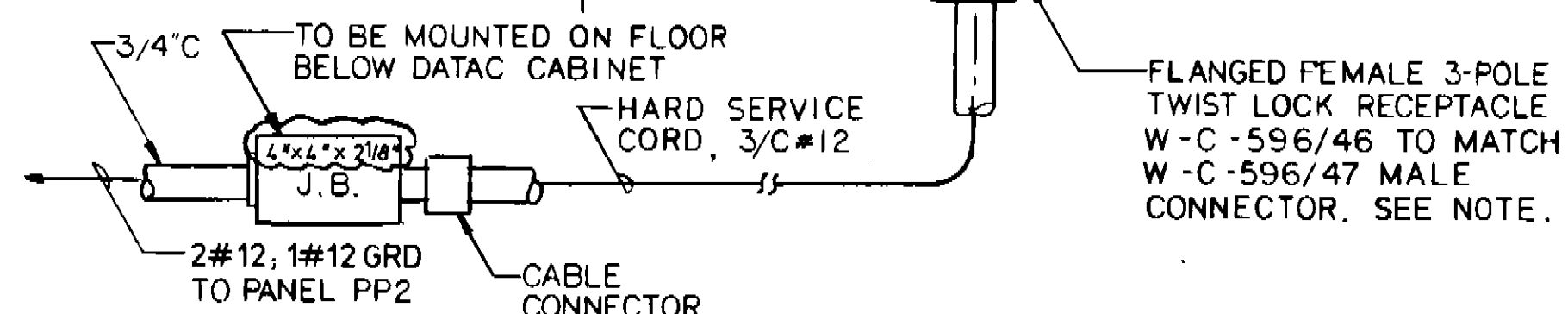
WIRING DIAGRAM
MULTIPLE SACRIFICIAL ANODE TEST STATION
SEE NOTE 3

CATHODIC PROTECTION NOTES

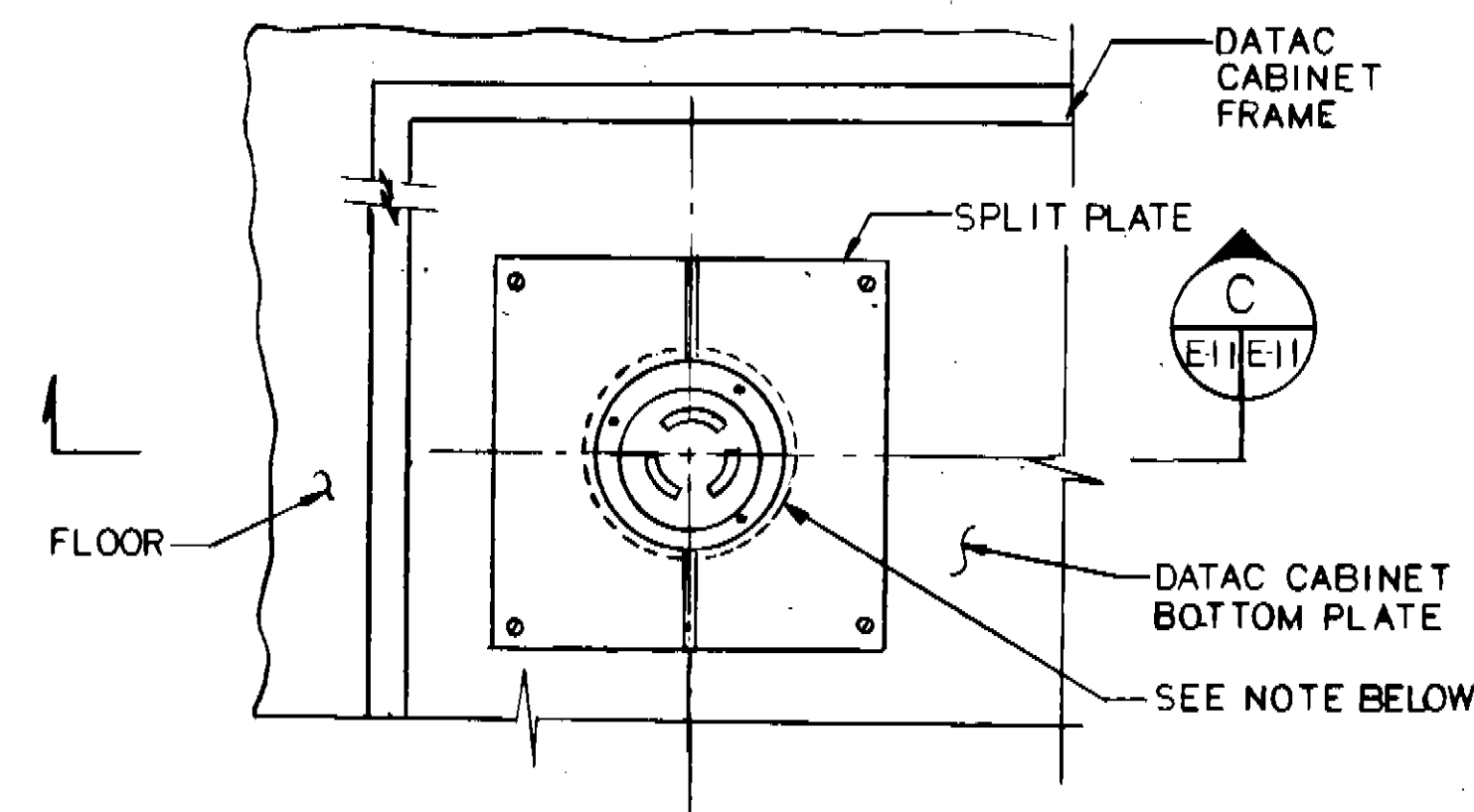
1. TEST STATION ENCLOSURE SHALL BE NEMA 4 MOLDED FIBERGLASS WITH EXTERNAL MOUNTING FEET. COVER SHALL BE ATTACHED WITH A CONTINUOUS STAINLESS STEEL HINGE AND SHALL BE GASKETED. TERMINALS SHALL ACCOMMODATE UP TO #4 AWG CABLE. TEST STATION SHALL BE SIZED TO SUIT INSTALLATION.
2. FOR DETAILS ON COPPER GROUND PLATE SEE SHEET E-10.
3. CONTRACTOR SHALL ROUTE ANODE CONNECTING WIRE THRU A SPARE 2 1/2" CONDUIT LOCATED IN 27" CONDUIT ENTRY CASING, INTO THE CABLE SPREADING AREA. FROM THE CABLE SPREADING AREA THE CONDUCTORS SHALL BE ROUTED IN A 1 1/2" CONDUIT TO THE CATHODIC PROTECTION TEST STATION. FOR LOCATION OF THE 27" CASING SEE THE EQUIPMENT POWER AND RACEWAY PLAN - CABLE SPREADING AREA SHEET E-2. FOR LOCATION OF THE CATHODIC PROTECTION TEST STATION SEE EQUIPMENT AND LIGHTING PLAN SHEET E-1.
4. SEALING BUSHING SHALL CONSIST OF PVC COATED STEEL DISCS, NEOPRENE SEALING RING, STAINLESS STEEL SOCKET CAP SCREWS AND WASHERS. BUSHING COLLAR SHALL BE HOT DIPPED GALVANIZED MALLEABLE IRON.



SENSOR CABLE TERMINATION
N.T.S.



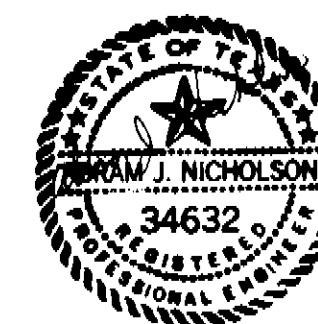
SECTION
C
N.T.S.



PART PLAN

6 DATAC POWER RECEPTACLE
N.T.S.

NOTE: ALL WORK SHOWN ON THIS DRAWING IS INCLUDED IN THIS CONSTRUCTION CONTRACT. ALL WORK SHOWN CLOUDED IS A REVISION BY BLACK & VEATCH.



NOTE: CONTRACTOR SHALL INSTALL DATAC POWER RECEPTACLE ASSEMBLIES AFTER DATAC CABINET FRAMES ARE SET IN THEIR FINAL POSITION ON THE SHED FLOOR. PENETRATIONS THROUGH CABINET BOTTOM PLATE AND FLOOR GRATING SHALL BE COORDINATED WITH CABINET LOCATIONS. PENETRATION DIAMETERS SHALL BE LARGE ENOUGH TO CLEAR RECEPTACLE FLANGE. CONTRACTOR SHALL PROVIDE BOTH FEMALE RECEPTACLE AND MALE CONNECTOR.

SATISFACTORY TO
DATE
SATISFACTORY TO
DATE
COMNAVSEASYSOCM

Frederic R. Harris, Inc.
300 East 42nd Street
New York, NY 10017
SUBMITTED BY
DATE
OCEAN ENGR. & CONST. PROJ. OFF. FPD-1
EIC

APPROVED HEAD FPD-1
DATE
APPROVED
DATE
EPD FOR COMMANDER, NAVFAC

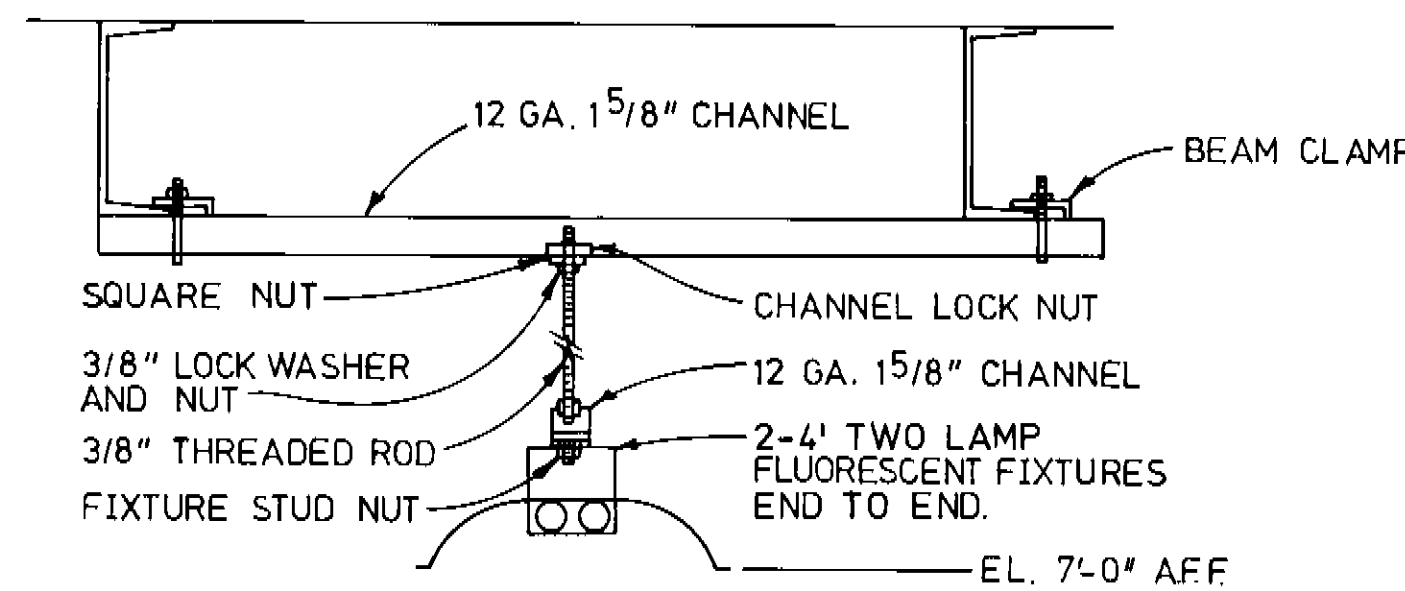
DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND
CHESAPEAKE DIVISION
WASHINGTON, D.C.

CATHODIC PROTECTION AND MISCELLANEOUS DETAILS

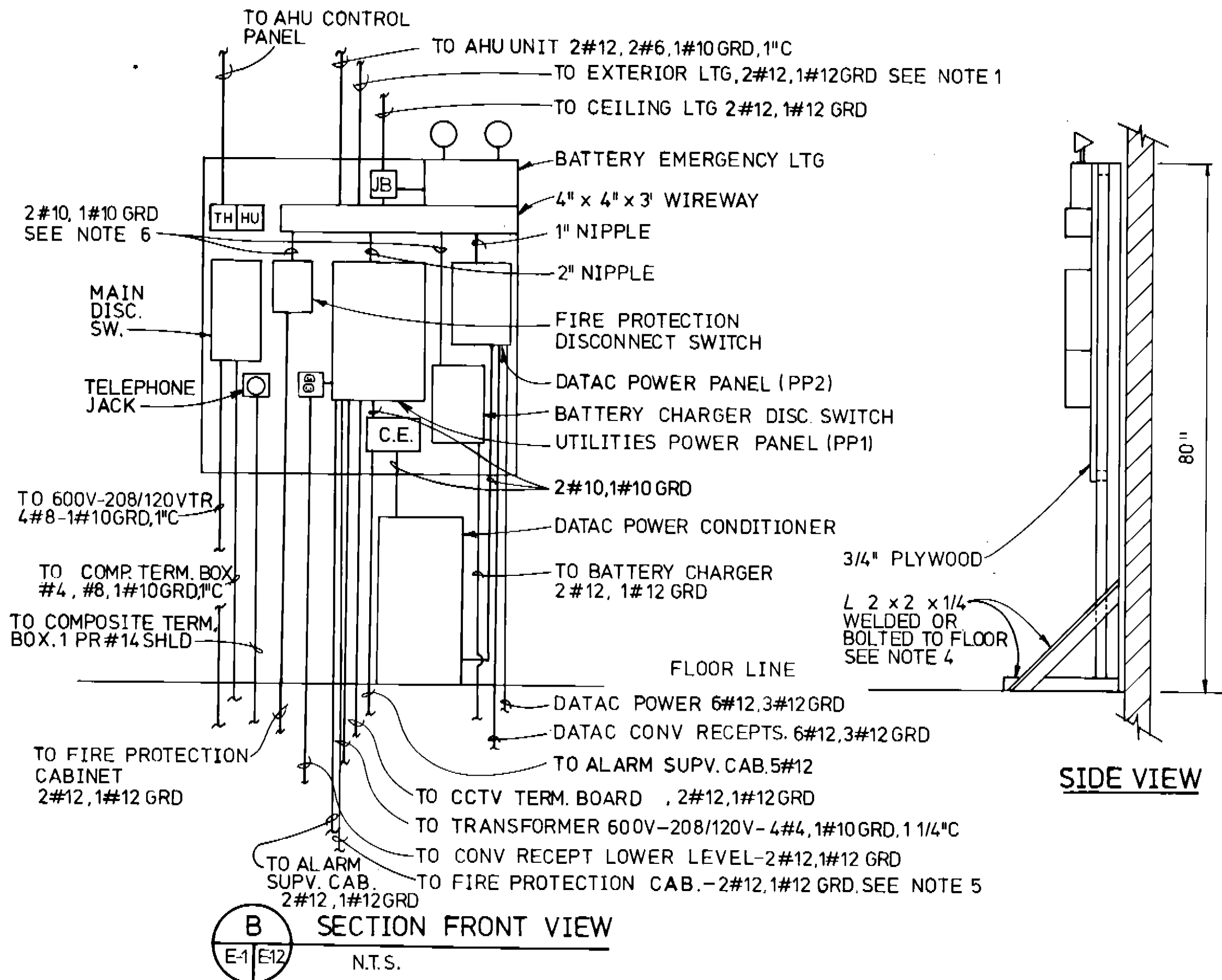
SIZE CODE IDENT. NO. NAVFAC DRAWING NO.
D 80091 3020704
CONSTR. CONTR. NO. N62477-89-C-0053
SCALE NOT TO SCALE SPEC. 21-89-0053 SHEET 31 OF 40

STANDARD 15 AMP
STRAIGHT BLADE
HOSPITAL GRADE
RECEPTACLE

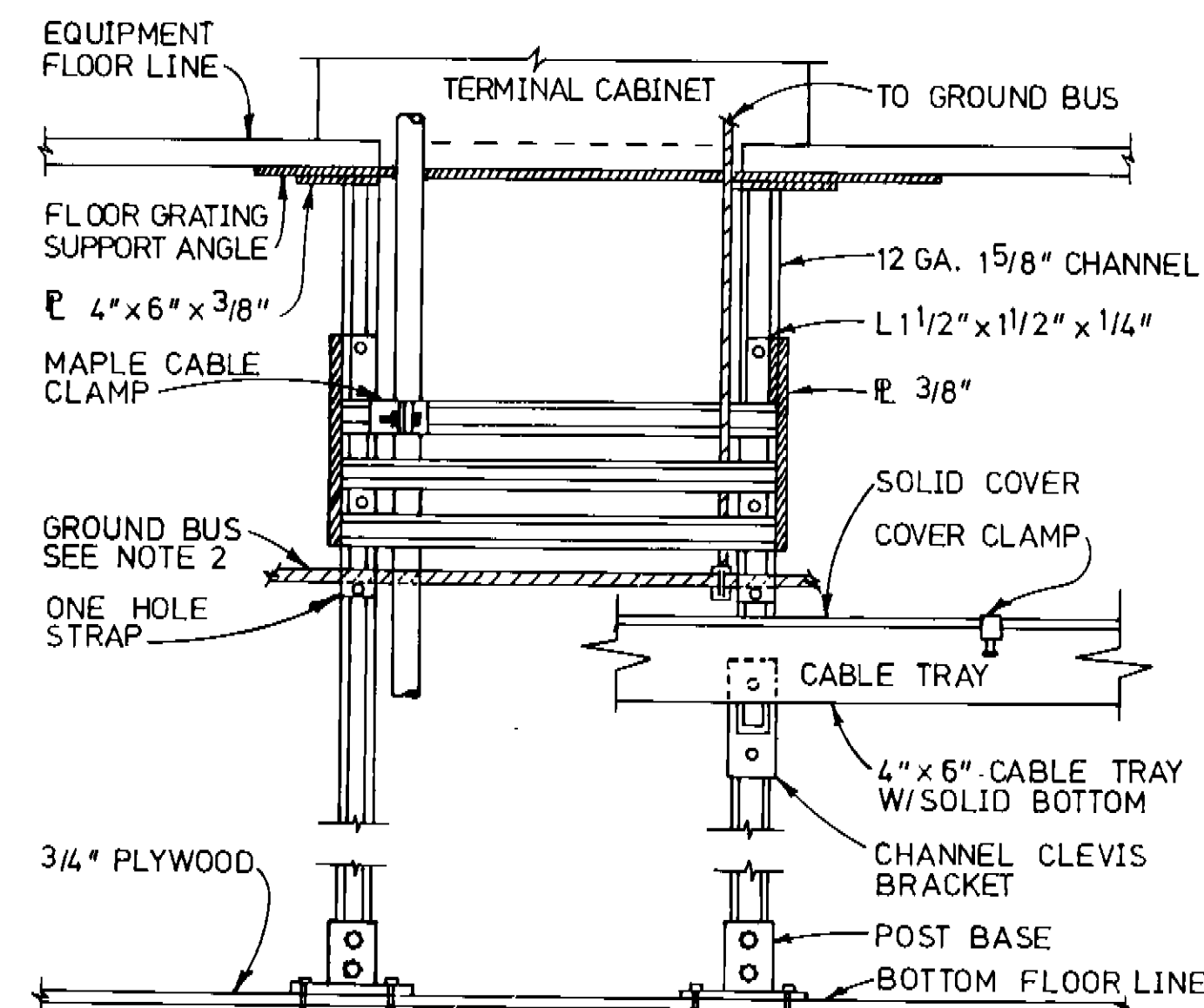
1 DETAIL
E-1 E-12
N.T.S.



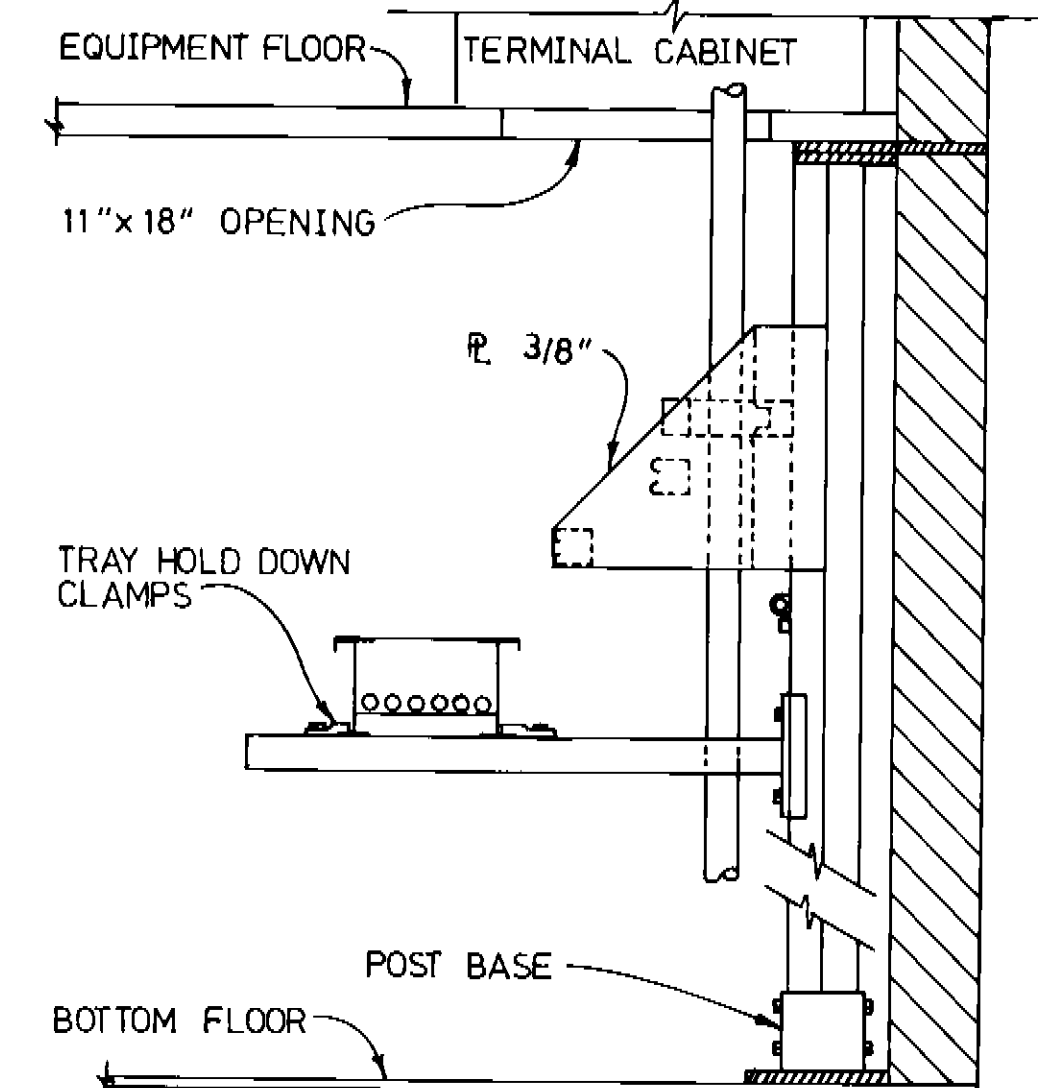
A SECTION END VIEW
E-1 E-12
N.T.S.



B SECTION FRONT VIEW
E-1 E-12
N.T.S.

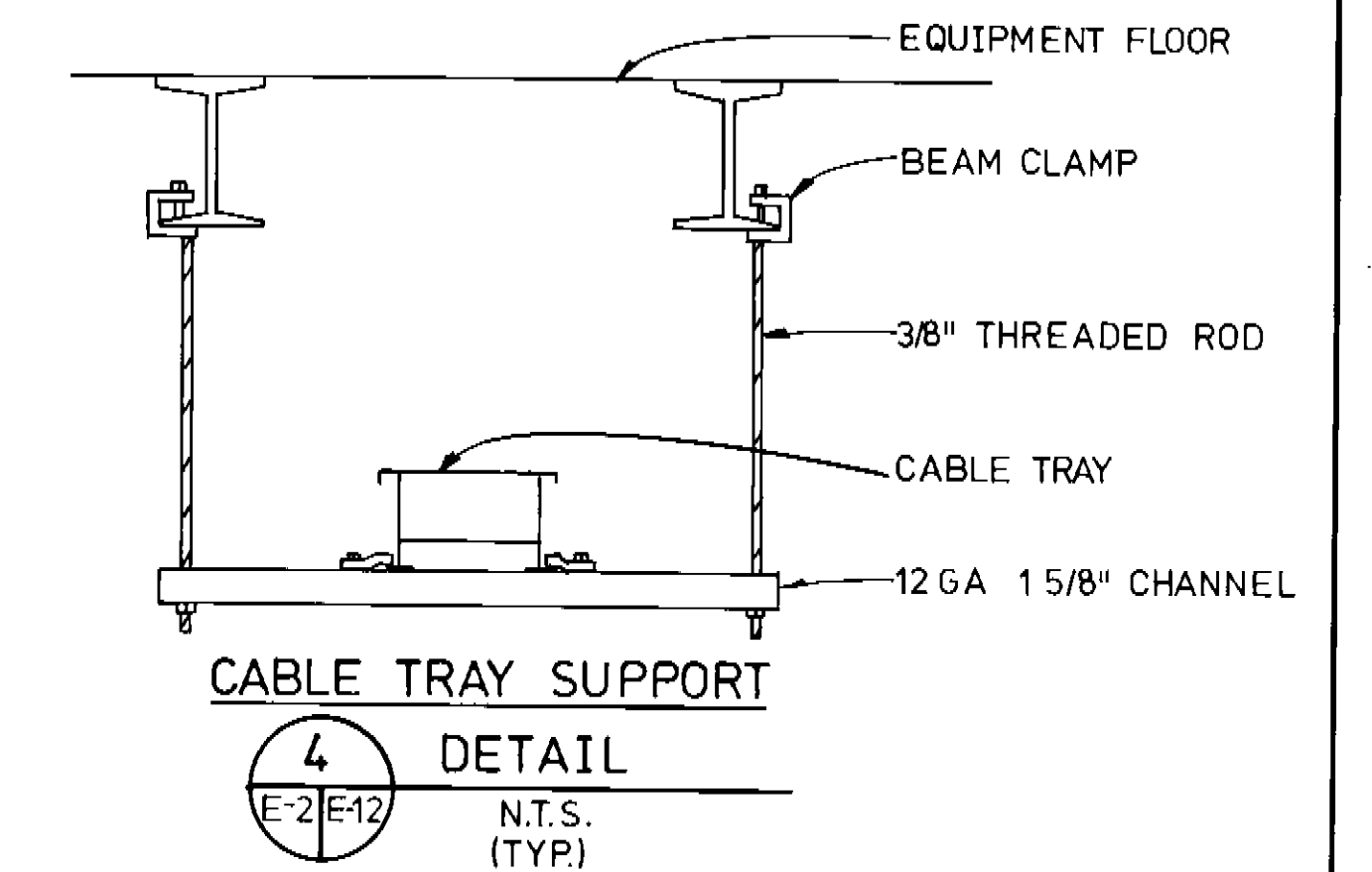


CABLE AND CABLE TRAY SUPPORT
C SECTION FRONT VIEW
E-2, E-10
SCALE: 1 1/2" = 1'-0"

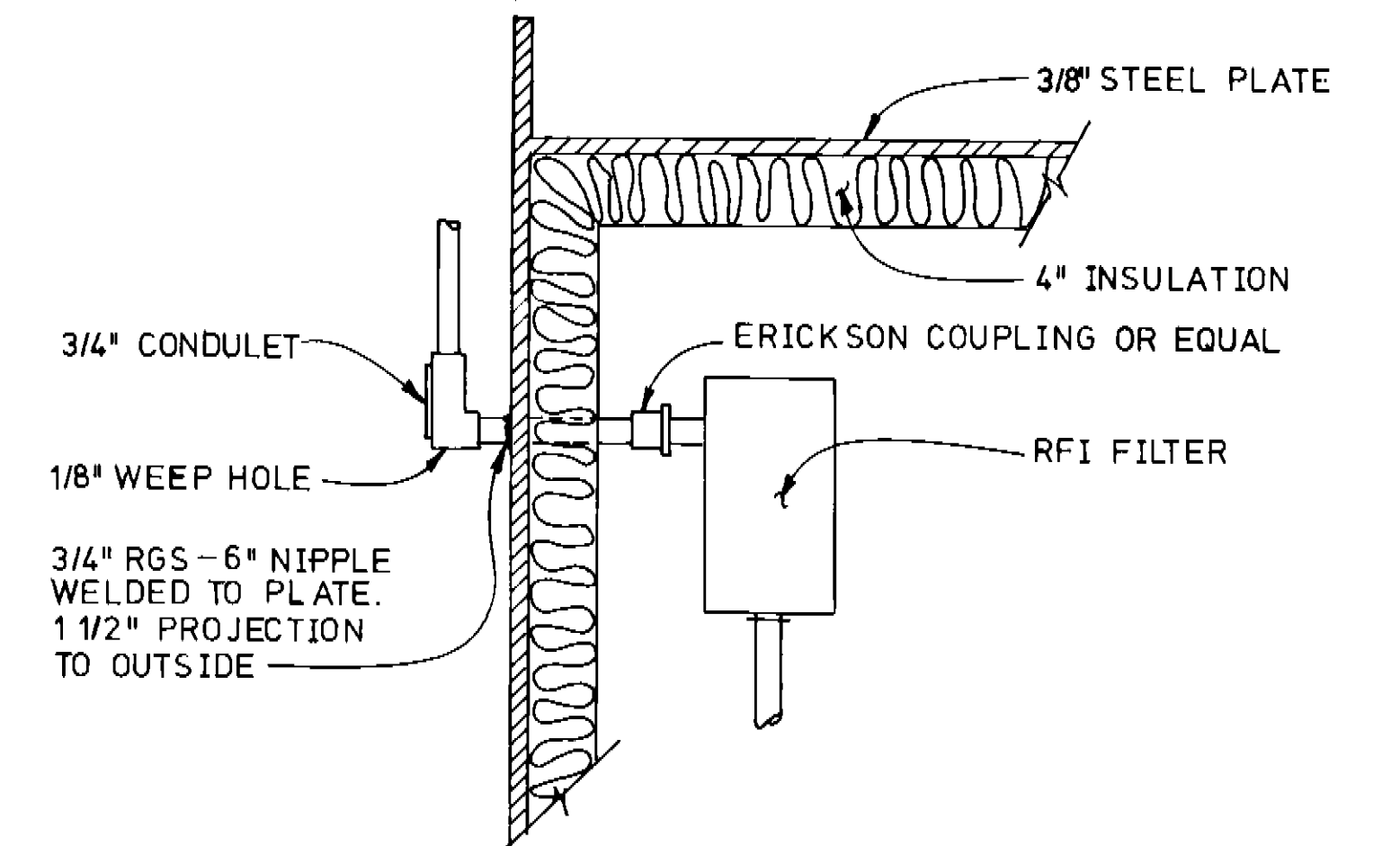


2 DETAIL SIDE VIEW
E-12 E-12
SCALE: 1 1/2" = 1'-0"

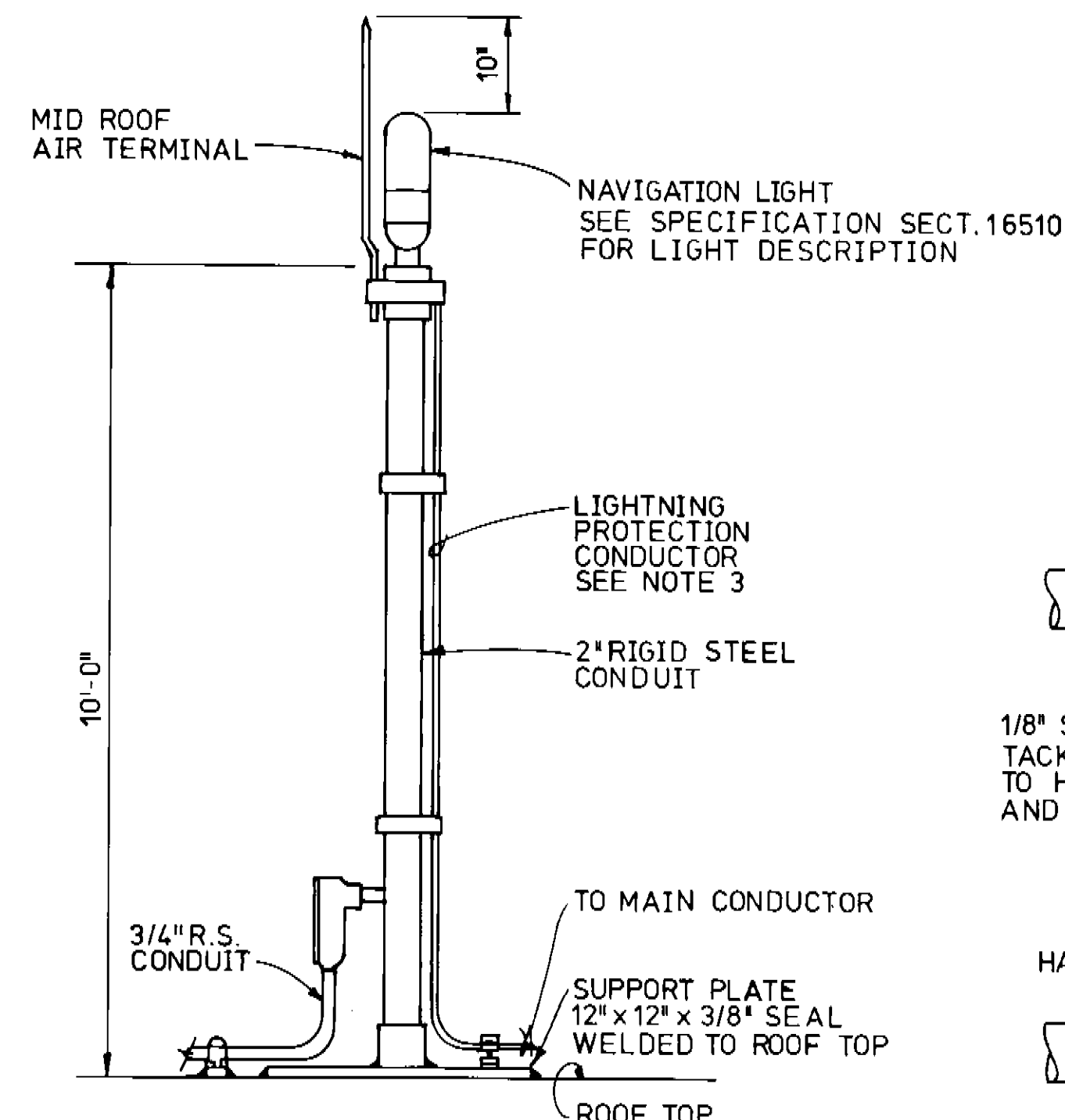
REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED



4 DETAIL
E-2 E-12
N.T.S.
(TYP.)

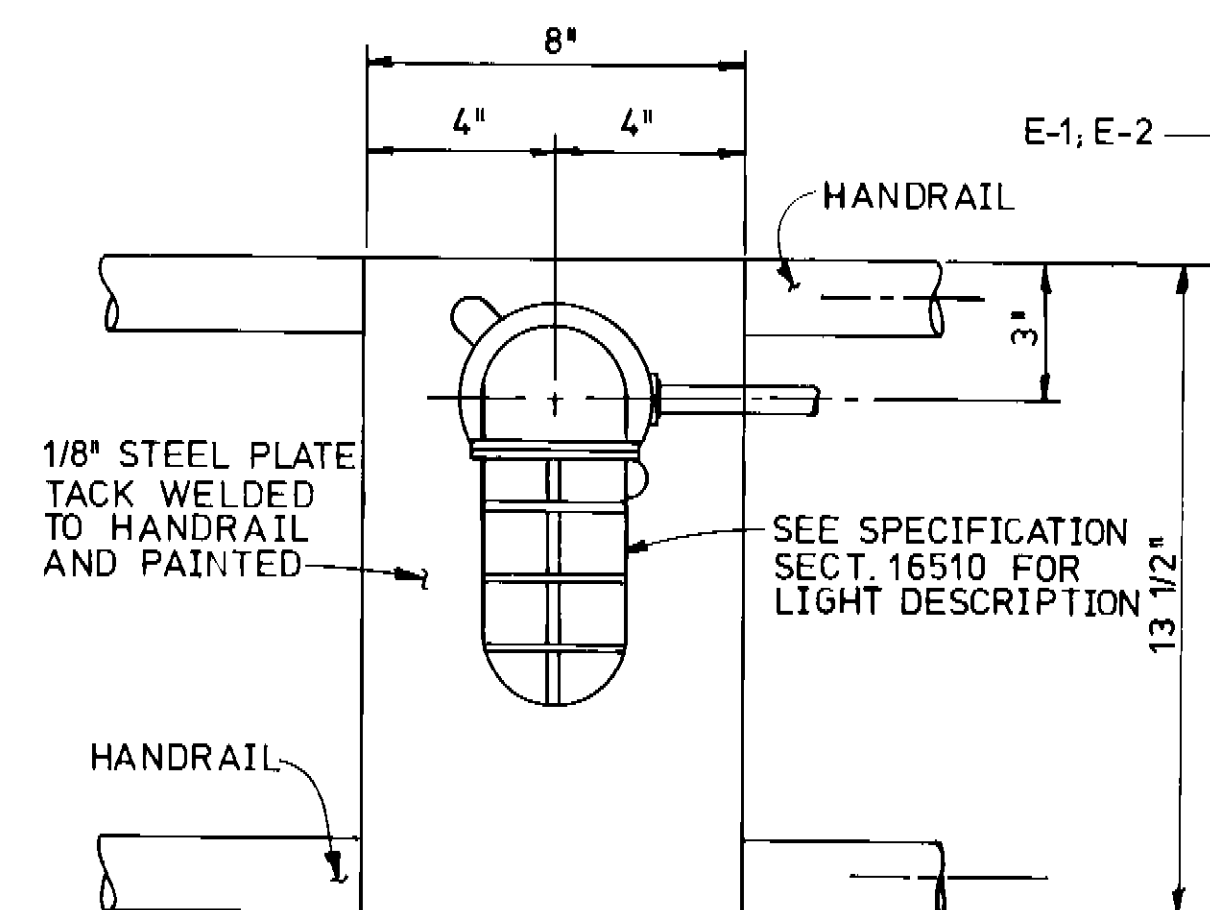


RFI WALL PENETRATION
5 DETAIL
E-1, E-2
N.T.S.
(TYP.)



NAVIGATION LIGHT AND
MID ROOF AIR TERMINAL

3 DETAIL
E-2 E-12
N.T.S.



EXTERIOR LIGHT MOUNTING

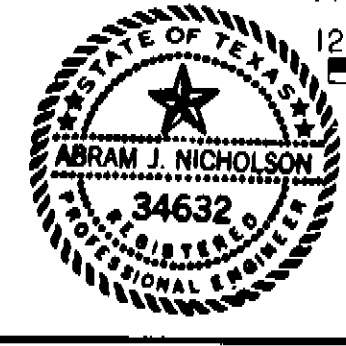
D SECTION
E-2 E-12
N.T.S.

GRAPHIC SCALE

WARNING: DRAWING MAY BECOME DISTORTED DURING REDUCTION OR OTHER REPRODUCTION PROCESS. USER IS CAUTIONED AGAINST SCALING.

12" 9" 6" 3" 0
1 1/2" = 1'-0"

SATISFACTORY TO
ROICC
DATE
SATISFACTORY TO
COMNAVSEASYS/COM
DATE



Black & Veatch
Engineers-Architects
DESIGNED BY DR MAB
CHECKED BY CHK HRA
SUBMITTED BY DATE
TITLE: PROGRAM. MOD.
OCEAN ENGR. & CONST. PROJ. OFF. EPO-1
EIC GAG & RVD RVD 5/1/89

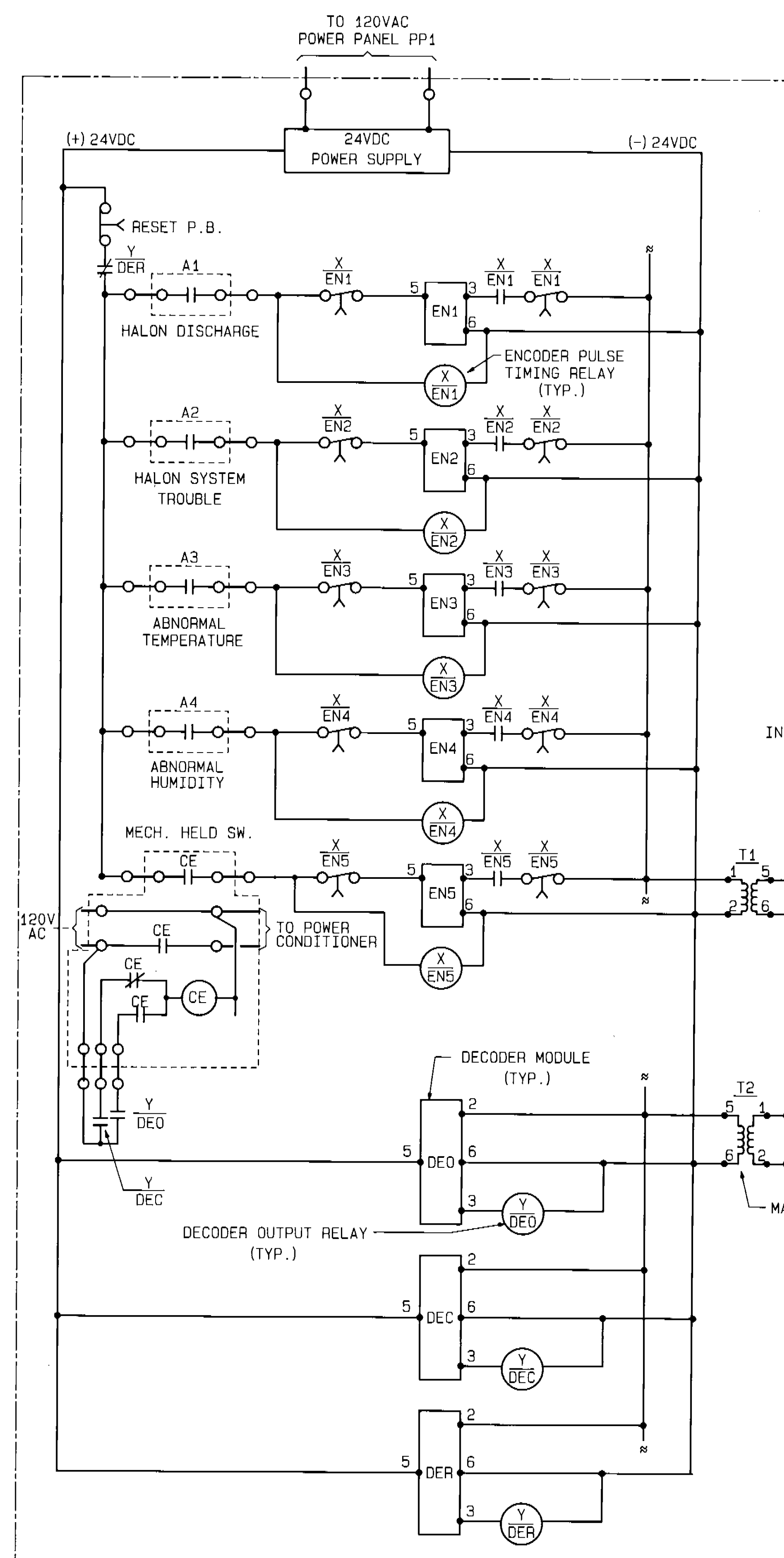
APPROVED HEAD EPO-1 DATE
APPROVED DATE
EFD FOR COMMANDER, NAVFAC

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
CHESAPEAKE DIVISION
WASHINGTON, D.C.

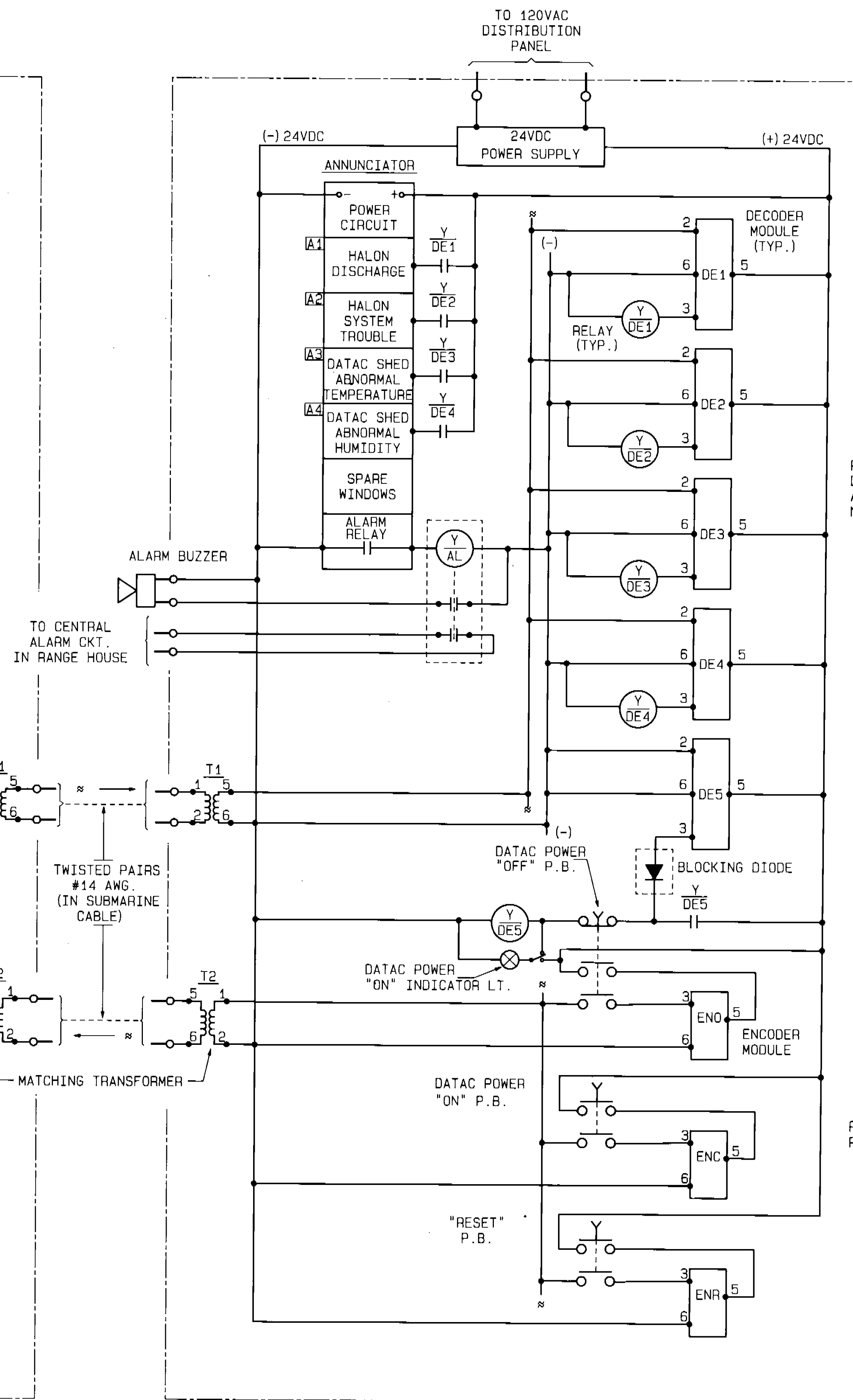
MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION

MISCELLANEOUS DETAILS

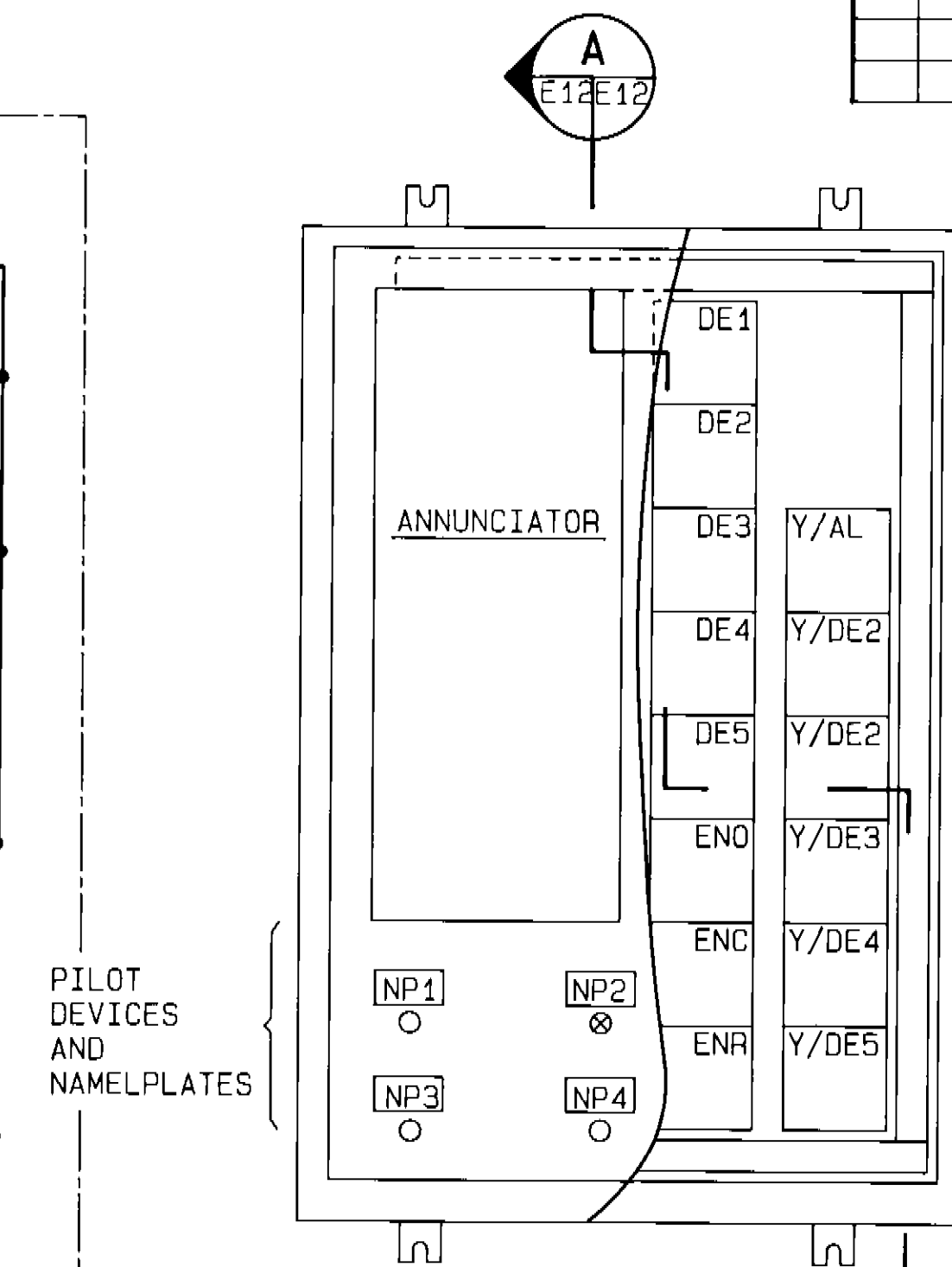
SIZE CODE IDENT. NO. NAVFAC DRAWING NO.
D 80091 3020705
CONSTR. CONTR. NO. N62477-89-C-0053
SCALE AS SHOWN SPEC 21-89-0053 SHEET 32 OF 40



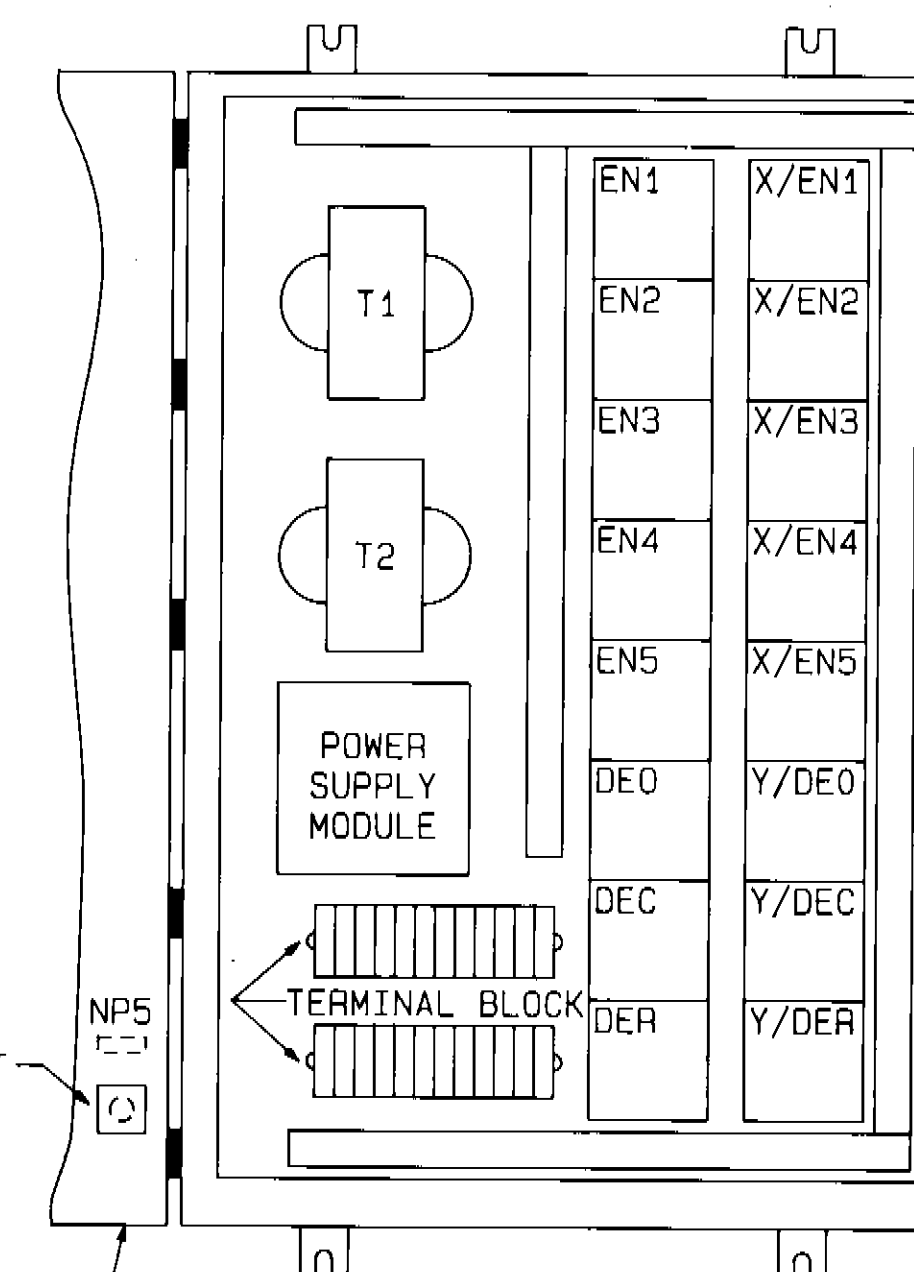
ALARM SUPERVISORY CIRCUIT
DATAC SHED UNIT



ALARM SUPERVISORY CIRCUIT
RANGE HOUSE UNIT

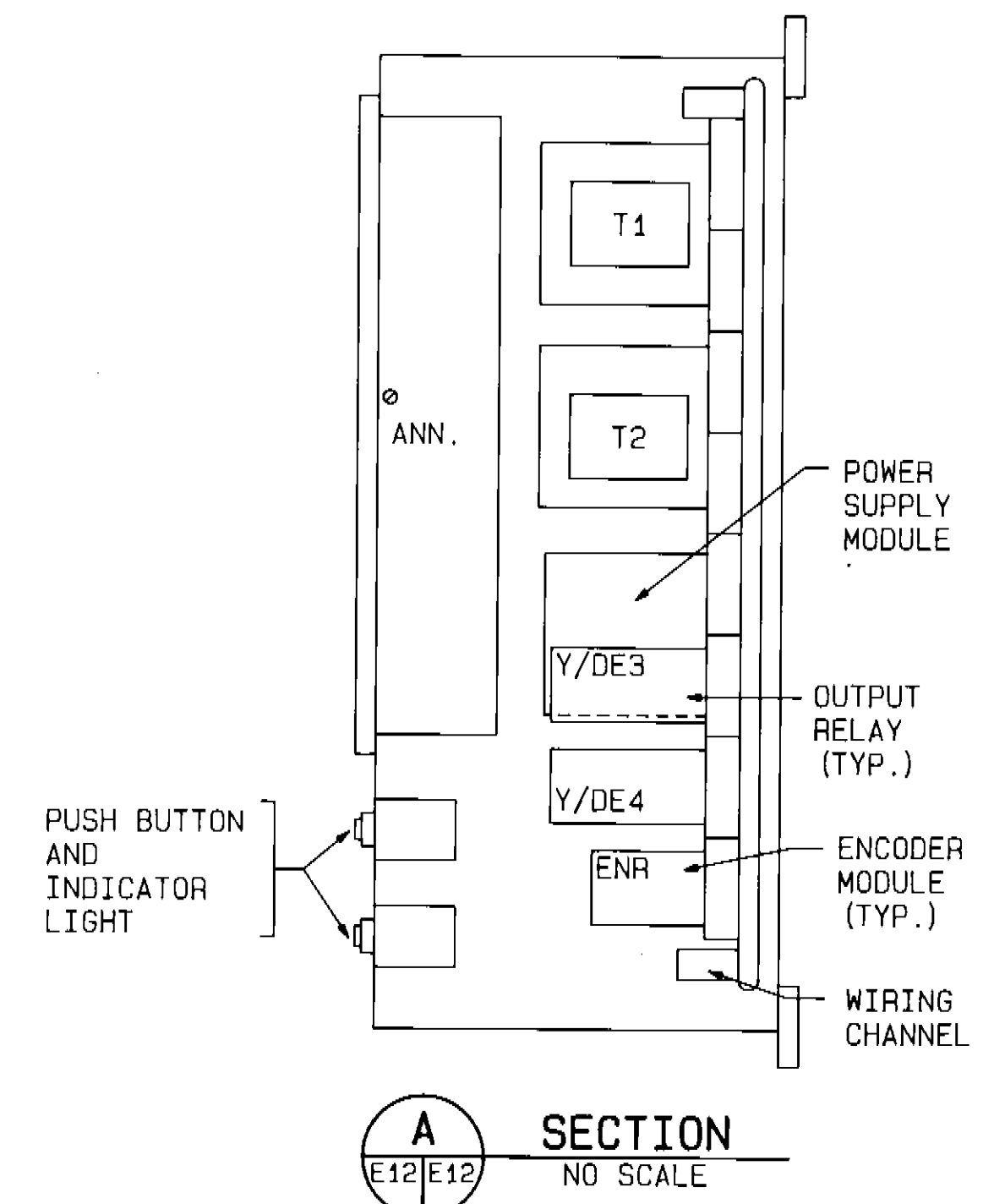


ALARM SUPERVISORY CABINET
RANGE HOUSE UNIT
N.T.S.



ALARM SUPERVISORY CABINET
DATAC SHED UNIT
N.T.S.

REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED



ENCODER DECODER	ALARM FUNCTION	CONTROL FUNCTION	INDICATION FUNCTION
EN1/DE1	HALON DISCHARGE	—	—
EN2/DE2	HALON SYS. TROUBLE	—	—
EN3/DE3	ABNORMAL TEMPERATURE	—	—
EN4/DE4	ABNORMAL HUMIDITY	—	—
EN5/DE5	—	—	DATAC POW "ON"
EN0/DE0	—	DATAC POW. "ON"	—
ENC/DEC	—	DATAC POW. "OFF"	—
ENR/DER	—	ALARM SUPV CKT. RESET	—

ENCODER/DECODER FUNCTIONS

No.	ENGRAVING
NP1	SUPERVISORY CKT. RESET
NP2	DATAC POWER ON
NP3	DATAC POWER OFF
NP4	DATAC POWER ON
NP5	RESET

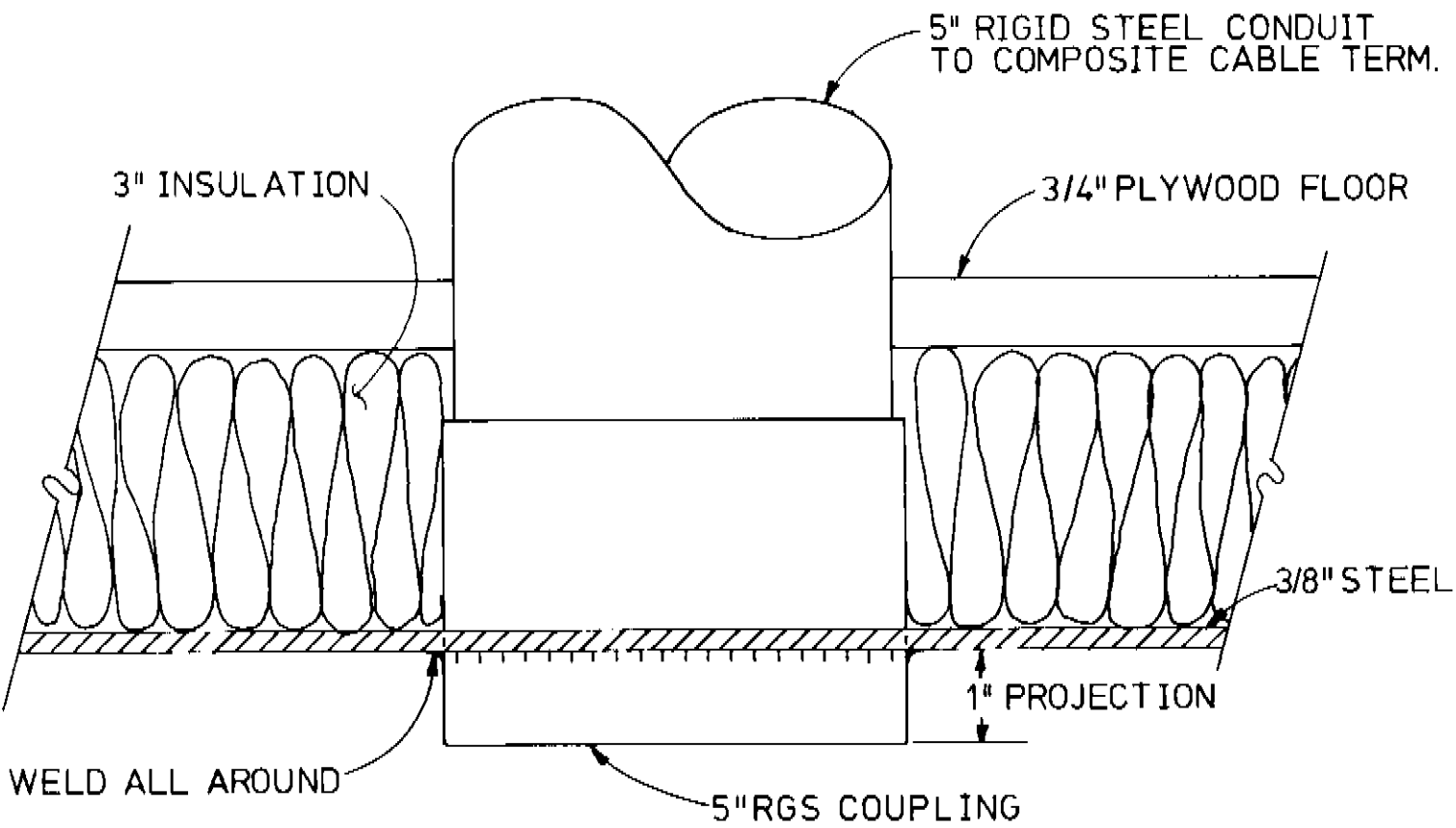
NAMEPLATES

Frederic R. Harris, Inc. 300 East 42nd Street New York, NY 10017		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
SUBMITTED BY: <i>E. SCHMELTZ</i> DATE: <i>4-15-88</i> OCEAN ENGR. & CONST. PROJ. OFF. FPO-1 ETC. <i>408 W. RVD TO 4/22/88</i>		ALARM SUPERVISORY CIRCUIT AND REMOTE DATAC POWER CONTROL	
SATISFACTORY TO: <i>2/1/88</i> ROICC DATE: <i>5-15-89</i> COMNAVSEASYS COM DATE:	APPROVED HEAD FPO-1: <i>[Signature]</i> DATE: <i>5-15-89</i> APPROVED: <i>[Signature]</i> DATE: <i>5/15/89</i> EFD FOR COMMANDER, NAVFAC	SIZE: D CODE: 80091 IDENT. NO.: 3020706 CONSTR. CONTR. NO.: N62477-89-C-0053 SCALE: NONE	NAVFAC DRAWING NO.: 3020706 SHEET: 33 OF 40

REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED

NOTES

- EXTERIOR CONDUITS AND FITTINGS SHALL BE PVC COATED RIGID STEEL WITH INTERIOR URETHANE COATING. CONDUIT COUPLINGS AND FITTINGS SHALL BE INSTALLED PER MANUFACTURERS RECOMMENDATIONS FOR SEAWATER APPLICATIONS.
- INSTALL CONDUIT SPACERS AT A MAXIMUM OF 6FT. INTERVALS. CASING SHALL BE FILLED WITH HIGH DENSITY FOAM IN STRAIGHT SECTION AFTER INSTALLATION OF PVC CONDUITS.
- SPARE CONDUITS SHALL BE SEALED WITH PVC THEADED PIPE PLUGS ON BOTH ENDS.
- ALL THREADED CONNECTIONS SHALL BE TAPED WITH TEFLON SEALING TAPE.
- RIGID COUPLING SHALL BE WELDED TO STRUCTURE AS SHOWN. EXTERIOR PROJECTION SHALL BE PROTECTED BY A COATING OF THE SAME MATERIAL REQUIRED FOR THE EXTERIOR OF THE DCS STRUCTURE.
- 5" RGS CHASE CONDUIT SHALL EXTEND DOWNWARD FROM COMPOSITE CABLE TERMINATION JUNCTION BOX INSIDE DCS TO A TERMINUS 3' BELOW AND PARALLEL TO MUD LINE. ANY BENDS IN CONDUIT SHALL HAVE MINIMUM RADIUS OF 48".

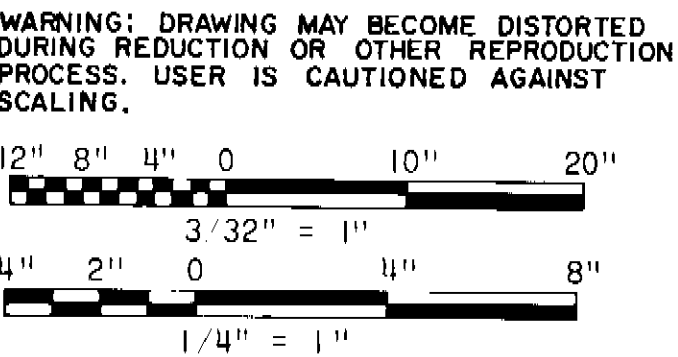


2 COMPOSITE CABLE CONDUIT DCS ENTRY DETAIL
E2/E14 N.T.S. SEE NOTE 5

NOTES (CONT'D)

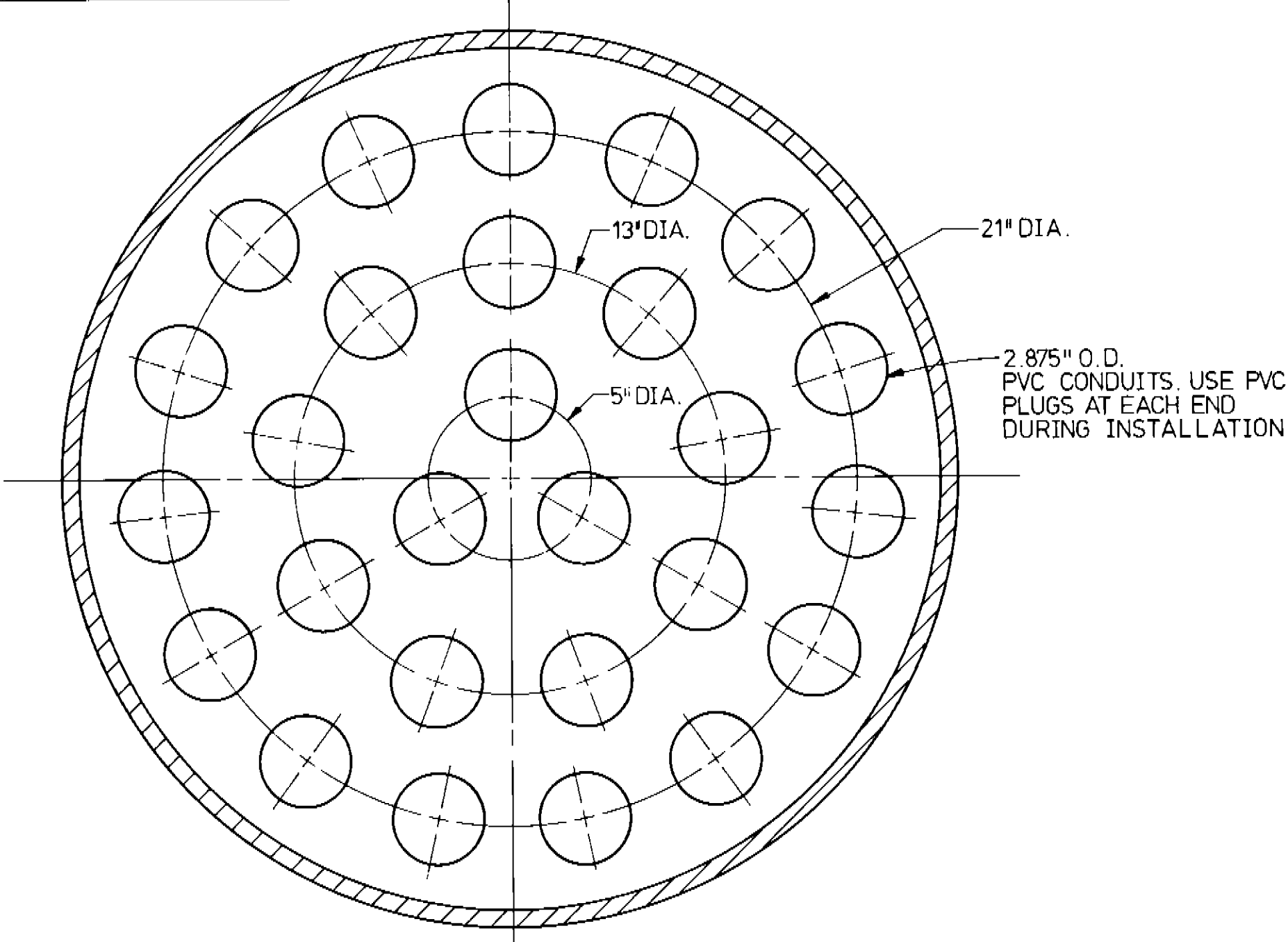
- SUPPORT GRIP SHALL BE STAINLESS STEEL SINGLE WEAVE WITH CLOSED OR SPLIT MESH FASTENED TO A SUPPORT RING. SUPPORT RINGS SHALL ALLOW THE THREADING OF COUPLINGS ON THE PVC TERMINAL ADAPTER WITH THE GRIP IN PLACE.
- CLOSURE PLATES SHALL BE WELDED TO CASING AND SHALL BE 12 GAGE STEEL MINIMUM. PROVIDE HOLES FOR CONDUIT AND FOAM FILLING.

GRAPHIC SCALE

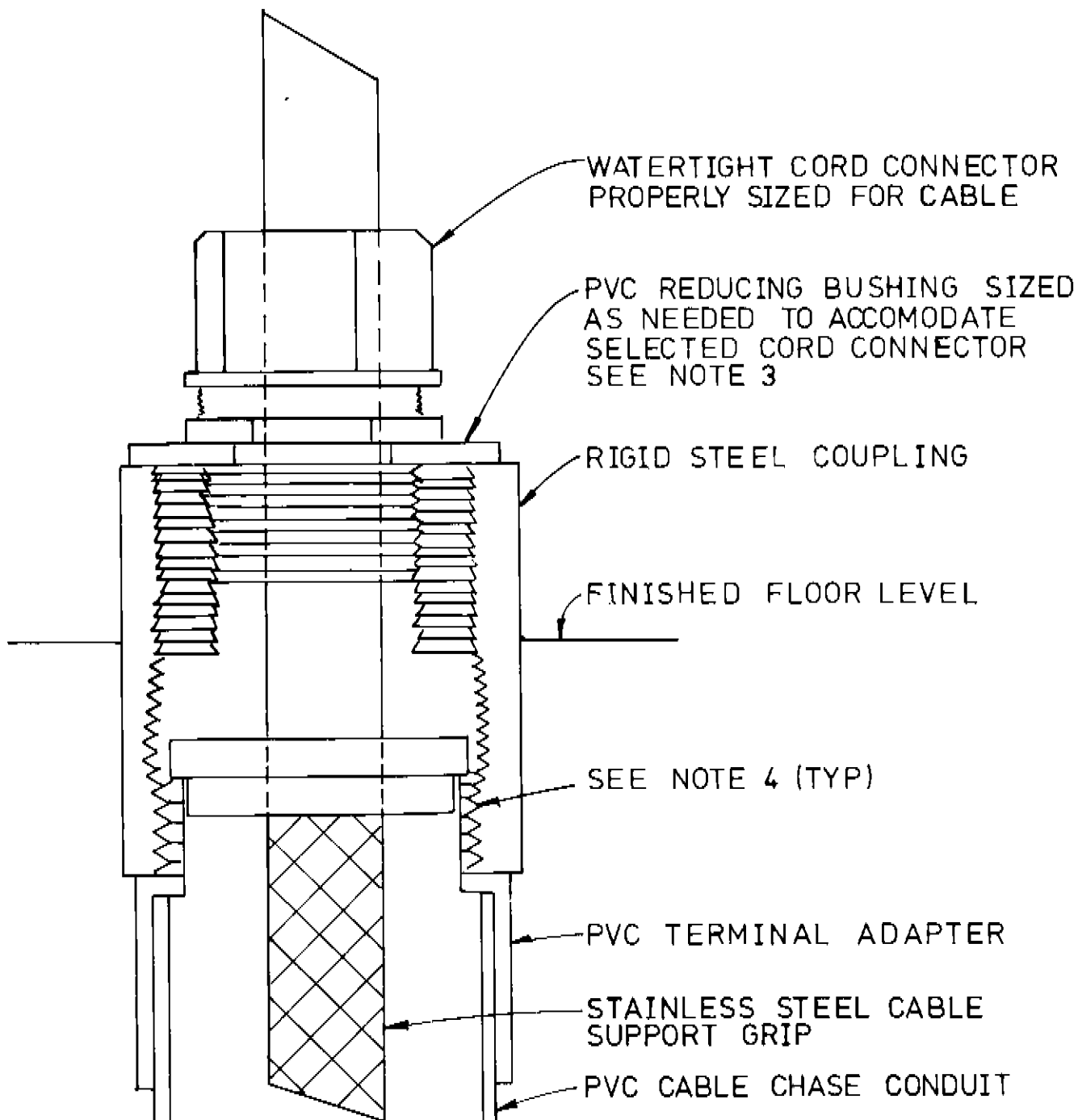


Black & Veatch Engineers-Architects		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DESIGN/AV DR DG CHK HRA SUBMITTED BY <i>[Signature]</i> DATE <i>02/01/82</i> OCEAN ENGR. B. CONST. PROJ. OFF. FPO-1 EIC. <i>[Signature]</i> RVD. <i>[Signature]</i> 3/4/82		MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION CONDUIT & CASING ASSEMBLY DETAILS	
SATISFACTORY TO <i>[Signature]</i> DATE <i>5/14/82</i> SATISFACTORY TO <i>[Signature]</i> DATE <i>5/17/82</i> COMNAVSEASYS.COM		SIZE D CODE IDENT. NO. 80091 NAVFAC DRAWING NO. 3020707	E-14
EFD FOR COMMANDER, NAVFAC		SCALE: AS SHOWN SPEC. 21-89-0053	SHEET 34 OF 40

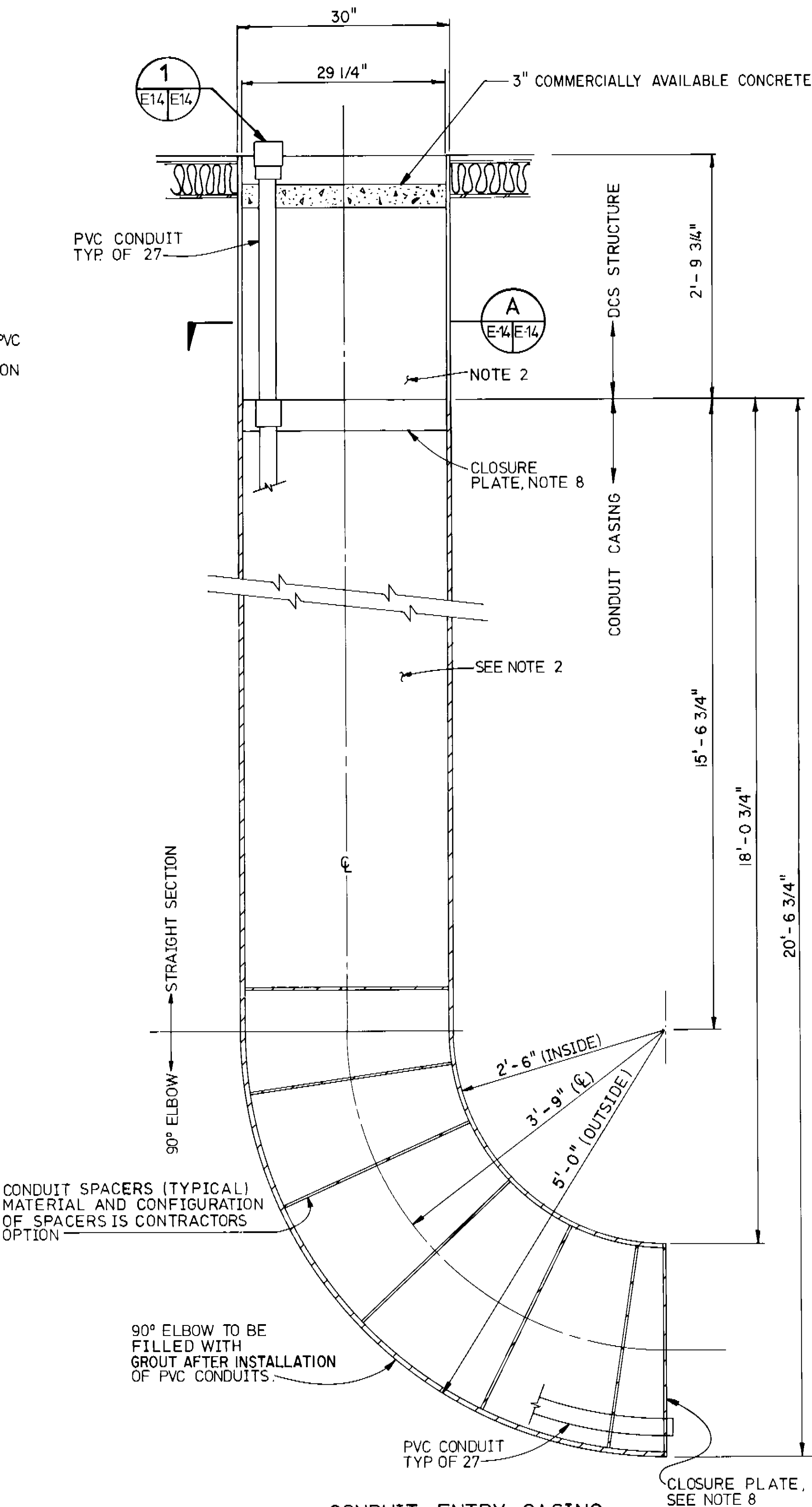
CONDUIT ENTRY CASING	
CASING OUTSIDE DIAMETER=	30 INCHES
CASING INSIDE DIAMETER=	29 1/4 INCHES
CONDUIT OUTSIDE DIAMETER=	2.875 INCHES
CONDUIT INSIDE DIAMETER=	2.5 INCHES



A SECTION
E14/E14
SCALE: 1/4" = 1"



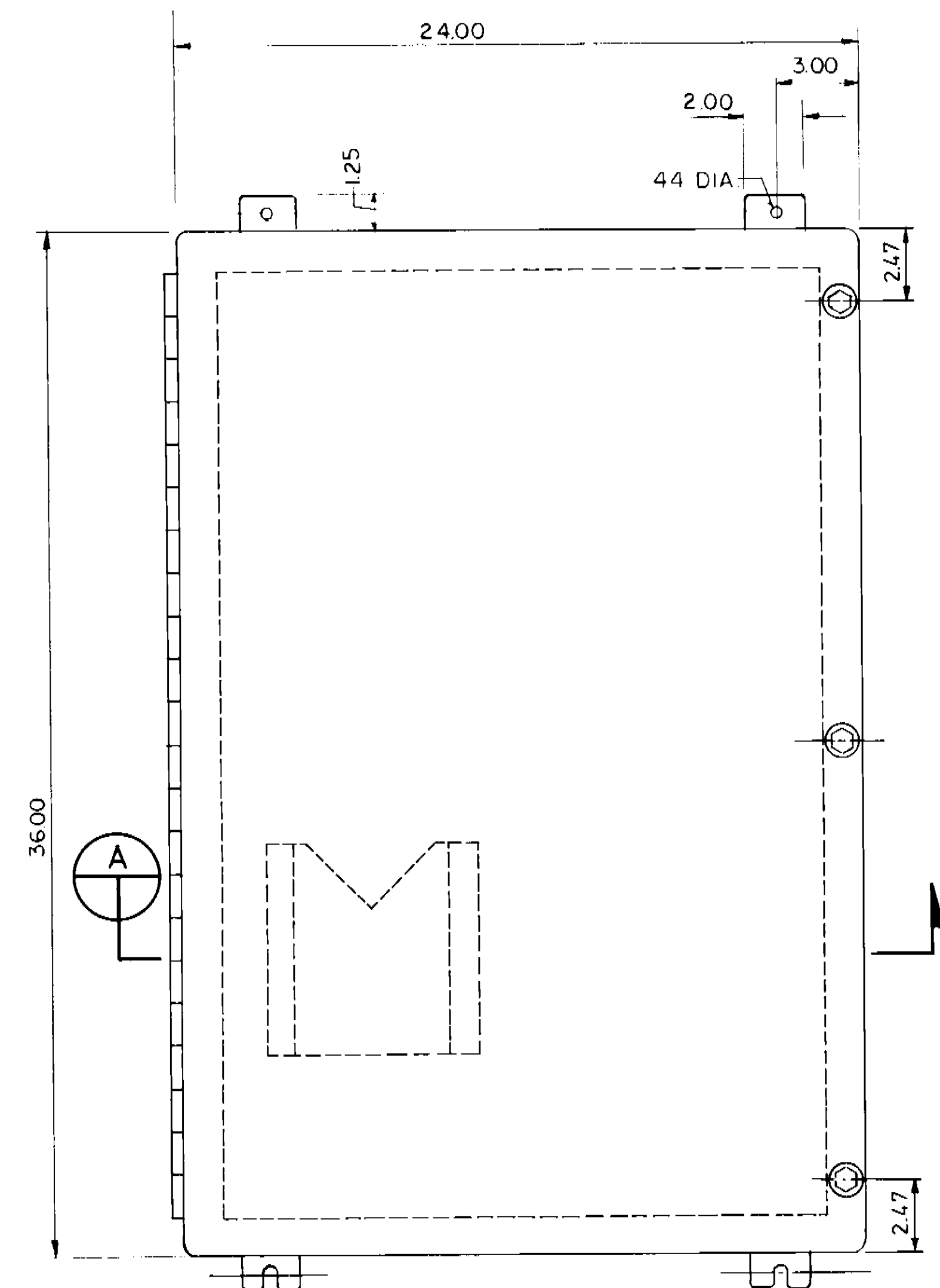
1 CABLE CHASE CONDUIT SEAL
E14/E14 N.T.S. TYPICAL EACH CONDUIT



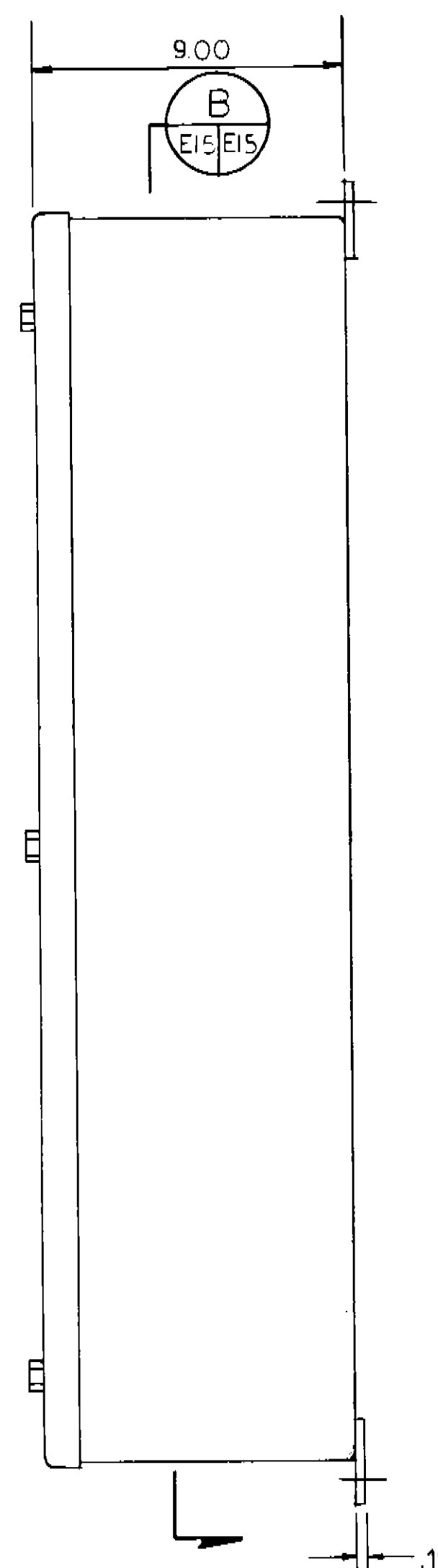
CONDUIT ENTRY CASING
N.T.S.



REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED



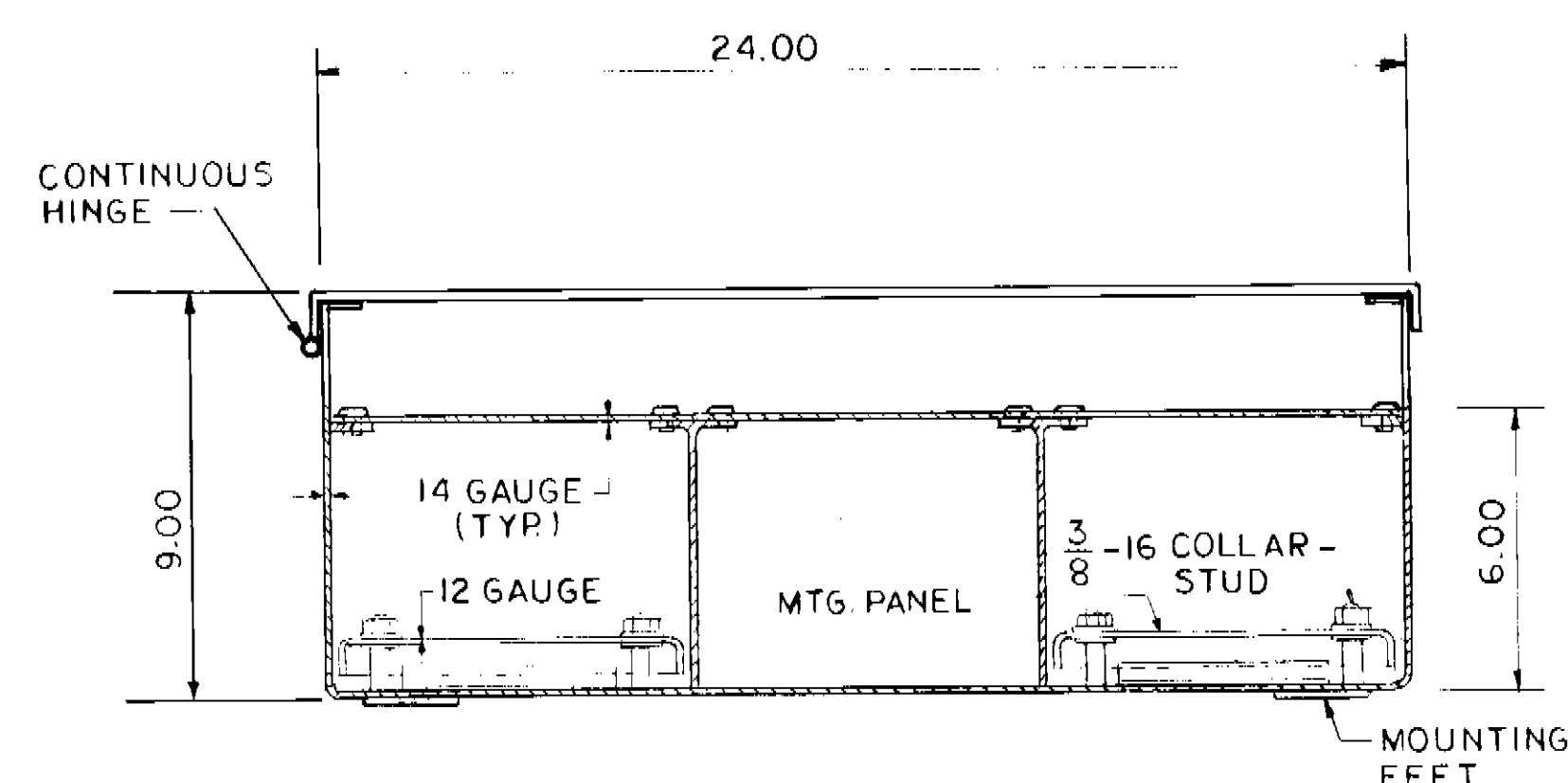
FRONT VIEW



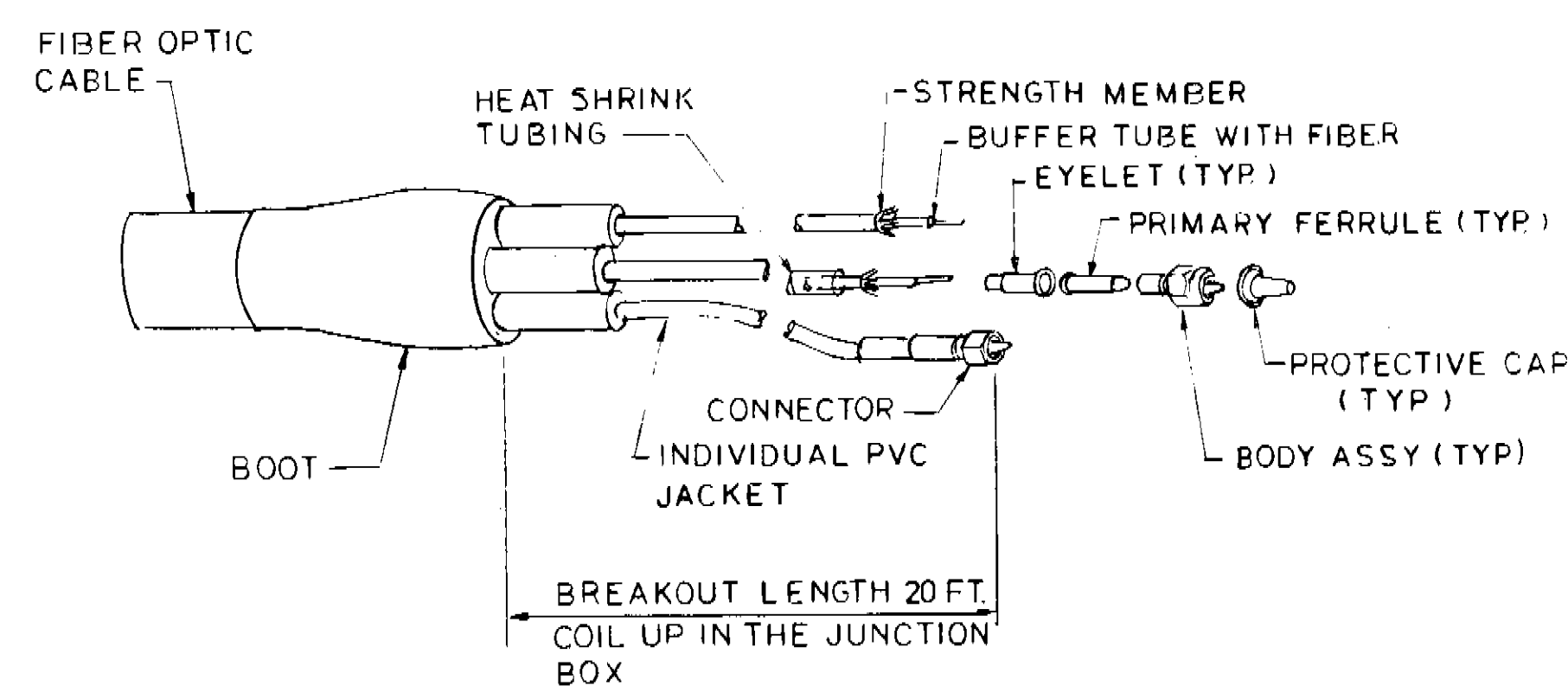
SIDE VIEW

RANGE HOUSE AND DCS COMPOSITE CABLE TERMINATION JUNCTION BOXES

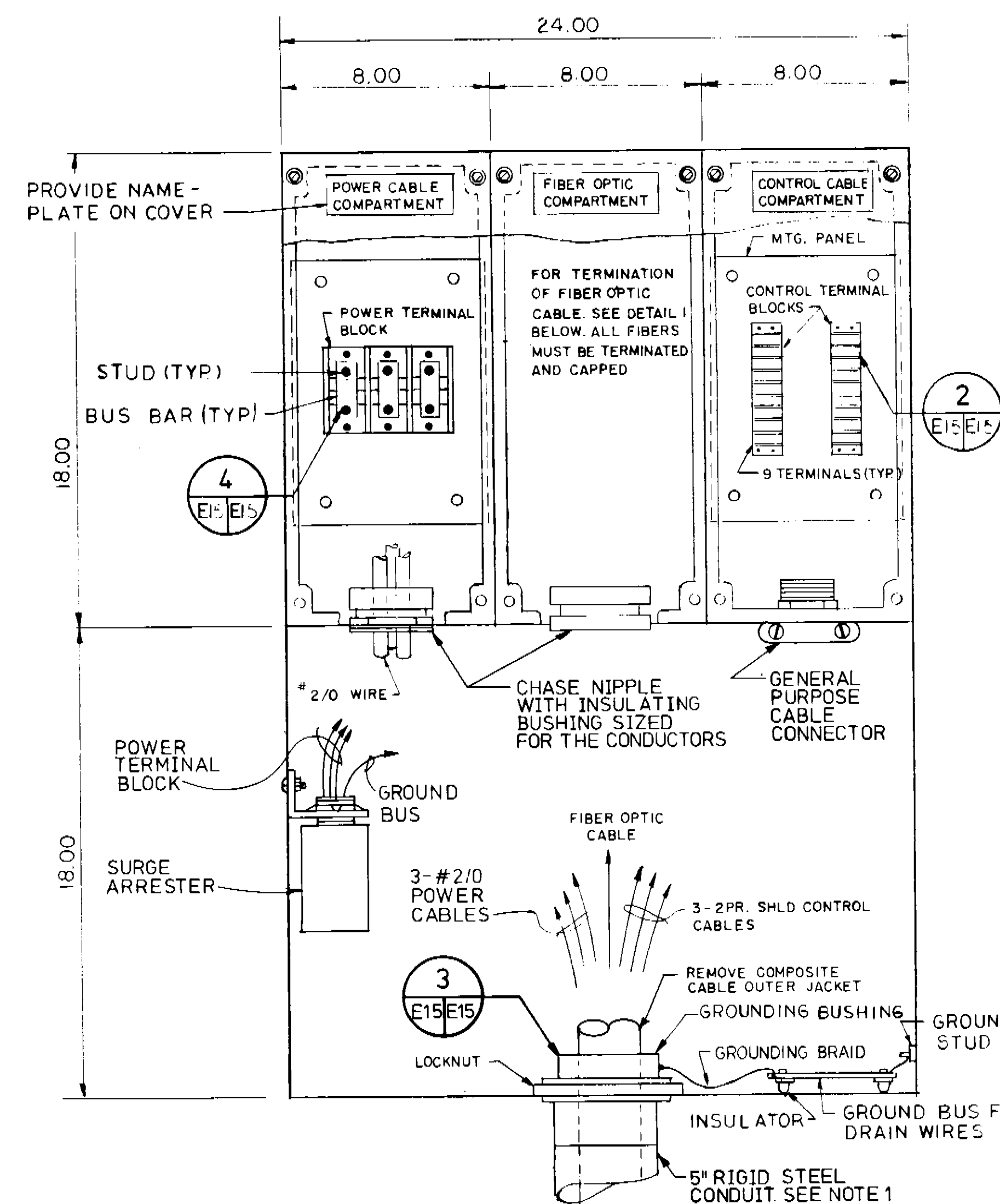
SCALE: 1/4" = 1"



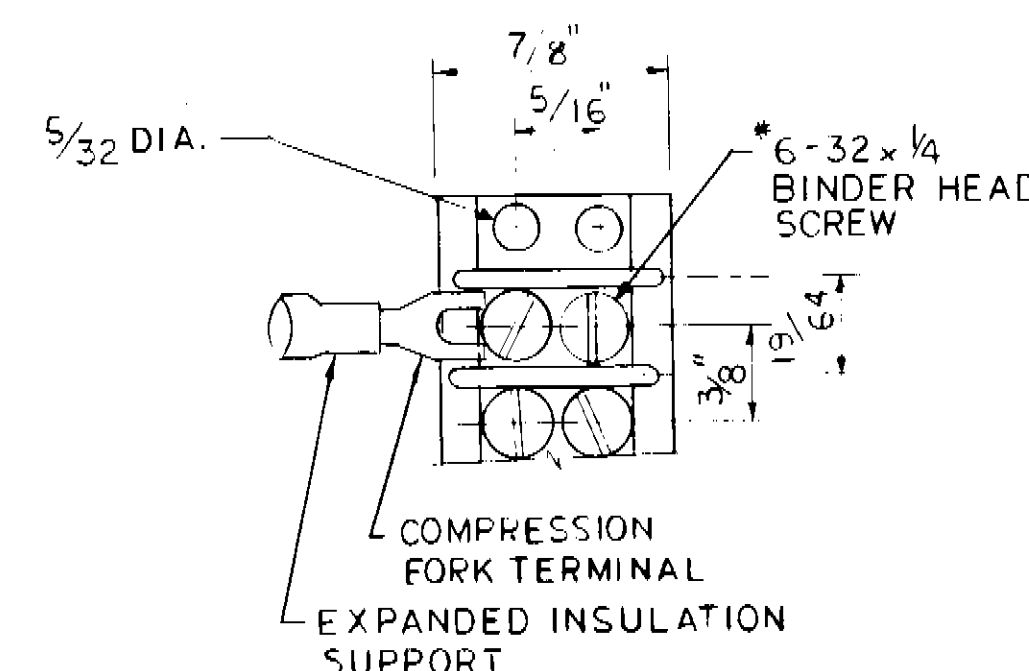
A SECTION
SCALE: 1/4" = 1"



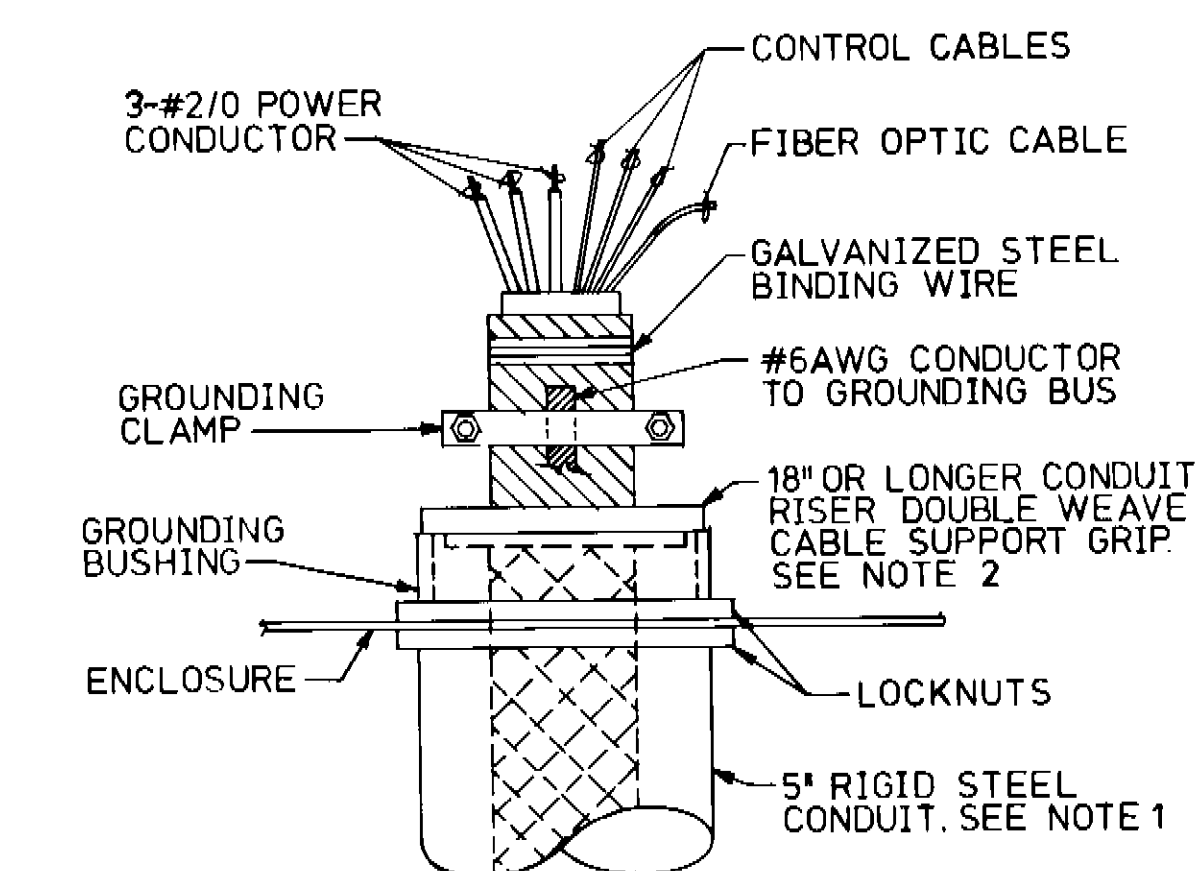
1 TERMINATION OF FIBER OPTIC CABLE
N.T.S.



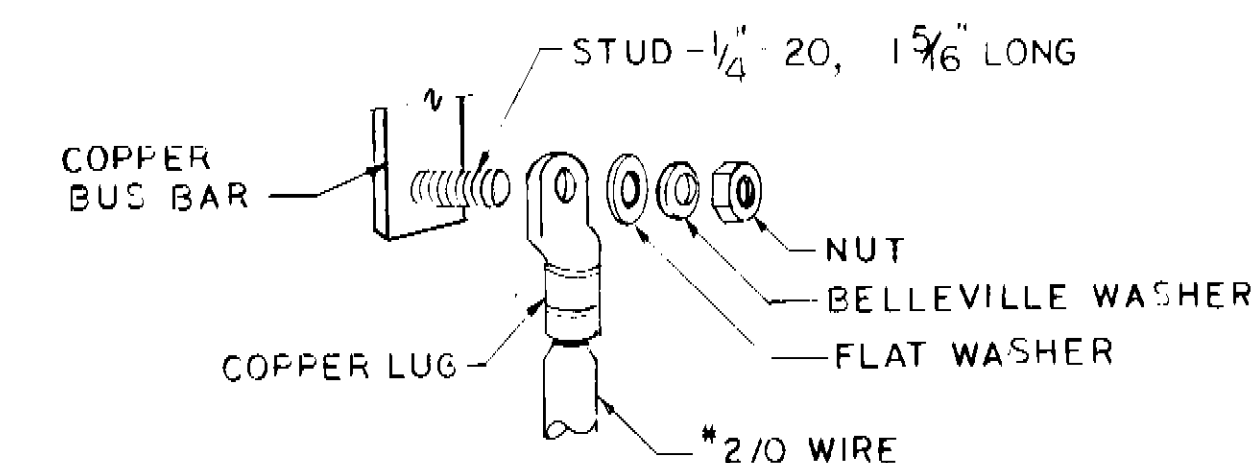
B SECTION
SCALE: 1/4" = 1"



2 CONTROL TERMINAL BLOCK
AND TERMINATION
N.T.S.

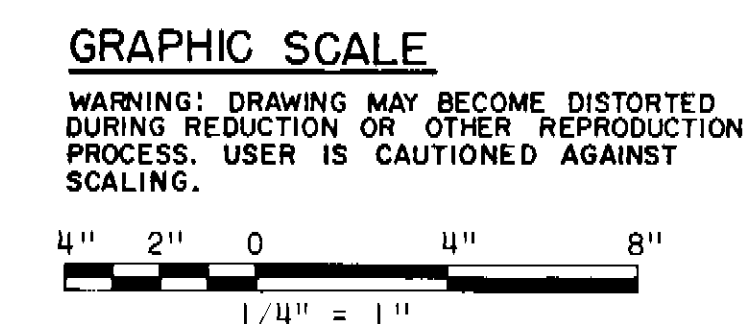


3 COMPOSITE CABLE TERMINATION
BOX ENTRY
N.T.S.

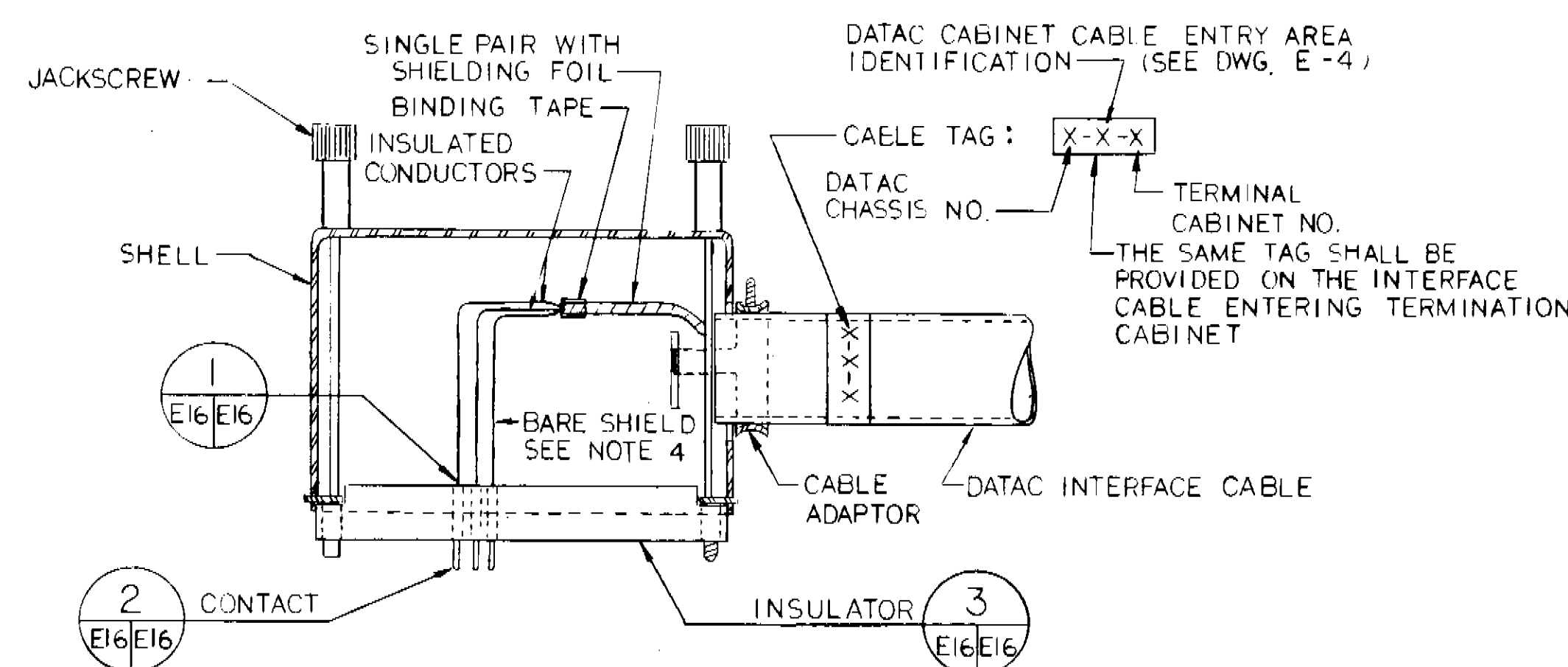


4 POWER CABLE TERMINATION
N.T.S.

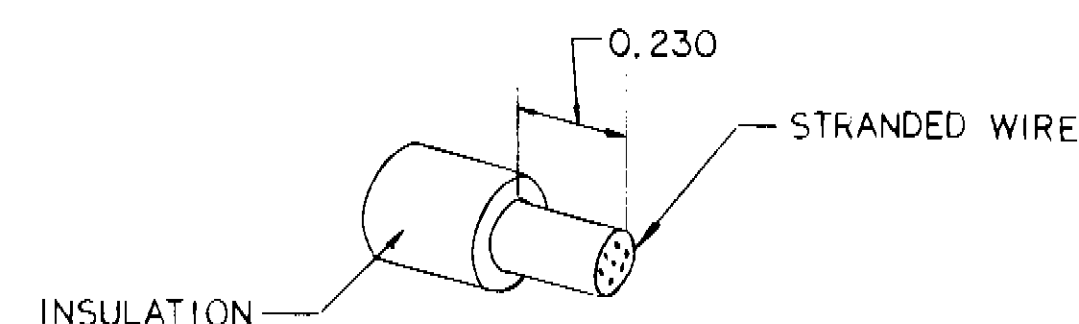
- NOTES:
1. FOR CONDUIT ENTRY DETAIL INTO DCS STRUCTURE, SEE DETAIL 2, DRAWING E-14.
 2. CABLE GRIP SUPPORT IS NOT REQUIRED ON COMPOSITE CABLE AT RANGE HOUSE TERMINATION.
 3. RANGE HOUSE AND DCS COMPOSITE CABLE TERMINATION BOXES SHALL BE PROPERLY GROUNDED. ALL THE POWER AND CONTROL CONDUCTORS AND SHIELDS SHALL BE TERMINATED AT THE TERMINAL BLOCKS.



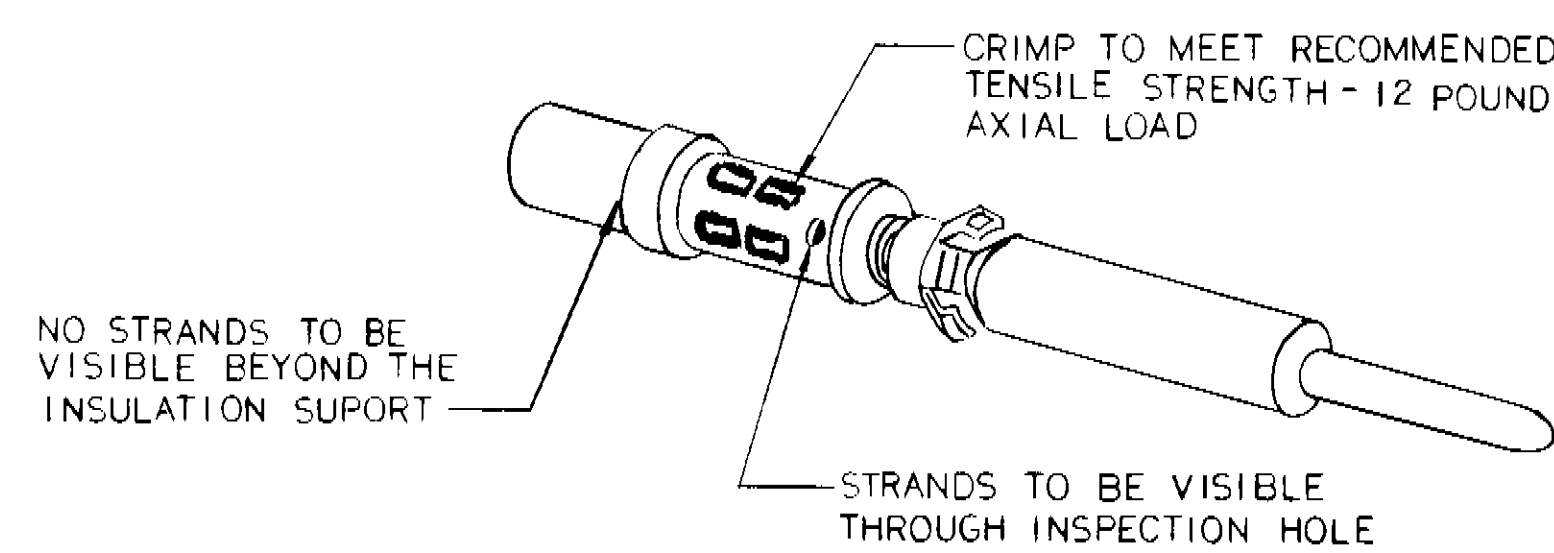
DESIGNED BY: DR. MAB SUBMITTED BY: DR. MAB DATE: 03/07/89 TITLE: PROGRAM MSH		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
OCEAN ENGR. & CONST. PROJ. OFF. FPO-1 EIC: 248 RVD: 10 4/22		MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION COMPOSITE CABLE TERMINATION DETAILS	
SATISFACTORY TO: 248 DATE: 5/15/89 SATISFACTORY TO: ROICC DATE: 5/15/89	APPROVED HEAD FPO-1: 248 DATE: 5/15/89 APPROVED: 248 DATE: 5/15/89	SIZE: D CODE IDENT. NO.: 80091 NAVFAC DRAWING NO.: 3020708	E-15
COMNAVJCS/SCOM DATE:		SCALE: AS SHOWN SPEC. 21-89-0053 SHEET 35 OF 40	



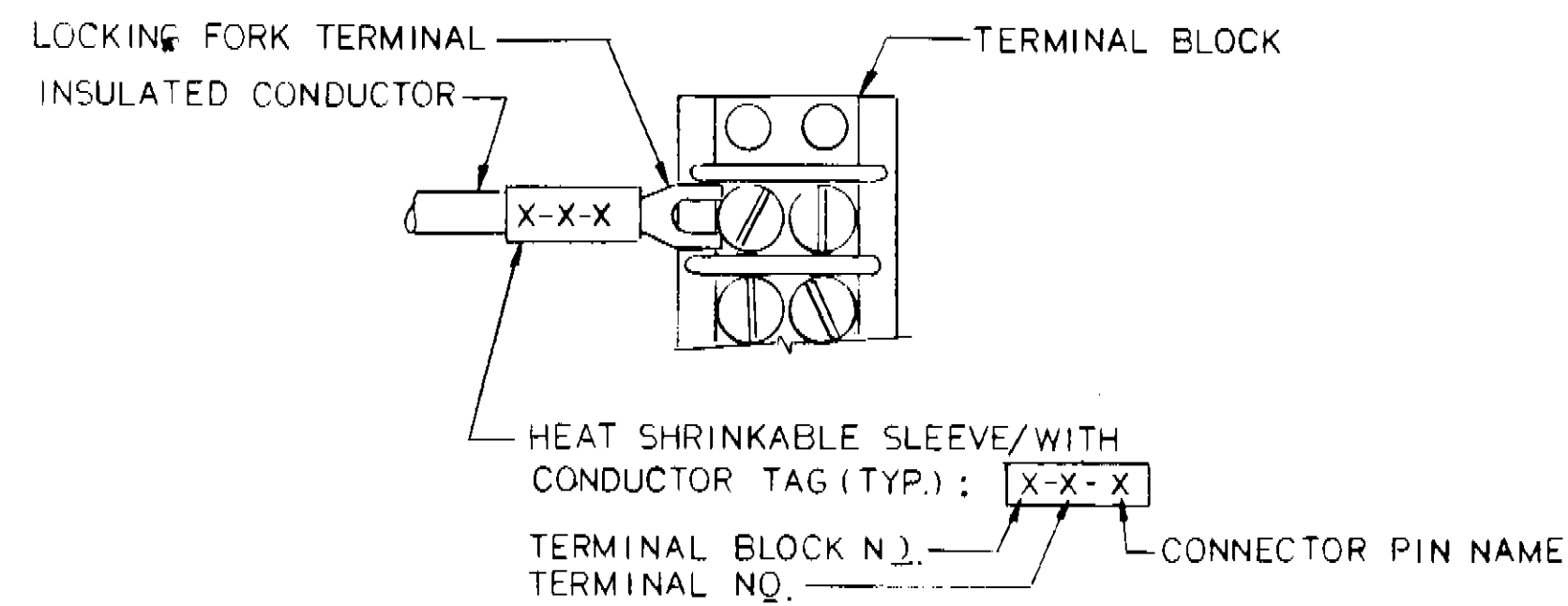
INTERFACE CABLE AND CONNECTOR ASSEMBLY
N.T.S.



DETAIL
N.T.S.

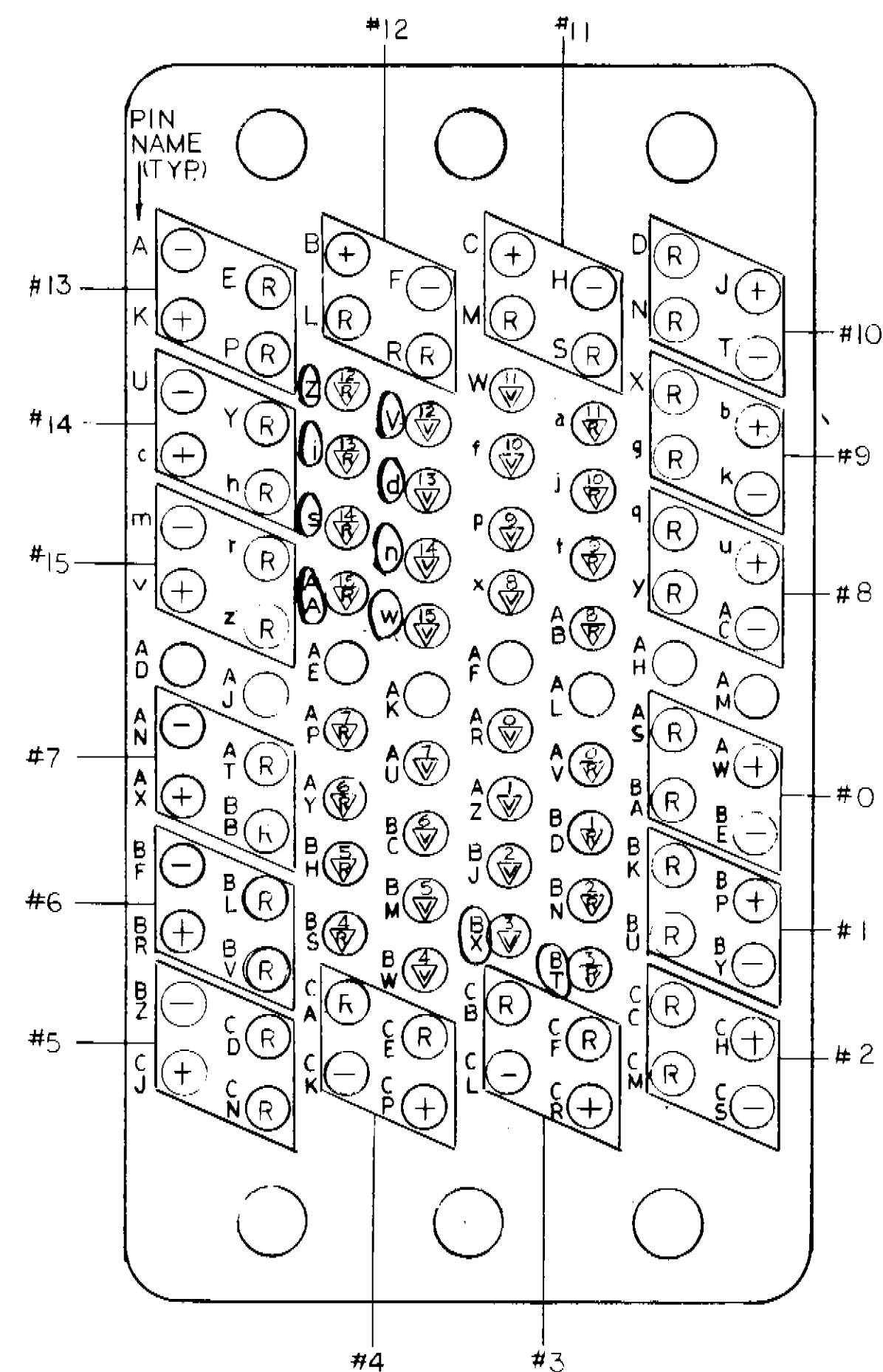


CONTACT - CRIMP TYPE
N.T.S.

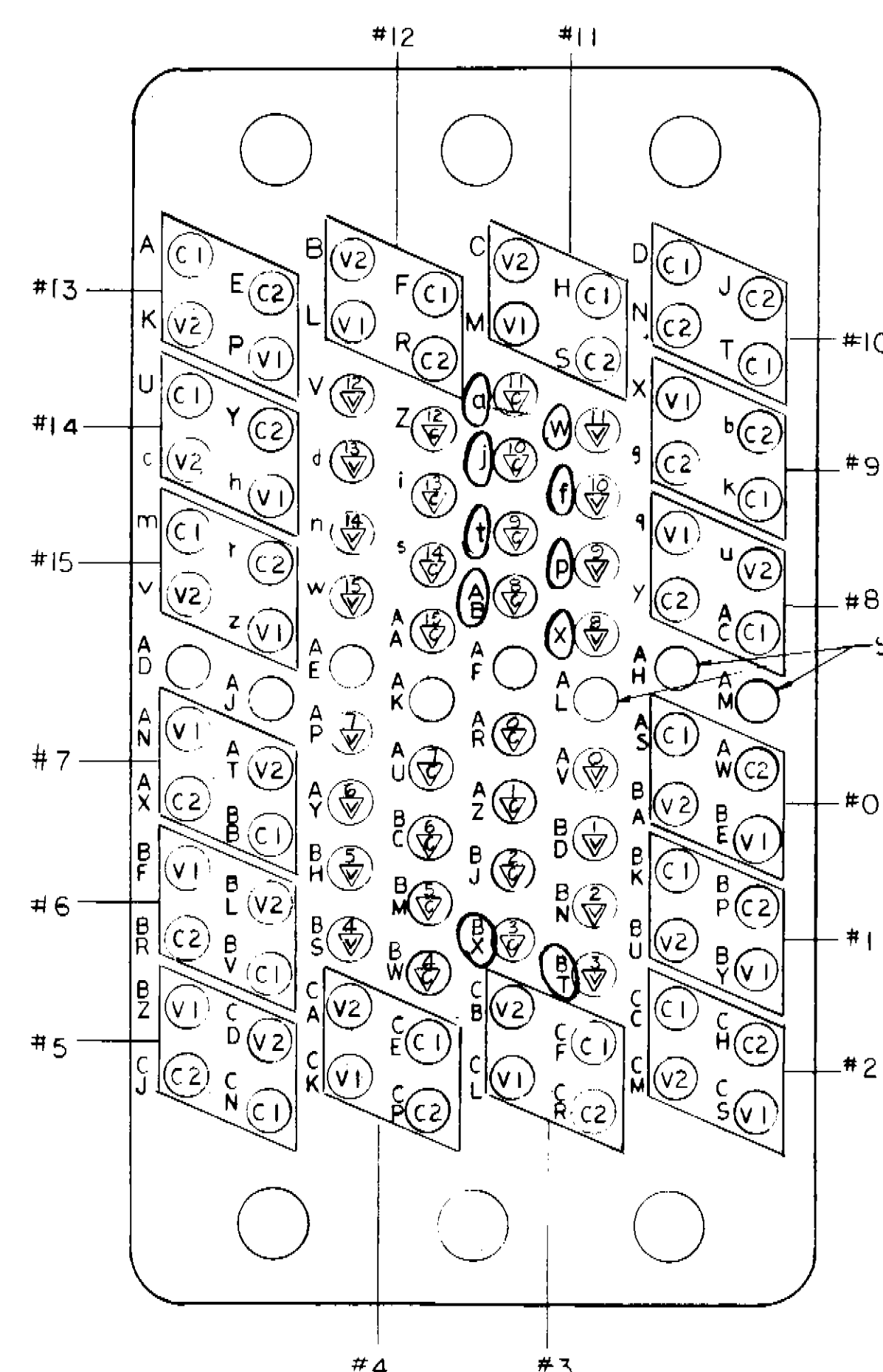


TYPICAL INTERFACE CABLE INDIVIDUAL PAIR CONDUCTOR TERMINATION AT TERMINAL CABINET
N.T.S.

▽ = SHIELD CONNECTION
= CHANNEL NUMBER
⊖ = -28V } TWISTED PAIR
⊕ = +28V } TWISTED PAIR
Ⓡ = ±28V RTN } TWISTED PAIR
Ⓢ = ±28V RTN } TWISTED PAIR

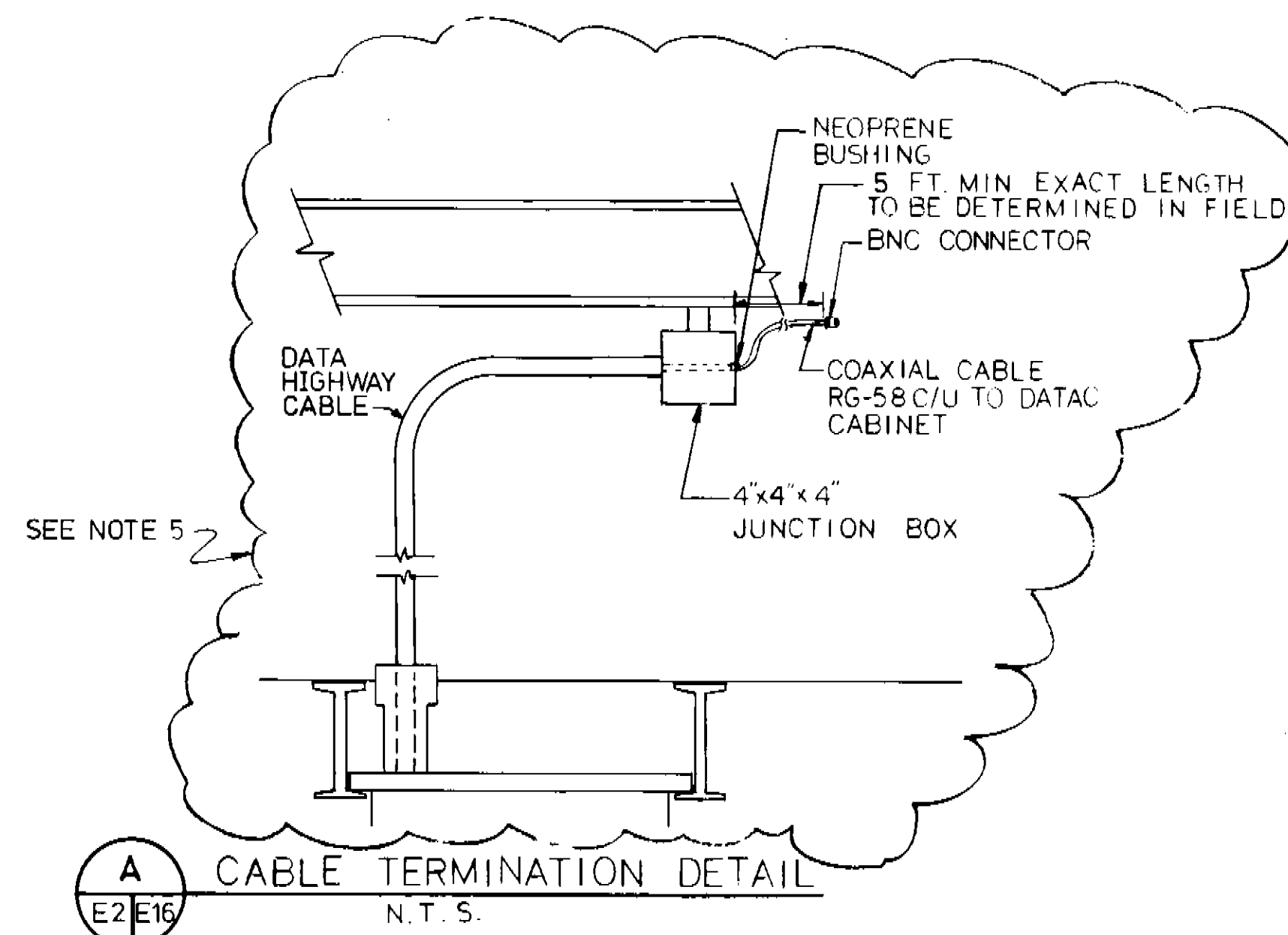


PIA CABLE CONNECTOR
REAR VIEW



SIA CABLE CONNECTOR
REAR VIEW

INSULATOR
N.T.S.



CABLE TERMINATION DETAIL
N.T.S.

DATA DATA HIGHWAY CABLE TERMINATION

REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED

NOTES

1. THE GOVERNMENT WILL PROVIDE THE RECTANGULAR INTERFACE CABLE CONNECTORS AND REMOVABLE CONTACT PINS MANUFACTURED BY POSITRONIC INDUSTRIES, INC. THE CONNECTOR AND TOOLS FOR INSTALLATION WILL BE AS FOLLOWS:

ITEMS	PART NO.
CONNECTOR ASSY.	GMCT104M0000V0
CONTACT PIN	MC 120N
HAND CRIMP TOOL/ WITH POSITIONER	9501/ WITH 9502 -1
CONTACT INSERT	9099
CONTACT EXTRACTION	9081

2. PIA AND SIA CABLE CONNECTORS ARE TYPICAL FOR ALL INTERFACE CABLES. FOR INTERFACE CABLE WIRING DIAGRAMS, COLOR CODING, DATA CHANNELS AND SENSOR CABLE IDENTIFICATIONS SEE DWG E-4 THRU E-9

3. INTERFACE CABLE SPARE AND UNUSED PAIRS SHALL BE WIRED TO DATA CABINET CONNECTOR CONTACT PINS AND SHALL BE TAPED, LABELED AND SECURED IN THE WIRING DUCT OF THE TERMINAL CABINET. LENGTH OF UNUSED PAIRS SHALL BE SUFFICIENT TO REACH ANY TERMINAL BLOCK FOR FUTURE USE.

4. A CLEAR, THIN WALL SLEEVE SHALL BE INSTALLED OVER EACH BARE SHIELD/ DRAIN WIRE

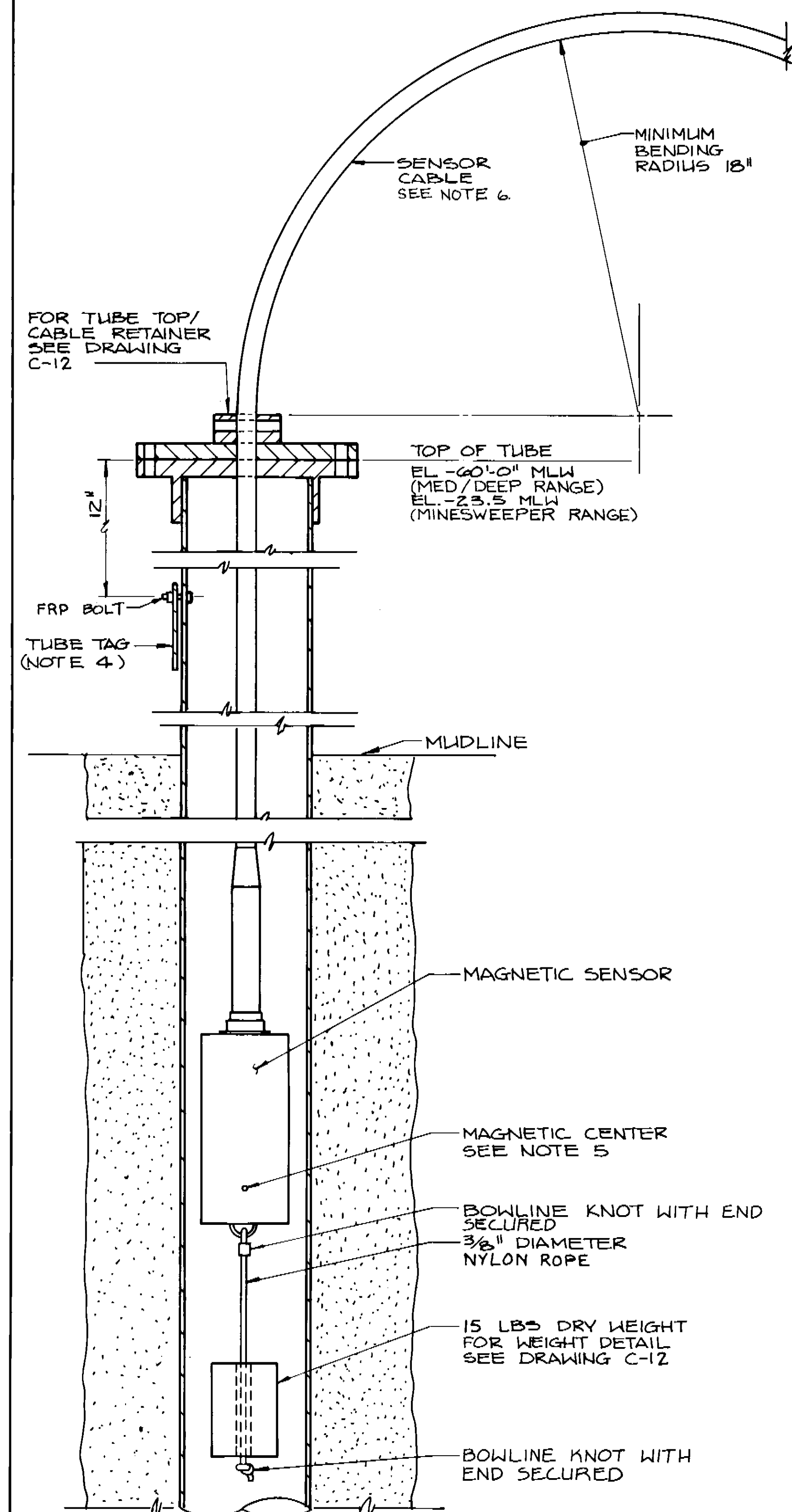
5. ALL WORK SHOWN ON THIS DRAWING IS INCLUDED IN THIS CONSTRUCTION CONTRACT. ALL WORK SHOWN CLOUDED IS A REVISION BY BLACK AND VEATCH.



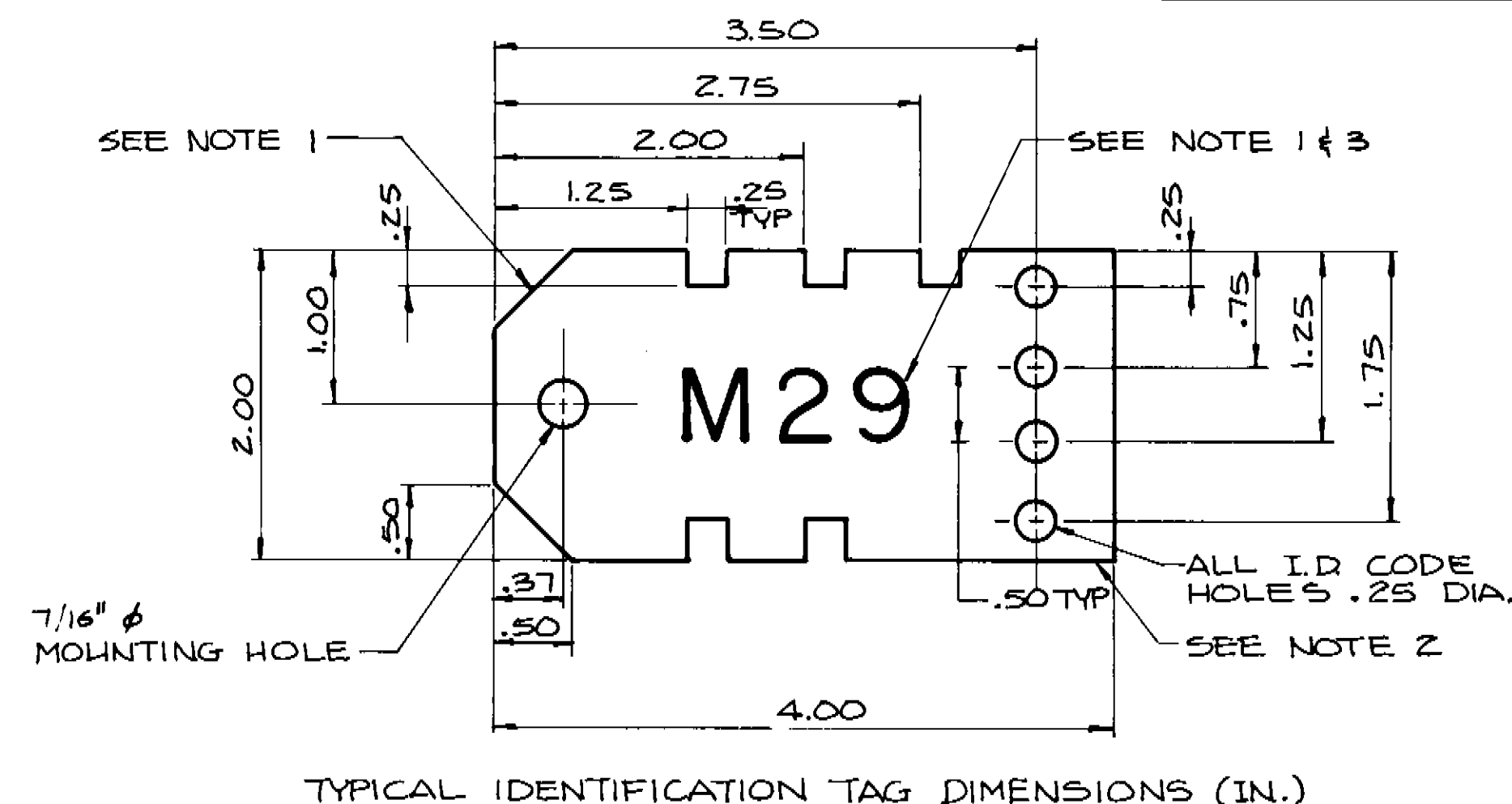
LEGEND

PIA — POWER INTERFACE ASSEMBLY
SIA — SIGNAL INTERFACE ASSEMBLY

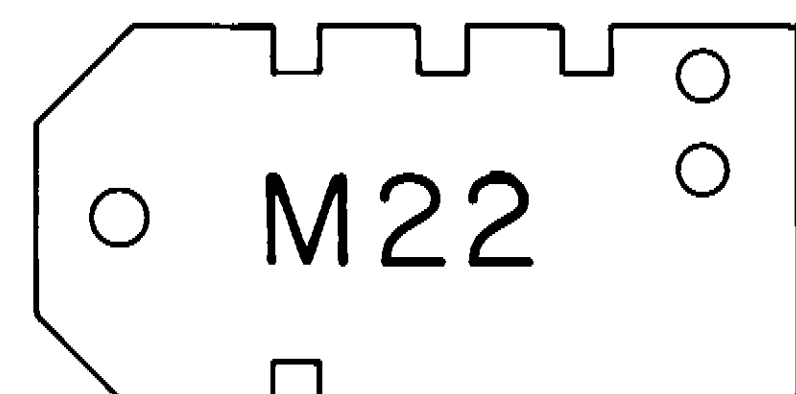
Frederic R. Harris, Inc. 300 East 42nd Street New York, NY 10017		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DESIGNED BY: H.S. DR. W.W. CHK. M.K. SUBMITTED BY: E. SCHWABER DATE: 1/15/80 TITLE: PROGRAM MGR. OCEAN ENGR. & CONST. PROJ. OFF. FPO-1 EIC: J. J. RVD 1/15/80		INTERFACE CABLE ASSEMBLY AND DATA HIGHWAY CABLE TERMINATIONS	
SATISFACTORY TO: ROICC DATE: 1/15/80 COMNAVSEASYS COM		APPROVED HEAD FPO-1: DATE: 1/15/80 APPROVED: DATE: 1/15/80 FFD FOR COMMANDER, NAVFAC	
SIZE: D CODE IDENT. NO.: 80091		NAVFAC DRAWING NO.: 3020709 CONSTR. CONTR. NO.: N62477-89-C-0053 SCALE: NONE SHEET: 36 OF 40	



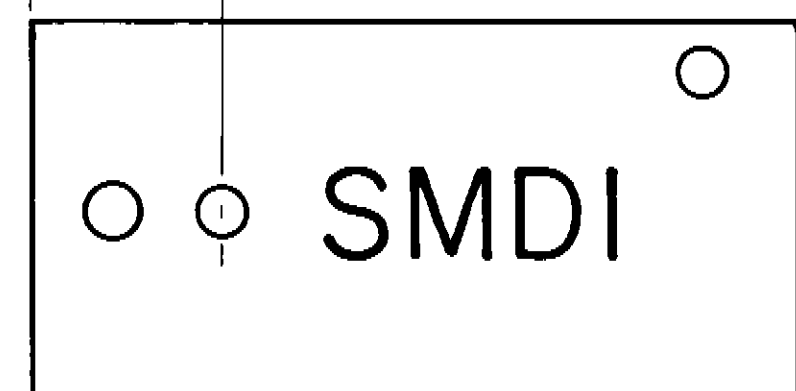
DETAIL
E-17/E-17



TYPICAL IDENTIFICATION TAG DIMENSIONS (IN.)

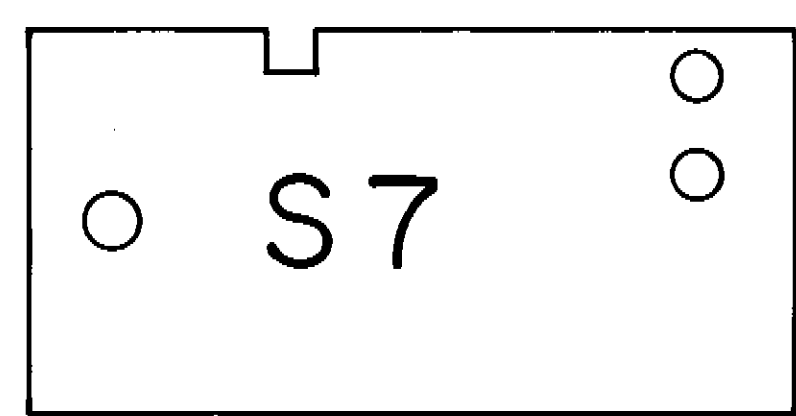


TYPICAL M-TUBE CABLE & SHIELD TAG

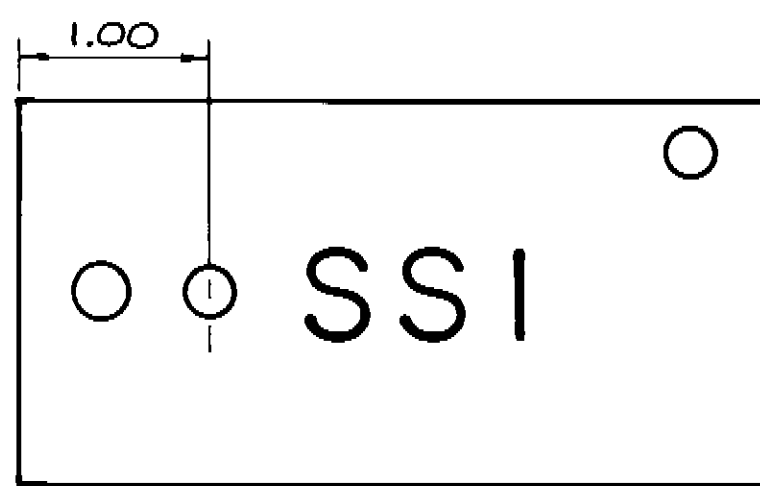


TYPICAL SPARE VERTICAL CABLE TAG

**MEDIUM / DEEP SENSOR RANGE
TUBE & CABLE IDENTIFICATION TAGS**



TYPICAL S-TUBE CABLE & SHIELD TAG



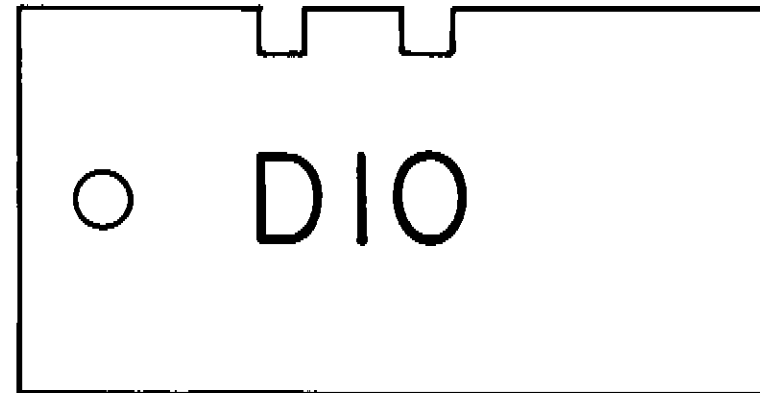
TYPICAL SPARE VERTICAL CABLE TAG

**MINESWEEPER (SHALLOW) SENSOR RANGE
TUBE'S CABLE IDENTIFICATION TAGS**
ACTUAL SIZE SHOWN

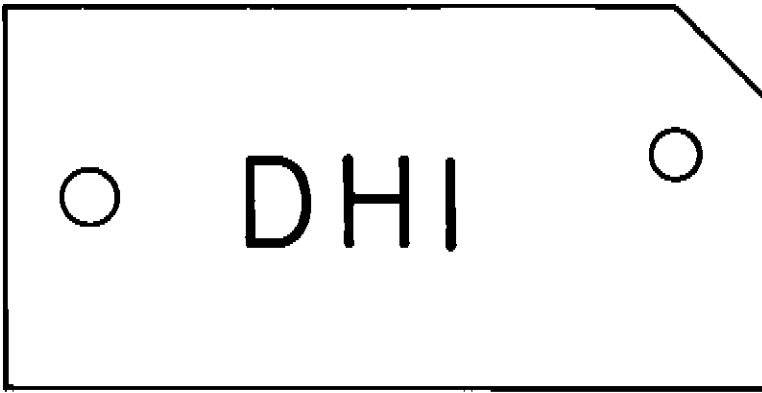
REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED
A	REVISED TO REFLECT AS BUILT CONDITION		

NOTES

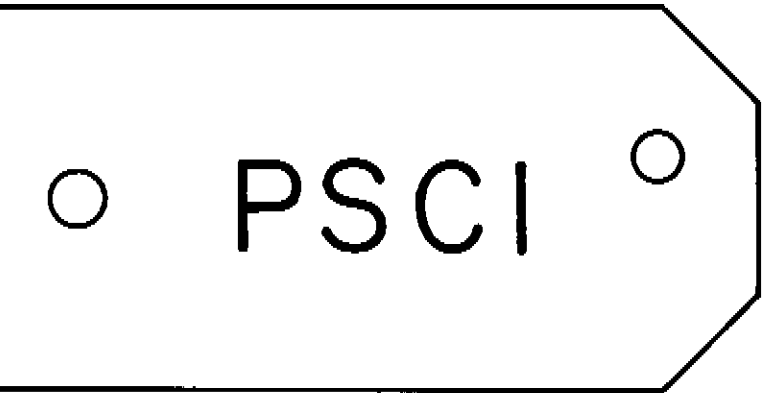
- M-TUBE AND CABLE TAGS SHALL BE CODED M1-M31 (DIMENSIONED M29 TAG SHOWN TO ILLUSTRATE ALL TAG IDENTIFICATION CODE FEATURES) WITH 45° CHAMFERS AT THE TOP AND BOTTOM LEFT CORNERS. D+5 TUBES AND CABLE TAGS SHALL BE CODED D1-D21 & S1-S21 AND SHALL BE RECTANGULAR IN SHAPE. SPARE VERTICAL SENSOR CABLE TAGS SHALL ALSO BE RECTANGULAR IN SHAPE. T-TUBE AND CABLE TAGS SHALL BE CODED T1-T5 WITH A 45° CHAMFER AT THE TOP LEFT CORNER ONLY. SPARE TRIAXIAL SENSOR CABLE TAGS SHALL ALSO BE CODED WITH A 45° CHAMFER AT THE TOP LEFT CORNER.
- ALL TUBE TAGS SHALL HAVE THE FOLLOWING PHYSICAL IDENTIFICATION CODE:
EACH .25 INCH BY .25 INCH NOTCH EQUALS FIVE UNITS.
EACH .25 INCH DIAMETER HOLE EQUALS ONE UNIT.
EACH TAG NUMBER PHYSICAL CODE SHALL USE THE MAXIMUM NUMBER OF NOTCHES WITH THE REMAINDER OF UNITS AS HOLES.
- TUBE AND CABLE TAGS SHALL BE .1875 INCHES IN THICKNESS.
- TUBE AND CABLE TAGS SHALL BE CONSTRUCTED OF WHITE POLYVINYL CHLORIDE (PVC) WITH CHARACTERS CARVED INTO THE SURFACE .0625 INCHES DEEP.
- SEE TUBE TAG CONNECTION ON DWG. C-12 AND ON DETAIL 1 THIS SHEET.
- MAGNETIC CENTER SHALL BE ELEV. -64 FEET MLW FOR TRIAXIAL (T1-T5) AND VERTICAL SENSORS (M1-M31) MAGNETIC CENTER FOR VERTICAL SENSORS (D1-D21) SHALL BE ELEV. -85 FEET MLW MAGNETIC CENTER FOR MINESWEEPER SENSORS S1-S21 SHALL BE ELEV. -30 FEET MLW. FOR SENSOR MAGNETIC CENTER SEE DWG. C-11.
- USE NYLON CABLE TIES AS REQUIRED TO FORM CABLE IN COIL AT MUDLINE.
- INSTALL CABLE IDENTIFICATION TAGS FOR EACH SENSOR CABLE AT THE SENSOR AND WITHIN 10 FEET OF THE UNDERWATER ENTRANCE TO THE CONDUIT CASING AT THE DCS. INSTALL CABLE IDENTIFICATION WITH HEAT SHRINKABLE SLEEVE INSIDE THE DCS WITHIN 6 FEET OF THE CONDUIT CASING ENTRANCE.



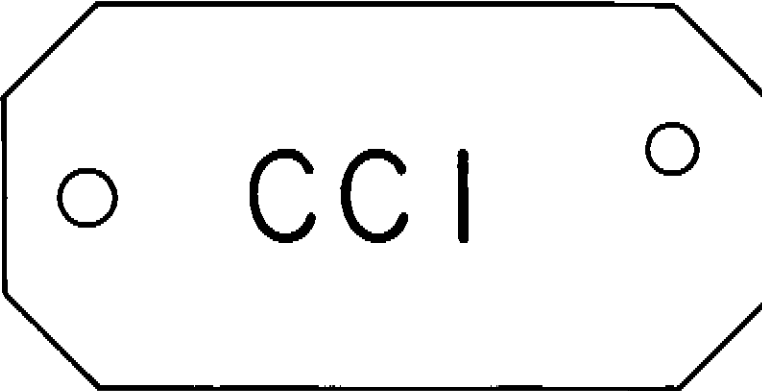
TYPICAL D-TUBE CABLE & SHIELD TAG



DATA HIGHWAY CABLE TAG
(FOR DHI AND DHZ CABLES)

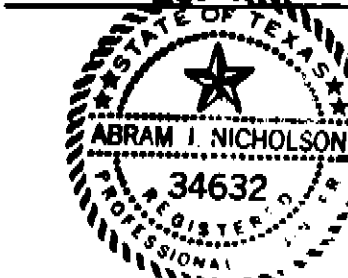


COMPOSITE CABLE TAG
(ONE CABLE)



CCTV CABLE TAG
(ONE CABLE)

CABLE IDENTIFICATION TAGS (CON'T)



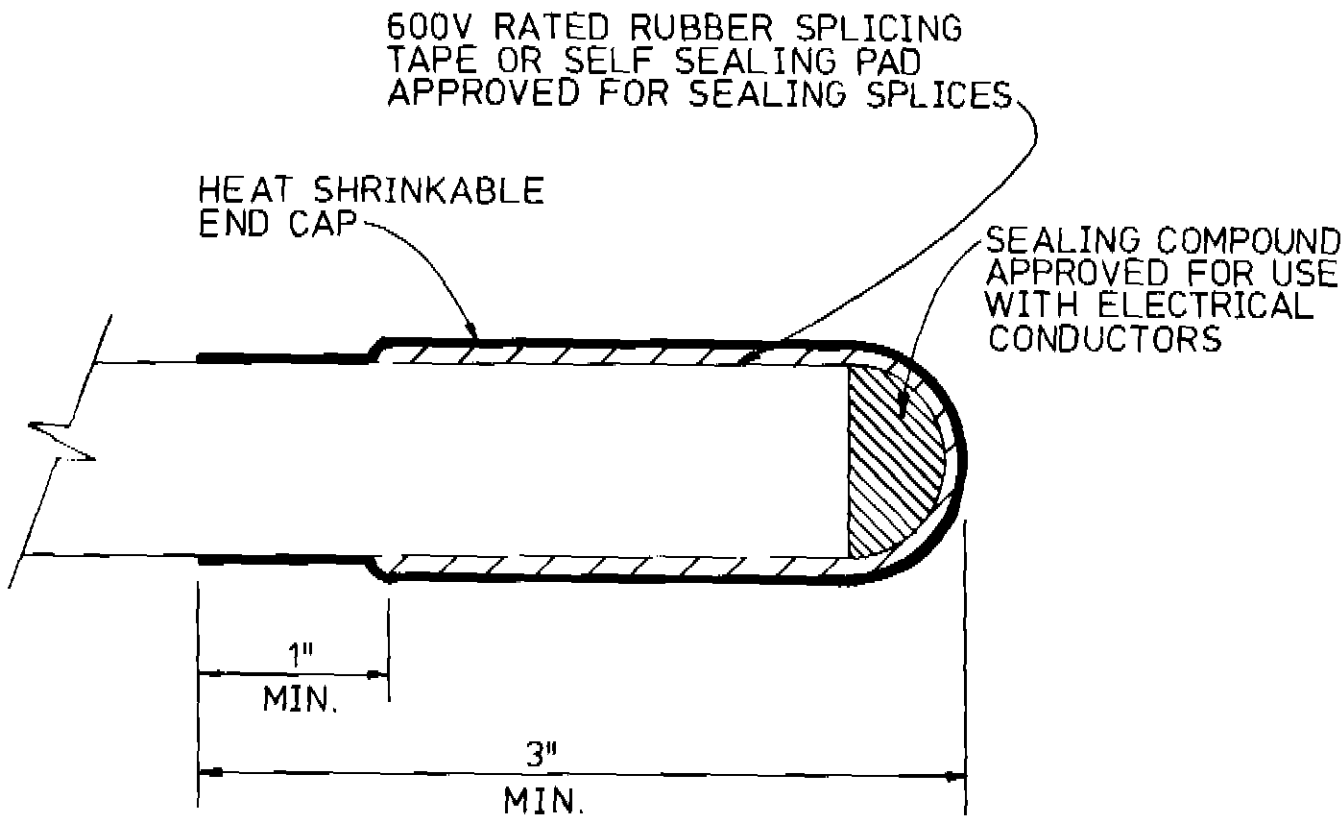
Black & Veatch Engineers-Architects		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DES ABV IDR LT CHK HRA SUBMITTED BY <i>Abraham J. Nicholson</i> DATE <i>5-15-81</i> TITLE: PROGRAM MGR OCEAN ENGR. & CONST. PROJ. OFF FPO-1 EIC <i>age</i> <i>P</i> RVD <i>10</i> <i>5/15/81</i>		MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION TUBE & CABLE TAGS AND SENSOR INSTALLATION DETAILS	
SATISFACTORY TO <i>27/10/81</i> DATE <i>5-15-81</i> ROICC <i>age</i>	APPROVED HEAD FPO-1 DATE <i>5-15-81</i> APPROVED <i>age</i> DATE <i>5/15/81</i> EFD FOR COMMANDER, NAVFAC	SIZE CODE IDENT. NO. NAVFAC DRAWING NO. D 80091 3020710	E-17
COMNAVSEASCOM DATE		CONSTR. CONTR. NO. N62477-89-C-0053 SCALE: NONE SPEC. 21-89-0053 SHEET 37 OF 40	

REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED
A	REVISED TO REFLECT AS BUILT CONDITION		

CABLE CUT LENGTHS

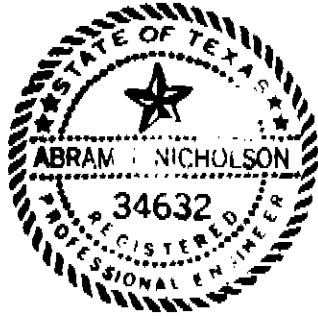
SENSOR	MEASURED LENGTH	CONTINGENCY LENGTH	SERVICE LOOP	CUT LENGTH	SENSOR	MEASURED LENGTH	CONTINGENCY LENGTH	SERVICE LOOP	CUT LENGTH
M-1	2270'	205'	75'	2550'					
M-2	2290'	205'		2570'					
M-3	2310'	215'		2600'					
M-4	2330'	215'		2620'					
M-5	2350'	215'		2640'					
M-6	2370'	225'		2670'					
M-7	2390'	225'		2690'	S-1	1085'	300'	45'	1430'
M-8	2410'	225'		2710'	S-2	1090'	300'		1435'
M-9	2430'	225'		2730'	S-3	1100'	300'		1445'
M-10	2450'	235'		2760'	S-4	1105'	300'		1450'
M-11	2470'	235'		2780'	S-5	1110'	305'		1460'
M-12	2490'	235'		2800'	S-6	1120'	305'		1470'
M-13	2510'	245'		2830'	S-7	1125'	305'		1475'
M-14	2530'	245'		2850'	S-8	1135'	305'		1485'
M-15	2550'	245'		2870'	S-9	1140'	310'		1495'
M-16	2570'	255'		2900'	S-10	1145'	310'		1500'
M-17	2590'	255'		2920'	S-11	1155'	310'		1510'
M-18	2610'	255'		2940'	S-12	1160'	310'		1515'
M-19	2630'	255'		2960'	S-13	1170'	310'		1525'
M-20	2650'	265'		2990'	S-14	1175'	315'		1535'
M-21	2670'	265'		3010'	S-15	1180'	315'		1540'
M-22	2690'	265'		3030'	S-16	1190'	315'		1550'
M-23	2710'	275'		3060'	S-17	1195'	315'		1555'
M-24	2730'	275'		3080'	S-18	1205'	315'		1565'
M-25	2750'	275'		3100'	S-19	1210'	320'		1575'
M-26	2770'	285'		3130'	S-20	1215'	320'		1580'
M-27	2790'	285'		3150'	S-21	1225'	320'		1590'
M-28	2810'	285'		3170'					
M-29	2830'	285'		3190'					
M-30	2850'	295'		3220'					
M-31	2870'	295'		3240'					
D-1	2285'	230'	55'	2570'	SMD-1	2375'	250'	55'	2680'
D-2	2315'	240'		2610'	SMD-2	2495'	270'		2820'
D-3	2345'	240'		2640'	SMD-3	2615'	280'		2950'
D-4	2375'	250'		2680'	SMD-4	2735'	300'		3090'
D-5	2405'	250'		2710'	SMD-5	2855'	320'		3230'
D-6	2435'	260'		2750'	SMD-6	2885'	330'		3270'
D-7	2465'	260'		2780'				45'	1510'
D-8	2495'	270'		2820'	SS-1	1155'	310'		1590'
D-9	2525'	270'		2850'	SS-2	1225'	320'	45'	
D-10	2555'	270'		2880'					
D-11	2585'	280'		2920'					
D-12	2615'	280'		2950'					
D-13	2645'	290'		2990'					
D-14	2675'	290'		3020'					
D-15	2705'	300'		3060'					
D-16	2735'	300'		3090'					
D-17	2765'	310'		3130'					
D-18	2795'	310'		3160'					
D-19	2825'	310'		3190'					
D-20	2855'	320'		3230'					
D-21	2885'	330'		3270'					

CABLE	CABLE CUT LENGTH
DH1	6900'
DH2	6900'
CCTV	6900'

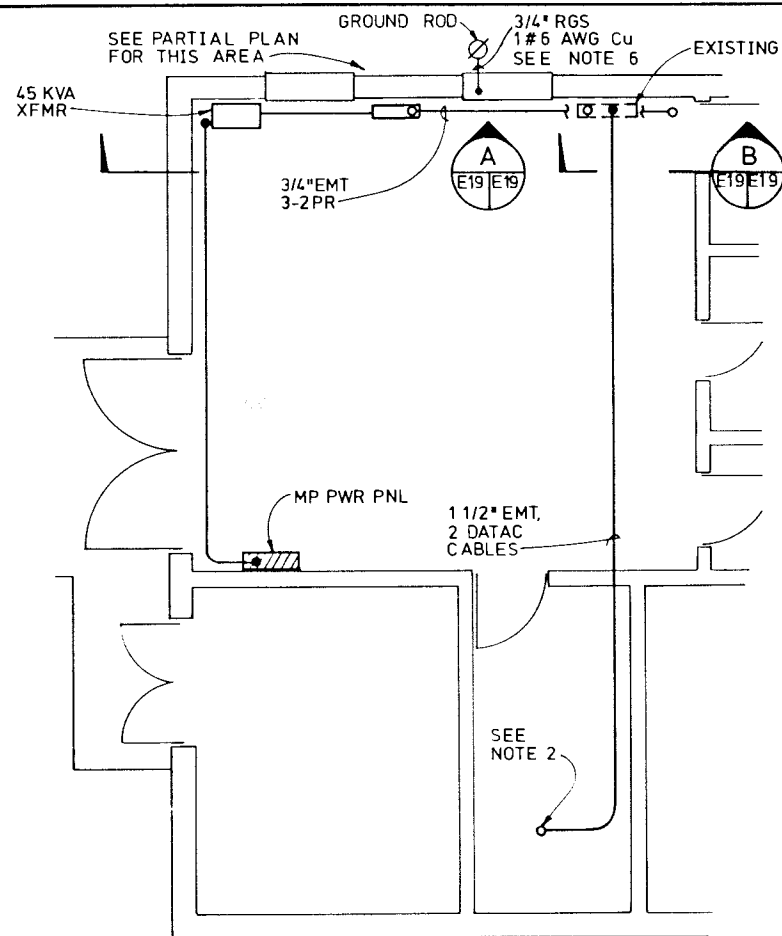


CABLE END SEAL
N.T.S.
SEE NOTES: TYPICAL ALL
SUBMERSIBLE CABLE INSTALLATIONS

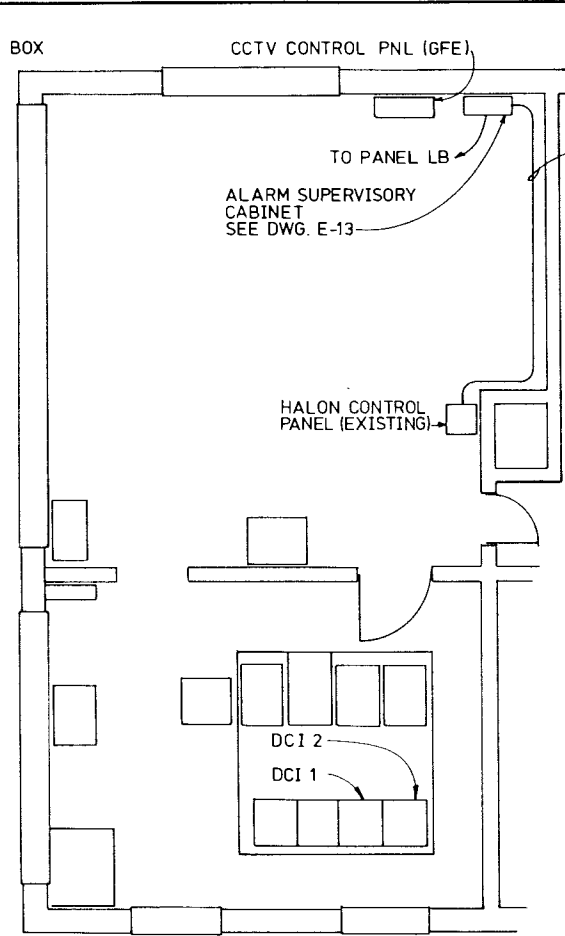
- NOTES:
- CABLE SHALL BE CUT AND CABLE END SEAL SHALL BE INSTALLED BEFORE END OF CABLE IS EMERSED IN WATER.
 - REFER TO SPECIFICATION SECTION 16680 FOR REQUIRED TEST SAMPLE APPROVAL BEFORE INSTALLATION IS PERMITTED.



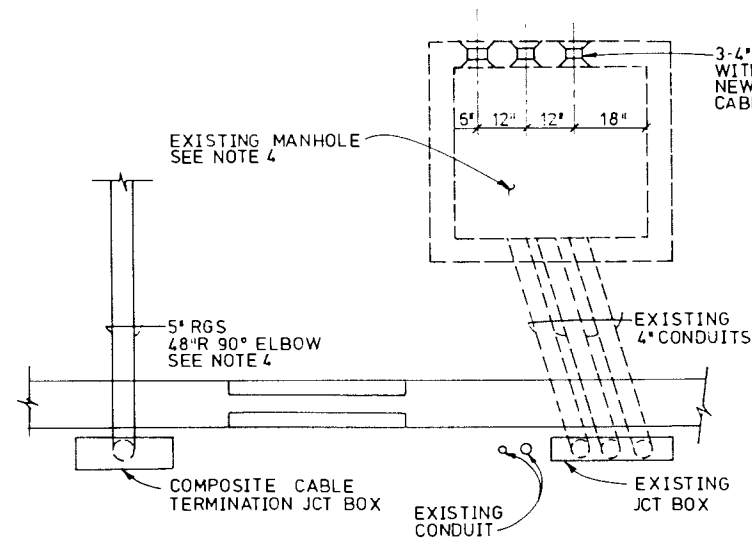
Black & Veatch Engineers-Architects		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DESIGNED BY: [Signature] DATE: 8/15/89 SUBMITTED BY: [Signature] DATE: 8/15/89 TITLE: PROGRAM MGR. OCEAN ENGR. B. CONST. PROJ. OFF. FPO-1 EIC: [Signature] RVD: [Signature] 8/15/89		MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION SENSOR CABLE LENGTHS	
SATISFACTORY TO: [Signature] DATE: 8/15/89 ROICC: [Signature] DATE: 8/15/89 COMNAVSEASYS/COM DATE: 8/15/89		APPROVED HEAD FPO-1 DATE: 8/15/89 APPROVED: [Signature] DATE: 8/15/89 EFD FOR COMMANDER, NAVFAC	
SIZE: D	CODE: 80091	IDENT. NO.: 3020711	E-18
SCALE: NONE		CONSTR. CONTR. NO. N62477-89-C-0053 SPEC. 21-89-0053 SHEET 38 OF 40	



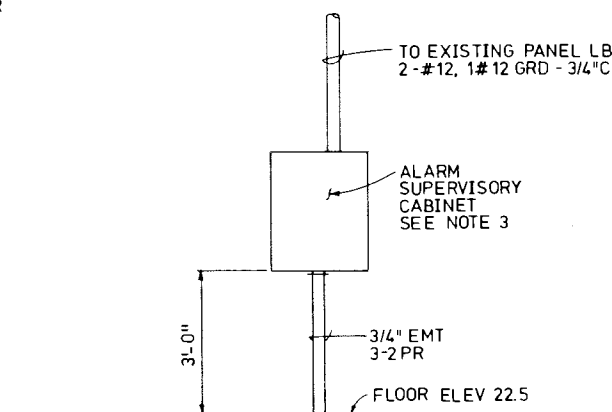
FIRST FLOOR PARTIAL PLAN
ELEV 12.5
SCALE: 1/4"=1'-0"



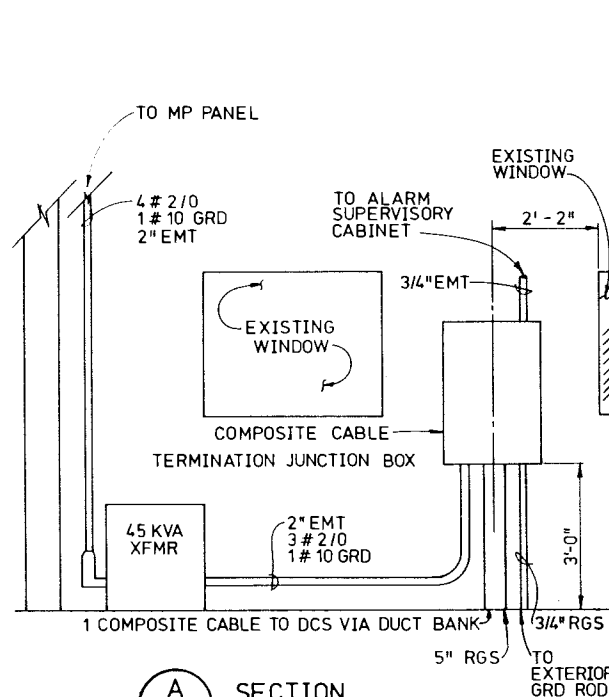
SECOND FLOOR PARTIAL PLAN
ELEV 22.5
SCALE: 1/4"=1'-0"



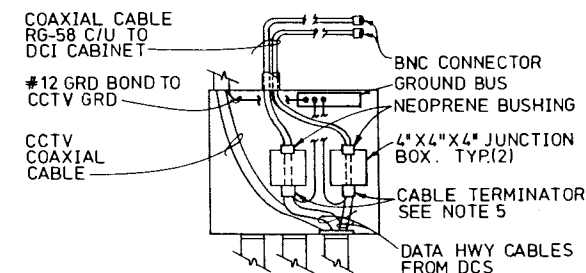
PARTIAL PLAN
SCALE: 1/2"=1'-0"



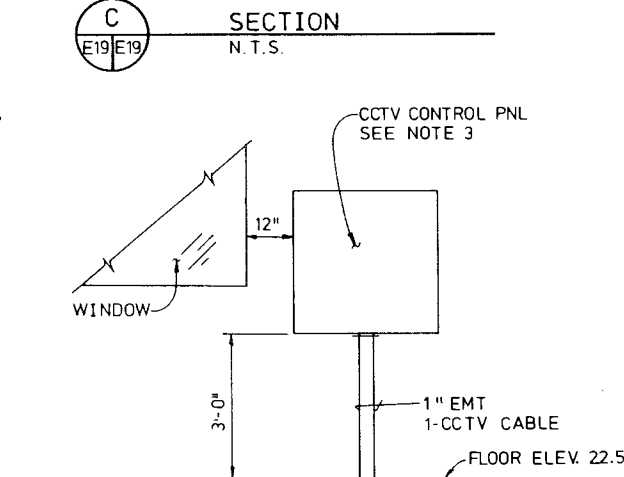
ALARM SUPERVISORY CABINET MOUNTING DETAIL
SCALE: 1/2"=1'-0"



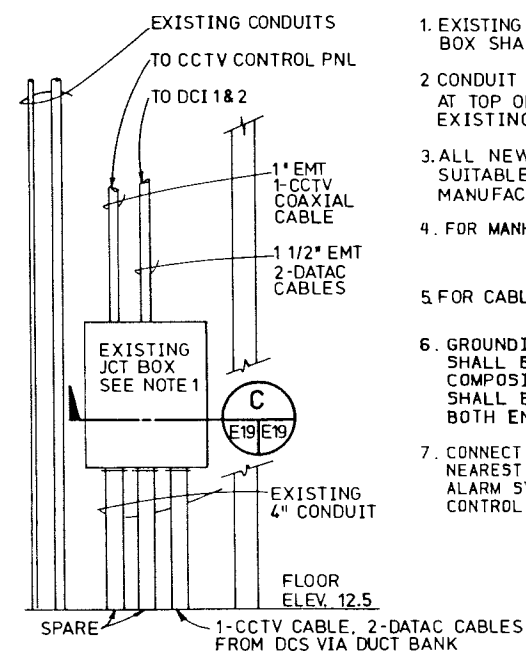
SECTION A
E-19/E-19 SCALE: 1/2"=1'-0"



SECTION B
E-19/E-19 SCALE: 1/2"=1'-0"



SECTION C
N.T.S.



SECTION D
E-19/E-19 SCALE: 1/2"=1'-0"

REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED

NOTES:

- EXISTING TERMINAL STRIPS AND EQUIPMENT IN JUNCTION BOX SHALL BE REMOVED.
- CONDUIT TO DCI 1 AND 2 EQUIPMENT SHALL TERMINATE AT TOP OF STRUCTURAL FLOOR ELEVATION 22.5, AT EXISTING SPARE FLOOR PENETRATION.
- ALL NEW PANELS SHALL BE MOUNTED TO WALLS WITH SUITABLE SIZE FRAMING CHANNEL USING PANEL MANUFACTURERS MOUNTING PROVISIONS.
- FOR MANHOLE LOCATIONS AND DETAILS SEE DRAWING C-8.
- FOR CABLE TERMINATOR DETAIL SEE DRAWING E-16, DETAIL 4.
- GROUNDING ELECTRODE AND GROUNDING ELECTRODE CONDUCTOR SHALL BE PROVIDED FOR SURGE ARRESTOR LOCATED IN THE COMPOSITE CABLE TERMINATION JUNCTION BOX. 3/4" RGS SHALL BE BONDED TO GROUNDING ELECTRODE CONDUCTOR AT BOTH ENDS BY MEANS OF THREADED GROUNDING BUSHINGS.
- CONNECT ALARM SUPERVISORY CABINET ALARM OUTPUT AT NEAREST AVAILABLE POINT TO BUILDING SINGLE ZONE ALARM SYSTEM. VERIFY LOCATION OF EXISTING HALON CONTROL PANEL.

LEGEND:

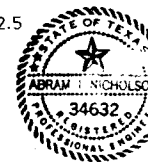
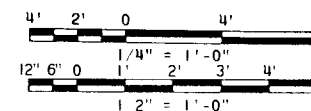
- CONDUIT TURNING UP
- CONDUIT TURNING DOWN
- ▨ POWER PANEL

ABBREVIATIONS:

- DCI DATAC CONCENTRATOR INTERFACE
- EMT ELECTRICAL METALLIC TUBING (CONDUIT)
- JCT JUNCTION
- PNL PANEL
- RGS RIGID GALVANIZED STEEL (CONDUIT)
- XFMR TRANSFORMER

GRAPHIC SCALE

WARNING: DRAWING MAY BECOME DISTORTED DURING REDUCTION OR OTHER REPRODUCTION PROCESS. USER IS CAUTIONED AGAINST SCALING.

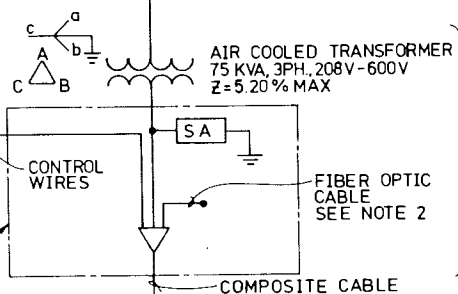
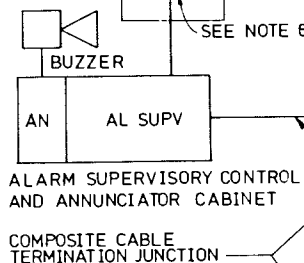


Black & Veatch Engineers-Architects		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DESIGNED BY: <i>John J. Hurler</i> SUBMITTED BY: <i>John J. Hurler</i> DATE: <i>2/15/89</i> TITLE: PROGRAM MOD OCEAN ENGR. B CONST PROJ OFF FPO-1 EIC: <i>John J. Hurler</i> RVD: <i>John J. Hurler</i>	CHECKED BY: <i>John J. Hurler</i> DATE: <i>2/15/89</i> APPROVED BY: <i>John J. Hurler</i> DATE: <i>2/15/89</i> APPROVED FOR COMMANDER, NAVFAC	SIZE: D CODE: 80091 IDENT NO.: 3020712 NAVFAC DRAWING NO.: 3020712 CONSTR. CONTR. NO.: N62477-89-C-0053	SHEET 39 OF 40

SATISFACTORY TO	DATE
ROICC	DATE
SATISFACTORY TO	DATE
COMNAVSEASYS COM	DATE

SUPERVISORY POINTS				
ALARMS	STATUS INDIC	CONTROL		
A1 DCS TEMP ABNORMAL	S1	DATA C PWR ON	C1	DATA C PWR ON
A2 DCS HUMIDITY ABNORMAL	S2	DATA C PWR OFF	C2	DATA C PWR OFF
A3 DATA SHED HALON DISCH				
A3 DATA SHED HALON SYS. TROUBLE				

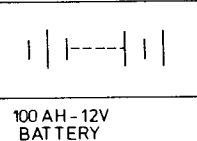
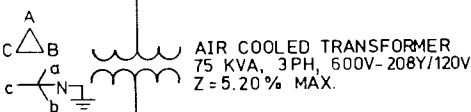
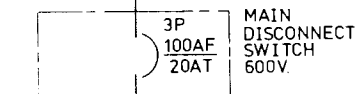
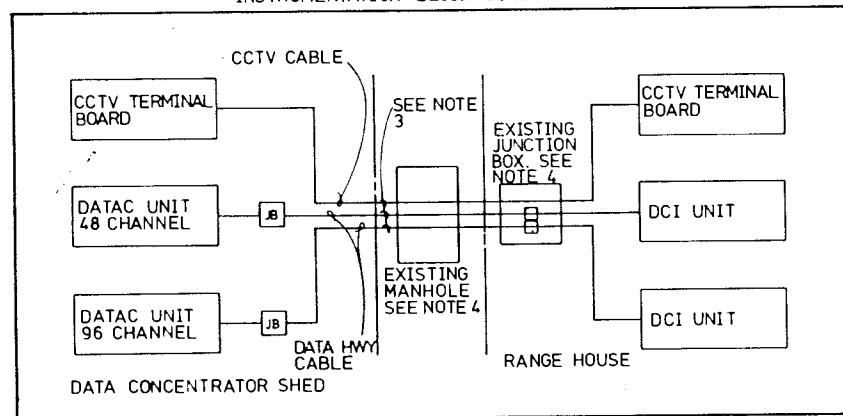
EXISTING PANEL LB



RANGE HOUSE
SEE NOTE 5

NAVIGATION LIGHT

INSTRUMENTATION BLOCK DIAGRAM



BATTERY CHARGER

BATTERY CHARGER
DISCONNECT SWITCH
1P 20A, 120VAC
30AMP RATED

FIRE PROTECTION
DISCONNECT SWITCH
120VAC 30A RATED
WITH 2A FUSE

POWER PANEL - PP1
208/120V - 3PH - 4W

SEE NOTE 1

3P
60A

1P
20A

1P
30A

1P
20A

1P
20A

1P
15A

SPARE

DCS
INTERIOR
DCS
EXTERIOR

CCTV
TERMINAL
BOARD

HUMIDIFIER
PUMP

SUPPLIED
WITH AHU UNIT

EXTERIOR LIGHTING

CCTV CAMERA

AHU CONTROL
PANEL

HU TH

EQPT FLOOR

CABLE ENTRY FLOOR

EL

EMERGENCY LIGHT UNIT
(WITH INTEGRAL BATTERY)

1P 20A
AC ONLY SWITCH

ALARM SUPERVISORY
MECH HELD SWITCH

CEILING LIGHTING
FIXTURES

120V N.C. CONTACT
IN FIRE PROTECTION CAB

1P 20A
A/C ONLY SWITCH

PC

ALARM DEVICES

POWER CONDITIONER

ALARM SUPERVISORY
CABINET

HU TH

FIRE PROTECTION
CABINET

HCV

ZONE 1 THRU 6

20A 20A 20A 20A 20A 20A

CONVENIENCE
OUTLETS IN
DATA CABS
FOR TEST
EQUIPMENT

DATA C PWR
RECEPTACLES

POWER PANEL - PP2
120V, 1PH, 3W

REVISIONS			
SYM	DESCRIPTION	DATE	APPROVED

PANEL ID PP1	AMPS 100	VOLTS 208/120 PH 3 WIRE 4	FED FROM: RANGE HOUSE THRU 600V 208/120V XFMR
LOCATION SEE DWG E-1	NEUT 100%		
CIRCUIT DESCRIPTION	LOAD/VA		CIRCUIT DESCRIPTION
AIR HANDLER	4244	1	
AIR HANDLER	4244	3	
LIGHTS AND EXH FAN	862	5	
		7	
		9	
CONVENIENCE RECEPT	540	11	
		13	
		15	
EXTERIOR LIGHTING	200	17	
		19	
SPARE		21	
		23	
SPARE		25	
		27	
		29	
SEE NOTE 1	4244 4244 1602	TOTAL LOAD PER PH	2530
		TOTAL CONNECTED LOAD	* 12620VA

* ESTIMATE

PANEL ID PP2	LOCATION: SEE DWG E-1	TYPE: SURFACE MTD
ISOLATED NEUTRAL BUS	VOLTS 120V PH 1 WIRE 3	FED FROM: PANEL PP1
ISOLATED GROUND BUS	AMPS 100 NEUT 100%	THRU POWER CONDITIONER
CIRCUIT DESCRIPTION	LOAD VA	CIRCUIT DESCRIPTION
DATA C CONV OUTLETS	67	1
DATA C CONV OUTLETS	67	3
DATA C CONV OUTLETS	67	5
		7
		9
		11
		12
	201	TOTAL CONNECTED LOAD
		* 2360 VA

* ESTIMATE

NOTES:

- ALL CIRCUITS EXCEPT THOSE SUPPLYING THE AIR HANDLING UNIT SHALL BE PLACED ON PHASE "C" AS SHOWN.
- FIBER OPTICS CABLE IS FOR FUTURE USE. TERMINATE FIBER OPTICS CABLE IN COMPOSITE CABLE TERMINATION BOX AS SHOWN ON DRAWING E-45, DETAIL 1.
- UTILIZE 2 1/2" PVC CONDUITS ENTERING THE DCS THROUGH THE CONDUIT CASING NEAREST THE POINT OF TERMINATION.
- SEE DRAWING E-19 FOR RANGE HOUSE RACEWAY AND CONTROL EQUIPMENT DETAILS.
- 3 POLE 60 AMP BREAKER WITH LUGS SIZED FOR 270 CU SHALL BE INSTALLED IN EXISTING PANEL MP. PANEL MP IS GENERAL ELECTRIC TYPE CCB 400 AMP 3 PH 4 WIRE WITH TYPE TJJ AND TJK CIRCUIT BREAKERS.
- 1 POLE 20 AMP BREAKER SHALL BE INSTALLED IN EXISTING PANEL LB. PANEL LB IS G.E. TYPE NLB 100 AMP 3 PH 4 WIRE WITH TYPE TQB CIRCUIT BREAKERS.



Black & Veatch Engineers-Architects		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND CHESAPEAKE DIVISION WASHINGTON, D.C.	
DESIGNED BY: J. M. Scholten SUBMITTED BY: J. M. Scholten DATE: 5-15-89 TITLE: PROGRAM MGR. OCEAN ENGR. & CONST. PROJ. OFF. FPO-1 EIC: J. M. Scholten REV: 1/1/89		MAGNETIC SILENCING FACILITY NORFOLK NAVAL STATION POWER AND CONTROL ONE-LINE DIAGRAM	
SATISFACTORY TO: ROICC DATE: 5-15-89 COMNAVSEASYS/COM	APPROVED HEAD FPO-1: J. M. Scholten DATE: 5-15-89 APPROVED: J. M. Scholten DATE: 5-15-89 EFD FOR COMMANDER, NAVFAC	SIZE: D CODE: 80091 SCALE: NONE	NAVFAC DRAWING NO.: 3020713 CONSTR. CONTR. NO. N62477-89-C-0053 SHEET 40 OF 40